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L.E

# *Fighter*

*Fight For Your Dream*



**All What You Need**

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- DEPARTMENT DEPARTMENT BANK
- PREVIOUS EXAMS

# **ANATOMY & HISTOLOGY**

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## **BOOKLET**

**2026**



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Fighter

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## 1. Terminology

46 MCQs | 23 written questions

**1. Which term describes the movement of the distal end of the limb in a circle of 360 degree? (2026, KNU 2026)**

- A) Circumduction
- B) Adduction
- C) Rotation
- D) Flexion

**2. In which anatomical location are the deep plantar vessels found? (2026, KNU 2026)**

- A) In the palm of the hand away from the skin.
- B) In the sole of the foot near the skin
- C) In the palm of the hand near the skin.
- D) In the sole of the foot away from the skin

**3. To put the person in anatomical position, what is the correct position of the hand and the foot? (2026, KNU 2026)**

- A. The palm faces anterior and the arm is adducted
- B. The palm faces anterior and the arm is abducted
- C. The palm faces posterior and the arm is adducted
- D. The palm faces posterior and the arm is abducted

**4. Movement of the shoulder superiorly is termed which of the following? (2026, KNU 2026)**

- A) Protraction
- B) Elevation
- C) Depression
- D) Retraction

**5. Which of the following best describes the anatomical position of the lower limbs and toes? (2025)**

- A) Knees flexed, the toes turned inward
- B) Legs abducted, feet rotated outward
- C) Legs extended and parallel, feet flat with toes directed forward
- D) Knees flexed, feet together with toes pointing sideways

**6. What is the meaning of superficial dissection of the dorsal aspect of the neck? (2025)**

- A) Dissection of the skin and front of the neck
- B) Dissection of the skin and back of the neck
- C) Dissection of the muscles and vessels of the back of the neck
- D) Dissection of the muscles and vessels of the front of the neck

**7. Moving the distal segment away from the proximal segment is called which of the following? (2025)**

- A) Adduction
- B) Abduction
- C) Flexion
- D) Extension

**8. Which of the following is one of the characteristics of anatomical position? (2024, 2023)**

- A) The arms are abducted



- B) The palm face anterior
- C) The lower limbs are abducted
- D) The toes directed to the left.

**9. Which of the following anatomical planes divided the body into anterior and posterior parts? (2024, 2023)**

- A) Horizontal plane.
- B) Coronal plane
- C) Midsagittal plane
- D) Parasagittal plane

**10. Which of the following movement of the limb upward? (2024, 2023)**

- A) Flexion
- B) Elevation
- C) Extension
- D) Depression

**11. in which of the following movements the palm face posterior? (2021, 2020)**

- A) Supination
- B) Pronation
- C) Eversion
- D) Inversion

**12. The term which means near to the trunk is: (Dr Rasha)**

- A) Medial
- B) superficial
- C) Proximal
- D) distal

**13. palmar surface means: (Dr Rasha)**

- A) the anterior surface of the hand
- B) the posterior surface of the foot
- C) the posterior surface of the hand
- D) the sole of the foot

**14. During dissection, the brain was sectioned into anterior and posterior parts. Which anatomical plane is used for the examination? (KNU 2026)**

- A) coronal plane
- B) parasagittal plane
- C) midsagittal plane
- D) horizontal plane

**15. To put the person in anatomical position, what is the correct position of the arm? (KNU 2026)**

- A) The palm faces anterior and the arm is adducted
- B) The palm faces anterior and the arm is abducted
- C) The palm faces posterior and the arm is adducted
- D) The palm faces posterior and the arm is abducted

**16. During dissection, the brain was sectioned into equal right and left parts. Which plane is used for this division? (KNU 2026)**



- A) Coronal plane
- B) Horizontal plane
- C) Midsagittal plane
- D) Parasagittal plane

**17. In which anatomical location are the deep palmar vessels found? (KNU 2026)**

- A) In the palm of the hand near the skin
- B) In the sole of the foot near the skin
- C) In the sole of the foot deep
- D) In the palm of the hand away from the skin

**18. In which movement does the palm face posterior? (KNU 2026)**

- A) Flexion
- B) Supination
- C) Pronation
- D) Protraction

**19. Which plane divides the body into anterior and posterior parts? (2025)**

- A) Midsagittal plane
- B) Parasagittal plane
- C) Coronal plane.
- D) Horizontal plane

**21. What is the meaning of deep dissection of the ventral aspect of the neck? (2025)**

- A) Dissection of the muscles and vessels and back of the neck
- B) Dissection of the muscles and vessels and front of the neck
- C) Dissection of the skin and back of the neck
- D) Dissection of the skin and front of the neck

**22. The plane which divides the body into unequal right and left parts is called which of the following? (2025)**

- A) Horizontal plane
- B) Midsagittal plane
- C) Parasagittal plane
- D) Coronal plane

**23. In anatomical position the palm of the hand faces (2023)**

- A) Anteriorly
- B) Posteriorly
- C) Superiorly
- D) Inferiorly

**24. Medial rotation means (2023)**

- A) Moving the limb externally around the long axis
- B) Moving the limb internally around the long axis
- C) Moving the limb upward around horizontal axis
- D) Moving the limb downward around horizontal axis

**25. Coronal plane divide the body into (2023)**

- A) Upper and lower halves
- B) Anterior and posterior halves



C) Right and left halves

D) Superior and inferior halves

**26. Planter aspect of the foot means (2023)**

A) Sole

B) Palm

C) Dorsum

**27. Which of the followings means movement of a part of the body around its long axis? (2022)**

A) Circumduction

B) Eversion

C) Pronation

D) Rotation

**28. which of the following plane divides the body into anterior and posterior parts? (2020)**

A) sagittal plane

B) coronal plane

C) parasagittal plane

D) horizontal plane

**29. which of the following terms means abduction of the fingers? (2020)**

A) Move the finger away from the median plane

B) Moving the finger away from the index finger

C) Moving the finger away from the middle finger

D) Moving the finger away from the little finger

**30. The anatomical position of the human body include the following except : (Dr Rasha)**

A) The person stands in erect position

B) The palm faces anterior

C) The arms are abducted

D) The lower limbs are close together

**31. Planter surface means : (Dr Rasha)**

A) The sole of the foot

B) The palm of the hand

C) The superior surface of the foot

D) The posterior surface of the hand

**32. Movement of a limb away from the midline in the coronal plane is: (Dr Rasha)**

A) Adduction.

B) Abduction.

C) Flexion.

D) Extension.

**33. Palm of the hand means: (Dr Rasha)**

A) the sole of the foot

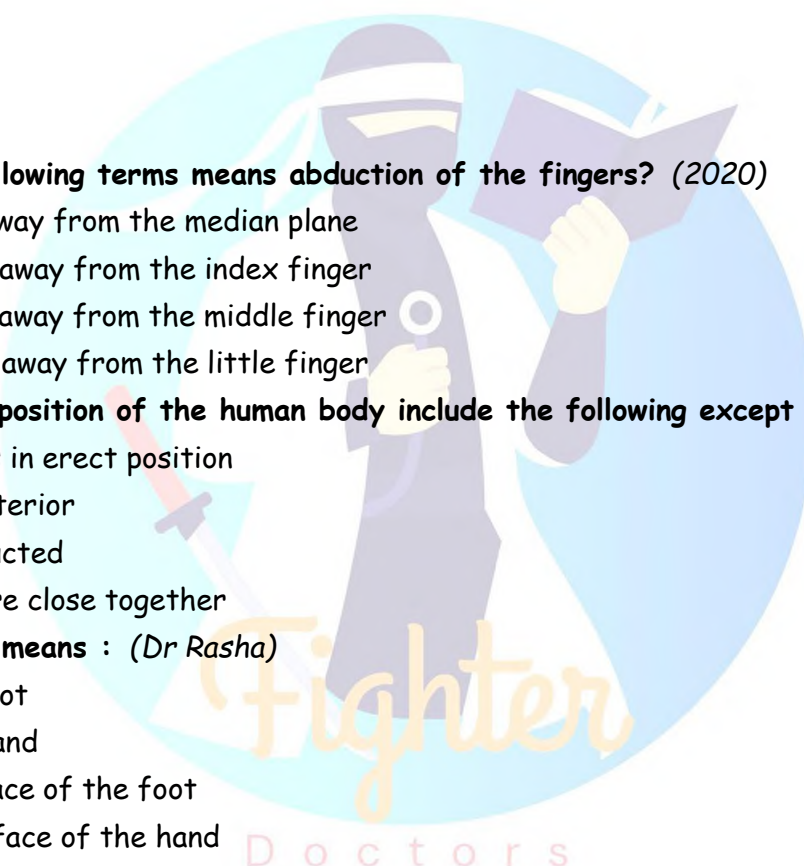
B) the anterior surface of the hand

C) the posterior surface of the hand

D) the posterior surface of the foot

**34. The anatomical position of the human body include the following except : (Dr Rasha)**

A) The person stands in erect position





- B) The palm faces anterior
- C) The arms are adjacent to the body
- D) The lower limbs are abducted

**35. Which is true about the parasagittal plane? (Dr Rasha)**

- A) It divides the body into equal upper and lower halves
- B) It divides the body into un equal anterior and posterior halves
- C) It divides the body into equal right and left halves
- D) It divides the body into un equal right and left halves

**36. The term lateral means: (Dr Rasha)**

- A) Toward the surface
- B) Near the attachment point
- C) On the same side
- D) Away from the midline

**37. Towards the front of the body means: (Dr Rasha)**

- A) Dorsal
- B) superior
- C) ventral
- D) inferior

**38. Which of the following terms means near the upper end? (Dr Rasha)**

- A) caudal
- B) cranial
- C) superficial
- D) ventral

**39. Movement of a part of the body around its long axis means: (Dr Rasha)**

- A) flexion
- B) extension
- C) rotation
- D) circumduction

**40. Elevation of the joint means: (Dr Rasha)**

- A) move apart superiorly
- B) move the part forward
- C) moving the part anteriorly
- D) moving the part inferiorly

**41. moving the apart away from the mid line is called..... (Dr Rasha)**

- A) flexion
- B) extension
- C) abduction
- D) adduction

**42. straightening of the joint is called.....: (Dr Rasha)**

- A) flexion
- B) extension
- C) adduction
- D) abduction

**43. Planter surface means: (Dr Rasha)**

- A) the anterior surface of the hand
- B) the posterior surface of the hand
- C) the posterior surface of the foot
- D) the sole of the foot

**44. Adduction of the fingers means: (Dr Rasha)**

- A) Moving the finger away from the body
- B) Moving the finger away from the middle finger



C) Moving the finger towards the middle finger

D) Moving the finger towards the body

**45. Regarding terms of movements one is wrong: (Dr Rasha)**

A) Flexion is forward movement except at knee joint

B) Adduction is movement away from the midline

C) Pivot joint is a uniaxial

D) Shoulder joint is polyaxial

**46. Anatomical position means all the following except: (Dr Rasha)**

A) Body is erect

B) Arms are adducted

C) Palms of the hands are facing backwards

D) Legs are close to each other

E) Face is directed forwards

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-A  | 2-D  | 3-A  | 4-B  | 5-C  | 6-B  | 7-B  | 8-B  | 9-B  | 10-B |
| 11-B | 12-C | 13-A | 14-A | 15-A | 16-C | 17-D | 18-C | 19-C |      |
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| 31-A | 32-B | 33-B | 34-D | 35-D | 36-D | 37-C | 38-B | 39-C | 40-A |
| 41-C | 42-B | 43-D | 44-C | 45-B | 46-C |      |      |      |      |

**Written questions**

1. Give a definition of: Coronal planes [repeated x3]

2. Give a definition of: Pronation [repeated x2]

3. Give a definition of: Sagittal planes

4. Give a definition of: Proximal and distal

5. Give a definition of: Flexion and extension

6. Give a definition of: Adduction and abduction

7. Give a definition of: Circumduction

8. .... refers to the structure that is nearer to the upper end of body, while .... refers to a structure that is nearer to the lower end of body.

9. Medial side of the forearm is called .... while the medial side of the leg is called .....

10. .... refers to the structure that is nearer to the median plane, while .... refers to the structure that is away from the median plane.

11. The term superficial means ..... while the term deep means .....

12. Anterior surface of the hand is called ...., while the other surface of hand is called .....

13. Flexion means ....., while extension means .....

14. .... means moving away from the median plane, while .... means moving towards the median plane.

15. Pronation means ....., while supination means .....

16. .... means to move apart superiorly, while .... means to moves apart inferiorly.

17. Give the definition of: Eversion

18. Give the definition of: Flexion

19. Give the definition of: Horizontal section



## 2. Skeletal System

50 MCQs 33 written questions

Doctors

1. Which of the following arteries can supply the bone marrow? (2026, 2025, KNU 2026)

- A) Periosteal
- B) Epiphyseal
- C) Metaphyseal
- D) Nutrient

2. Regarding the characters of the cartilage one is incorrect: (Dr Rasha)

- A) It is firm in consistency
- B) It is vascularised
- C) It form the temporary skeleton of the fetus
- D) It retained throughout the life in air passage

3. The following is true about the compact bone: (Dr Rasha)

- A) It form the interior of all bones



B) It consists of numbers of Haversian system

C) It is formed of trabeculae

D) It is spongy

**4. which of the following arteries supply the bone marrow? (2021, 2020, 2019)**

A) epiphyseal arteries

B) metaphyseal arteries

C) nutrient arteries

D) periosteal arteries

**5. Costal cartilage is: (2019)**

A) Hyaline cartilage.

B) Yellow elastic cartilage.

C) White fibrocartilage

D) Reticular cartilage

**6. The amount of compact bone is increased in the middle of shaft of long bone why? (2026, KNU 2026)**

A) To increase the length.

B) As it looks like a sponge to sustain the pressure

C) As it makes bone narrow in the middle.

D) As it provides strength for weight

**7. During the X-ray examination of a child, the bones appeared incomplete. The physician explained that the bones are still in the process of growing. Which of the following structures is responsible for increase the length of the bone? (2026, KNU 2026)**

A) Periosteum

B) Endosteum

C) Epiphyseal plate (growth plate)

D) Articular cartilage

**8. Which of the following is the character of the compact bone? (2025)**

A) Contains large spaces filled with bone marrow

B) Composed of loosely arranged trabeculae

C) Dense and organized into a number of cylindrical units

D) Found only in the epiphysis of long bones

**9. Which of the following is considered a short bone? (2025)**

A) Carpal bone

B) Phalanges

C) Patella

D) Clavicle

**10. which of the following arteries supply the outer compact bone? (2025)**

A) Nutrient artery

B) Periosteal artery

C) Epiphyseal artery

D) metaphyseal artery

**11. Which of the following cartilage disappears in the adult? (2024, 2023)**

A) Growing plate



- B) Articular cartilage
- C) Costal cartilage
- D) Cartilage of the auricle

**12. There are two forms of bone, which of the following is the character of spongy bone?** (2024, 2023)

- A) It consists of number of cylindrical units called haversian system
- B) It is formed of trabeculae
- C) It forms the surface layer of all bones
- D) It provides strength for weight bearing

**13. Carpal bones are:** (Dr Rasha)

- A) Small long bones
- B) Irregular bones
- C) Short bones
- D) Sesamoid bones

**14. The axial skeleton includes all of the following, EXCEPT:** (Dr Rasha)

- A) The skull.
- B) The clavicle.
- C) The sternum.
- D) The vertebra.

**15. Which one of the following is a flat bone?** (Dr Rasha)

- A) Clavicle.
- B) Humerus.
- C) Scapula.
- D) Vertebra.

**16. The phalanges are:** (Dr Rasha)

- A) Flat bones
- B) Irregular bones
- C) Long bones
- D) Short bones

**17. Which one of the following is irregular bone?:** (Dr Rasha)

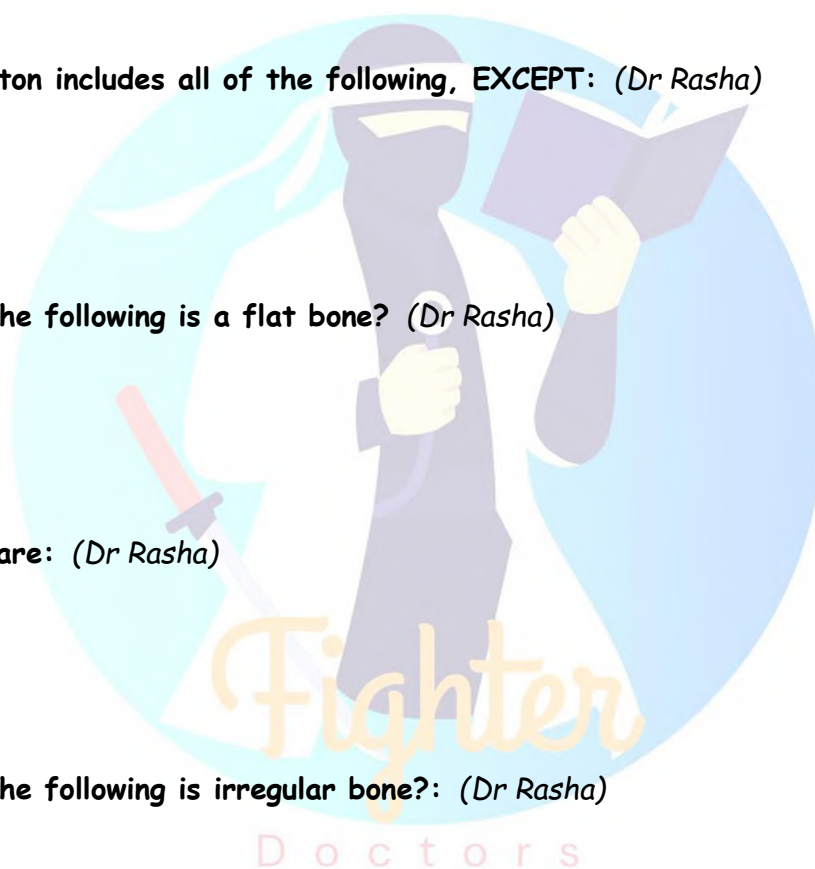
- A) Clavicle
- B) Humerus
- C) Scapula
- D) Vertebra

**18. The amount of compact bone is increased in the middle of shaft of long bones because:** (KNU 2026)

- A) As it contains bone marrow in its meshes
- B) As it looks like a sponge to sustain the pressure
- C) As it provides strength for weight bearing
- D) To increase its length

**19. Which of the following arteries can supply the outer compact bone?** (KNU 2026)

- A) Epiphyseal
- B) Metaphyseal





- C) Nutrient
- D) Periosteal

**20. Which of the following bones are ossified by membranous ossification? (KNU 2026)**

- A) Vault of the skull
- B) Humerus
- C) Femur
- D) Radius

**21. Which of the following is the characteristic of the compact bone? (2025)**

- A) It provides strength for weight bearing
- B) It is formed of trabeculae
- C) It looks like a sponge usually filled with red bone marrow
- D) It is found inside the hard bone

**22. Which of the following arteries supplies bone marrow? (2024)**

- A) Periosteal artery
- B) articular artery
- C) Epiphyseal artery
- D) Nutrient artery

**23. Which of the following structures is avascular? (2024)**

- A) Articular cartilage
- B) intercalated disc
- C) ribs
- D) muscle

**24. which of the followings is responsible for the growth of the length of the bone (2023)**

- A) periosteum
- B) growing plate
- C) arterial supply
- D) nerve supply

**25. Bone marrow is supplied by (2023)**

- A) Periosteal artery
- B) Epiphyseal artery
- C) Nutrient artery

**26. Which of the following is axial skeleton (2023)**

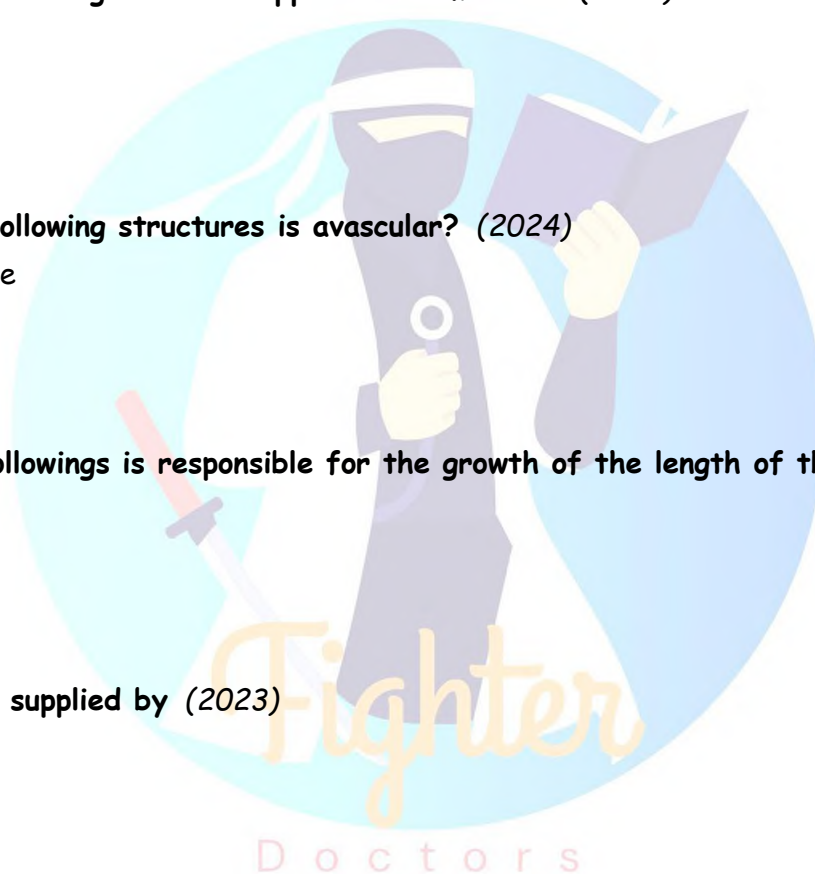
- A) Femur
- B) Humerus
- C) Skull
- D) Hip

**27. Carpal bones are (2023)**

- A) Long bone
- B) Short bones
- C) Irregular bones
- D) Pneumatic bone

**28. which of the following is the character of the spongy bone (2023)**

- A) it forms the out part of any bone





- B) it is consists of number of cylindrical units called Haversian system
- C) It is formed of trabeculae, the lamellae are branching and anastomose with

**29. Which of the following arteries supplies the bone marrow? (2023)**

- A) Periosteal artery
- B) The articular artery
- C) Epiphyseal artery
- D) Nutrient artery

**30. middle phalanx of the little finger is example to which type of bone ? (2023)**

- A) Long
- B) Short
- C) Irregular
- D) Flat

**31. Which of the following bones is a part of axial skeleton? (2022)**

- A) Femur
- B) humerus
- C) Rib cage
- D) Metacarpal bone

**32. The amount of the hard bone increased at which of the followings? (2022)**

- A) Lower end of the long bone
- B) Medullary cavity
- C) Middle of the shaft of the bone
- D) Upper end of the long bone

**33. Which of the following bones contain red bone marrow in the adult? (2020)**

- A) Shaft of femur
- B) Shaft of radius
- C) sternum
- D) shaft of tibia

**34. which of the following arteries supply the spongy bone? (2020)**

- A) epiphyseal arteries
- B) metaphyseal arteries
- C) nutrient arteries
- D) periosteal arteries

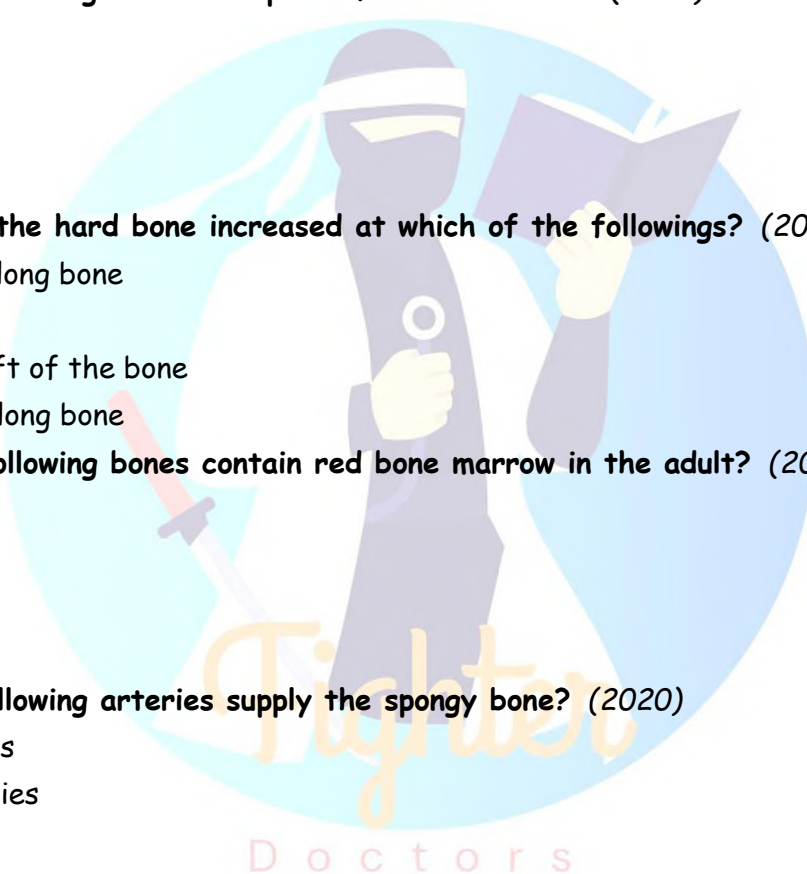
**35. which of the following bones contain yellow bone marrow in the adult ?: (2019)**

- A) Hip bone
- B) shaft of tibia
- C) sternum
- D) upper end of femur

**36. which of the following arteries supply the inner spongy bone ? (2019)**

- A) epiphyseal arteries
- B) metaphyseal arteries
- C) nutrient arteries
- D) periosteal arteries

**37. The artery that supplies the bone marrow is..... (2019)**





- A) Periosteal
- B) Epiphyseal
- C) Metaphyseal
- D) Nutrient

**38. following bones contain red bone marrow in the adult except: (2019)**

- A) Hip bone.
- B) shaft of tibia.
- C) sternum.
- D) upper end of femur.

**39. which of the following arteries supply the outer compact bone? (2019)**

- A) epiphyseal arteries.
- B) metaphyseal arteries.
- C) nutrient arteries.
- D) periosteal arteries.

**40. Which of the following bones ossifies in a membrane? (2019)**

- A) Radius
- B) Tibia
- C) Vault of the skull
- D) Vertebrae

**41. The type of cartilage found in the external ear is called: (2019)**

- A) Hyaline cartilage.
- B) Yellow elastic cartilage.
- C) White fibrocartilage.
- D) Articular cartilage.

**42. All the following bones are appendicular skeleton except: (Dr Rasha)**

- A) Scapula
- B) Tibia
- C) Radius
- D) Hyoid bone

**43. the axial skeleton is formed of the following bone except (Dr Rasha)**

- A) Vertebral column
- B) Femur
- C) Skull
- D) Sternum

**44. regarding the vertebral column. All of the followings are true except (Dr Rasha)**

- A) Thoracic vertebrae are 7 in number
- B) It belongs to the axial skeleton
- C) Sacral vertebrae are fused
- D) Cervical vertebrae are 7 in number

**45. the following is an example of pneumatic bone (Dr Rasha)**

- A) Humerus
- B) skull
- C) Tibia





D) sternum

**46. concerning the bones, all the following are true except (Dr Rasha)**

A) Long bones of the limb are developed by membranous ossification

B) A sesamoid bone is a small nodule inside the tendon

C) Growth which takes place in the epiphyseal plate is responsible for the increase in the length of the bone

**47. The artery that supplies the bone marrow is: (Dr Rasha)**

A) Periosteal artery.

B) Epiphyseal artery.

C) Metaphyseal artery.

D) Nutrient artery.

**48. Pneumatic bones are present in the: (Dr Rasha)**

A) Upper limb.

B) Skull.

C) Lower limb.

D) Thorax.

**49. Vertebrae are considered as: (Dr Rasha)**

A) Long bones.

B) Flat bones.

C) Short bones.

D) Irregular bones.

**50. The following are long bones, EXCEPT: (Dr Rasha)**

A) Humerus.

B) Scapula.

C) Femur.

D) Tibia.

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-B  | 3-B  | 4-C  | 5-A  | 6-D  | 7-C  | 8-C  | 9-A  | 10-B |
| 11-A | 12-B | 13-C | 14-B | 15-C | 16-C | 17-D | 18-C | 19-D | 20-A |
| 21-A | 22-D | 23-A | 24-B | 25-C | 26-C | 27-B | 28-C | 29-D | 30-A |
| 31-C | 32-C | 33-C | 34-A | 35-B | 36-C | 37-D | 38-B | 39-D | 40-C |
| 41-B | 42-D | 43-B | 44-A | 45-B | 46-A | 47-D | 48-B | 49-D | 50-B |

*Note on Q34: Debatable: epiphyseal vs metaphyseal arteries.*

*Note on Q36: Debatable: nutrient artery (inner spongy bone / marrow) vs epiphyseal/metaphyseal arteries.*

## Written questions

1. Name three sites where cartilage is found and three sites where red bone marrow is located in the adult. [repeated x2]
2. Describe blood supply of the bones? [repeated x2]
3. Describe the bone marrow (definition, function, forms according to the age) [repeated x2]
4. Give a definition of: Spongy bone [repeated x2]
5. Write short note about: Functions of the skeleton [repeated x2]



6. Describe the bone marrow?
7. Give short notes on: Forms of bone
8. Give short notes on: Growth of bone
9. Give short notes on: Functions of the skeleton
10. Give short notes on: Blood supply of bone
11. Give a definition of: Compact bone
12. Give a definition of: Short bone
13. Give a definition of: Pneumatic bone
14. Give a definition of: Periosteum
15. Give a definition of: Axial skeleton
16. Give a definition of: Appendicular skeleton
17. The cartilage remain in certain sites of the body such as .....and.....
18. Types of ossification of bone are .....and.....
19. ....and.....are flat bones while.....is a sesamoid bone.
20. In the adult, the red bone marrow is present in .....and.....
21. The skeleton is divided into .....and.....
22. The number of cervical vertebrae is ....., while the number of thoracic vertebrae is .....
23. The parts of the skeletal system are..... And.....
24. The axial skeleton is formed of the following bone ....., ....., ..... and.....
25. The types of bone according to the shape is ..... example..... example..... example.....
26. Write short note about: Shapes of the bone
27. Write short note about: Arterial supply of long bone
28. Write short note about: Bone marrow in different stage of life
29. Bones containing air spaces are called .....
30. .... is a structure essential for bone growth in length.
31. Forms of bone are ..... and.....

### 3. Joints

50 MCQs (top 50 of 56 available, most repeated / most recent first) | 20 written questions

**1. Presence of joint cavity is one of the characters of which of the following joint? (2026, 2024, 2023, KNU 2026)**

- A) Fibrous joint
- B) Primary cartilaginous joint
- C) Secondary cartilaginous joint
- D) Synovial joint

**2. One of the following is poly axial joint: (Dr Rasha)**

- A) Shoulder joint
- B) Knee joint
- C) Elbow joint
- D) Wrist joint

**3. The type of joint between the bodies of the vertebrae is: (Dr Rasha)**

- A) Synovial joint.



- B) Fibrous joint.
- C) Primary cartilaginous joint.
- D) Secondary cartilaginous joint.

**4. The joints that move freely are: (Dr Rasha)**

- A) Fibrous joints.
- B) Primary cartilaginous joints.
- C) Synovial joints.
- D) Secondary cartilaginous joints.

**5. The joint between the tooth and its socket is: (Dr Rasha)**

- A) Sutural joint.
- B) Ball and socket.
- C) Syndesmosis.
- D) Gomphosis.

**6. The sutures of the skull are considered as: (Dr Rasha)**

- A) Fibrous joints
- B) Primary cartilaginous joints
- C) Secondary cartilaginous joints
- D) Synovial joints

**7. Which of the following joints are formed when two bones are connected by fibrocartilage? (2026, KNU 2026)**

- A) Symphysis pubis
- B) Shoulder joint
- C) Skull sutures
- D) Hip joint

**8. Which of the following is a secondary cartilaginous joint? (2025)**

- A) Intervertebral disc
- B) Suture
- C) Elbow.
- D) Sternocostal junction

**9. The opposed bony surfaces connected by an interosseous ligament are found in which of the following? (2025)**

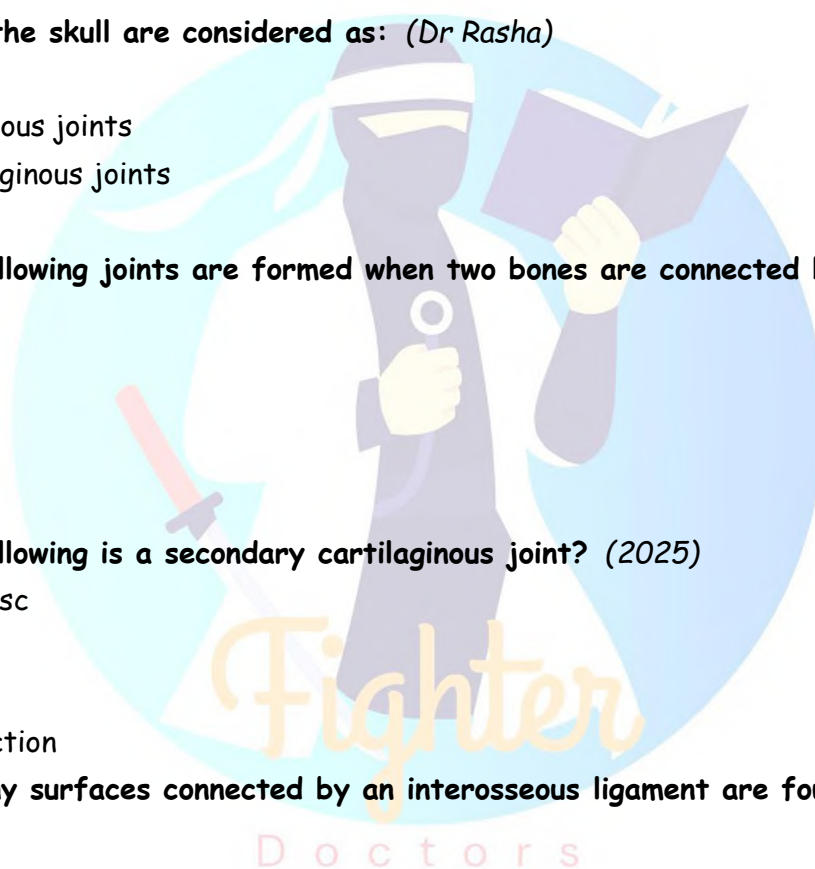
- A) Syndesmosis
- B) Suture
- C) Hinge joint.
- D) Plane synovial joint

**10. Primary cartilaginous joints may be temporary, like which of the following? (2025)**

- A) Costochondral junction
- B) Growing plate
- C) Intervertebral disc
- D) Symphysis pubis

**11. What type of joint connects the bones of the vault of the skull? (2025)**

- A) Fibrous joint
- B) Primary cartilaginous joint





C) Secondary cartilaginous joint

D) Synovial joint

**12. Which of the following is fibrous joint? (2024, 2023)**

A) Elbow joint

B) Shoulder joint

C) Inferior tibiofibular joint

D) Intervertebral disc

**13. What is the type of costochondral junction? (2024, 2023)**

A) Fibrous joint

B) Primary cartilaginous joint

C) Secondary cartilaginous joint

D) Synovial joint

**14. Which of the following is an example of secondary cartilaginous joint? (2021, 2020)**

A) Shoulder joint

B) Radio-ulnar joint

C) costo chondral joint

D) Symphysis pubis

**15. Symphysis pubis is: (Dr Rasha)**

A) Secondary cartilaginous joint

B) Primary cartilaginous joint

C) Plane synovial joint

D) Fibrous joint

**16. the following is a fibrous joint in which the margins are serrated (Dr Rasha)**

A) Syndesmosis

B) Sutures

C) secondary cartilagenous joint

D) gomphosis

**17. Regarding the different types of joints, the following statements are correct EXCEPT: (Dr Rasha)**

A) The sutures between bones of the skull form fibrous joints.

B) The joints between vertebral bodies are of synovial type.

C) In the synovial joints the articular surfaces are covered with hyaline cartilage.

D) The ligaments and tone of muscle around the joints play an important role in the stability of joints.

**18. Hip and shoulder joints allow movements in (Dr Rasha)**

A) One direction

B) Two directions

C) Three directions

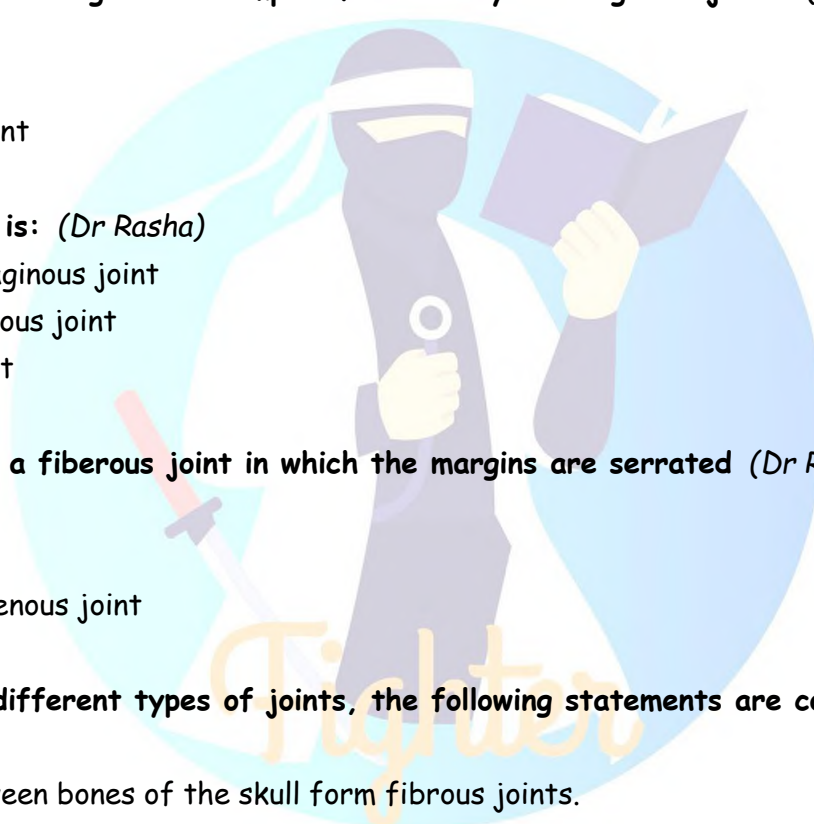
D) All directions

**19. A child presents with cranial deformities due to premature fusion of skull bones. Which joint type connected the skull bones? (KNU 2026)**

A) primary cartilaginous joint

B) secondary cartilaginous joint

C) fibrous joint





D) synovial joint

**20. Presence of joint cavity is seen in: (KNU 2026)**

A) Primary cartilaginous joint

B) Synovial joint

C) Secondary cartilaginous joint

D) Fibrous joint

**21. The type of joint between the teeth and the jaw is: (KNU 2026)**

A) Synovial joint

B) Suture

C) Gomphosis

D) Syndesmosis

**22. What is the type of the joint between the teeth and the mandible ? (2025)**

A) Fibrous joint

B) Primary cartilaginous joint

C) Secondary cartilaginous joint

D) Synovial joint

**23. Which of the following represents a primary cartilaginous joint made of direct contact between bone and thin cartilage? (2025)**

A) Intervertebral disc

B) Costochondral junction

C) Growing plate

D) Symphysis pubis

**24. What is the type of symphysis pubis? (2024)**

A) Fibrous joint

B) Primary cartilaginous joint

C) Secondary cartilaginous joint

D) Synovial joint

**25. Which of following is an example of secondary cartilaginous joint? (2023)**

A) Shoulder joint

B) radio-ulnar joint

C) costo-chondral

D) intervertebral disc

**26. Which of the followings is immovable joint? (2022)**

A) Fibrous joint

B) Primary cartilaginous joint

C) Secondary cartilaginous joint

D) Synovial joint

**27. Which of the following joints connect between serrated margins? (2022)**

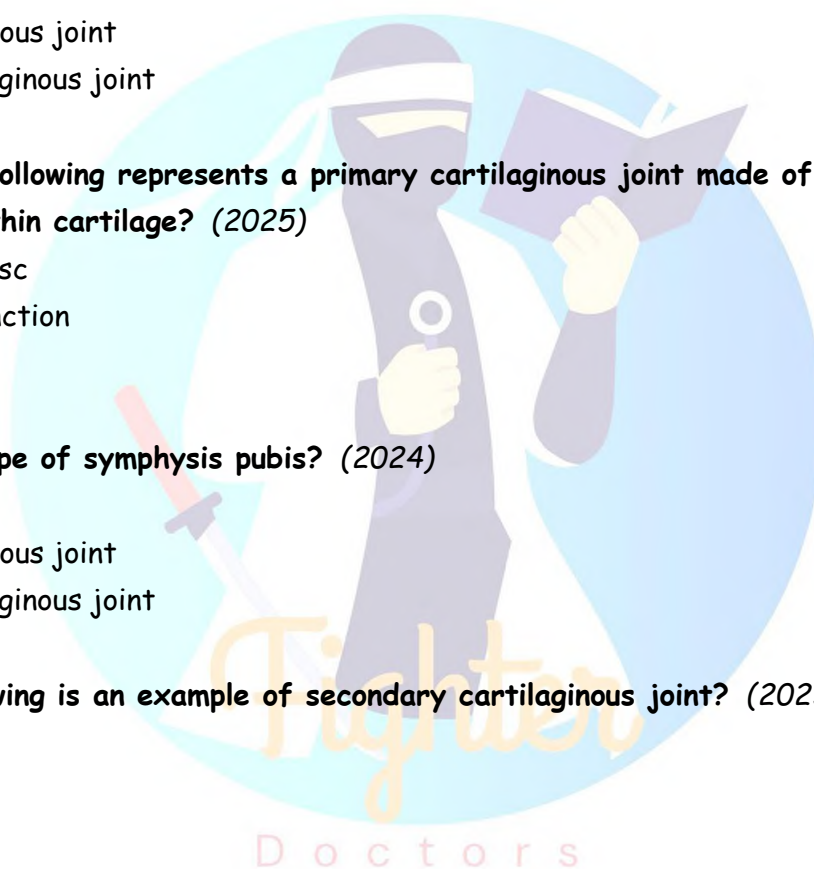
A) Gomphosis

B) Suture

C) Symphysis pubis

D) Syndesmosis

**28. Which of the following joint can do circumduction? (2022)**





- A) Elbow joint
- B) Intercarpal joint
- C) Interphalangeal joint
- D) Wrist joint

**29. Which of the following is an example of pivot joint? (2020)**

- A) Shoulder joint
- B) radio-ulnar joint
- C) costo chondral joint
- D) intervertebral discs

**30. Which of the following is an example of the poly-axial joint? (2020)**

- A) Acromio-clavicular joint joint
- B) hip joint
- C) metacarpo-phalangeal joint
- D) inferior radio-ulnar joint

**31. Fibrous joints are present in: (2019)**

- A) Costochondral junction
- B) Intervertebral discs
- C) Sutures
- D) Symphysis pubis

**32. What is the type of the joint between the tooth and its socket? (2019)**

- A) Ball and socket
- B) Gomphosis
- C) Sutural joint
- D) Syndesmosis

**33. Primary cartilaginous joints are present in: (2019)**

- A) Costochondral junction
- B) Intervertebral discs
- C) Sutures
- D) Symphysis pubis

**34. Pivot joints move around: (2019)**

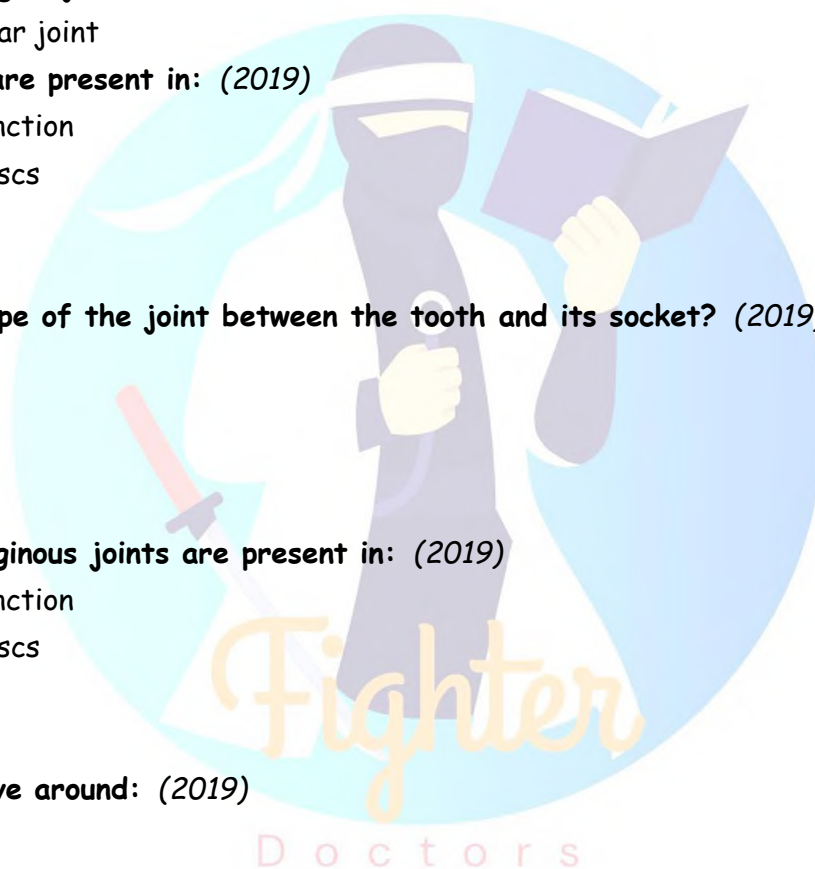
- A) Vertical axis
- B) Horizontal axis
- C) Two axis
- D) More than two axis

**35. the intervertebral disc is (Dr Rasha)**

- A) Primary cartilagenous
- B) Secondary cartilagenous
- C) Fiberous joint
- D) Synovial joint

**36. the followings are synovial joint except (Dr Rasha)**

- A) Hip joint
- B) Inferior tibiofibular joint
- C) Wrist joint





D) Elbow joint

E) Shoulder joint

**37. the followings are true about the joint except (Dr Rasha)**

A) Synovial joint contains fluid

B) Sutures between skull are examples of fibrous joint

C) Secondary cartilaginous joint disappears by age

D) Syndesmosis is present between the lower ends of tibia and fibula

**38. The ulna shares in the following joints, EXCEPT: (Dr Rasha)**

A) The elbow joint.

B) The superior radio-ulnar joint.

C) The wrist joint.

D) The inferior radio-ulnar joint.

**39. The following are characteristics of the synovial joints, EXCEPT: (Dr Rasha)**

A) The articular surfaces of bones are covered by hyaline cartilage.

B) Permit a great degree of freedom of movement.

C) All synovial joints have intra articular ligaments.

D) All synovial joints have fibrous capsule.

**40. The ellipsoid joint allows all of the following movements, EXCEPT: (Dr Rasha)**

A) Abduction.

B) Rotation.

C) Flexion.

D) Extension.

**41. Regarding the classification of synovial joints, mark ONE WRONG answer: (Dr Rasha)**

A) The hinge joints.

B) The ellipsoid joints.

C) The sutural joints.

D) The ball and socket joints.

**42. The intervertebral disc is considered as: (Dr Rasha)**

A) Fibrous joint.

B) Primary cartilaginous joint.

C) Secondary cartilaginous joint.

D) Synovial joint.

**43. Pronation and supination of the forearm occurs at: (Dr Rasha)**

A) Elbow joint.

B) Superior and inferior radio-ulnar joints.

C) Shoulder joint.

D) Radio-carpal joint.

**44. The radius shares in the following joints, EXCEPT: (Dr Rasha)**

A) Elbow joint.

B) Shoulder joint.

C) Superior radio-ulnar joint.

D) Inferior radio-ulnar joint.

**45. The fibula shares in the following joints ,EXCEPT: (Dr Rasha)**



- A) Ankle joint.
- B) Knee joint.
- C) Superior tibiofibular joint.
- D) Inferior tibiofibular joint.

**46. connective tissue which connect the bone with bone is called : (Dr Rasha)**

- A) tendon
- B) ligament
- C) aponeurosis
- D) fascia

**47. Types of joints are classified on basis of (Dr Rasha)**

- A) number of cartilages
- B) degree of movement
- C) weight strength
- D) number of bones

**48. Location or a point where two bones contact each other is called (Dr Rasha)**

- A) joint
- B) rib
- C) vertebrae
- D) pectoral girdle

**49. One of the following is poly axial joint: (Dr Rasha)**

- A) Shoulder joint
- B) Knee joint
- C) Elbow joint
- D) d- Wrist joint

**50. One of the following is a primary cartilaginous joint.. (Dr Rasha)**

- A) elbow joint
- B) costochondral junction
- C) syndesmosis
- D) hip joint

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-A  | 3-D  | 4-C  | 5-D  | 6-A  | 7-A  | 8-A  | 9-A  | 10-B |
| 11-A | 12-C | 13-B | 14-D | 15-A | 16-B | 17-B | 18-D | 19-C | 20-B |
| 21-C | 22-A | 23-B | 24-C | 25-D | 26-A | 27-B | 28-D | 29-B | 30-B |
| 31-C | 32-B | 33-A | 34-A | 35-B | 36-B | 37-C | 38-C | 39-C | 40-B |
| 41-C | 42-C | 43-B | 44-B | 45-B | 46-B | 47-B | 48-A | 49-A | 50-B |

## Written questions

1. Give short notes on: Characters of the synovial joints [repeated x2]
2. Describe with drawing the Characters of synovial joints.
3. Give short notes on: Fibrous joint
4. Give short notes on: Cartilaginous joints
5. Give a definition of: Syndesmosis
6. Give a definition of: Gomphosis



7. Give a definition of: Hinge joint
8. Give a definition of: Pivot joint
9. Give a definition of: Saddle joint
10. The sutures are .....joints
11. Intra articular structures can be seen in different joints as.....,..... and.....
12. The elbow joint is.....while the hip joint is .....
13. Hilton `s law is .....
14. Give the definition of: Joint
15. Write short note about: Characters of synovial joints
16. Write short note about: Types of fibrous joints
17. Elbow joint is .....joint while symphysis pubis is..... joint.



#### 4. Muscular System

50 MCQs (top 50 of 54 available, most repeated / most recent first) | 26 written questions

1. Any muscle oppose the action of prime mover is named: (2019, Dr Rasha)

- A) antagonist
- B) action of paradox
- C) synergist
- D) fixator

2. Which of the following structures is connect the fibers of cardiac muscles? (2025)

- A) connective tissue
- B) intercalated disc
- C) ligament
- D) tendon

3. regarding the smooth muscle, all of the following is true except: (Dr Rasha)

- A) consists of long spindle shaped cells
- B) controlled by autonomic nervous system
- C) have the property of spontaneous and rhythmic contraction



D) found in the wall of the viscera and blood vessels

**4. Any muscle acts as anti gravity muscle is named: (Dr Rasha)**

A) Antagonist

B) Action of paradox

C) Synergist

D) Fixator

**5. Which of muscle has central tendon and the muscle fibers attached to the both sides of the tendon? (2026, KNU 2026)**

A) Triangular muscle.

B) Spiral muscle

C) Unipennate muscle.

D) Bipennate muscle

**6. During the flexion of the elbow, the triceps act as (2026, KNU 2026)**

A) Synergist

B) Prime mover

C) Fixator

D) Antagonist

**7. During the flexion of the elbow, the biceps acts as: (2026, KNU 2026)**

A) Prime mover

B) Synergist

C) Antagonist

D) Fixator

**8. The attachment of a muscle to the bone that moves when the muscle contracts is called which of the following? (2026, KNU 2026)**

A) Origin

B) Insertion

C) Belly

D) Ligament

**9. Which of the following muscles is voluntary non striated? (2025, 2024)**

A) Muscles of the upper limb

B) Muscles of the trunk

C) Muscles of the heart

D) Muscle of the wall of blood vessel

**10. During elbow flexion, which muscle acts as the prime mover and which as the antagonist? (2025)**

A) Biceps brachii as antagonist, triceps brachii as prime mover

B) Biceps brachii as prime mover, triceps brachii as antagonist

C) Deltoid as prime mover, biceps brachii as antagonist

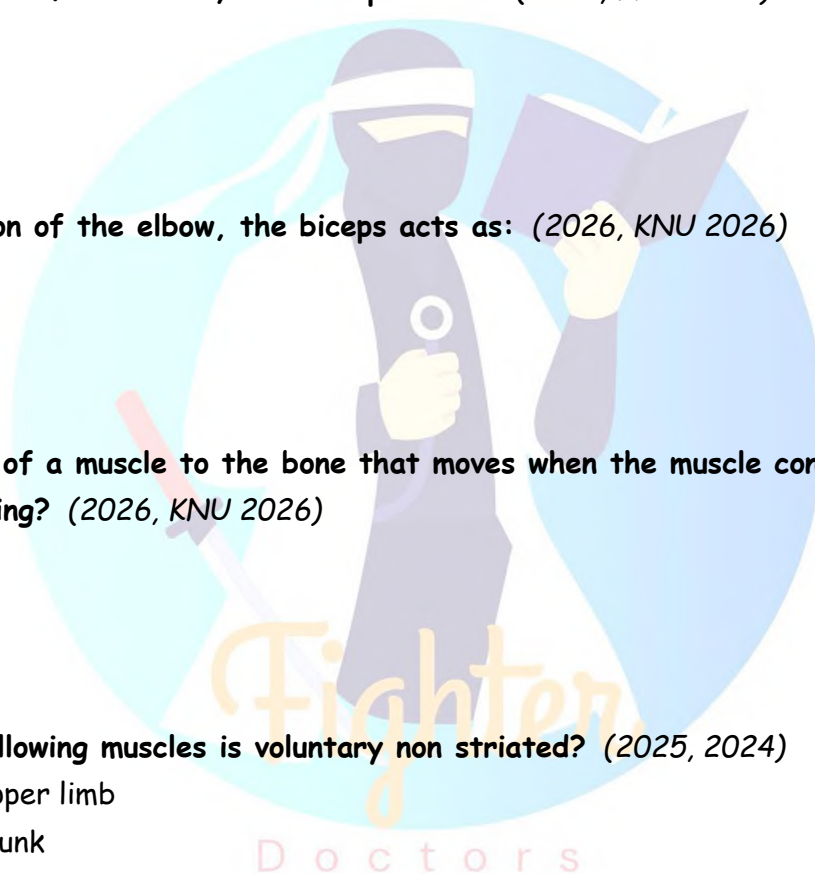
D) Triceps brachii as prime mover, deltoid as antagonist

**11. What is the role of a synergist muscle during movement? (2025)**

A) It opposes the action of the prime mover

B) It initiates the movement as the main force

C) It stabilizes the origin of the prime mover





D) It assists the prime mover by reducing unnecessary movement

**12. Which of the following muscles is involuntary striated ? (2025)**

A) Muscles of the upper limb

B) Muscles of the trunk

C) Muscles of the heart

D) Muscle of the wall of blood vessel

**13. Which of the following is the character of the smooth muscle? (2024, 2023)**

A) it is voluntary non striated muscle

B) it is attached to the skeleton.

C) it is non striated involuntary muscle

D) it is connected by intercalated disc

**14. When the muscle removes unwanted movement in the proximal joint and the prime mover act on the distal, it is called \_\_\_\_\_: (2024, 2023)**

A) antagonist

B) prime mover

C) synergist

D) Fixator

**15. Muscles with fibers oblique to the line of pull is present in which of the following? (2021, 2020)**

A) Trapezius muscle

B) Temporalis muscle

C) Extensor digitorum longus muscle

D) Rectus femoris muscle

**16. Bipennate muscles are present in: (2019)**

A) Extensor digitorum longus muscle

B) Rectus abdominus muscle

C) Rectus femoris

D) Temporalis muscle

**17. deep fascia is absent in: (Dr Rasha)**

A) abdomen

B) palms

C) leg

D) forearm

**18. One of the following about cardiac muscle is true : (Dr Rasha)**

A) It consists of non striated muscle fibers that branch

B) It is voluntary muscle

C) It has property for spontaneous contraction

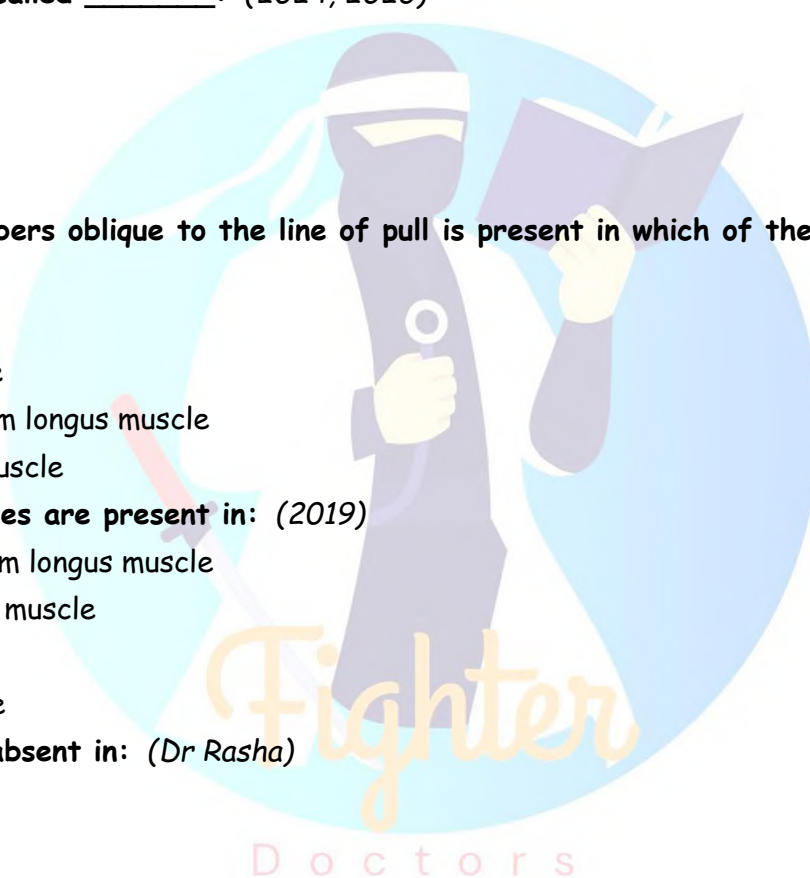
D) It has peripheral nucleus

**19. Which form of muscle has central tendon and the muscle fibers attached to one side of the tendon? (KNU 2026)**

A) Bipennate muscle

B) Spiral muscle

C) Triangular muscle





D) Unipennate muscle

**20. During the flexion of the elbow, the muscles of the shoulder act as: (KNU 2026)**

A) Antagonist

B) Fixator

C) Prime mover

D) Opposing

**21. Which of the following hold the tendons in place to facilitate action of muscles? (KNU 2026)**

A) Aponeurosis

B) Intermuscular septa

C) Retinacula

D) Superficial fascia

**22. A firm grip requires the wrist to be held in extension. In this position, the wrist extensors are acting as: (KNU 2026)**

A) Synergist

B) Antagonist

C) Prime mover

D) Fixator

**23. the deep fascia is well developed in some regions and thin in others, where it is thin? (2025)**

A) Upper limb

B) Thorax

C) Lower limb

D) neck

**24. What do you know about the plantar aponeurosis? (2025)**

A) It is thick deep fascia in the neck

B) It is thick deep fascia around the joint

C) It is thick deep fascia in the palm

D) It is thick deep fascia in the sole

**25. The fibers converge from the origin to the insertion with considerable obliquity to the long axis in which of the following? (2025)**

A) Spiral muscle

B) Triangular muscle

C) Unipennate muscle

D) Bipennate muscle

**26. When muscle fibers are attached to one side of tendon, it is called : (2024)**

A) triangular

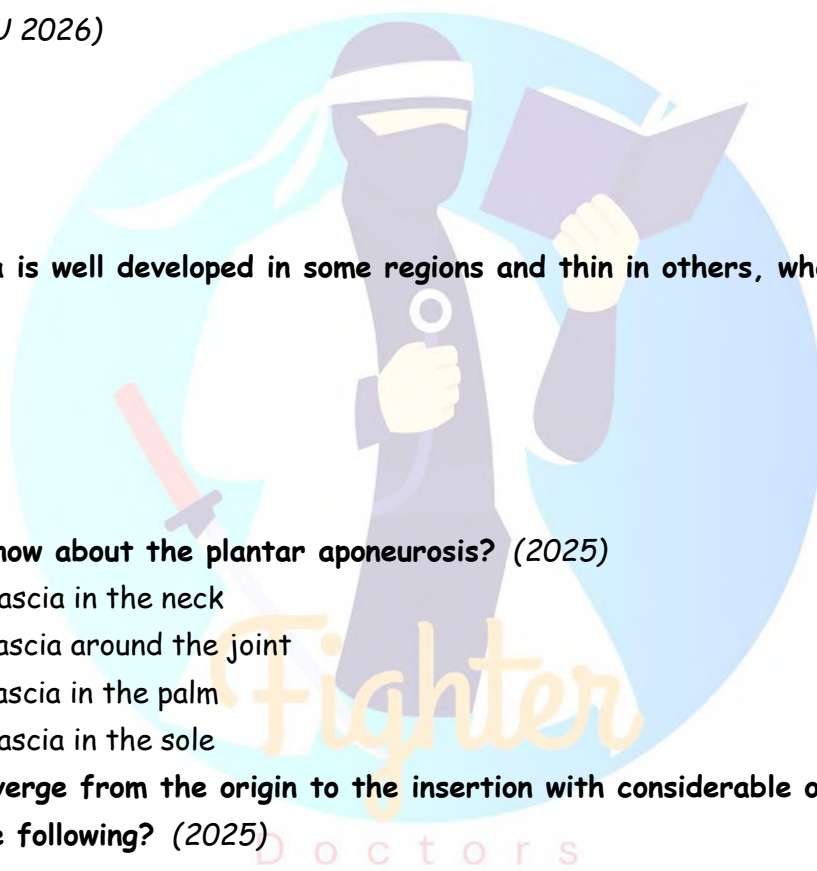
B) unipennate

C) bipennate

D) spiral

**27. When the muscle fibers are attached to one side of the tendon, it is called which of the following? (2023)**

A) triangular





- B) unipennate
- C) bipennate
- D) spiral

**28. Which of the followings is the character of the cardiac muscle fiber? (2022)**

- A) It has multiple nuclei and connected by connective tissue
- B) It has central nucleus and connected by intercalated disc
- C) It is polygonal and has voluntary control
- D) It is nonstriated and has voluntary control

**29. Fixator muscle is..... ? (2022)**

- A) Eliminate unwanted movement of the proximal joint while the prime mover act on the distal joints
- B) Form a firm base to the prime mover at the origin
- C) Has antigravity muscle
- D) Oppose the action of prime mover

**30. Tendon means which of the followings? (2022)**

- A) bands of connective tissue which connect muscles with bones.
- B) bands of connective tissue which connect bones with bones.
- C) bands of connective tissue which connect muscles with muscles.
- D) bands of connective tissue which connect muscles with skin.

**31. Unipennate muscles form is present in which of the following? (2020)**

- A) trapezius muscle
- B) temporalis muscle
- C) extensor digitorum longus muscle
- D) rectus femoris muscle

**32. Which of the following sentence describe the Spiral muscle? (2019)**

- A) It is twisted muscle from the origin to its insertion
- B) Its fibers are arranged parallel to the line of pull of the muscle
- C) Its fibers converge from all sides on a central tendon
- D) Its tendon lies at the apex of the triangle

**33. Circumpennate muscles are present in: (2019)**

- A) Biceps muscle
- B) Deltoid muscle
- C) Rectus femoris
- D) Tibialis anterior muscle

**34. flat tendon is named: (Dr Rasha)**

- A) aponeurosis
- B) muscle belly
- C) endomysium
- D) all of the above

**35. all of the following about the muscles are true except (Dr Rasha)**

- A) The smooth muscles are present in the wall of blood vessels
- B) The skeletal muscles are attached to the skeleton
- C) The cardiac muscle is present only in the heart
- D) The cardiac muscle is supplied by somatic nerves



**36. The skeletal muscle action includes all of the following, EXCEPT: (Dr Rasha)**

- A) Prime mover.
- B) Peristalsis.
- C) Fixator.
- D) Antagonistic.

**37. Regarding skeletal muscle, the following statements are correct, EXCEPT: (Dr Rasha)**

- A) They are voluntary muscles.
- B) They have two or more attachments.
- C) A prime mover is the muscle responsible for a particular movement.
- D) A fixator muscle is the muscle that opposes the action of a flexor muscle.

**38. Regarding skeletal muscle, the following statements are correct, EXCEPT: (Dr Rasha)**

- A) They are voluntary muscles.
- B) It is supplied by autonomic nerves.
- C) Maintain the body posture.
- D) Striated with peripheral nuclei.

**39. Regarding the varieties of the muscular tissue, the following are correct EXCEPT: (Dr Rasha)**

- A) Cardiac muscle.
- B) Aponurosis.
- C) Skeletal muscles.
- D) Smooth muscles.

**40. The actions of the skeletal muscle include the following, EXCEPT: (Dr Rasha)**

- A) Fixator.
- B) Antagonist.
- C) Synergist.
- D) Peristalsis.

**41. In the triangular muscle : (Dr Rasha)**

- A) the muscle fibers slope into the two sides of the central tendon
- B) The muscle fibers gaining two attachment sites
- C) the entire fibers slope into one side of the tendon
- D) The fibers converge from the origin to the insertion

**42. Any muscle has antigravity muscle is named: (Dr Rasha)**

- A) antagonist
- B) action of paradox
- C) synergist
- D) fixator

**43. which of the following is the character of cardiac muscle ?: (Dr Rasha)**

- A) It consists of long spindle shaped cells
- B) It has voluntary control
- C) have the property of spontaneous and rhythmic contraction
- D) found in the wall of the viscera and blood vessels

**44. Which muscle can eliminate unwanted movement on the proximal joint while the main muscle act on distal joint ? (Dr Rasha)**



- A) antagonist
- B) action of paradox
- C) synergist
- D) fixator

**45. which of the following muscles has striated long fibers with voluntary control ? (Dr Rasha)**

- A) muscles of the heart
- B) muscles of the upper limbs
- C) muscles of blood vessels
- D) muscles of the intestine

**46. the visceral muscles are : (Dr Rasha)**

- A) striated and voluntary
- B) striated and involuntary
- C) non striated and involuntary
- D) non striated and involuntary

**47. the muscle fibers which connected by intercalated disc is present in....: (Dr Rasha)**

- A) wall of the heart
- B) body wall
- C) wall of blood vessels
- D) upper limb

**48. which of the following muscles contracting to give the working muscles a firm base? (Dr Rasha)**

- A) prime mover
- B) antagonist
- C) fixator
- D) synergist

**49. part of muscle attached with some movable bone is called (Dr Rasha)**

- A) Origin
- B) Moveable muscles
- C) Insertion
- D) Muscle belly

**50. The muscle that perform a movement is called: (Dr Rasha)**

- A) Prime mover
- B) Antagonist
- C) Fixator
- D) Synergic muscle

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-A  | 2-B  | 3-C  | 4-B  | 5-D  | 6-D  | 7-A  | 8-B  | 9-D  | 10-B |
| 11-D | 12-C | 13-C | 14-D | 15-A | 16-C | 17-A | 18-C | 19-D | 20-B |
| 21-C | 22-D | 23-B | 24-D | 25-D | 26-B | 27-B | 28-B | 29-A | 30-A |
| 31-C | 32-A | 33-B | 34-A | 35-D | 36-B | 37-D | 38-B | 39-B | 40-D |
| 41-D | 42-D | 43-C | 44-C | 45-B | 46-C | 47-A | 48-C | 49-C | 50-A |

## Written questions



1. Describe the forms of the skeletal muscle [repeated x2]
2. Give short notes on: Forms of the skeletal muscle [repeated x2]
3. Describe the skeletal muscles forms with example of each type
4. Enumerate forms of skeletal muscles and example of each type?
5. Give short notes on: Superficial fascia
6. Give short notes on: Deep fascia
7. Give short notes on: The differences between the skeletal, cardiac and smooth muscle
8. Give a definition of: Retinacula
9. Give a definition of: Aponeurosis
10. Give a definition of: Tendon
11. Give a definition of: Ligaments
12. Give a definition of: Prime mover
13. Give a definition of: Synergists
14. Give a definition of: Fixator muscle
15. The fascia is divided into.....and.....
16. Superficial fascia is ..... While deep fascia is .....
17. The deep fascia is well developed in ..... and.....while it is thin in.....and.....
18. the attachment of the muscle to the bone that remains stationary is the ..... while the attachment of the muscle to the bone that moves is the .....
19. The fleshy part of the muscle between the tendons of origin and/or insertion is called.....
20. The skeletal muscle is supplied by..... while the smooth muscle is supplied by.....
21. Write short note about: Differentiate between types of muscles
22. Write short note about: Forms of the skeletal muscles
23. Cardiac muscle is found in ..... and supplied by .....
24. The muscle which fibers converge from the origin to the insertion is called .....

## 5. Circulatory System

40 MCQs | 12 written questions

1. Which of the following has valves and constricted at their sites showing beaded appearance? (2026, KNU 2026)
  - A) Capillaries
  - B) Veins
  - C) Lymph vessels
  - D) Arteries
2. Which of the following can withstand the blood pressure created by the heart's beating? (2026, KNU 2026)
  - A) Large conducting arteries
  - B) Small distributing arteries
  - C) Large veins
  - D) Blood capillaries
3. During the pulmonary circulation, the deoxygenated blood is pumped from the right ventricle to the lungs through which of the following? (2026, KNU 2026)
  - A) Pulmonary veins



- B) Pulmonary trunk
- C) Aorta
- D) Inferior vena cava

**4. Which of the following structures receives oxygenated blood through the pulmonary veins?** (2026, KNU 2026)

- A) Right atrium
- B) Left atrium
- C) Right ventricle
- D) Pulmonary trunk

**5. During the portal circulation, the blood from the digestive system passes to which of the following?** (2025)

- A) From portal vein to the heart
- B) From the portal vein to the liver
- C) From inferior vena cava to the liver
- D) From the portal vein to the inferior vena cava

**6. What is the primary function of pulmonary circulation?** (2025)

- A) Regulate blood pressure
- B) Exchange gases between blood and lungs
- C) Deliver nutrients to tissues
- D) Transport oxygenated blood to the body

**7. Where does the superficial palmar arch lie?** (2024, 2023)

- A) In the anterior surface of the hand
- B) In the posterior surface of the hand
- C) In the superior surface of the foot
- D) In the inferior surface of the foot

**8. In portal circulation, which of the following ends in the inferior vena cava?** (2024, 2023)

- A) portal vein
- B) hepatic vein
- C) left gastric vein
- D) superior pancreatico duodenal vein

**9. Which of the following vessel is fenestrated?** (2024, 2023)

- A) large artery
- B) small artery
- C) vein
- D) blood capillary

**10. which of the followings contain deoxygenated blood?** (2021, 2020)

- A) Internal carotid artery
- B) Pulmonary artery
- C) Pulmonary vein
- D) External carotid artery

**11. In the systemic circulation:** (2019)

- A) The oxygenated blood is pumped from the heart to the body
- B) The deoxygenated blood is pumped from the heart to the lung



- C) The oxygenated blood is pumped from the heart to the lung
- D) The deoxygenated blood is pumped from the heart to the body

**12. One of the following is lymphoid organ (2019, Dr Rasha)**

- A) Liver
- B) Pancrease
- C) Spleen
- D) Lung

**13. The blood vessels that carry blood to the heart and have thin walls are called: (2025)**

- A) Arteries
- B) Veins
- C) Capillaries
- D) Lymph vessels

**14. blood vessels that carry blood to heart, have thin wall is called which of the following? (2025)**

- A) Artery
- B) vein
- C) capillary
- D) lymph vessel

**15. in systemic circulation the blood pass to the body, how it can reach the heart again? (2025)**

- A) Through the inferior and superior vena cava which open into the left atrium
- B) Through the inferior and superior vena cava which open into the right atrium
- C) Through the pulmonary veins which open into the left atrium
- D) Through the pulmonary arteries which open into the right ventricles

**16. Which blood vessel carries blood under high pressure and has strong, thick walls? (2025)**

- A) Lymph vessel
- B) Artery
- C) Vein
- D) Capillary

**17. Which of the following is the character of the artery? (2024)**

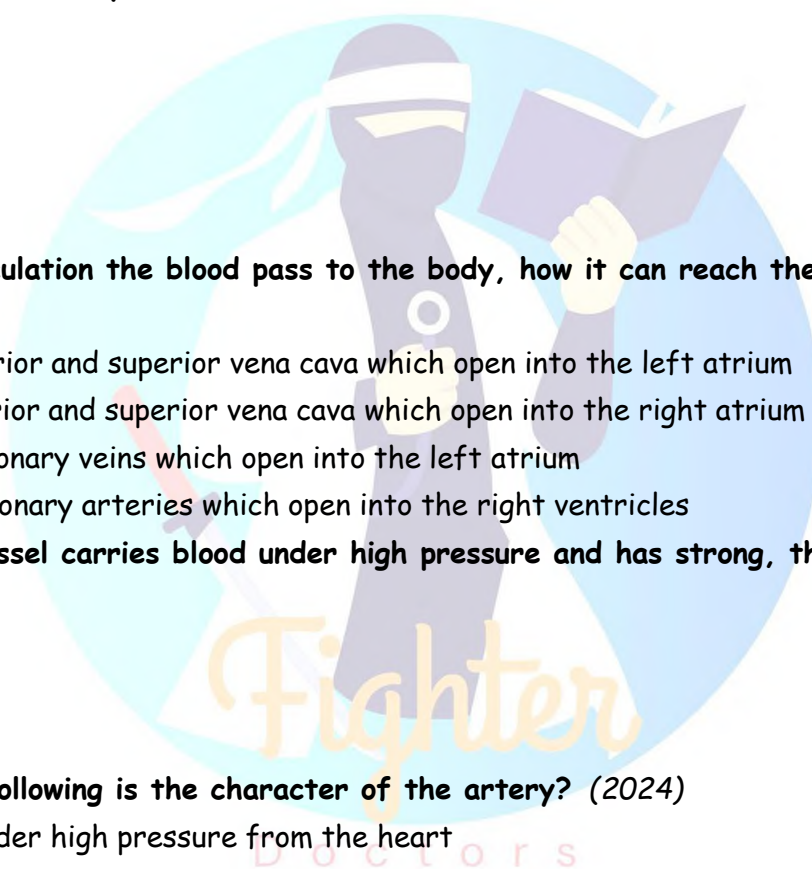
- A) It Carry blood under high pressure from the heart
- B) It has valves
- C) Its wall is semi-permeable
- D) Its wall is thin

**18. The vessel which carries fluid from the tissue to be filtered is called (2022)**

- A) Artery
- B) Vein
- C) Lymph vessel
- D) Blood capillary

**19. Superficial palmer arch lies in which of the followings? (2022)**

- A) Foot
- B) Forearm
- C) Hand





D) Leg

**20. In systemic circulation..... ? (2022)**

- A) The deoxygenated blood pass from the left atrium to the left ventricle
- B) The deoxygenated blood pass from the right ventricle to the lung
- C) The oxygenated blood pass from the left ventricle to the aorta
- D) The oxygenated blood pass from the right atrium to the right ventricle

**21. which of the following is lymphoid tissue? (2019)**

- A) Liver
- B) Lung
- C) Pancreas
- D) Tonsils

**22. In the pulmonary circulation: (2019)**

- A) The oxygenated blood is pumped from the heart to the body
- B) The deoxygenated blood is pumped from the heart to the lung
- C) The oxygenated blood is pumped from the heart to the lung
- D) The deoxygenated blood is pumped from the heart to the body

**23. the lymph vessel is absent in following organ: (2019)**

- A) Brain
- B) Liver
- C) Lung
- D) Pancreas

**24. As regarding to the spleen which of the following statement is wrong? (2019)**

- A) It destroys worn-out RBCs
- B) It is Largest lymphoid organ
- C) It lies below right cupola of diaphragm
- D) It produces lymphocytes

**25. Regarding medium veins one is wrong: (Dr Rasha)**

- A) They drain venous plexus
- B) Most of them have valves
- C) They accompany medium arteries
- D) Superior and inferior vena cava are medium veins

**26. The main lymphatic vessels in the body: (Dr Rasha)**

- A) Thoracic duct
- B) Right lymphatic duct
- C) Cystic duct
- D) Both (a) and (b )

**27. One of the following arteries supplies the head and neck: (Dr Rasha)**

- A) Arch of aorta
- B) Brachial artery
- C) Common carotid artery
- D) Popliteal artery

**28. With the heart in its normal position in the body: (Dr Rasha)**

- A) The left ventricle forms most of the anterior surface.



- B) The right border is formed by right ventricle.
- C) No part of the left ventricle can be seen on the diaphragmatic surface.
- D) The apex lies in the 5th left intercostals space at midclavicular line

**29. The largest portion of the sternocostal surface of the heart is formed by: (Dr Rasha)**

- A) Left atrium
- B) Right atrium
- C) Right ventricle
- D) Left ventricle

**30. Which group of structures empties directly into the right atrium? (Dr Rasha)**

- A) Superior vena cava, coronary sinus and hemiazygos vein
- B) Coronary sinus and pulmonary vein
- C) Pulmonary and bronchial veins
- D) Superior and inferior venae cavae and coronary sinus

**31. The anterior interventricular artery is accompanied by: (Dr Rasha)**

- A) Anterior cardiac vein.
- B) Great cardiac vein
- C) Coronary sinus
- D) Middle cardiac vein

**32. Which is the largest branch of the right coronary artery? (Dr Rasha)**

- A) The anterior interventricular artery
- B) The nodal artery
- C) The posterior interventricular artery.
- D) The right marginal artery

**33. The vein that accompanies circumflex artery of the heart and ends in right atrium is: (Dr Rasha)**

- A) Great cardiac vein.
- B) Middle cardiac vein.
- C) Coronary sinus.
- D) Anterior cardiac vein

**34. The component of conductive system that is called the pacemaker of the heart (Dr Rasha)**

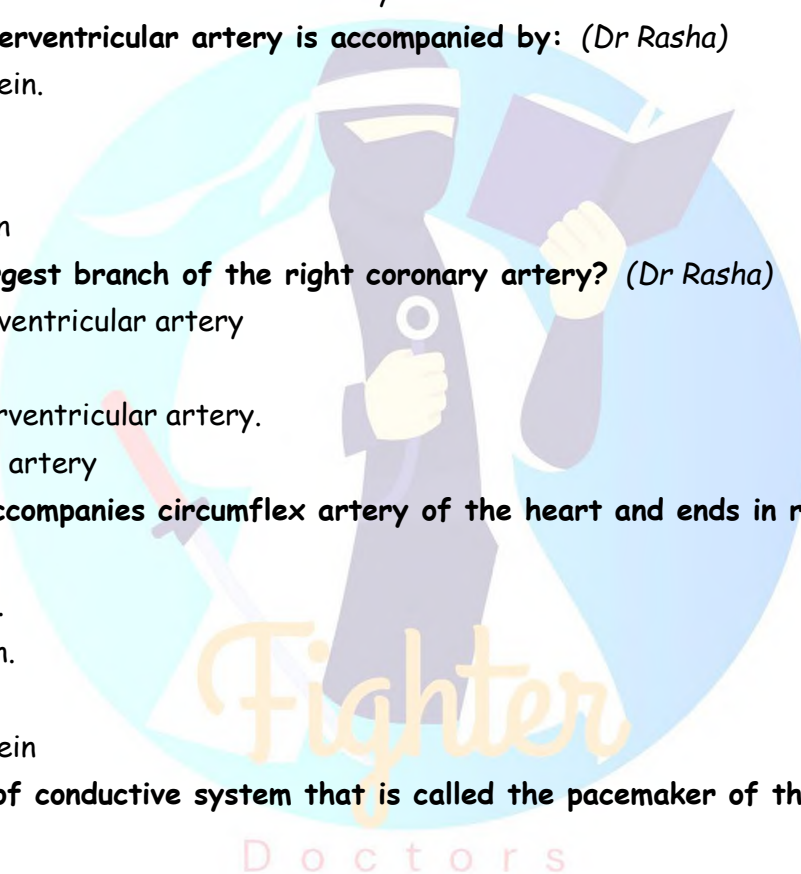
- A) sino-atrial node .
- B) Atrioventricular node.
- C) Atrioventricular bundle.
- D) Purkinje fibers

**35. Chamber of the heart that forms the apex and most of the left border is : (Dr Rasha)**

- A) Right ventricle.
- B) Right atrium.
- C) Left ventricle.
- D) Left atrium .

**36. Chamber of the heart that forms the right border of the heart (Dr Rasha)**

- A) Right ventricle.
- B) Right atrium.
- C) Left ventricle.





D) Left atrium

**37. As regard the right coronary artery, one is false: (Dr Rasha)**

- A) Is branch of the ascending aorta
- B) Is one of blood supply of the heart
- C) Carry deoxygenated blood

**38. Pulmonary valve lies between : (Dr Rasha)**

- A) Right atrium and right ventricle
- B) right ventricle and pulmonary trunk
- C) Left ventricle and pulmonary trunk
- D) None of the above

**39. One of the following has not lymph vessel : (Dr Rasha)**

- A) Heart
- B) brain
- C) stomach
- D) spleen

**40. the main artery of the arm is : (Dr Rasha)**

- A) femoral artery
- B) brachial artery
- C) common carotid artery
- D) dorsalis pedis artery

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-A  | 3-B  | 4-B  | 5-B  | 6-B  | 7-A  | 8-B  | 9-D  | 10-B |
| 11-A | 12-C | 13-B | 14-B | 15-B | 16-B | 17-A | 18-C | 19-C | 20-C |
| 21-D | 22-B | 23-A | 24-C | 25-D | 26-D | 27-C | 28-D | 29-C | 30-D |
| 31-B | 32-D | 33-C | 34-A | 35-C | 36-B | 37-C | 38-B | 39-B | 40-B |

Note on Q23: Fighter key lists "Liver"; kept Brain (lymph vessels are absent in the brain).

**Written questions**

1. Mention the difference between the artery, vein and lymph vessel. [repeated x2]
2. Give a definition for each of the following: Portal vein, Inferior vena cava, Lymphatic capillaries, Pulmonary vein, Right ventricle, Aorta.
3. What do you know about the portal and pulmonary circulation?
4. Give short notes on: Types of arteries
5. Give short notes on: Anatomy of the right atrium of the heart and openings in it.
6. Give short notes on: Mention the types of blood circulation.
7. Give short notes on: Give a short note on the nerve supply and conducting system of the heart.
8. Give a definition of: Artery
9. Give a definition of: lymph
10. The main artery in the arm is ..... While in the thigh is.....
11. Superficial veins of the upper limb includes.....and.....
12. Discuss types of blood circulation and enumerate parts of conducting system of the heart (10marks)



## 6. Nervous System

32 MCQs | 12 written questions

1. Which of the following contains the motor area in the brain? (2026, KNU 2026)

- A) Temporal lobe
- B) Frontal lobe
- C) Parietal lobe
- D) Occipital lobe

2. Which part of spinal nerve is mixed and form the plexuses in the body? (2026, KNU 2026)

- A) Ventral rami
- B) Dorsal root
- C) Ventral root
- D) Dorsal rami

3. Which of the following provides nutrients and acts as a shock absorber in the brain? (2026, KNU 2026)

- A) Cerebrospinal fluid (CSF)
- B) Blood plasma
- C) Synovial fluid
- D) Lymph



4. Which of the following contains the visual area in the brain? (2025)
  - A) Frontal lobe.
  - B) Occipital lobe
  - C) Temporal lobe.
  - D) Parietal lobe
5. Which layer of the meninges is the outermost and toughest?? (2025)
  - A) arachnoid mater
  - B) pia mater
  - C) dura mater.
  - D) subarachnoid space
6. The occipital lobe of the brain contains: (2021, 2020)
  - A) motor area
  - B) sensory area
  - C) auditory area
  - D) visual area
7. The number of cervical segments of the spinal cord are: (KNU 2026)
  - A) Eight segments
  - B) Five segments
  - C) Seven segments
  - D) Twelve segments
8. From outward to inward, the layers of meninges are arranged as: (KNU 2026)
  - A) Pia, arachnoid, dura
  - B) Dura, arachnoid, pia
  - C) Arachnoid, dura, pia
  - D) Pia, dura, arachnoid
9. The two cerebral hemispheres are connected together by: (KNU 2026)
  - A) Cerebellum
  - B) Corpus callosum
  - C) Brain stem
  - D) Spinal cord
10. Which of the following can Maintain posture and balance? (2025)
  - A) frontal lobe
  - B) Mid brain
  - C) Cerebellum
  - D) occipital lobe
11. Which of the following is part of peripheral nervous system? (2024)
  - A) Medulla
  - B) Mid brain
  - C) Cerebellum
  - D) Somatic nerves
12. Which of the followings is correct about the sympathetic innervation? (2022)
  - A) it increases the blood pressure and heart rate
  - B) it is active at the rest



C) it is active during the digestion

D) it is craniosacral in origin

**13. which of the following lobes of the brain contains the auditory area? (2019)**

A) frontal lobe

B) occipital lobe

C) parietal lobe

D) temporal lobe

**14. The frontal lobe of the brain contains..... (2019)**

A) Auditory area

B) Motor area

C) Sensory area

D) Visual area

**15. What is the effect of stimulation of lesser Splanchnic nerve? (2019)**

A) Defecation.

B) Ejaculation.

C) Erection.

D) Micturition.

**16. Which of the following refers to autonomic nervous system? (2019)**

A) Having one efferent neuron.

B) Efferent nerves have either excitatory or inhibitory effects on the effector organs.

C) Postganglionic neurons are located in the lateral horn of spinal cord

D) Preganglionic neurons are located in the anterior horn of the spinal cord.

**17. What is the effect of Generalized sympathetic stimulation? (2019)**

A) Decreases blood sugar levels.

B) Inhibition of sweating.

C) Increase gastric motility.

D) Rise arterial blood pressure.

**18. which of the following lobes of the brain contains the sensory area? (2019)**

A) frontal lobe

B) occipital lobe

C) parietal lobe

D) temporal lobe

**19. Which of the following lobes of the brain contains the visual area? (2019)**

A) Frontal lobe

B) Occipital lobe

C) Parietal lobe

D) Temporal lobe

**20. What is the result of the Sympathetic stimulation ? (2019)**

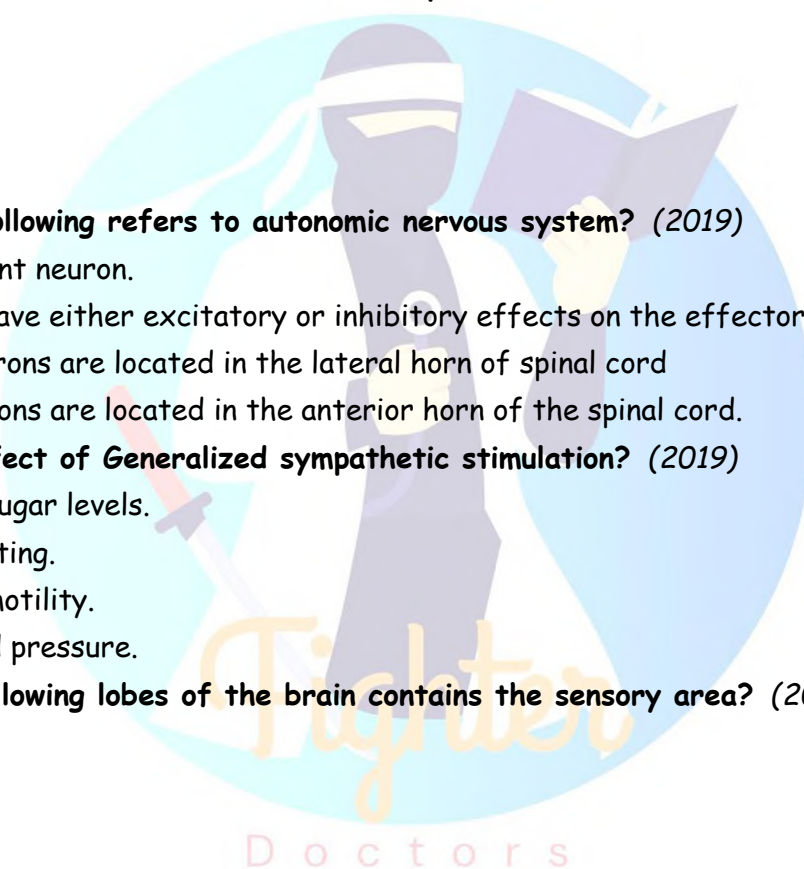
A) Decrease in the heart rate

B) Decrease in the respiratory rate

C) Increase in the adrenaline release

D) Increase the digestion

**21. What is the effect of stimulation of greater splanchnic nerve? (2019)**





- A) Decreased blood volume.  
 B) Increased blood glucose level.  
 C) Increased gastrointestinal motility .  
 D) Secretion of epinephrine 20% & norepinephrine 80% from adrenal medulla.
- 22. Which character of the followings as regard autonomic nervous system ? (2019)**  
 A) Efferent nerves have either excitatory or inhibitory effects on the effector organs.  
 B) Having one efferent neuron.  
 C) Preganglionic neurons are located in the anterior horn of the spinal cord.  
 D) Postganglionic neurons are located in the lateral horn of spinal cord.
- 23. What is the effect of activation of sympathetic system? (2019)**  
 A) Bradycardia.  
 B) Cutaneous vasodilatation.  
 C) Pupillary constriction.  
 D) Sweat secretion.
- 24. Which of the following symptoms appear in Horner's syndrome? (2019)**  
 A) Dilatation of pupil .  
 B) Dropping of the upper eye lid  
 C) Increased sweating.  
 D) Vasoconstriction of skin blood vessels
- 25. Which character belongs to parasympathetic nerves? (2019)**  
 A) Are important in emergency conditions e.g. fight & flight reactions.  
 B) Are catabolic in their actions.  
 C) Generally exert opposite effects to those of the symp .Nerves  
 D) Tend to have longer postgang. fibres than pregang .Fibers.
- 26. Regarding the divisions of the brain, one is wrong: (Dr Rasha)**  
 A) Cerebrum: formed of two cerebral hemispheres  
 B) Diencephalon: formed of thalamus, hypothalamus epithalamus and subthalamus  
 C) Cerebellum: responsible for the muscle coordination and equilibrium  
 D) Brain stem: formed of medulla oblongata and spinal cord
- 27. Regarding the sympathetic system, select one correct statement: (Dr Rasha)**  
 A) It has cranio-sacral outflow.  
 B) It has no distribution to the limbs.  
 C) It causes vasoconstriction of blood vessels.  
 D) It causes slowness of the heart.
- 28. the part of the brain responsible for equilibrium (Dr Rasha)**  
 A) Cerebrum  
 B) Brain stem  
 C) Cerebellum
- 29. the brain stem is formed of (Dr Rasha)**  
 A) Medulla oblongata  
 B) Midbrain, medulla oblongata and pons  
 C) Cerebellum  
 D) Pons and medulla



**30. The optic nerve is responsible for : (Dr Rasha)**

- A) Vision
- B) Olfaction
- C) Hearing
- D) Taste

**31. The spinal cord end at the level of : (Dr Rasha)**

- A) First lumbar vertebrae
- B) fifth lumbar vertebrae
- C) first sacral vertebrae
- D) fifth sacral vertebrae

**32. The spinal nerves are : (Dr Rasha)**

- A) 31 pairs
- B) 31 nerves
- C) 12 pairs
- D) 12 nerves

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-B  | 2-A  | 3-A  | 4-B  | 5-C  | 6-D  | 7-A  | 8-B  | 9-B  | 10-C |
| 11-D | 12-A | 13-D | 14-B | 15-B | 16-B | 17-D | 18-C | 19-B | 20-C |
| 21-B | 22-A | 23-D | 24-B | 25-C | 26-D | 27-C | 28-C | 29-B | 30-A |
| 31-A | 32-A |      |      |      |      |      |      |      |      |

**Written questions**

1. The parts of the brain are ....., ..... And..... [repeated x2]
2. Compare between sympathetic and parasympathetic fibers?
3. Name the parts of the brain and give the function of each.
4. Give short notes on: Mention the length, location, beginning, end , segments and enlargements of the spinal cord
5. Give short notes on: Meninges
6. Give short notes on: Parts of cerebrum and their functions
7. Give short notes on: Parts of brain stem
8. The parts of the nervoys system are ..... and.....
9. The parts of the central nervous system are .....and .....
10. The types of the peripheral nerves are ..... And.....
11. The types of the meninges are ....., ..... And.....
12. Enumerate the parts of the brain and the functions of each one (10 marks)

## 7. Scalp

24 MCQs / 2 written questions

**1. The occipital belly is supplied by which of the following? (2021, 2020)**

- A) Cervical branch of facial nerve
- B) Posterior auricular nerve
- C) Mandibular branch of facial nerve
- D) Occipital branch of facial nerve

**2. Which of the followings is the dangerous area of the scalp? (2022, KNU 2026)**



- A) Connective tissue layer
- B) Periosteum
- C) Skin layer
- D) Subaponeurotic space

**3. An 8-year-old child presents with fever and swelling of the scalp after a minor injury. The physician cautions that the infection may spread to the brain. Through which layer of the scalp can the infection reach the intracranial structures? (2026, KNU 2026)**

- A) Skin of the scalp
- B) Connective tissue layer (dense subcutaneous tissue)
- C) Loose areolar tissue layer
- D) Pericranium

**4. Why is the loose areolar tissue considered dangerous? (2025)**

- A) It is highly vascularized
- B) It contains sweat glands
- C) It allows infection to spread to the brain via emissary veins
- D) It is the thickest layer of the scalp

**5. Which of the following is the dangerous area of the scalp? (2021, 2020)**

- A) The connective tissue layer
- B) The pericranium
- C) Loose areolar tissue layer
- D) Skin layer

**6. The dangerous area of the scalp is: (2019)**

- A) connective tissue layer
- B) Epicranial aponeurosis
- C) Skin layer
- D) Subaponeurotic space

**7. The occipital belly of occipitofrontalis is supplied by: (2019)**

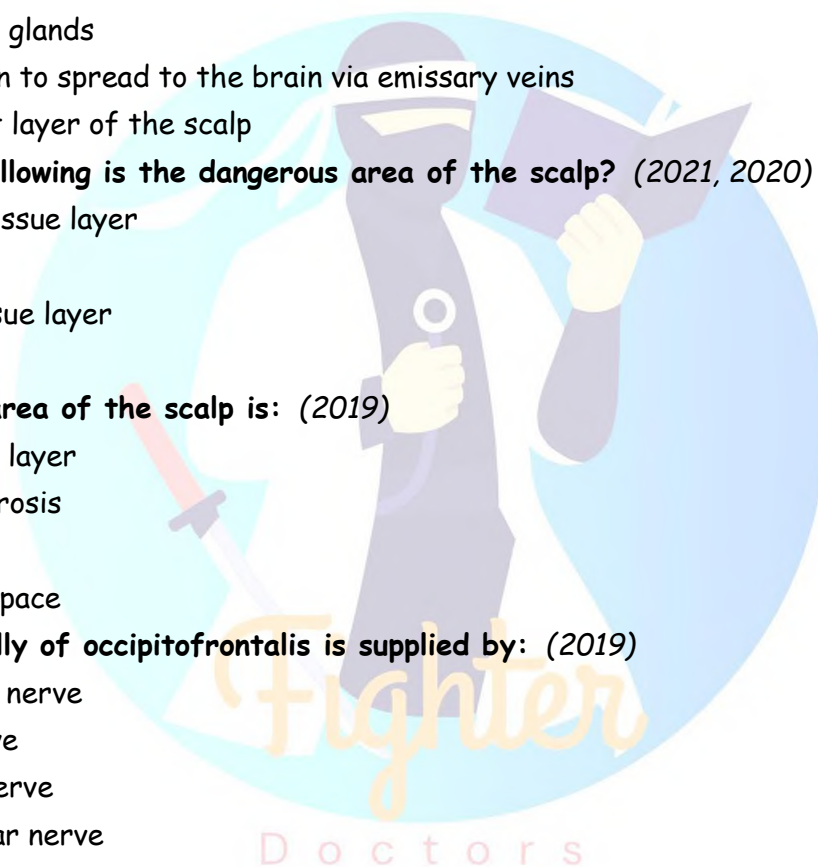
- A) Auriculotemporal nerve
- B) 3rd occipital nerve
- C) Supratrochlear nerve
- D) Posterior auricular nerve

**8. The occipital belly of occipitofrontalis is supplied by which of the following? (KNU 2026)**

- A) Greater occipital nerve
- B) Posterior auricular branch of facial nerve
- C) Supratrochlear nerve
- D) Temporal branch of facial nerve

**9. A 40 years old male patient came to the hospital with bleeding in the scalp region on examination the wound showing a large gap, what is the cause of this gap? (2025)**

- A) cutting of the epicranial aponeurosis
- B) cutting of the second layer of the scalp
- C) cutting of the pericranium
- D) the wound is superficial





10. A 13 years old male patient came to the hospital with bleeding in the scalp region on examination the wound was without gapping, this wound may be in which of the following layers of scalp? (2024)
- first and second
  - third
  - fourth
  - Fifth
11. The frontal belly of occipitofrontalis is supplied by which of the following? (2022)
- Cervical branch of facial nerve
  - posterior auricular nerve
  - mandibular branch of facial nerve
  - temporal branch of facial nerve
12. The occipital belly of occipitofrontalis is supplied by which of the following? (2022)
- Buccal branch of facial nerve.
  - posterior auricular nerve.
  - mandibular branch of facial nerve.
  - temporal branch of facial nerve
13. The lateral part of the scalp is drained into which of the following? (2022)
- Parotid lymph nodes.
  - Mastoid lymph nodes.
  - Occipital lymph nodes.
  - Submandibular lymph nodes
14. Frontal belly of occipitofrontalis muscle is supplied by which of following nerve? (2019)
- Auriculotemporal
  - Temporal branch of facial
  - Zygomatic branch of facial
  - Zygomaticofacial
15. The posterior aspect of the scalp drains into: (2019)
- Mastoid lymph nodes
  - Parotid lymph nodes
  - Submandibular lymph nodes
  - Submental lymph nodes
16. Which one of the following is the dangerous layer of the scalp ? (2019)
- Connective subcutaneous tissue
  - Loose areolar tissue
  - Pericranium .
  - Skin
17. The motor nerve supply of the scalp behind the auricle is : (2019)
- Great auricular nerve
  - Greater occipital nerve
  - Lesser occipital nerve
  - Posterior auricular branch of facial nerve
18. Regarding Scalp, all are true Except: (Dr Rasha)



- A) Its wounds bleed profusely.
- B) Its wounds gape if the aponeurosis is cut.
- C) It has no sebaceous glands.
- D) Its veins communicate with the intracranial venous sinuses.
- E) It contains a muscle of facial expression.

**19. choose one correct statement regarding the scalp: (Dr Rasha)**

- A) Its wounds gape if superficial to aponeurosis.
- B) Its blood vessels and nerves lie in the loose areolar connective tissue layer.
- C) Its wounds do not bleed easily.
- D) Bleeding beneath the periosteum usually extends under its whole extent.
- E) Its infections can spread through emissary veins to cranial cavity.

**20. Regarding the scalp the following statements are true Except: - (Dr Rasha)**

- A) Infection from scalp wounds can spread to superior sagittal sinus.
- B) Its blood vessels run in the loose subaponeurotic space.
- C) Its skin, superficial fascia and aponeurosis move as one layer.
- D) Frontal bellies of occipito-frontalis can corrugate skin of forehead.
- E) Occipital bellies are supplied by facial nerve.

**21. The plane of movement of the scalp is between: (Dr Rasha)**

- A) Skin and epicranial aponeurosis.
- B) Epicranial aponeurosis (galea aponeurotica) and pericranium.
- C) Skin and subcutaneous
- D) Pericranium and skull.
- E) None or the above.

**22. Bleeding beneath the pericranium of the scalp: (Dr Rasha)**

- A) Spreads underneath all of the scalp.
- B) Makes a swelling that takes the shape of the underlying bone.
- C) Reaches to the upper eyelids.
- D) Reaches to the back of neck.
- E) Reaches the extradural space.

**23. which of the following layers of the scalp does infection tend to spread: (Dr Rasha)**

- A) Skin.
- B) Connective tissue layer.
- C) Aponeurotic layer
- D) Loose areolar tissue layer.
- E) Periosteal layer.

**24. The large blood vessels of the scalp found in the: (Dr Rasha)**

- A) Skin.
- B) Subcutaneous tissue.
- C) Epicranial aponeurosis.
- D) Subaponeurotic tissue.
- E) Pericranium.

**Answer Key**

|     |     |     |     |     |     |     |     |     |      |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| 1-B | 2-D | 3-C | 4-C | 5-C | 6-D | 7-D | 8-B | 9-A | 10-A |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|



|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 11-D | 12-B | 13-A | 14-B | 15-A | 16-B | 17-D | 18-C | 19-E | 20-B |
| 21-B | 22-B | 23-D | 24-B |      |      |      |      |      |      |

### Written questions

- Enumerate the layers of the scalp and describe the dangerous area of the scalp.
- Draw a diagram for the venous drainage of the scalp and face.

## 8. Face

28 MCQs / 6 written questions

**1. The anterior facial vein is formed by which of the following? (2023, 2022, 2020)**

- A) Superficial temporal + maxillary veins
- B) Occipital vein + posterior auricular vein
- C) Supra-orbital+ supra-trochlear veins
- D) Anterior facial+ posterior facial veins.

**2. The common facial vein is formed by the union of which veins? (2026, KNU 2026)**

- A) Anterior Facial vein and anterior jugular vein
- B) Anterior Facial vein and anterior division of retromandibular vein
- C) Posterior division of posterior Facial vein and external jugular vein
- D) anterior division of retromandibular vein and posterior auricular vein

**3. A 50-year-old man complains of headache and pain around the eye. Which of the following nerves supply this region? (2026, KNU 2026)**

- A) Maxillary nerve (V2)
- B) Mandibular nerve (V3)
- C) Ophthalmic nerve (V1)
- D) Greater occipital nerve

**4. Which veins unite to form the external jugular vein? (2025)**

- A) Superficial temporal and maxillary veins
- B) anterior division of retromandibular vein and anterior facial vein
- C) posterior division of retromandibular vein and posterior auricular vein
- D) union of posterior auricular with the occipital vein

**5. A 35-year-old female complains of pain in the middle third of her face. Which of the following nerves is responsible for this pain? (2025)**

- A) mandibular
- B) maxillary
- C) ophthalmic
- D) facial

**6. Which of the following closes the eye tightly? (KNU 2026)**

- A) Frontal belly of occipitofrontalis
- B) Lacrimal part of orbicularis oculi
- C) Orbital part of orbicularis oculi
- D) Palpebral part of orbicularis oculi

**7. The retromandibular vein is formed by union of which of the following? (KNU 2026)**

- A) Anterior and posterior facial veins
- B) Common facial and occipital veins



C) Maxillary and superficial temporal veins

D) Supraorbital and supratrochlear veins

**8. The common facial vein is one of veins of the face and scalp, how is formed? (2025)**

A) by the union of superficial temporal and maxillary veins

B) by the union of anterior division of retromandibular vein with anterior facial vein

C) by the union of posterior division of retromandibular vein with posterior auricular vein

D) union of posterior auricular with the occipital vein

**9. A 35-year-old female complains of pain in the upper third of her face. Which of the following nerves is responsible for this pain? (2025)**

A) mandibular

B) maxillary

C) ophthalmic

D) facial

**10. The external jugular vein is one of veins of the face and scalp, how is it formed? (2024)**

A) by the union of superficial temporal and maxillary veins

B) by the union of anterior division of retromandibular vein with anterior facial vein

C) by the union of posterior division of retromandibular vein with posterior auricular vein

D) union of posterior auricular with the occipital vein

**11. The dangerous area of the face is triangular in shape include which of the following? (2024)**

A) nose and upper lip

B) cheek

C) lower lip

D) zygomatic region

**12. Which of the following nerves carry sensory innervation from the lower third of the face? (2024)**

A) Maxillary nerve

B) Ophthalmic nerve

C) Mandibular nerve

D) Facial nerve

**13. All the followings are branches of facial nerve except: (2023)**

A) Mental

B) Cervical

C) Buccal

D) Temporal

**14. The common facial vein drained into which of the followings? (2022)**

A) External jugular vein

B) Internal jugular vein

C) Subclavian vein

D) Superior vena cava

**15. temporal nerve arises from: (2019)**

A) Facial nerve

B) Mandibular nerve



- C) Maxillary nerve
- D) Ophthalmic nerve

**16. What is the action of orbital part of orbicularis oculi? (2019)**

- A) Blinking
- B) Compressing the contents of orbit
- C) Direction of tears to lacrimal sac
- D) Supporting the lower eyelid against gravity

**17. Which of the following lymph nodes drains the central part of the face? (2019)**

- A) Buccal
- B) Submandibular
- C) Preauricular (Parotid)
- D) Submental

**18. The zygomatico temporal nerve arises from: (2019)**

- A) Facial nerve
- B) Mandibular nerve
- C) Maxillary nerve
- D) Ophthalmic nerve

**19. All are true regarding orbicularis oculi Except: (Dr Rasha)**

- A) Orbital part closes eye tightly.
- B) Palpebral part closes eye gently.
- C) Lacrimal part helps flow of tears.
- D) Palpebral part lies in the eyelid.
- E) Orbital part has no bony attachment medially.

**20. All are true regarding Buccinator muscle Except: (Dr Rasha)**

- A) Its middle fibers decussate.
- B) Helps to form orbicularis oris.
- C) Keeps food within the grinding force of the molars.
- D) If paralysed food dribbles from corner of mouth.
- E) Supplied by buccal branch of mandibular nerve.

**21. Buccinator muscle is supplied by which nerve: (Dr Rasha)**

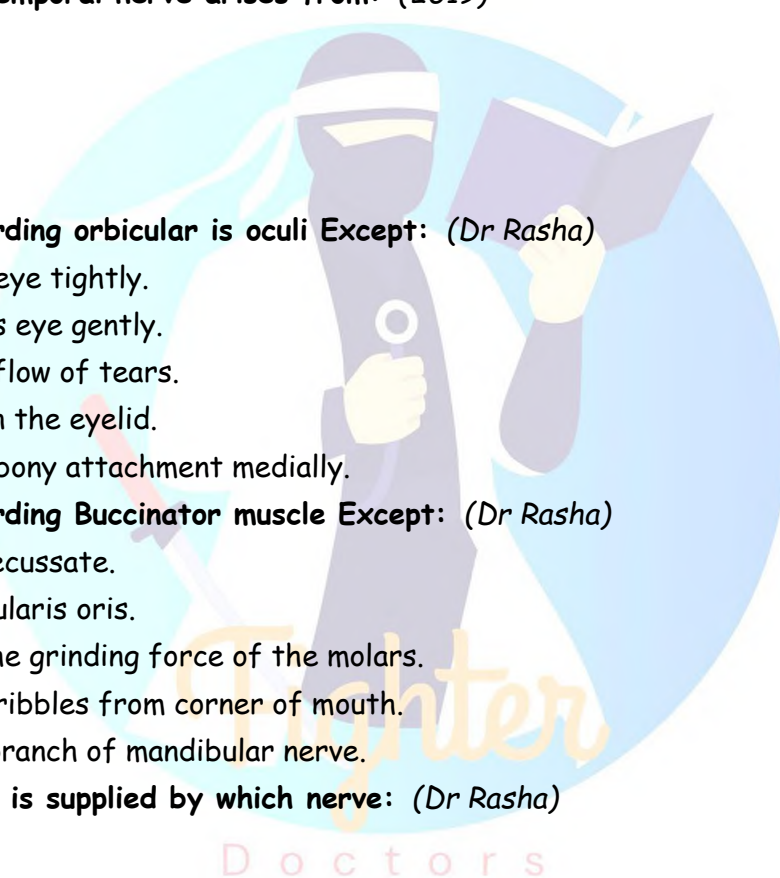
- A) Infraorbital.
- B) Facial.
- C) Mental.
- D) Great auricular.
- E) Auriculotemporal.

**22. Choose one correct combination: of a muscle a muscle and its action: (Dr Rasha)**

- A) Lateral pterygoid ----- opens mouth.
- B) Medial pterygoid ----- depresses mandible.
- C) Masseter ----- retracts mandible.
- D) Mylohyoid ----- elevates mandible.
- E) Temporalis ----- protracts mandible

**23. All are true regarding the upper eyelid Except: (Dr Rasha)**

- A) Is strengthened by the tarsus.





- B) Is lowered by orbicularis oculi.
- C) - Is elevated by levator palpebrae superioris.
- D) Is supplied by sensory fibers from lacrimal nerve medially.
- E) Contains part of the lacrimal gland.

**24. Regarding muscles of facial expression all are true Except: (Dr Rasha)**

- A) Levator labii superioris can elevate lower lip.
- B) Risorius is a dilator of lips.
- C) Procerus produces horizontal wrinkles at root of nose.
- D) frontalis produces horizontal wrinkles on skin on forehead.
- E) Corrugator supercilii produces vertical wrinkles in skin of forehead.

**25. all the following regarding orbicularis oris muscle is correct Except: (Dr Rasha)**

- A) It is a muscle of facial expression.
- B) It is supplied by facial nerve.
- C) Its fibers form complete circles around the mouth.
- D) It can close the oral (labial) fissure.
- E) It helps in speech.

**26. Buccinator's (Dr Rasha)**

- A) Is one of the muscles of mastication.
- B) Is supplied by buccal branch of mandibular nerve.
- C) Arises from both mandible and maxilla.
- D) Is covered by investing layer of fascia.
- E) Accumulates food in vestibule of mouth.

**27. Regarding buccinator's muscle, all the following statements are true Except: (Dr Rasha)**

- A) It arises from both maxilla and mandible.
- B) The pterygomandibular raphe is a common origin for this muscle and the middle constrictor.
- C) Its middle fibers decussate at the modiolus.
- D) It is covered by buccopharyngeal fascia.
- E) It compresses the cheeks against the teeth

**28. Buccinator : - (Dr Rasha)**

- A) Is pierced by parotid duct.
- B) Lies at a more superficial plane than other muscles of facial expression.
- C) Arises from alveolar margin of mandible ,and maxilla only.
- D) Is covered by pharyngobasilar fascia.
- E) Its uppermost fibers decussate.

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-B  | 3-C  | 4-C  | 5-B  | 6-C  | 7-C  | 8-B  | 9-C  | 10-C |
| 11-A | 12-C | 13-A | 14-B | 15-C | 16-B | 17-B | 18-C | 19-E | 20-E |
| 21-B | 22-A | 23-D | 24-A | 25-C | 26-C | 27-B | 28-A |      |      |

## Written questions

- Describe the origin and the nerve supply of the following muscles: a-Buccinator b-Orbicularis oculi c-Sternomastoid [repeated x2]
- Mention the origin, insertion, nerve supply and action of buccinator muscle [repeated x2]



3. Describe with drawing the venous drainage of the face? [repeated x2]
4. Describe the blood supply of the face.
5. Describe the dangerous area of the face and its clinical significance.
6. Describe the dangerous area of the face and explain why it is dangerous?



## 9. Neck

10 MCQs / 3 written questions

1. Which of the following structures is a content of the digastric triangle? (2026, KNU 2026)

- A) Submandibular gland
- B) Parotid gland
- C) Common carotid artery
- D) External jugular vein

2. Which of the following muscles is present in floor of posterior triangle? (2019)

- A) Levator scapulae muscle
- B) Omohyoid muscle
- C) Sternomastoid muscle
- D) Trapezius muscle

3. The posterior triangle is divided into two triangles by which of the following? (KNU 2026)

- A) Inferior belly of omohyoid
- B) Levator scapulae
- C) Sternomastoid muscle



D) Trapezius muscle

4. Which of the following is one of the content of carotid sheath? (2025)

A) sternomastoid muscle

B) vagus nerve

C) external jugular vein

D) facial artery

5. Which of the following is one of the content of carotid sheath? (2025)

A) sternomastoid muscle

B) vagus nerve

C) vagus nerve

D) facial artery

6. Which of the following Form the floor of submental triangle? (2025)

A) sternomastoid muscle

B) Prevertebral Fascia

C) mylohyoid

D) The carotid sheath

7. Which of the following fascia Form the floor of posterior triangle? (2024)

A) Investing layer

B) Prevertebral Fascia

C) Pretracheal Fascia

D) The carotid sheath

8. The sternomastoid is supplied by which of the following nerves? (2022)

A) Accessory nerve

B) Facial nerve

C) Great auricular nerve

D) Trigeminal nerve

9. Where is the insertion of the sternomastoid muscle ? (2019)

A) lateral surface of mastoid process and lateral part of superior nuchal line

B) lateral surface of mastoid process and medial part of superior nuchal line

C) Medial surface of mastoid process and lateral part of superior nuchal line

D) Medial surface of mastoid process and medial part of superior nuchal line

10. The following structure is not a content of the carotid triangle : (2019)

A) Anterior jugular vein

B) Internal carotid artery

C) Internal jugular vein

D) Vagus nerve

**Answer Key**

|     |     |     |     |     |     |     |     |     |      |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| 1-A | 2-A | 3-A | 4-B | 5-B | 6-C | 7-B | 8-A | 9-A | 10-A |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|

**Written questions**

1. List the anatomical contents of the posterior triangle and the carotid triangle of the neck
2. Draw a coloured labelled diagram of the posterior triangle and the carotid triangle of the neck.
3. Enumerate the contents of the carotid triangle.

10. Other Anatomy (Respiratory, Digestive, Urinary, Genital & Endocrine)



6 MCQs / 24 written questions

1. The right lung has : (Dr Rasha)

- A) 8 broncho-pulmonary segments
- B) 2 lobes
- C) 10 broncho-pulmonary segments
- D) One fissure

2. Regarding the prostate gland one is wrong: (Dr Rasha)

- A) It surrounds the proximal part of the urethra
- B) It secretes an alkaline fluid
- C) It lies in the scrotum
- D) The ejaculatory duct traverse it to open in the urethra

3. The organ which produce male sex hormones: (Dr Rasha)

- A) Testis
- B) Seminal vesicle
- C) Prostate
- D) Bulbourethral gland

4. All the followings are parts of biliary system except: (Dr Rasha)

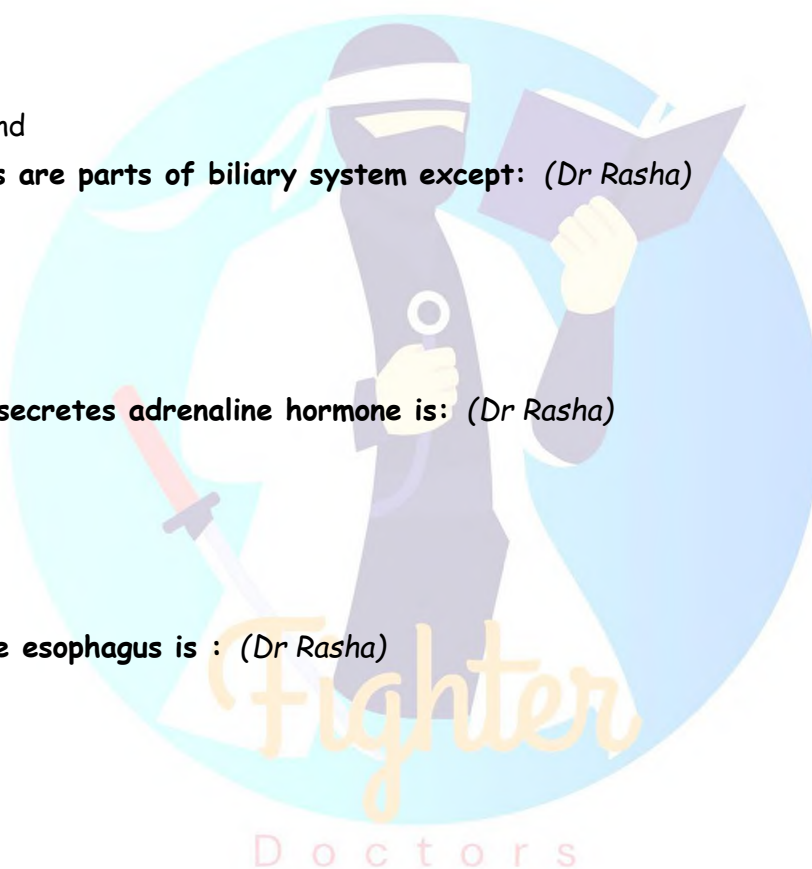
- A) Cystic duct
- B) Common bile duct
- C) Parotid gland
- D) Gall bladder

5. the gland which secretes adrenaline hormone is: (Dr Rasha)

- A) supra renal gland
- B) parathyroid gland
- C) parotid gland
- D) pituitary gland

6. the length of the esophagus is : (Dr Rasha)

- A) 15 cm
- B) 25 cm
- C) 13 cm
- D) 6cm



**Answer Key**

|     |     |     |     |     |     |  |  |  |  |
|-----|-----|-----|-----|-----|-----|--|--|--|--|
| 1-C | 2-C | 3-A | 4-C | 5-A | 6-B |  |  |  |  |
|-----|-----|-----|-----|-----|-----|--|--|--|--|

**Written questions**

1. Give short notes on: The main differences between right and left bronchi [repeated x2]
2. Give short notes on: Structure of kidney
3. Give short notes on: Nephron
4. Give short notes on: The difference between male and female urethra
5. Give short notes on: Fallopian tube
6. Give short notes on: Give short notes about parts, characters and functions of the large intestine?
7. Give short notes on: Discuss the anatomy of the ureter
8. Give short notes on: Give the main differences between the small and large intestine



9. Give short notes on: Write a brief account of the digestive glands
10. Give short notes on: Write a brief account of the Biliary system
11. Give short notes on: Mention the function and the blood supply of the liver
12. Give short notes on: Describe the anatomy of the stomach and its function
13. Give a definition of: Ejaculatory duct
14. Give a definition of: Vestibule of vagina
15. The trachea is divided into.....and.....
16. The urinary system is consists of ....., ....., ..... and.....
17. The testis lies in the ..... as it needs .....
18. The epididymis is formed of .....,..... and.....
19. The uterus is divided into .....,..... and.....
20. Female genital system is formed of .....,.....,..... and .....
21. Give short notes on the kidney and mention the functions of renal system (10 marks)
22. Compare between the followings: a. Male and female urethra (5marks) b. Small intestine and large intestine (5marks)
23. Describe the female internal genital organs (10marks)
24. Mention the site and functions of the followings: (20 marks) a. Pancreas b. Liver c. Pituitary gland d. Thyroid gland e. Prostate gland

## 1. Reproductive Cycle

16 MCQs | 1 written questions

1. If Sarah doesn't conceive, what event marks the end of the luteal phase? (2025)

- A) Ovulation
- B) Menstruation
- C) LH surge
- D) Implantation

2. If Sara's cycle is 38 days, on which day is she most likely to ovulate? (2025)

- A) Day 14
- B) Day 24
- C) Day 28
- D) Day 35

3. What hormone prepares the endometrium for implantation? (2025)

- A) Estrogen
- B) Follicle-stimulating hormone (FSH)
- C) Progesterone
- D) Luteinizing hormone (LH)



4. A 28-year-old woman has regular 28-day cycles. On day 21, serum progesterone is markedly elevated. Which ovarian structure is primarily responsible for this hormone level? (KNU 2026)
  - A) Primary follicle
  - B) Secondary follicle
  - C) Corpus luteum
  - D) Corpus albicans
5. A woman undergoing infertility evaluation shows failure of the LH surge despite adequate estrogen levels. Which phase of the reproductive cycle is most directly affected? (KNU 2026)
  - A) Menstrual phase
  - B) Follicular phase
  - C) Ovulation
  - D) Luteal phase
6. Endometrial biopsy taken on day 23 of the cycle (28 days cycle) shows straight glands. This finding suggests deficiency of which hormone? (KNU 2026)
  - A) Estrogen
  - B) Progesterone
  - C) LH
  - D) FSH
7. Which hormone surges right before ovulation, triggering the release of an egg? (2025)
  - A) Progesterone
  - B) Estrogen
  - C) Luteinizing hormone (LH)
  - D) Follicle-stimulating hormone (FSH)
8. Which hormone is released after implantation and is used to confirm pregnancy? (2025)
  - A) LH (Luteinizing hormone)
  - B) FSH (Follicle-stimulating hormone)
  - C) HCG (Human chorionic Gonadotropin)
  - D) Estrogen
9. Which hormone surges right before ovulation, triggering the release of an egg? (2025)
 

|                 |             |
|-----------------|-------------|
| A) Progesterone | B) Estrogen |
| C) (LH)         | D) (FSH)    |
10. Which hormone is released after implantation and is used to confirm pregnancy? (2025)
 

|                                       |        |
|---------------------------------------|--------|
| A) LH                                 | B) FSH |
| C) HCG (Human chorionic Gonadotropin) |        |
| D) Estrogen                           |        |
11. What hormone prepares the endometrium for implantation? (2025)
 

|                 |          |
|-----------------|----------|
| A) Estrogen     | B) (FSH) |
| C) Progesterone | D) (LH)  |
12. Which hormone is released after implantation and is used to confirm pregnancy? (2025)
 

|             |        |
|-------------|--------|
| A) FSH      | B) LH  |
| C) Estrogen | D) hCG |
13. At the time of ovulation the endometrium is in the: (2024)



- A) Menstrual phase
- B) Early proliferative phase
- C) Late proliferative phase
- D) Late secretory phase

**14. the secretory phase of the uterus is characterized by which of the following (2023)**

- A) it is under the influence of progesterone
- B) it starts from the 5th day till 14th day of the cycle
- C) initiate the formation of a new layer of endometrium in the uterus
- D) Thickness of endometrium in this stage is 0.5 mm

**15. the estrogen is secreted from (2023)**

- A) granulosa cell
- B) corpus luteum
- C) cells of anterior lobe of pituitary gland
- D) endometrial cells

**16. Pregnancy tests generally are based upon the detection in maternal urine of: (2019)**

- A) Estrogen
- B) Human chorionic corticotropin
- C) Human chorionic gonadotrophin
- D) Progesterone

## Answer Key

|      |      |      |      |      |      |     |     |     |      |
|------|------|------|------|------|------|-----|-----|-----|------|
| 1-B  | 2-B  | 3-C  | 4-C  | 5-C  | 6-B  | 7-C | 8-C | 9-C | 10-C |
| 11-C | 12-D | 13-C | 14-A | 15-A | 16-C |     |     |     |      |

## Written questions

1. Discuss the menstrual cycle and its hormonal control

## 2. Gametogenesis

14 MCQs / 2 written questions

**1. The ovulated mammalian oocyte is arrested at: (2024, 2023, Dr Mona)**

- A) prophase of meiosis I
- B) metaphase of meiosis I
- C) prophase of meiosis II
- D) metaphase of meiosis II

**2. Which term describes complete absence of sperm in semen analysis? (2025)**

- A) Azoospermia
- B) Oligospermia
- C) Teratozoospermia
- D) Asthenozoospermia

**3. The time taken for a spermatogonium to develop into a mature spermatozoon is approximately: (2019)**

- A) 64 days
- B) 60 days
- C) 24 days



D) 40 days

**4. Which of the following undergoes differentiation process: (2019)**

A) Primary spermatocyte

B) Secondary spermatocyte

C) Spermatid

D) Spermatogonia

**5. A 25-year-old male presents with infertility. Semen analysis shows azoospermia. Testicular biopsy reveals the presence of Sertoli cells but absence of germ cells. Which stage of gametogenesis is most likely affected? (KNU 2026)**

A) Meiosis I in primary spermatocytes

B) Mitosis of spermatogonia

C) Spermiogenesis

D) Migration of primordial germ cells

**6. Which of the following is a condition characterized by absent sperms? (KNU 2026)**

A) Teratozoospermia

B) Asthenozoospermia

C) Oligospermia

D) Azoospermia

**7. Which of the following processes involves morphological transformation of spermatids into mature sperm? (KNU 2026)**

A) Spermatogenesis

B) Spermiogenesis

C) Spermiation

D) Sperm capacitation

**8. Which term describes a sperm concentration below 15 million/ml? (2025)**

A) Asthenozoospermia

B) Oligospermia

C) Azoospermia

D) Teratozoospermia

**9. The secondary oocyte undergoes completion of the second meiotic division: (2024)**

A) Before ovulation

B) At fertilization

C) Prenatal

D) At puberty

**10. which of the followings is the character of oogenesis (2023)**

A) starts at puberty and extend till old age

B) it takes about 64 days

C) Primary oocyte complete its first meiotic division at puberty

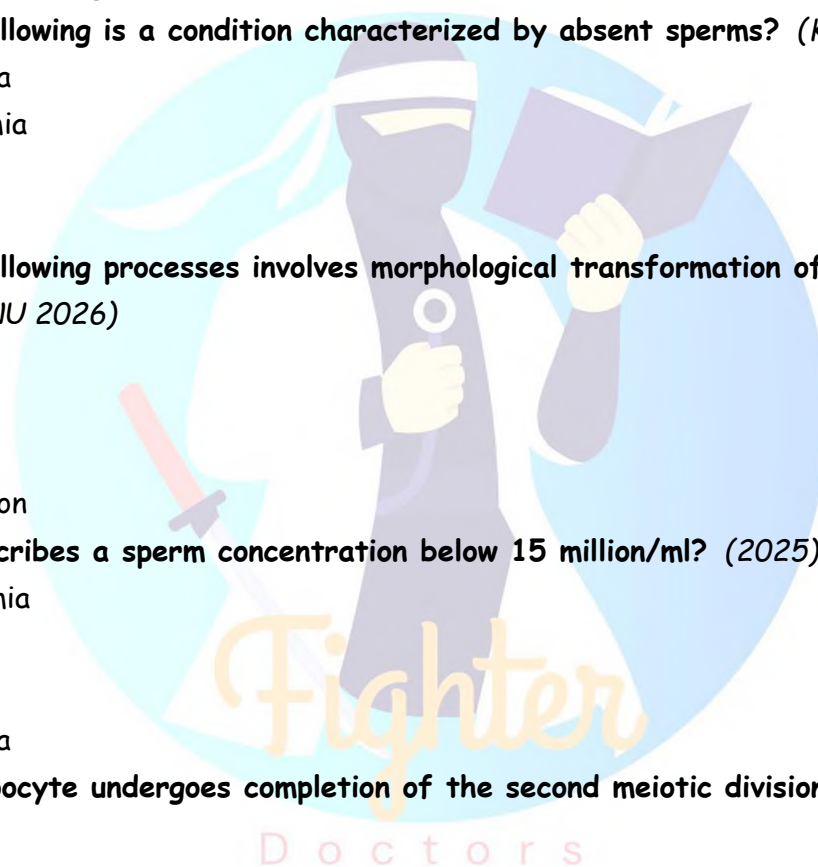
D) Secondary oocyte completes its 2nd meiotic division at the time of ovulation

**11. morphological changes of spherical spermatids to form elongated sperm is called (2023)**

A) spermatogenesis

B) Spermiogenesis

C) acrosomal reaction





12. the spermylin lies in which part of the sperm (2023)

- A) Head
- B) Neck
- C) Middle piece
- D) Tail

13. As regarding Anatomy of the Sperm which one of the following is WRONG? (2019)

- A) Its total length is about 50  $\mu\text{m}$
- B) It consists of: head, neck, middle piece and tail
- C) The head contains a nucleus, a pool of actin
- D) The acrosome is a vesicle contains the genetic material of the sperm

14. the site of storage of sperm for 6 weeks is : (Dr Rasha)

- A) testis
- B) prostate
- C) epididymis
- D) seminal vesicles

## Answer Key

|      |      |      |      |     |     |     |     |     |      |
|------|------|------|------|-----|-----|-----|-----|-----|------|
| 1-D  | 2-A  | 3-A  | 4-C  | 5-D | 6-D | 7-B | 8-B | 9-B | 10-C |
| 11-B | 12-A | 13-D | 14-C |     |     |     |     |     |      |

Note on Q4: Fighter key lists "Spermatogonia"; kept Spermatid (spermiogenesis = differentiation).

## Written questions

1. Describe maturation of spermatozoa
2. Give definition OF gametogenesis

## 3. Fertilization & Cleavage

27 MCQs / 8 written questions

1. What is the primary mechanism that prevents polyspermy after the first sperm enters the egg? (2025)

- A) Cortical reaction
- B) Decidual reaction
- C) Acrosome reaction
- D) Cleavage division

2. On day 5 post-fertilization, an embryo develops a fluid-filled cavity and differentiates into two cell types. Which gives rise to embryo proper? (2026, KNU 2026)

- A) Syncytiotrophoblast
- B) Cytotrophoblast
- C) Trophoblast
- D) Inner cell mass

3. After fertilization, the zygote begins dividing in a process known as cleavage. 16-cell stage is? (2026, KNU 2026)

- A) Blastocyst



- B) Zygote  
C) Morula  
D) Embryoblast
4. During fertilization a sperm secretes enzymes to penetrate the oocyte's outer layer. Which structure prevents polyspermy? (2026, KNU 2026)
- A) Zona pellucida  
B) Cell membrane  
C) Endometrium  
D) Corona radiata
5. A 30-year-old woman undergoes in vitro fertilization (IVF). Eggs are retrieved after hormonal stimulation and are found to be arrested in metaphase II. What triggers the completion of meiosis II in the oocyte? (2026, KNU 2026)
- A) Sperm penetration  
B) LH surge  
C) Estrogen peak  
D) FSH rise
6. All but one of following statements about the cleavage are true. EXCEPT? (2024, 2023)
- A) Cleavage occurs when the embryo travels through the Fallopian tube  
B) During cleavage, the amount of DNA inside the embryo increases  
C) Embryo is closed inside zona pellucida during cleavage  
D) Cleavage continues after implantation
7. The first week of human development is characterized by formation of the: (2023, Dr Mona)
- A) inner cell mass  
B) trophoblast  
C) blastocyst  
D) all of the above
8. What is the most correct sequence of early development following fertilization? (2022, 2019)
- A) zygote, blastomeres, morula, blastocyst  
B) oocyte, zygote, morula, blastocyst  
C) zygote, conceptus, blastocyst  
D) polar bodies, zygote, conceptus, blastocyst
9. Capacitation of the sperm: (2019, Dr Mona)
- A) is caused by the zona pellucida  
B) occurs in the male  
C) prevents polyspermy  
D) is essential for fertilization  
E) removes the head of the sperm
10. A genetic defect prevents cortical granule release after fertilization. Which consequence is most likely? (KNU 2026)
- A) Failure of sperm capacitation  
B) Polyspermy  
C) Failure of the acrosome reaction  
D) Inability of the zygote to implant



**11. Where the fertilization occurs usually? (2024)**

- A) In infundibular part of Fallopian tube
- B) In ampullary part of Fallopian tube
- C) Horn of the uterus
- D) Cavity of the uterus

**12. What is a capacitation? (2024)**

- A) Capacitation is a process which the glycoprotein layer over acrosome is dissolved
- B) Capacitation is a process in which sperm penetrates zona pellucida.
- C) Capacitation is a process in which tail of sperm is inactivated
- D) Capacitation is a process in which tail of sperm is activated

**13. Capacitation means which of the followings (2023)**

- A) adaptation of sperms in female genital tract
- B) Acrosome release proteolytic enzymes such as trypsin like substances.
- C) It is induced by zona pellucida proteins
- D) release of lysosomal enzymes from cortical granules

**14. which of the followings is true about the fertilization (2023)**

- A) it occurs usually in the infundibulum of the fallopian tube
- B) chromosomal sex of the embryo is determined at fertilization.
- C) the oocyte remains viable for 5 days
- D) all the sperms can penetrate the zona pellucida

**15. compaction means which of the followings (2023)**

- A) the blastomere forming a compact ball of cells held together by tight junctions
- B) changes in the zona reaction to prevent further binding and penetration of the sperm
- C) release of the acrosomal enzymes helps penetration of the sperm to the zona pellucida
- D) the ability of the sperm to fertilize the ovum

**16. Where the fertilization occurs usually? (2023)**

- A) In the infundibular part of the Fallopian tube
- B) In the ampullary part of the Fallopian tube
- C) Horn of the uterus (cornua uteri)
- D) Cavity of the uterus

**17. What is a capacitation? (2023)**

- A) Capacitation is a process in which the glycoprotein layer over the acrosome is dissolved
- B) Capacitation is a process in which sperm penetrates zona pellucida
- C) Capacitation is a process in which the tail of the sperm is inactivated
- D) Capacitation is a process in which the tail of the sperm is activated

**18. The early stages of cleavage are characterized by which of the following? (2022)**

- A) formation of a hollow ball of cells
- B) formation of the zona pellucida
- C) increase in the size of the cells in the zygote
- D) increase in the number of cells in the zygote

**19. Where does capacitation of the sperm take place? (2022)**

- A) Epididymis
- B) female genital tract



C) prostatic gland.

D) testis

**20. Where does fertilization of the egg occur? (2022)**

A) Ovary

B) Internal Os of the Uterus

C) Fimbriae

D) Ampulla

**21. Which sentence of the following does NOT true for fertilization? (2019)**

A) It is the union of a human ovum and sperm

B) It is occurring in the isthmus of the fallopian tube.

C) The result of this union is the production of a zygote cell

D) It can occur 12-24 hours after ovulation

**22. Acrosome Reaction is..... (2019)**

A) Alteration of the molecular structure of the glycoproteins in zona pellucida

B) Removal of the surface-coating proteins derived from seminal fluid

C) The digestive acrosomal enzymes at the tip of the sperm's head allow the sperm to burrow through the corona radiata

D) The enzymatic content of acrosome, which can dissolve the glycoproteins in zona pellucida, is thus released.

**23. The part of the sperm that contains proteolytic enzymes to digest the zona pellucida is the: (2019)**

A) Acrosome

B) Capacitator

C) Corona

D) Head

**24. The part of the sperm containing proteolytic enzymes to digest the zona pellucida is the: (Dr Mona)**

A) capacitor

B) head

C) corona

D) acrosome

E) cumulus

**25. Haploid nuclei that fuse at fertilization are called: (Dr Mona)**

A) homunculi

B) mitotic figures

C) centrioles

D) nucleoli

E) pronuclei

**26. The early stages of cleavage are characterized by: (Dr Mona)**

A) formation of a hollow ball of cells

B) formation of the zona pellucida

C) increase in the size of the cells in the zygote

D) increase in the number of cells in the zygote



E) none of the above

27. With the light microscope, the zona pellucida appears as a translucent membrane surrounding the: (Dr Mona)

A) primary oocyte

B) zygote

C) morula

D) very early blastocyst

E) all of the above are correct

#### Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-A  | 2-D  | 3-C  | 4-A  | 5-A  | 6-D  | 7-D  | 8-A  | 9-D  | 10-B |
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| 21-B | 22-D | 23-A | 24-D | 25-E | 26-D | 27-E |      |      |      |

#### Written questions

1. WHAT ARE THE STEPS OF FERTILIZATION [repeated x2]
2. Enumerate the results of fertilization?
3. Define Fertilization
4. Discuss Cleavage of fertilized ovum
5. Give definition to Capacitation
6. Give definition to Compaction in early embryo
7. Steps of fertilization
8. Describe in detail the process of embedding of fertilized ovum

### 4. Second Week

39 MCQs / 6 written questions

1. Where originates the extraembryonic mesoderm? (2025, 2024, 2023)

A) It is a result of the migration of cells from the neuroectoderm

B) In a site between trophoblast and embryoblast

C) In a site between hypoblast and epiblast

D) In a site between endoderm and ectoderm

2. The amniotic cavity develops: (2019, Dr Mona)

A) in extraembryonic mesoderm

B) on the ventral surface of embryonic disc

C) within the inner cell mass near the cytotrophoblast

D) within the outer cell mass

3. At what stage does the fertilized egg typically implant in the uterus? (2025)

A) Zygote

B) Morula

C) Blastocyst

D) Embryo

4. What major event typically occurs at the start of the second week of development? (2025)



- A) Gastrulation
- B) Fertilization
- C) Implantation of the blastocyst
- D) Formation of the neural tube

**5. Which part of the blastocyst directly interacts with the endometrial lining during implantation? (2025)**

- A) Inner cell mass (Embryoblast)
- B) Syncytiotrophoblast
- C) Amniotic cavity
- D) Cytotrophoblast

**6. The most common site for implantation in ectopic pregnancy is: (2024, 2023, Dr Mona)**

- A) internal os of the uterus
- B) mesentery
- C) ovary
- D) uterine tube

**7. Implantation is complete by day 10-12, and the conceptus is fully embedded in the endometrium. Which layer of maternal trophoblast is directly responsible for eroding maternal capillaries to access blood? (2026, KNU 2026)**

- A) Syncytiotrophoblast
- B) Cytotrophoblast
- C) Epiblast
- D) Extraembryonic mesoderm

**8. In an embryo at day 12 post-fertilization lacunar spaces appear within syncytiotrophoblast filled with maternal blood. What is this process critical for? (2026, KNU 2026)**

- A) Neurulation process
- B) Gastrulation process
- C) Uteroplacental circulation
- D) Cleavage process

**9. An embryo shows failure of formation of the amniotic cavity. Which layer of the bilaminar disc is primarily affected? (2026, KNU 2026)**

- A) Hypoblast
- B) Epiblast
- C) Cytotrophoblast
- D) Syncytiotrophoblast

**10. Failure of formation of extraembryonic mesoderm would prevent development of which of the following? (2026, KNU 2026)**

- A) Amniotic cavity
- B) Chorionic cavity
- C) Bilaminar germ disc
- D) Trophoblastic shell

**11. What cavity begins to develop during the end of the second week? (2025)**

- A) Chorionic cavity
- B) Amniotic cavity



C) Yolk sac.

D) Blastocoele

**12. Which layer of the bilaminar disc contributes to the formation of the primitive yolk sac? (2025)**

A) Hypoblast

B) Epiblast

C) Cytotrophoblast.

D) Syncytiotrophoblast

**13. During which week of development does the chorionic cavity form? (2025)**

A) First week

B) Second week

C) Third week

D) Fourth week

**14. Which structure does not take place in the late blastocyst? (2024, 2023)**

A) Zona pellucida

B) Trophoblast

C) Embryoblast

D) Blastocoele

**15. Which statement is true about the trophoblast? (2024, 2023)**

A) It differentiates into outer cytotrophoblast and inner syncytiotrophoblast

B) It differentiates into inner cytotrophoblast and outer syncytiotrophoblast

C) Structure called "trophoblast" does not exist during the development of human

D) It does not differentiate further

**16. When starts the implantation? (2024, 2023)**

A) Immediately after fertilization

B) 3rd day after fertilization

C) 6th day after fertilization

D) 9th day after fertilization

**17. When does Implantation of human embryos typically occurs? (2021, Dr Mona)**

A) about 1 day after fertilization

B) about one week (i.e. 7 days) after fertilization

C) about two weeks (i.e. 14 days) after fertilization

D) after gastrulation

**18. A researcher examines an embryo at day 9 post-fertilization and notes fluid accumulation between the epiblast and trophoblast. The cavity will develop into which structure? (KNU 2026)**

A) Chorionic cavity

B) Amniotic cavity

C) Yolk sac

D) Blastocyst cavity

**19. A patient shows abnormal development of the yolk sac during the second week. Which embryonic layer directly contributes to the formation of the primary yolk sac? (KNU 2026)**

A) Epiblast

B) Hypoblast



C) Cytotrophoblast

D) Extraembryonic mesoderm

**20. Histological examination of an implanted blastocyst shows persistence of the exocoelomic (Heuser's) membrane with inadequate hypoblast proliferation. Which structure will be affected? (KNU 2026)**

A) Amniotic cavity

B) Yolk sac

C) Chorionic cavity

D) Connecting stalk

**21. Which event marks the transition from week 1 to week 2 of human development? (KNU 2026)**

A) Appearance of the primitive streak

B) Completion of implantation

C) Formation of trilaminar disc

D) Closure of neural tube

**22. A teratogenic insult specifically damages the extraembryonic mesoderm during the second week. Which of the following structures is most likely to be malformed? (KNU 2026)**

A) Neural tube

B) Chorionic cavity

C) Epiblast

D) Primitive node

**23. A mutation in the embryo prevents the differentiation and formation of the syncytiotrophoblast. Which of the following stages will be directly blocked? (KNU 2026)**

A) Fertilization

B) Cleavage

C) Implantation

D) Pronuclear formation

**24. What fluid-filled cavity begins to develop within the epiblast during the second week of development? (2025)**

A) Amniotic cavity

B) Blastocoel

C) Yolk sac

D) Chorionic cavity

**25. Which cells form the outer layer of the chorionic vesicle? (2025)**

A) Syncytiotrophoblast

B) Germ cells

C) Epiblast cells

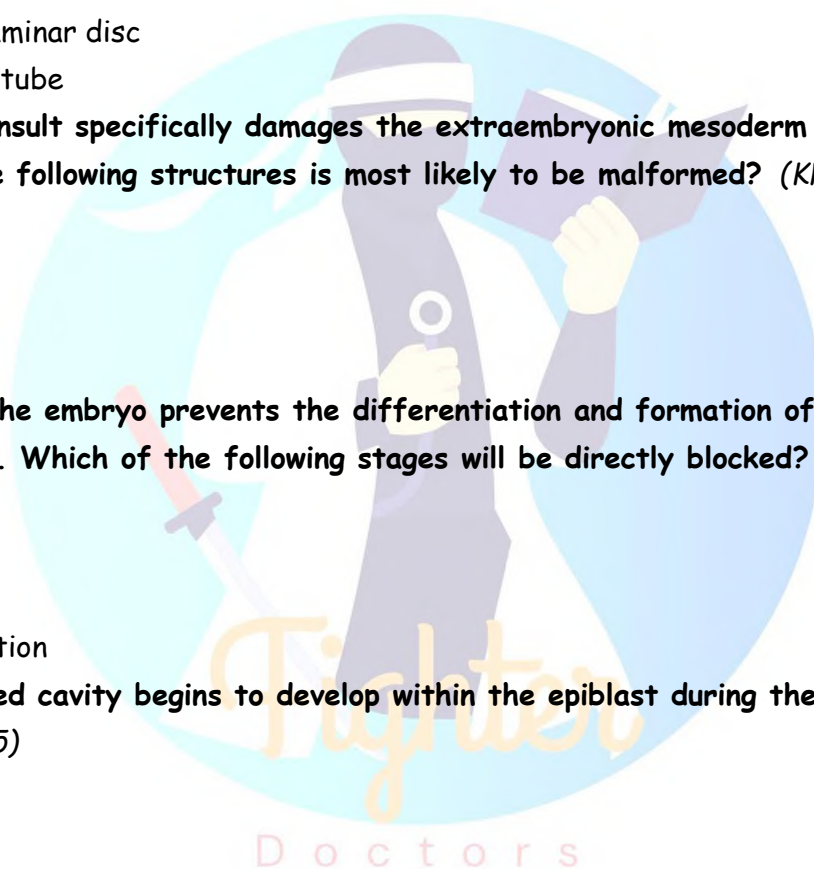
D) Hypoblast cells

**26. During which week of development does the extraembryonic mesoderm primarily form? (2025)**

A) Fourth week

B) First week

C) Third week





D) Second week

**27. first week of human development is characterized by formation of: (2024)**

A) inner cell mass

B) trophoblast

C) blastocyst

D) all of above

**28. At what stage of embryonic development does an embryo normally begin to implant in the endometrium? (2022)**

A) Blastocyst

B) Four-cell stage

C) Morula

D) Trilaminar embryo

**29. The blastocyst is called late blastocyst after which of the following: (2022)**

A) Implantation

B) Fertilization

C) Hatching

D) Cleavage

**30. The cells by which the embryo nourished are which of the following: (2022)**

A) Cytotrophoblast

B) Decidual

C) Amnioblast

D) Blastomeres

**31. After the blastocytic cavity is surrounded by exocoelomic membrane it is called: (2022)**

A) Amniotic cavity

B) Chorionic cavity

C) Primary yolk sac

D) Extraembryonic mesoderm

**32. What implants into the nutrient-rich endometrium (i.e. decidua) about 8-9 days after ovulation? (2022)**

A) Blastocyst

B) Morula

C) Epiblast

D) Hypoblast

**33. Which two embryological structures together form the bilaminar disk? (2022)**

A) Cytotrophoblast and Hypoblast

B) Hypoblast and Epiblast

C) Epiblast and Cytotrophoblast

D) Syncytiotrophoblast and Inner cell mass

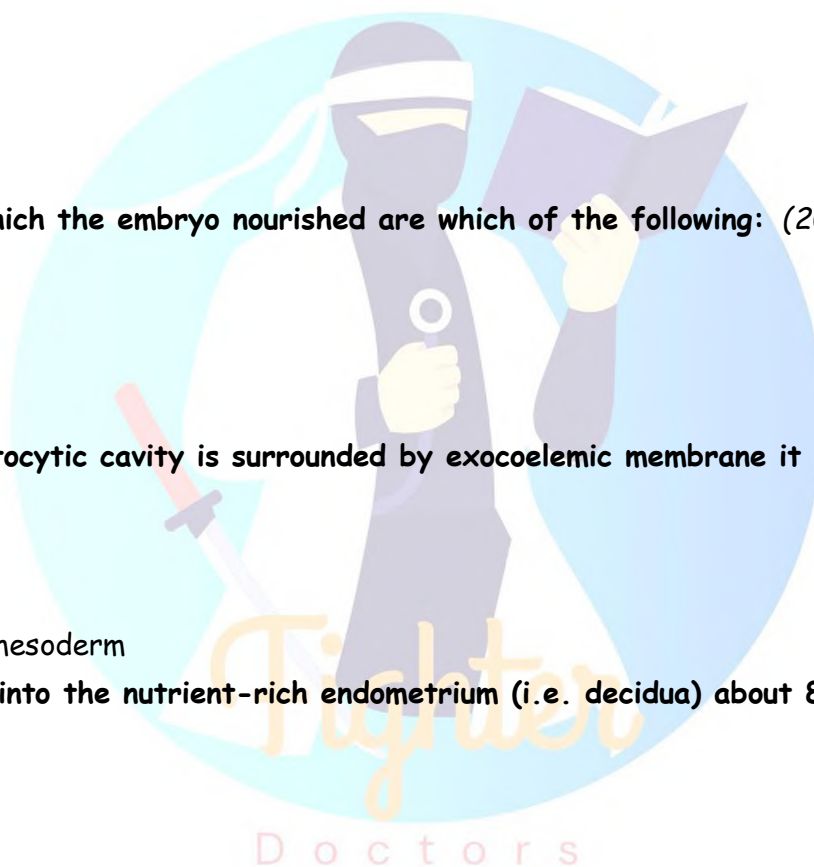
**34. The seven-day blastocyst: (Dr Mona)**

A) has a single layer of trophoblast at the embryonic pole

B) has an amniotic cavity

C) is attached to the endometrial epithelium

D) is surrounded by a degenerating zona pellucida





E) is called the hypoblast

**35. During implantation, the blastocyst:** (Dr Mona)

- A) implants in the endometrium
- B) usually attaches to endometrial epithelium at its embryonic pole
- C) usually implants in the posterior wall of the body of the uterus
- D) causes change in the endometrial tissues
- E) all of the above are correct

**36. During the second week of development, the trophoblast differentiates into:** (Dr Mona)

- A) syncytiotrophoblast
- B) ectoderm
- C) intraembryonic mesoderm
- D) yolk sac (secondary)

**37. The first two intraembryonic germ layers to differentiate are the:** (Dr Mona)

- A) ectoderm and hypoblast
- B) epiblast and hypoblast
- C) ectoderm and endoderm
- D) ectoderm and mesoderm

**38. The blastocoele becomes the:** (Dr Mona)

- A) amniotic cavity
- B) extraembryonic coelom
- C) primary yolk sac
- D) chorionic cavity
- E) secondary cavity

**39. The bilaminar germ disc:** (Dr Mona)

- A) consists of epiblast and mesoblast
- B) is derived from the outer cells of the morula
- C) forms the embryo proper
- D) synthesizes human chorionic gonadotropin, HCG

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
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| 31-C | 32-A | 33-B | 34-C | 35-E | 36-A | 37-B | 38-C | 39-C |      |

## Written questions

1. Mention normal and abnormal sites of implantation?
2. The second week is a period of two. Describe the cavities which appear in the second week of development?
3. Discuss Implantation
4. Discuss Blastocyst formation
5. Define the Bilaminar embryonic disc
6. Ectopic pregnancy



### 5. Third Week

35 MCQs | 9 written questions

1. What is the primary role of the notochord, which forms during gastrulation? (2025)

- A) To serve as a precursor to the heart
- B) To form the body cavity
- C) To induce the formation of the neural tube
- D) To give rise to the digestive tract

2. A 28-year-old woman is in her first trimester of pregnancy and is informed by her doctor that gastrulation is about to occur. What is the primary developmental process that occurs during gastrulation? (2025)

- A) Formation of the neural tube
- B) Establishment of the three germ layers
- C) Development of the heart
- D) Formation of the placenta

3. Which embryonic structure is formed adjacent to the notochord? (2025)

- A) Endoderm
- B) Somites
- C) Amnion



D) Allantois

**4. Cells of primitive streak do not become: (2019, Dr Mona)**

A) Endoderm

B) Intermediate mesoderm

C) Paraxial mesoderm

D) Aminoblasts

**5. Which system is the first organ system to be visually evident in the human embryonic development: (2019)**

A) Circulatory

B) Digestive

C) Musculoskeletal

D) Nervous

**6. A researcher observes an embryo with a visible primitive streak on its epiblast surface.**

**Which of the following best describes the significance of the structure? (2026, KNU 2026)**

A) Initiation of heart tube formation

B) Formation of the yolk sac

C) Origin of the three germ layers

D) Site of neural tube closure

**7. During gastrulation, a subset of epiblast cells moves through the primitive streak and displaces the hypoblast. These cells will become which germ layer? (2026, KNU 2026)**

A) Ectoderm

B) Hypoblast

C) Mesoderm

D) Endoderm

**8. A pathology report describes an embryo with a tumor in the sacrococcygeal region composed of tissues from all three germ layers. What is the embryological origin of this tumor? (2026, KNU 2026)**

A) Abnormal yolk sac differentiation

B) Persistence of primitive streak remnants

C) Neural crest cell migration defect

D) Notochord remnants

**9. A newborn presents with a sacrococcygeal mass containing tissues derived from all three germ layers. The most likely embryological defect is failure of which of the following? (2026, KNU 2026)**

A) Regression of the primitive streak

B) Formation of the primitive node

C) Fusion of the neural folds

D) Differentiation of paraxial mesoderm

**10. An embryo has 7 cervical, 12 thoracic, 5 lumbar, and 3 sacral somites identified. What is the most likely embryonic age? (2026, KNU 2026)**

A) 21 days

B) 23 days

C) 25 days



D) 29 days

**11. At approximately what embryonic age does an embryo typically have about 27 pairs of somites? (2025)**

A) 21 days

B) 29 days

C) 30 days

D) 40 days

**12. Primitive streak originates: (2024, 2023)**

A) In lateral parts of hypoblast

B) In lateral parts of epiblast

C) In the medial line of hypoblast

D) In the medial line of epiblast

**13. How to describe the origin of the intraembryonic mesoderm? (2024, 2023)**

A) Cells from the primitive node migrate between ectoderm and endoderm

B) Cells from the wall of the amnionic sack migrate between ectoderm and endoderm

C) Cells from the wall of the yolk sack migrate between ectoderm and endoderm

D) Cells from primitive streak migrate between ectoderm and endoderm

**14. Which statement about 'formation of somites is correct? (2024, 2023)**

A) Formation of somites is gradual: Cranial somites start, caudal somites continue later on

B) Formation of somites is gradual: Caudal somites start, cranial somites continue later on

C) Timing of formation of somites is less or more chaotic

D) All somites are formed at the same time

**15. Which embryonic structure gives rise to cells in our bodies? (2024, 2023)**

A) epiblast and hypoblast

B) Hypoblast

C) Epiblast

D) Extraembryonic mesoderm

**16. Which of the following structure is the postnatal remnant of the embryonic notochord? (2019)**

A) Annulus fibrosus

B) Nucleus pulposus

C) Rib cage

D) Spinal cord

**17. The intraembryonic coelom is formed within: (2019)**

A) Intermediate mesoderm

B) Lateral plate

C) Neural tube

D) Somites

**18. Somites are developmental structures that contribute the following adult structures: (2019)**

A) kidney, body wall connective tissue, sensory ganglia

B) kidney, gastrointestinal tract smooth muscle, mesentery

C) vertebra, intervertebral discs, dermis, skeletal muscle

D) vertebra, notochord, dermis, skeletal muscle



**19. The primitive streak: (2019, Dr Mona)**

- A) is derived from the outer cells of the morula
- B) is formed during the second week in development
- C) persists as the cloacal membrane
- D) is the site of involution of epiblast cells to form mesoderm

**20. A researcher observes failure of epiblast cell migration through the primitive streak in an early embryo. Which of the following structures will be most immediately affected? (KNU 2026)**

- A) Amniotic cavity
- B) Trilaminar germ disc
- C) Chorionic cavity
- D) Secondary yolk sac

**21. A mutation causes abnormal regression of the primitive streak after gastrulation. Which congenital anomaly is most likely to develop? (KNU 2026)**

- A) Neural tube defect
- B) Sacroccocygeal teratoma
- C) Absent connecting stalk
- D) Absent chorionic cavity

**22. A teratogen disrupts the function of the primitive node (Hensen's node) during week 3. Which structure normally originates from this region? (KNU 2026)**

- A) Neural crest cells
- B) Notochord
- C) Somites
- D) Lateral plate mesoderm

**23. At approximately what embryonic age does an embryo typically have about 30 pairs of somites? (2025)**

- A) 21 days
- B) 27 days
- C) 30 days
- D) 40 days

**24. The embryonic epiblast gives rise to the following: (2024)**

- A) Ectoderm
- B) Mesoderm
- C) Endoderm
- D) All of the above

**25. Somites appear in the third week, they form which of the followins ? (2023)**

- A) Muscles of chest wall
- B) Pleural cavity
- C) Kidney
- D) Bone and cartilage

**26. The process by which the bilaminar disc is transformed into trilaminar disc is called: (2022)**

- A) Neurulation
- B) Gastrulation
- C) Implantation
- D) Folding

**27. Which of the following cells form endoderm and mesoderm? (2022)**

- A) Migrating epiblast cells
- B) Migrating hypoblast cells
- C) Non-migrating epiblast cells
- D) Non-migrating hypoblast cells

**28. What is the MOST LIKELY cause for sacroccocygeal teratoma?: (2021)**

- A) Abnormal migration and proliferation of lumbosacral neural crest.



- B) Abnormal persistence of primitive streak tissue after completion of gastrulation.
- C) Premature regression of the primitive streak during gastrulation.
- D) Abnormal closure of the posterior (caudal) neuropore of neural tube.

**29. What is the primitive streak ? (2019)**

- A) It is derived from the outer cells or the morula
- B) It is formed during the second week in development
- C) It is called the cloacal membrane
- D) It is the site of involution of epiblast cells to form mesoderm

**30. What is correct about the notochordal process ? (2019)**

- A) It arises from involuting endodermal Cells
- B) It extends from the prochordal plate to the primitive node
- C) It is involved in the induction of the primitive gut
- D) It becomes the appendicular skeleton

**31. The primitive streak first appears at the beginning of the \_\_\_\_\_ week. (Dr Mona)**

- A) first
- B) second
- C) third
- D) fourth
- E) fifth

**32. In the third week of human embryonic development: (Dr Mona)**

- A) the amnion appears
- B) a bilaminar embryonic disc is formed
- C) the body stalk moves ventrally and joins with the yolk sac stalk to form the umbilical cord
- D) the neural plate is induced by the notochordal process and associated mesoderm
- E) the uteroplacental circulation is established

**33. During development, the notochordal process: (Dr Mona)**

- A) arises from involuting endodermal cells
- B) extends from the prochordal plate to the primitive node
- C) is involved in the induction of the primitive gut
- D) becomes the appendicular skeleton

**34. Somites: (Dr Mona)**

- A) differentiate into myotomes which give rise to skeletal muscle in trunk and limbs
- B) differentiate into sclerotomes which give rise to vertebrae
- C) arise from segmentation of the paraxial mesoderm
- D) differentiate into myotomes which give rise to skeletal muscle of the limbs
- E) all of the above are correct

**35. The notochord is replaced by the: (Dr Mona)**

- A) ependyma
- B) nucleus pulposus
- C) spinal canal
- D) dorsal roots
- E) spinal cord

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
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|      |      |      |      |      |  |  |  |  |  |
|------|------|------|------|------|--|--|--|--|--|
| 31-C | 32-D | 33-B | 34-E | 35-B |  |  |  |  |  |
|------|------|------|------|------|--|--|--|--|--|

Note on Q7: Fighter key lists "Mesoderm"; kept Endoderm (cells that displace the hypoblast form definitive endoderm).

Note on Q17: Fighter key lists "Intermediate mesoderm"; kept Lateral plate (the intraembryonic coelom forms in lateral plate mesoderm).

Note on Q18: Fighter key lists "notochord" option; kept C (the notochord is not a somite derivative).

Note on Q25: Debatable: both Muscles of chest wall (myotome) and Bone and cartilage (sclerotome) are somite derivatives.

## Written questions

1. Describe the steps of gastrulation [repeated x2]
2. What is the notochord? How is this structure formed? [repeated x2]
3. Enumerate the steps of notochord formation.
4. Describe the steps of notochord.
5. Illustrate differentiation of intraembryonic mesoderm
6. What is gastrulation?
7. What are somites?
8. Discuss Differentiation of the intraembryonic mesoderm
9. Describe the developmental events taking place during the third week of gestation.

## 6. Folding & Neurulation

17 MCQs | 6 written questions

1. Absence of the notochord would primarily interfere with development of which of the following? (2026, KNU 2026)

- A) Epidermis
- B) Neural tube
- C) Amnion
- D) Extraembryonic mesoderm

2. A fetus is diagnosed on ultrasound with absence of the cranial vault and cerebral hemispheres. What is the embryological defect? (2026, KNU 2026)

- A) Neural plate induction
- B) Neural fold elevation and closure
- C) Notochord formation
- D) Neural crest cell differentiation

3. During the 4th week of embryonic development, lateral folding occurs due to rapid growth lateral body folds. Which structure is most affected by lat folding? (2025)

- A) Yolk sac
- B) Heart tube
- C) Septum transversum
- D) Allantois



**4. Which statement about the neural crest is NOT true? (2024, 2023)**

- A) Pia mater can develop from the neural crest
- B) Neural crest gives rise to the cortex of the suprarenal gland
- C) Melanocytes originate from the neural crest
- D) Ganglionic cells originate from neural crest cells

**5. A specimen shows a normal trilaminar disc, but absence of proper notochord formation.**

**Which developmental event will be most severely impaired? (KNU 2026)**

- A) Somitogenesis
- B) Neural plate induction
- C) Formation of lateral plate mesoderm
- D) Formation of intermediate mesoderm

**6. A severe embryonic malformation shows failure of neural tube closure with abnormal ventral body wall formation. The most likely underlying issue is disruption of: (KNU 2026)**

- A) Ectoderm differentiation only
- B) Mesoderm differentiation only
- C) Coordinated neurulation and embryonic folding
- D) Endodermal differentiation

**7. During the 4th week of embryonic development, cranial folding occurs due to rapid growth of the neural tube and somites. This folding shifts the developing heart tube into the thoracic region. Which structure is most affected by cranial folding? (2025)**

- A) Yolk sac
- B) Amniotic cavity
- C) Septum transversum
- D) Allantois

**8. Which of the following is derived from neural crest cells: (2024)**

- A) Dura mater
- B) Stomach
- C) Urinary bladder.
- D) Adrenal medulla

**9. The germ layer that gives origin to the central nervous system is: (2024)**

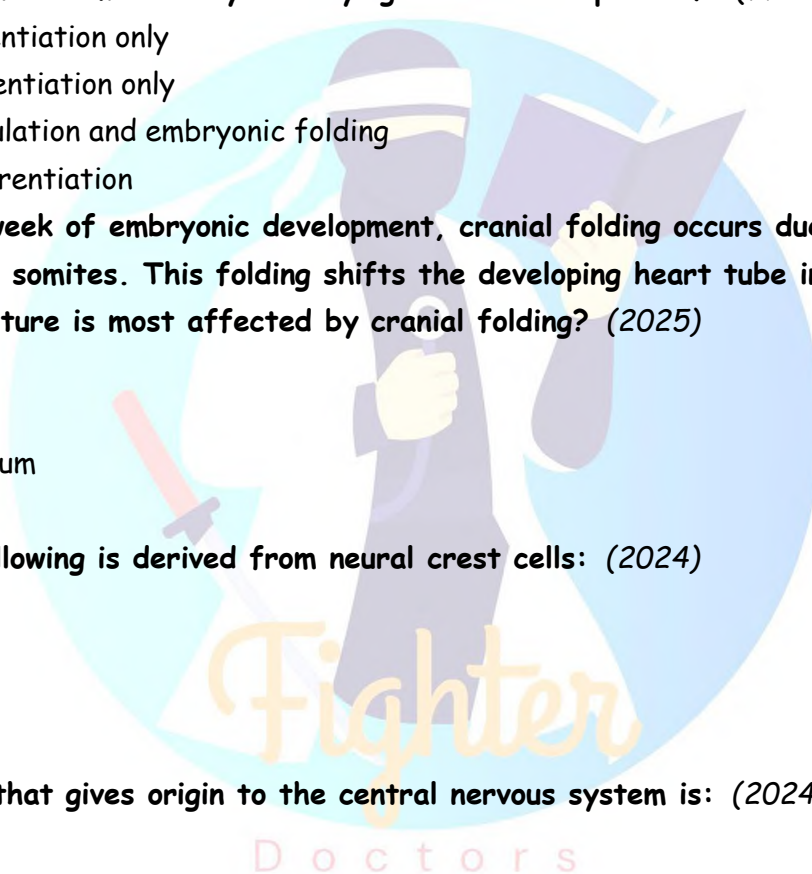
- A) Ectoderm
- B) Mesoderm
- C) Endoderm
- D) Neural crest

**10. Which of the followings is true about the neural tube formation? (2023)**

- A) It appears in the 4th week of development.
- B) It is endodermal in origin.
- C) It forms the central nervous system.
- D) Its caudal end divided into three vesicles

**11. Which of the followings is not formed from neural crest cells: (2022)**

- A) Central nervous system
- B) Pigment cells
- C) Adrenal gland medulla





D) Peripheral ganglia

**12. neural tube and plate forms from: (2022)**

A) Intermediate mesoderm

B) Paraxial mesoderm

C) Neural crest cells

D) Neuroectoderm

**13. Which of the following cavities expands and surrounds the embryo as a result of folding? (2021)**

A) Amniotic cavity

B) Yolk sac

C) Chorionic cavity

D) Blastocystic cavity

**14. Which of the following structures is derived from neural crest cells? (2020)**

A) Dorsal root ganglion

B) Dura mater

C) Epidermis covering the back

D) Intervertebral disk

**15. What is the sequence of the development of the central nervous system? (2020)**

A) neural plate to neural crest to neural zone

B) neural plate to neural groove to neural tube

C) neural tube to neural plate to neural groove

D) notochord to neural groove to neural tube

**16. What is the incidence of abnormality the folate supplementation used to lower? (2020)**

A) abortion

B) cretinism

C) Ectopic pregnancy

D) neural tube defects

**17. The central nervous system forms in the sequence: (2019)**

A) Neural plate to neural crest to neural zone

B) Neural plate to neural groove to neural tube

C) Neural tube to neural plate to neural groove

D) Notochord to neural plate to neural tube

**Answer Key**

|      |      |      |      |      |      |      |     |     |      |
|------|------|------|------|------|------|------|-----|-----|------|
| 1-B  | 2-B  | 3-A  | 4-B  | 5-B  | 6-C  | 7-C  | 8-D | 9-A | 10-C |
| 11-A | 12-D | 13-A | 14-A | 15-B | 16-D | 17-B |     |     |      |

**Written questions**

- How does the embryo turn into the neurula from gastrula? [repeated x2]
- Enumerate the results of embryonic folding.
- Describe neurulation, definition and steps.
- How is the neural tube formed?
- What is the embryonic origin of the nervous system in human embryo?
- Derivatives of Neural crest cells



## 7. Placenta, Decidua & Chorion

27 MCQs | 8 written questions

1. By the 4th month the placental barrier is formed of which of the following: (2025, 2024)

- A) Syncytiotrophoblast and endothelium
- B) Cytotrophoblast and syncytiotrophoblast
- C) Cytotrophoblast and endothelium
- D) Endothelium and extraembryonic mesoderm

2. The wall of the chorionic sac is composed of: (2024, 2022, 2019, Dr Mona)

- A) cytotrophoblast and syncytiotrophoblast
- B) two layers of trophoblast lined by extraembryonic somatic mesoderm
- C) trophoblast and exocoelomic membrane
- D) extraembryonic splanchnic mesoderm and both layers of trophoblast

3. Which of the following is not a component of the mature placental barrier: (2019, Dr Mona)

- A) The basement membrane of the fetal capillaries
- B) The cytotrophoblast
- C) The endothelial lining of fetal capillaries
- D) The syncytiotrophoblast

4. A placenta later shows placenta previa. The abnormality most likely originated from implantation of the blastocyst in which of the followings? (2026, KNU 2026)

- A) Fundus of uterus
- B) Posterior uterine wall
- C) Lower uterine segment
- D) Myometrium

5. Which cells form the inner layer of the chorion? (2025)

- A) Epiblast cells
- B) Syncytiotrophoblast
- C) Hypoblast cells
- D) Primary mesoderm

6. Which structure is responsible for the establishment of uteroplacental circulation? (2025)

- A) Blastocyst
- B) Amnion
- C) Yolk sac
- D) Syncytiotrophoblast

7. on examination, you noticed that the the placenta extended into urinary bladder. What is the possible cause of this case (2025)

- A) failure of formation of cytotrophoblastic shell
- B) Changes in trophoblast
- C) Congenital anomalies in the heart
- D) Placental insufficiency

8. When chorionic villi become vascularized they are called \_\_\_\_\_ villi: (2024, Dr Mona)

- A) branch
- B) stem
- C) tertiary



D) anchoring

**9. What is the site of blastocyst implantation in placenta previa? (2023, 2022)**

A) Ovary

B) rectouterine pouch

C) the tube

D) uterine cavity

**10. When the chorionic villi are called secondary chorionic villi? (2019)**

A) Contact the decidua basalis

B) Develop a mesenchymal core

C) Give rise to branch villi

D) Are covered by syncytiotrophoblast

**11. What is region of the decidua which is essential for nourishment of fetus? (2019)**

A) Basalis

B) Capsularis

C) Laeve

D) Parietalis

**12. A 32-year-old woman with a history of two previous cesarean sections presents with severe postpartum hemorrhage. Histology shows absence of decidua basalis with direct attachment of chorionic villi to the myometrium. The most likely diagnosis is: (KNU 2026)**

A) Placenta previa

B) Placenta accreta

C) Placenta increta

D) Placenta percreta

**13. Which component of the placenta is derived from extra-embryonic mesoderm? (KNU 2026)**

A) Syncytiotrophoblast

B) Cytotrophoblast

C) Chorionic villous core

D) Decidua basalis

**14. Post-delivery bleeding occurs despite apparent complete placental expulsion. Ultrasound reveals retained placental tissue. This complication is most commonly associated with: (KNU 2026)**

A) Placenta circumvallata

B) Placenta succenturiata

C) Placenta membranacea

D) Placenta fenestrata

**15. Which layer is lost by late pregnancy, significantly thinning the placental barrier? (KNU 2026)**

A) Endothelium of fetal capillaries

B) Syncytiotrophoblast

C) Cytotrophoblast

D) Basement membrane

**16. Which structure is responsible for the establishment of the fetoplacental circulation during implantation? (2025)**



- A) Blastocyst
- B) Syncytiotrophoblast
- C) Yolk sac
- D) Amnion

**17. Which part of decidua share in formation of the placenta? (2022)**

- A) chorion leave
- B) decidua basalis
- C) decidua capsularis
- D) decidua parietalis

**18. Placental barrier at the second half of pregnancy (from 4th month onward), consists of which of the following layers? (2022)**

- A) The Iry mesodermal core of the villus and cytotrophoblast
- B) The Iry mesodermal core of the villus and endothelium of fetal vessels
- C) The cytotrophoblast and syncytiotrophoblast
- D) The endothelial lining of fetal vessels and syncytiotrophoblast

**19. The placenta is part of the chorion. The chorion consists of which of the following? (2022)**

- A) Extraembryonic mesoderm
- B) Trophoblast
- C) Hypoblast
- D) A & B

**20. Placenta previa means implantation in which of the following sites? (2021)**

- A) Abdominal cavity
- B) Uterine tube
- C) Ovary
- D) Near the cervix

**21. The most distinctive character of a primary chorionic villus is its: (2019)**

- A) Cytotrophoblastic core
- B) Cytotrophoblastic shell
- C) Extraembryonic visceral mesodermal core
- D) Extraembryonic somatic mesodermal core

**22. The most distinctive characteristic of a primary chorionic villus is its: (Dr Mona)**

- A) cytotrophoblastic shell
- B) extraembryonic somatic mesodermal core
- C) bushy appearance
- D) cytotrophoblastic core

**23. Chorionic villi are designated as secondary chorionic villi when they: (Dr Mona)**

- A) contact the decidua basalis
- B) are covered by syncytiotrophoblast
- C) develop a mesenchymal core
- D) give rise to branch villi
- E) none of the above

**24. The most important region of the decidua for the nourishment of the conceptus is the decidua (Dr Mona)**



- A) frondosum
- B) capsularis
- C) parietalis
- D) basalis
- E) laeve

**25. The intervillous space contains all of the following substances EXCEPT: (Dr Mona)**

- A) oxygen
- B) carbon dioxide
- C) maternal blood cells
- D) fetal blood
- E) electrolytes

**26. The portion of the decidua which does not survive until the end of pregnancy is the: (Dr Mona)**

- A) capsularis
- B) basalis
- C) laeve
- D) parietalis
- E) frondosum

**27. During the development of the placenta: (Dr Mona)**

- A) cytotrophoblast cells divide and their progeny fuse to form syncytiotrophoblast cells.
- B) syncytiotrophoblast cells invade the uterine wall.
- C) fetal blood vessels attach directly to maternal vessels such that there is mixing of fetal and maternal blood.
- D) the placenta develops from the chorionic plate.
- E) most of the placenta is derived from embryonic intermediate mesoderm.
- F) NONE of the above

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-A  | 2-B  | 3-B  | 4-C  | 5-D  | 6-D  | 7-A  | 8-C  | 9-D  | 10-B |
| 11-A | 12-B | 13-C | 14-B | 15-C | 16-B | 17-B | 18-D | 19-D | 20-D |
| 21-A | 22-D | 23-C | 24-D | 25-D | 26-A | 27-A |      |      |      |

## Written questions

1. Explain the congenital anomalies of placenta [repeated x3]
2. A patient experiences severe postpartum hemorrhage due to placenta accreta. Discuss the normal development of chorionic villi and decidua basalis, and explain embryological base of placenta accreta. : [repeated x2]
3. compare between chorionic villi as regarding their structure [repeated x2]
4. Compare between different types of chorionic villi.
5. Discuss placental function
6. Describe human placental structure
7. Anomalies of placenta
8. Placental barrier



## 8. Fetal Membranes

14 MCQs | 6 written questions

1. An embryo is found attached to the chorionic plate by a thick band of tissue. This structure will later develop which of the followings? (2026, KNU 2026)

- A) Allantois
- B) Vitelline duct
- C) Umbilical cord
- D) Amnion

2. On examination, you noticed that the amount of amniotic fluid in a pregnant female is more than 2 litres. What is the possible cause of this case? (2025, 2024)

- A) Oesophageal atresia
- B) Congenital anomalies in the heart
- C) Placental insufficiency
- D) Changes in trophoblast

3. Which event forms the definitive yolk sac? (2025)

- A) Gastrulation
- B) Neurulation
- C) Folding
- D) Fertilization

4. A 22-year-old pregnant woman comes for her first-trimester ultrasound at 6 weeks of gestation. The scan shows a gestational sac with an embryo and a well-defined yolk sac. The yolk sac is connected to the midgut of the embryo by which structure? (KNU 2026)

- A) Allantois
- B) Vitello-intestinal duct
- C) Umbilical vein
- D) Chorionic villi

5. A 30-year-old pregnant woman at 28 weeks of gestation complains of rapid abdominal enlargement and discomfort. On ultrasound, the fetus is seen floating freely with an excessive amount of amniotic fluid. Which fetal abnormality is commonly associated with this condition? (KNU 2026)

- A) Renal agenesis
- B) Esophageal atresia
- C) Potter sequence
- D) Pulmonary hypoplasia

6. As regarding the urinary bladder development, the apex of the bladder is formed by: (KNU 2026)

- A) Mesonephric duct
- B) Metanephric cap
- C) Allantois
- D) Primitive urogenital sinus

7. When amount of amniotic fluid is less than half liter, condition is called? (2023)

- A) oligohydramnios

B) polyhydramnios or hydramnios

C) amnionitis

D) bag of water

**8. When the amount of amniotic fluid exceeds two liters, the condition is called: (2022)**

A) oligohydramnios

B) polyhydramnios

C) amnionitis

D) bag of waters hydrogravidia

**9. Secondary yolk sac can be seen in which of the following stages? (2022)**

A) after folding

B) blastocyst stage

C) chorionic vesicle stage

D) morula stage

**10. connecting stalk that attaches the early embryo to placenta is precursor to the: (2019)**

A) Umbilical cord

B) Chorionic villi

C) Yolk sac

D) Umbilical vein

**11. When the amount of amniotic fluid exceeds two liters, the condition is called: (Dr Mona)**

A) oligohydramnios

B) polyhydramnios or hydramnios

C) amniotitis

D) bag of waters

E) hydrogravidia

*The following three questions refer to the image below. For each of these questions, choose the SINGLE BEST answer.*

**12. The label indicating a space containing fluid that is produced by the kidneys in the developing fetus is (Dr**

**Mona)**

A) A

B) B

C) C

D) D

**13. The label indicating a space lined by endoderm is: (Dr Mona)**

A) A

B) B

C) C

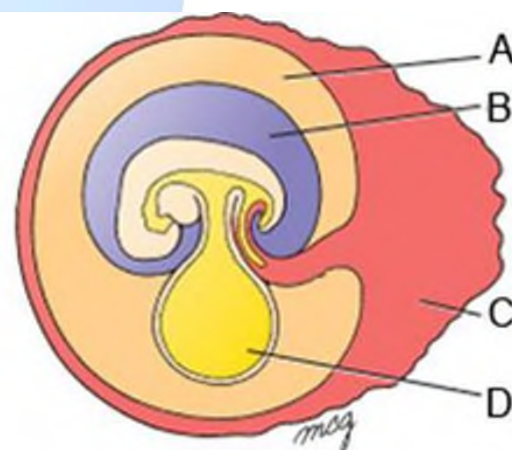
D) D

**14. The label indicating the amniotic cavity is: (Dr Mona)**

A) A

B) B

C) C





D) D

**Answer Key**

|      |      |      |      |     |     |     |     |     |      |
|------|------|------|------|-----|-----|-----|-----|-----|------|
| 1-C  | 2-A  | 3-C  | 4-B  | 5-B | 6-C | 7-A | 8-B | 9-C | 10-A |
| 11-B | 12-B | 13-D | 14-B |     |     |     |     |     |      |

**Written questions**

1. A fetus develops unexplained intrauterine growth restriction, and delivery reveals a true knot in the umbilical cord. Discuss the formation and growth of the umbilical cord:... [repeated x2]
2. A 26-year-old primigravida at 20 weeks of gestation undergoes a routine antenatal ultrasound. The scan reveals a single umbilical artery and mild intrauterine growth restriction. Give a short note about the normal development of the umbilical cord, its contents, and associated congenital anomalies.
3. Describe the congenital anomalies of the yolk sac.
4. Mention the congenital anomalies of umbilical cord.
5. Give short note on umbilical cord development and its anomalies?
6. Describe the development umbilical cord in mammals

**9. Twinning**

0 MCQs / 1 written questions

**Written questions**

1. Describe the types and congenital anomalies of twins. [repeated x2]

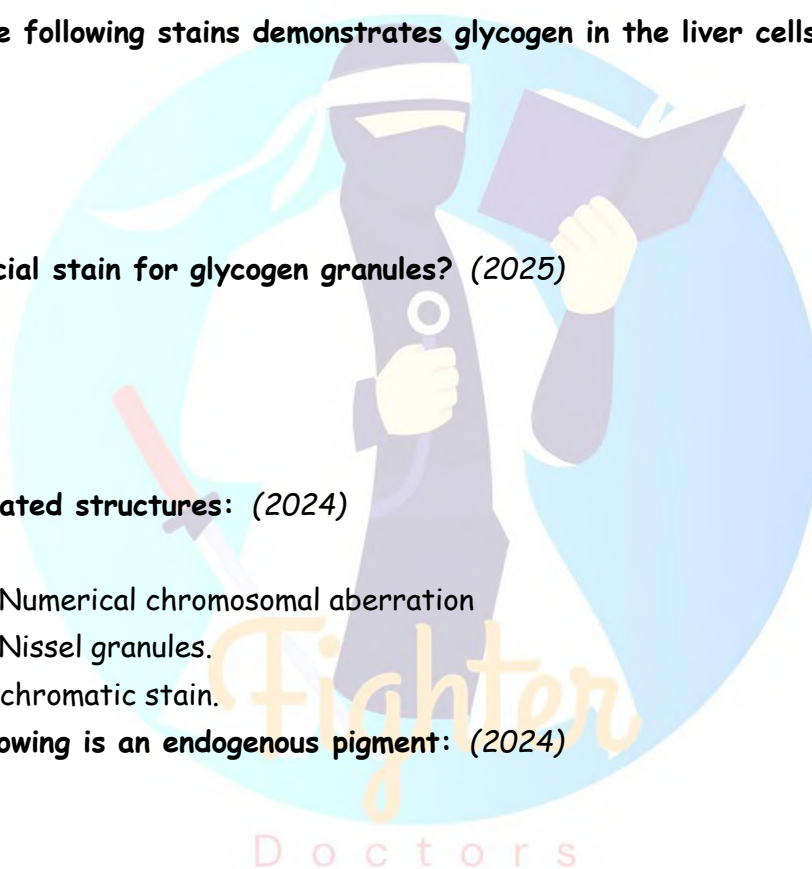
**Histology****1. Histology Introduction**

13 MCQs / 2 written questions

1. Which of the following best suited to visualize lipids: (2022, 2019)
  - A) H&E
  - B) silver
  - C) Mallory
  - D) sudan III
2. What is the typical description of glycogen granules? (2026, KNU 2026)
  - A) Alpha glycogen forms rosette aggregation in the liver
  - B) Appears red with H&E stain
  - C) Can be stained with osmic acid
  - D) Beta glycogen forms rosette aggregation in the liver
3. Which of the following is true regarding stains? (2025)
  - A) Collagen fibers → Pink with H & E.
  - B) Mast cells → Blue with Toluidine blue.
  - C) Mitochondria → Blue with Janus green.
  - D) Reticular fibers → Red with silver.
4. Which one is regarded as an exogenous pigment? (2025)
  - A) Haemoglobin



- B) Melanin  
C) Carotene  
D) Lipofuscin
- 5. Which one of the following stains demonstrates lipid in the liver cells? (2025)**  
A) Sudan III  
B) PAS  
C) Silver  
D) trypan blue
- 6. Which of the following would be best suited to visualize lipid? (2021, Fighter Histology bank)**  
A) Hematoxylin and eosin stain  
B) Sudan stain  
C) Silver impregnation  
D) Masson's trichrome stain
- 7. Which one of the following stains demonstrates glycogen in the liver cells? (2025)**  
A) Sudan III  
B) PAS  
C) Silver  
D) Sudan black
- 8. What is the special stain for glycogen granules? (2025)**  
A) Sudan III  
B) Silver  
C) Iron HX  
D) Best's Carmine
- 9. Choose the unrelated structures: (2024)**  
A) Cilia / Centriole  
B) Down syndrome / Numerical chromosomal aberration  
C) Nerve cell axon / Nissel granules.  
D) Mast cells / Metachromatic stain.
- 10. One of the following is an endogenous pigment: (2024)**  
A) Lead  
B) Carotene  
C) Tattoo  
D) Hemoglobin
- 11. Which one is an endogenous pigment? (2019)**  
A) Lead  
B) Carotenes  
C) Tattoo  
D) Hemoglobin
- 12. All of the below are exogenous pigments expect (Fighter Histology bank)**  
A) Carotene  
B) Carbon  
C) Lipofuscin  
D) All of the above are exogenous pigments





13. Fat can be demonstrated by using (Fighter Histology bank)

- A) Best's carmine
- B) PAS
- C) Sudan black stain
- D) None of the above

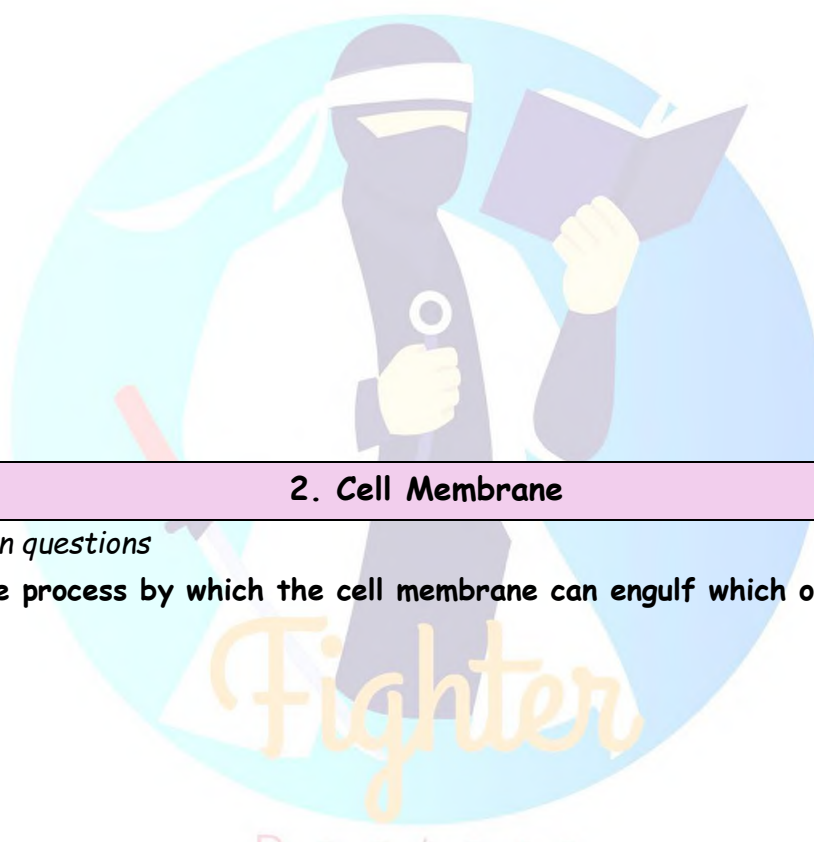
**Answer Key**

|      |      |      |     |     |     |     |     |     |      |
|------|------|------|-----|-----|-----|-----|-----|-----|------|
| 1-D  | 2-A  | 3-A  | 4-C | 5-A | 6-B | 7-B | 8-D | 9-C | 10-D |
| 11-D | 12-C | 13-C |     |     |     |     |     |     |      |

Note on Q3: Debatable: Janus green (mitochondria) also true.

**Written questions**

1. Give a comment on :- A) Cell inclusions :
2. (A) Terminology Structure means



## 2. Cell Membrane

15 MCQs | 1 written questions

1. Pinocytosis is the process by which the cell membrane can engulf which of the following? (2025)

- A) Fluid droplets.
- B) Solid particles
- C) Bacteria
- D) Foreign bodies

2. Which of the following best describes the phospholipid molecules of cell membrane? (2025)

- A) The head is non-charged and hydrophobic
- B) The head is directed inwards away from the aqueous solutions
- C) The fatty acid tail is charged and hydrophilic
- D) The head is directed towards the aqueous solutions

3. What process is nicknamed "cell drinking"? (2019, Fighter Histology bank)

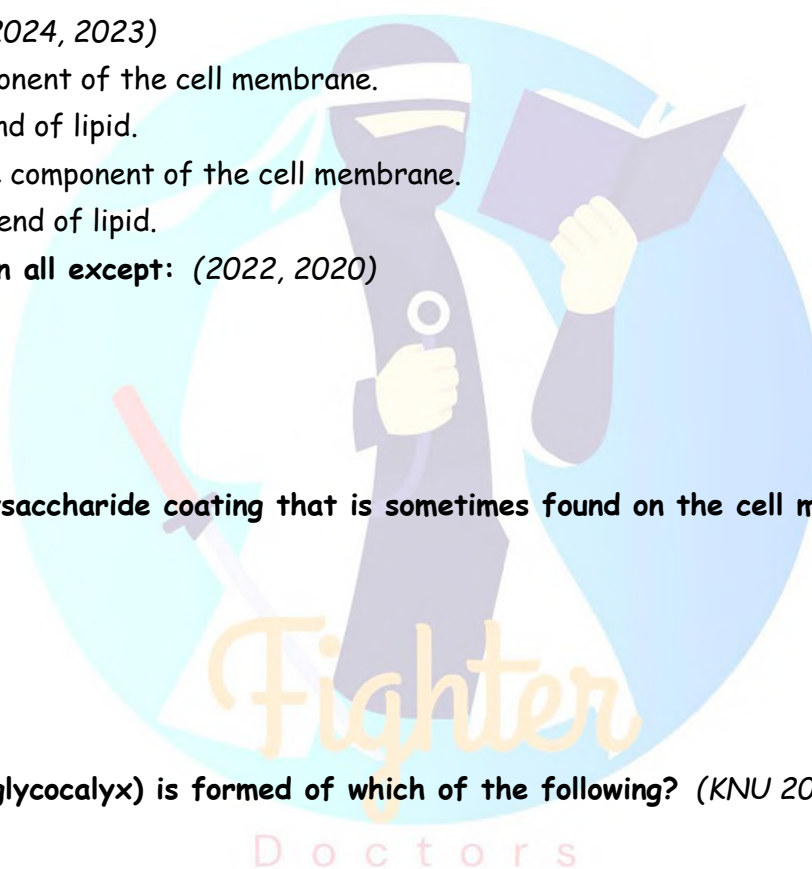
- A) Endocytosis
- B) Exocytosis
- C) Pinocytosis
- D) Phagocytosis

4. What is concerned with Glycocalyx? (2026, KNU 2026)

- A) The protein component of the cell membrane



- B) The hydrophobic end of lipid  
C) The hydrophilic end of lipid  
D) The carbohydrate component of the cell membrane
- 5. Which of the following stains can demonstrate the plasma membrane? (2026, KNU 2026)**  
A) PAS  
B) Osmic acid  
C) Sudan II  
D) Orcein
- 6. What is the term for the process by which the cells transport solids? (2025)**  
A) Endocytosis  
B) Exocytosis  
C) Pinocytosis  
D) Phagocytosis
- 7. Glycocalyx is: (2024, 2023)**  
A) The protein component of the cell membrane.  
B) The hydrophilic end of lipid.  
C) The carbohydrate component of the cell membrane.  
D) The hydrophobic end of lipid.
- 8. Cell coat function all except: (2022, 2020)**  
A) Cell adhesions.  
B) Cell allergy.  
C) Cell recognition.  
D) Cell transport.
- 9. What is the polysaccharide coating that is sometimes found on the cell membrane? (2022, 2019)**  
A) Plasmalemma  
B) Glycocalyx  
C) Protoplasm  
D) Cristae
- 10. The cell coat (glycocalyx) is formed of which of the following? (KNU 2026)**  
A) Integral protein  
B) Peripheral proteins  
C) Glycosaminoglycans  
D) Glycoproteins and glycolipids
- 11. The membrane of cytoplasmic vesicles can be added to the cell membrane in which of the following processes? (KNU 2026)**  
A) Phagocytosis  
B) Endocytosis  
C) Pinocytosis  
D) Exocytosis
- 12. What is the term for the process by which the cells transport fluids? (2025)**  
A) Endocytosis  
B) Exocytosis





- C) Pinocytosis
- D) Phagocytosis

**13. The cell membrane can be demonstrated by which staining technique? (2025)**

- A) PAS
- B) Sudan III
- C) H&E
- D) Orcein

**14. In the molecular structure of the cell membrane, which of the following structures fill the gap between the fatty acid tails in a ratio one by one?\* (2023)**

- A) Cholesterol
- B) Lipid bilayer
- C) Phospholipid
- D) Integral protein

**15. What is the term for the process used by cells for the transport solids? (2022)**

- A) Endocytosis
- B) Exocytosis
- C) Pinocytosis
- D) Phagocytosis

## Answer Key

|      |      |      |      |      |     |     |     |     |      |
|------|------|------|------|------|-----|-----|-----|-----|------|
| 1-A  | 2-D  | 3-C  | 4-D  | 5-B  | 6-D | 7-C | 8-D | 9-B | 10-D |
| 11-D | 12-C | 13-B | 14-A | 15-D |     |     |     |     |      |

## Written questions

1. Mention one important function for each of the following: Cell coat [repeated x2]

## 3. Organelles

50 MCQs (top 50 of 78 available, most repeated / most recent first) | 9 written questions

**1. What structure contributes to the cells cytoskeleton? (2022, 2021, 2020, 2019)**

- A) Ribosomes
- B) Peroxisome
- C) Microfilaments
- D) Centrioles

**2. In H&E stained sections of plasma cell, How can Golgi apparatus appears? (2022, 2019)**

- A) Perinuclear.
- B) Apical.
- C) Unstained.
- D) Basal.

**3. Which Two organelles share in the formation of lysosomes? (2022, 2019, Fighter Histology bank)**

- A) rER & sER.
- B) sER & Golgi.
- C) Golgi & cell membrane.
- D) rER & Golgi.

**4. Which organelle is responsible for maintenance of cell health? (2026, 2025, KNU 2026)**

- A) Rough endoplasmic reticulum
- B) Golgi apparatus



C) Smooth endoplasmic reticulum

D) Lysosome

**5. Localized cytoplasmic basophilia indicates the presence of: (2019)**

A) Mitochondria.

B) DNA.

C) sER.

D) rER

**6. Which of the following statements is incorrect? (2023, KNU 2026)**

A) Microtubules are involved in ciliary movement and intracellular transport.

B) The cis (forming) face of Golgi apparatus is where secretory granules are formed.

C) The microtubules of the centrioles and in the basal body of cilia have a 9x3 arrangement.

D) Ribosomes, made of rRNA and protein, are the site of protein synthesis.

**7. Which of the following describes Multivesicular bodies? (2026, KNU 2026)**

A) Primary lysosomes

B) Autophagic vacuoles

C) Formed as a result of digestion of solid material

D) Formed as a result of digestion of fluid material

**8. What is about smooth endoplasmic reticulum? (2026, KNU 2026)**

A) Areas of localized basophilia indicate its presence

B) Is rarely found in skeletal muscle cells

C) Is present in cells where drug detoxification is taking place

D) Sometimes has ribosomes attached to its membrane

**9. What is the function of rough endoplasmic reticulum? (2026, KNU 2026)**

A) Cytoplasmic protein necessary for the cells own existence

B) Carbohydrate

C) Lipid

D) Protein that will be secreted by the cell

**10. Regarding the centrioles, which of the following describes it? (2026, KNU 2026)**

A) Appear as two dark bodies with H&E

B) Share in the formation of cilia

C) Present as two pairs in non-dividing cells

D) Formed by dynamic microtubules

**11. Diffuse cytoplasmic basophilia caused by ribosome can be found in which cells? (2026, KNU 2026)**

A) Embryonic cells

B) Pancreatic cells

C) Nerve cells

D) Glial cells

**12. Which of the following is true as regard Ribosomes? (2025)**

A) Appears basophilic when stained with H & E.

B) Stains red when stained with silver.

C) It is found supranuclear in nerve cells.

D) Appears as unstained area in deeply basophilic cytoplasm.



**13. Which of the following is absent from matrix of mitochondria? (2025)**

- A) Circular DNA & RNA.
- B)  $\text{Ca}^{2+}$  acting as catalysts condensed within dense granules.
- C) Oxidative enzymes of Krebs's cycle.
- D) Hydrolytic digestive enzymes.

**14. What is the special stain of Golgi apparatus? (2025)**

- A) Silver
- B) Best's Carmine
- C) Sudan III
- D) Iron HX

**15. Which organelle is responsible for cell digestion? (2025)**

- A) rER
- B) Golgi apparatus
- C) sER
- D) Lysosome

**16. Regarding rER, which of the following describes it? (2025)**

- A) Form proteins secreted by the cells
- B) Form proteins used within the cytosol
- C) Linked together by tRNA forming polysomes.
- D) Bind to the outer surface of SER

**17. Which feature of the followings describes centrioles? (2025)**

- A) They are a membranous organelle
- B) Can be demonstrated with silver stain
- C) Formed of nine triplets of microtubules
- D) Have the same number of microtubules as the shaft of cilia

**18. The function of rough endoplasmic reticulum is to synthesize: (2024, 2023)**

- A) Lipid.
- B) Protein that will be secreted by the cell.
- C) Cytoplasmic protein necessary for the cells' own existence.
- D) Carbohydrates.

**19. Smooth endoplasmic reticulum: (2024, 2023)**

- A) Sometimes has ribosomes attached to its membrane
- B) Is rarely found in skeletal muscle cells
- C) Areas of localized basophilic indicate its presence
- D) Is present in cells where drug detoxification is taking place

**20. The Golgi apparatus is: (2024, 2023)**

- A) The site of packaging of secretory products of the cell.
- B) The site of protein synthesis.
- C) Continuous with smooth endoplasmic reticulum.
- D) Capable of providing energy.

**21. Multivesicular bodies are: (2024, 2023)**

- A) Primary lysosomes.
- B) Formed as a result of digestion of solid material.



C) Formed as a result of digestion of fluid material.

D) Autophagic vacuoles.

**22. Microtubules have important role in: (2024, 2023)**

A) Carbohydrate metabolism.

B) Oxidative phosphorylation.

C) Water balance.

D) Formation of the mitotic spindle during mitosis.

**23. DNA could be found in: (2024, 2023)**

A) Ribosomes.

B) Golgi apparatus.

C) Mitochondria.

D) Lysosomes.

**24. Which of the following is concerned with formation of ribosomes?\* (2023)**

A) Golgi

B) Centriole

C) RER

D) Nucleolus

**25. The cytoplasmic side of the coated vesicle is surrounded by: (2022, 2020)**

A) Clathrin.

B) Tubulin.

C) Actin.

D) Spectrin.

**26. Which organelle is involved in lipid metabolism? (2022, 2021)**

A) RER

B) Golgi apparatus

C) SER

D) Lysosome

**27. Detoxification of drugs in liver cells is the function of which organelle? (KNU 2026)**

A) rER

B) SER

C) Golgi apparatus

D) Lysosome

**28. Which of the following represents a special stain for mitochondria? (KNU 2026)**

A) PAS stain

B) Janus green

C) Sudan black

D) Hematoxylin

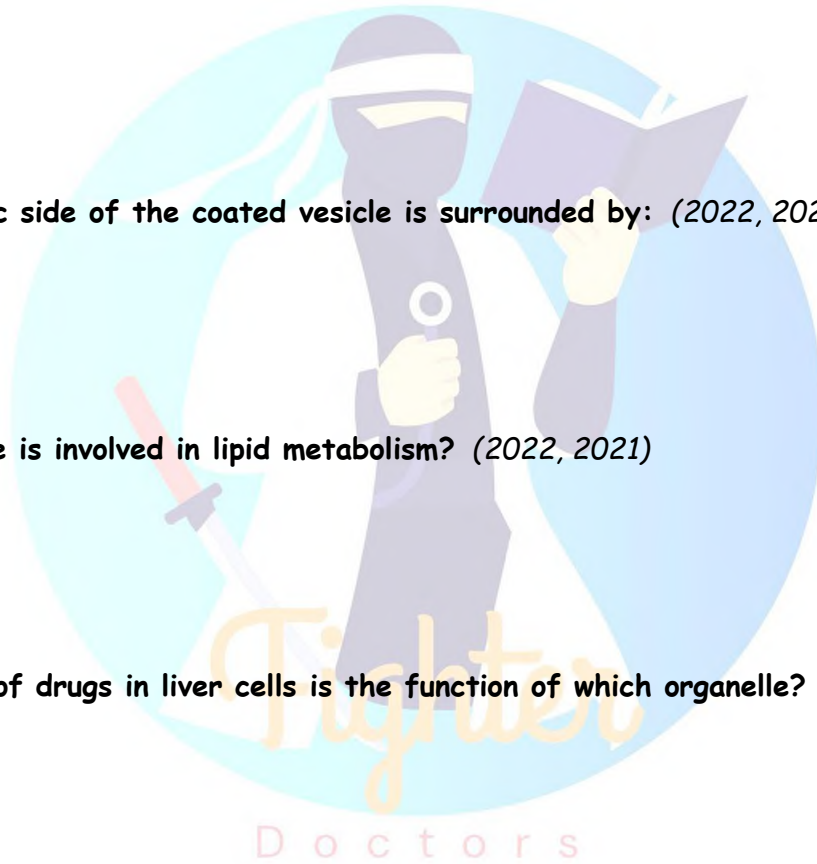
**29. Type of intermediate filaments found in the epithelium: (KNU 2026)**

A) Cytokeratin

B) Vimentin

C) Neuro filaments

D) Desmin





**30. In a patient with a genetic defect in tubulin, which of the following cellular structures would be most directly affected? (KNU 2026)**

- A) Formation of the nuclear envelope
- B) Assembly of the mitotic spindle
- C) Organization of microfilaments
- D) Synthesis of ribosomal RNA

**31. Which of the following describes the histological structure of Golgi apparatus? (KNU 2026)**

- A) Produces cytoplasmic basophilia with H&E
- B) Appears as a network of brown granules with silver stain
- C) Appears as a basophilic area around the nucleus in plasma cell
- D) The Cis face represents the mature surface

**32. Which of the following is truly matching as regard each organelle? (2025)**

- A) Inner mitochondrial membrane → Cristae.
- B) Multivesicular bodies → Engulfed solid foreign bodies.
- C) Protein synthesis → SER.
- D) Euchromatin → Nucleolus associated.

**33. Which of the following is mis-matching as regard each organelle? (2025)**

- A) Attached ribosomes → Synthesis of cytosol proteins.
- B) Primary lysosomes → Originate from trans-face of Golgi apparatus.
- C) Initial protein modifications → RER.
- D) Transfer vesicles → Transport proteins from RER to cis-face.

**34. What is the origin of secretory vesicles of Golgi apparatus? (2025)**

- A) Smooth endoplasmic reticulum
- B) Rough endoplasmic reticulum
- C) The trans face of Golgi apparatus
- D) The cell membrane

**35. Which of the following is truly- matching as regard each organelle? (2025)**

- A) Cristae: inner mitochondrial membrane
- B) Multivesicular bodies: Engulfed solid foreign bodies
- C) SER: protein synthesis
- D) Euchromatin: Nucleolus associated

**36. Which of the following is mis matching as regard each organelle? (2025)**

- A) Attached ribosomes: Synthesis of cytosol proteins
- B) Primary lysosomes: Originate from trans- face of Golgi apparatus
- C) RER: Initial protein modifications
- D) Transfer vesicles: Transport proteins from RER to cis-face

**37. What is the origin of secretory vesicles of Golgi apparatus? (2025)**

- A) SER
- B) RER
- C) trans face of Golgi apparatus.
- D) cell membrane

**38. Regarding free ribosomes, which of the following describes it? (2025)**

- A) Form proteins secreted by the cells



- B) Form proteins used within the cytosol
- C) Linked together by tRNA forming polysomes
- D) Bind to the outer surface of rER

**39. Which feature of the followings describes centrioles? (2025)**

- A) They are a membranous organelle
- B) Can be demonstrated with iron hematoxylin stain
- C) Formed of eight bundles of microtubules
- D) Have the same number of microtubules as the shaft of cilia

**40. What is the origin of transfer vesicles of Golgi apparatus? (2025)**

- A) Smooth endoplasmic reticulum
- B) Rough endoplasmic reticulum
- C) The trans face of Golgi apparatus
- D) The cell membrane

**41. Which of the following is mismatching as regarded each organelle? (2025)**

- A) Attached ribosomes - Synthesis of secretory proteins
- B) Lysosomes - Originated from cis-face of the Golgi apparatus
- C) Initial protein modification - RER
- D) Transfer vesicle - Transfer protein from RER to cis-face

**42. Which of the following is true regarding Golgi apparatus? (2025)**

- A) Appears as unstained area in deeply basophilic cytoplasm
- B) Stains red when stained with silver
- C) Appears basophilic when stained with H & E
- D) Is found supranuclear in nerve cells

**43. Which of the following is truly matching as regards each organelle? (2025)**

- A) Outer mitochondrial membrane: Cristae
- B) Multivesicular bodies: Engulfed solid foreign bodies
- C) Euchromatin: Nucleolus associated
- D) Phospholipid and cholesterol synthesis: SER

**44. Which of the following statements is incorrect? (2024)**

- A) Microtubules are involved in ciliary movement and intracellular transport.
- B) cis (forming) face of Golgi apparatus is where secretory granules are formed.
- C) microtubules of the centrioles and in the basal body of cilia have a 9×3 arrangement.
- D) Ribosomes, made of rRNA and protein, are site of protein synthesis.

**45. All of the following are characters of Golgi apparatus except: (2024)**

- A) Shows -Ve image in deeply basophilic cytoplasm.
- B) Stained with silver.
- C) Supranuclear in nerve cells.
- D) Has mature trans- face & immature cis- face.

**46. The basal body of a cilium has a microtubular arrangement which is identical to the pattern found in: (2024)**

- A) microvilli
- B) centrioles
- C) spindle fibers



D) stereocilia.

**47. The organelle most prominent in steroid secreting cells is: (2024)**

- A) RER
- B) SER
- C) Lysosomes
- D) Cilia

**48. As regards ribosomes: (2024)**

- A) Their subunits are formed in the nucleolus.
- B) Cause cytoplasmic acidophilia.
- C) Are formed of DNA associated with proteins.
- D) Are formed of two subunits of equal size.

**49. Enzymes associated with drug detoxification are associated with: (2024)**

- A) SER
- B) RER
- C) Golgi
- D) Mitochondria

**50. When the protein secreting cells stained with Hematoxylin and Eosin, There is a clear unstained area near the nucleus. This pale area is called:\* (2023)**

- A) Brown network or granules
- B) Cisternae
- C) Negative Golgi image
- D) Cristae

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-C  | 3-D  | 4-D  | 5-D  | 6-B  | 7-D  | 8-C  | 9-D  | 10-B |
| 11-A | 12-A | 13-D | 14-A | 15-D | 16-A | 17-C | 18-B | 19-D | 20-A |
| 21-C | 22-D | 23-C | 24-D | 25-A | 26-C | 27-B | 28-B | 29-A | 30-B |
| 31-B | 32-A | 33-A | 34-C | 35-A | 36-A | 37-C | 38-B | 39-B | 40-B |
| 41-B | 42-A | 43-D | 44-B | 45-C | 46-B | 47-B | 48-A | 49-A | 50-C |

## Written questions

1. Mention: C. 3 functions for sER (1.5 marks) [repeated x2]
2. Discuss the following: C. The electron microscopic picture of the types of Endoplasmic Reticulum and their functions. (2 marks)
3. Enumerate: g. Two functions for Golgi apparatus. (1 mark)
4. Give the scientific name for the following statements: b) They have the same structure as cilia but are much longer. The only example in human is sperm.
5. Give a comment on :- B) Centriole :
6. Describe EM appearance of Gogi apparatus & reference to its function.
7. Describe the types of filaments.
8. Mention one important function for each of the following: Lysosomes
9. Mention the fate of primary lysosomes.



#### 4. Nucleus

20 MCQs | 7 written questions

1. Which of the following statements describes active cells? (2025)

- A) Dark nucleus
- B) Heterochromatic nucleus
- C) Euchromatic nucleus
- D) Few mitochondria

2. Which of points related to Heterochromatin? (2026, KNU 2026)

- A) Special type of stain
- B) Present in the nuclei of metabolically active cells
- C) Type of cell inclusions
- D) Present in the nuclei of metabolically inactive cells

3. What about the nuclear pore? (2026, KNU 2026)

- A) Allows for communication between the nuclei & cytoplasm
- B) Is a transient structure
- C) Is at regular intervals
- D) Is bridged by a unit membrane

4. Regarding nuclear chromatin, what is the typical character? (2026, KNU 2026)

- A) Extended type appears dark basophilic with LM
- B) Condensed type appears pale basophilic with LM
- C) Extended type contains the active genes
- D) Euchromatin appears as coarse granules with EM

5. Which of the following best describes peripheral chromatin? (2025)

- A) Connected to inner border of nuclear envelope
- B) Connected to nucleoli
- C) Clumped and scattered in the nuclear sap
- D) Fine granules and scattered in the nuclear sap

6. What about heterochromatin? (2025)

- A) Makes the nucleus pale by LM.
- B) Appears as coarse granules in the nucleus.
- C) Contains active genes.
- D) Represents extended parts of chromatin.

7. What is the typical feature of condensed chromatin? (2025)

- A) Appears dark acidophilic by LM
- B) Appears as fine granules by EM
- C) Contains the inactive genes
- D) Appears pale basophilic by LM

8. Heterochromatin is: (2024, 2023)

- A) Present in the nuclei of metabolically active cells.
- B) Special type of stain.
- C) Present in the nuclei of metabolically inactive cells
- D) Type of cell inclusions.

9. The nucleolus is formed of: (2024, 2023)



- A) Heterochromatin and euchromatin.
- B) Nucleolonema and pars amorpha.
- C) Nucleolus associated and peripheral chromatin.
- D) Condensed chromatin.

**10. The nuclear pore: (2024, 2023)**

- A) Is at regular intervals.
- B) Is bridged by a unit membrane.
- C) Is a transient structure.
- D) Allows for communication between the nucleus and the cytoplasm

**11. What is the nuclear envelope? (KNU 2026)**

- A) A single membrane
- B) A double membrane with pores
- C) Without pores
- D) Continuous with the Golgi apparatus

**12. Euchromatin is characterized by which of the following? (KNU 2026)**

- A) Dark staining and coiled DNA
- B) Light staining and uncoiled DNA
- C) Being transcriptionally inactive
- D) Located only at the nuclear periphery

**13. The nucleolus is involved in which of the following? (KNU 2026)**

- A) DNA replication
- B) rRNA synthesis and ribosomal subunit assembly
- C) Protein degradation
- D) Lipid metabolism

**14. The perinuclear space is continuous with which of the following? (KNU 2026)**

- A) Golgi cisternae
- B) Rough endoplasmic reticulum
- C) Mitochondrial intermembrane space
- D) Lysosomal lumen

**15. Which of the following is true regarding heterochromatin? (KNU 2026)**

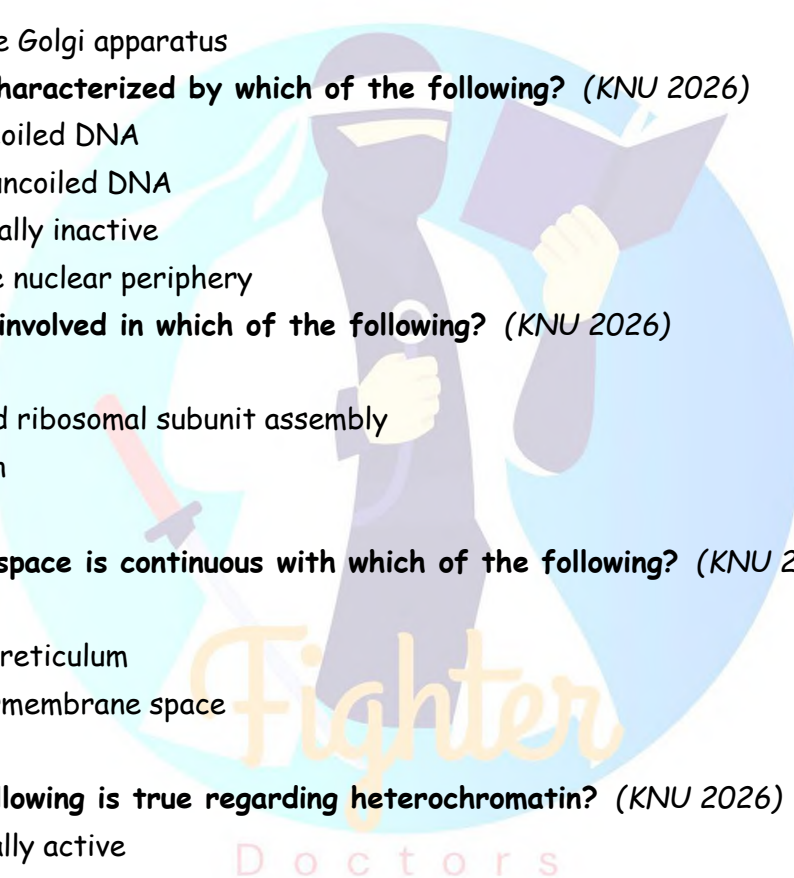
- A) It is transcriptionally active
- B) It is predominant in protein-synthesizing cells
- C) It is concentrated at the nuclear periphery
- D) It stains lightly with basic dyes

**16. What is the typical feature of extended chromatin? (2025)**

- A) Appears pale basophilic by LM
- B) Appears as coarse granules by EM
- C) Contains the inactive genes
- D) Appears dark basophilic by LM

**17. Which of the following best describes chromatin islands? (2025)**

- A) Connected to nucleoli
- B) Clumped and scattered in the nuclear sap
- C) Fine granules and scattered in the nuclear sap





D) Connected to the inner border of the nuclear envelope

**18. Which of the following best describes euchromatin? (2025)**

A) Appears as condensed masses in the nucleus

B) Represents extended parts of chromatin

C) Contains inactive genes

D) Makes the nucleus dark by LM

**19. Which of the following statements is incorrect? (2024)**

A) Heterochromatin, as seen in EM, is lighter staining DNA.

B) The nucleolus is the region within the nucleus where rRNA is formed.

C) The function of SER includes lipid synthesis.

D) RER and SER can be continuous

**20. The nucleus is surrounded by which of the following? (2019)**

A) A double membrane

B) A single membrane

C) No membrane around nucleus

D) Proteins

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-D  | 3-A  | 4-C  | 5-A  | 6-B  | 7-C  | 8-C  | 9-B  | 10-D |
| 11-B | 12-B | 13-B | 14-B | 15-C | 16-A | 17-B | 18-B | 19-A | 20-A |

**Written questions**

- Describe the followings C. Histological structure of nuclear envelop and its function (2 marks) [repeated x2]
- Mention: A. 3 parts of nucleolus (1.5 marks) [repeated x2]
- Enumerate the following: C. Types of Nuclear Chromatin. (1.5 marks)
- Discuss the following: A. Nucleolus (EM structure, function, and staining). (3 marks)
- Enumerate: e. Sites of heterochromatin. (1.5 marks)
- State the following:- A. Sites of heterochromatin in nucleus
- State the histological structure of outer & inner nuclear membrane.

## 5. Cytogenetics

35 MCQs / 9 written questions

**1. Regarding metaphase I in second meiotic division, which of the following occurs? (2026, KNU 2026)**

A) Crossing over occurs at this stage

B) The d-chromosome splits at the centromere into two chromatids

C) The 23-bivalent d-chromosomes become aligned in the equatorial plane

D) The 23 d-chromosomes are aligned at the equatorial plate

**2. Regarding the Gap 1 phase of interphase, which of the following describes it? (2026, KNU 2026)**

A) Each cell contains 92 s chromosomes

B) The nucleus of the daughter cell contains 46 s chromosomes

C) The more specialized the cell, the shorter the G1-phase

D) Each cell contains 46 d chromosome



**3. Down syndrome: (2022, 2019)**

- A) due to duplication of chromosome 22.
- B) due to X chromosome monosomy.
- C) due to multiple X syndrome.
- D) caused by chromosome 21 trisomy.

**4. Which of the following cells are continuously turn over cells? (2026, KNU 2026)**

- A) Heart muscles
- B) Blood cells
- C) Nerve cells
- D) Liver cells

**5. Isochromosome occurs mostly in which type of chromosomes? (2026, KNU 2026)**

- A) Acrocentric
- B) Metacentric
- C) Telocentric
- D) Submetacentric

**6. A 15-year-old girl presents with primary amenorrhea, short stature, and webbed neck. What is Her karyotype? (2026, KNU 2026)**

- A) 46,XX
- B) 46,XY
- C) 45,X
- D) 47,XXX

**7. Regarding the prophase of the first meiotic division, crossing over occurs in which of the following stages? (2026, KNU 2026)**

- A) Leptotene
- B) Zygotene
- C) Pachytene
- D) Diplotene

**8. Which of the following describes the chromosomal microtubules? (2026, KNU 2026)**

- A) They are continuous
- B) Arise from MTOC surrounding the centrioles
- C) Responsible for elongation of the cell.
- D) Responsible for pulling the sister chromatids

**9. Which of the following is considered as aneuploidy? (2026, KNU 2026)**

- A) Triploid
- B) Trisomy
- C) Tetraploid
- D) Polyploidy

**10. In which of the following chromosomal aberrations two copies of the same gene can be found on the same chromosome? (2026, KNU 2026)**

- A) Isochromosome
- B) Ring chromosome
- C) Addition
- D) translocation



**11. Duplication of DNA occurs in: (2024, 2019)**

- A) G1 phase
- B) G2 phase
- C) S phase
- D) Mitosis

**12. Which of the following describes Barr body? (KNU 2026)**

- A) The Y chromosome in male
- B) The X chromosome in males
- C) The XX chromosomes in females
- D) The inactive X chromosome in females

**13. What is the typical feature of submetacentric chromosome? (KNU 2026)**

- A) The centromere is terminal
- B) The centromere is at the center of the chromosome
- C) The centromere is midway between the center and upper end
- D) The centromere is very close to upper end

**14. The bivalent chromosomes separate from one another at the chiasma in which stage of meiosis? (KNU 2026)**

- A) Anaphase 2
- B) Metaphase 1
- C) Metaphase 2
- D) Anaphase 1

**15. Which of the following represents a structural chromosomal aberration? (KNU 2026)**

- A) Trisomy
- B) Monosomy
- C) Turner's syndrome
- D) Deletion

**16. Apoptosis, which of the following describes it? (KNU 2026)**

- A) It is a physiological condition
- B) The cells add their organelles and nucleus
- C) Occurs normally at the end of life span of the cell
- D) May result from exposure to toxins and anemia

**17. Which of the following represents a potentially renewable cell? (KNU 2026)**

- A) Nerve cell
- B) Liver cell
- C) Cardiac muscle cell
- D) Blood cell

**18. What is the nuclear description of the interphase in cell cycle? (KNU 2026)**

- A) It lasts for a short period
- B) The daughter cell contains 92 chromosomes in the G1 phase
- C) The changes are visible with the microscope
- D) The daughter cell contains 92 chromosomes in the G0 phase

**19. Which of the following occurs during the prophase of mitosis? (KNU 2026)**

- A) Disappearance of nuclear membrane



- B) The chromosomes are arranged in the equatorial plate
- C) The chromosomes split longitudinally at the centromere
- D) The nucleolus reappears

**20. As regarding to Sex chromosome abnormalities, monosomy means: (KNU 2026)**

- A) Klinefelter's syndrome
- B) Super female syndrome
- C) Turner's syndrome
- D) Down's syndrome

**21. The cleavage furrow develops at which of the following stages of mitosis? (KNU 2026)**

- A) Anaphase
- B) Telophase
- C) Prophase
- D) Metaphase

**22. What is the position of centromere in submetacentric chromosomes? (KNU 2026)**

- A) At the center of the chromosome
- B) Midway between the centre and upper end
- C) Very close to upper end
- D) Terminal

**23. In which of the following chromosomal aberrations, the chromosome divides transversely at the centromeres instead of longitudinally during anaphase of mitosis? (KNU 2026)**

- A) Inversion
- B) Isochromosomes
- C) Ring Chromosome
- D) Reciprocal translocation

**24. One of the following is a numerical aberration of sex chromosomes: (2024)**

- A) Turner's syndrome
- B) Down's syndrome
- C) Translocation
- D) Isochromosome

**25. The nucleus of which of the following cells shows Barr body in females (drum stick appearance):\* (2023)**

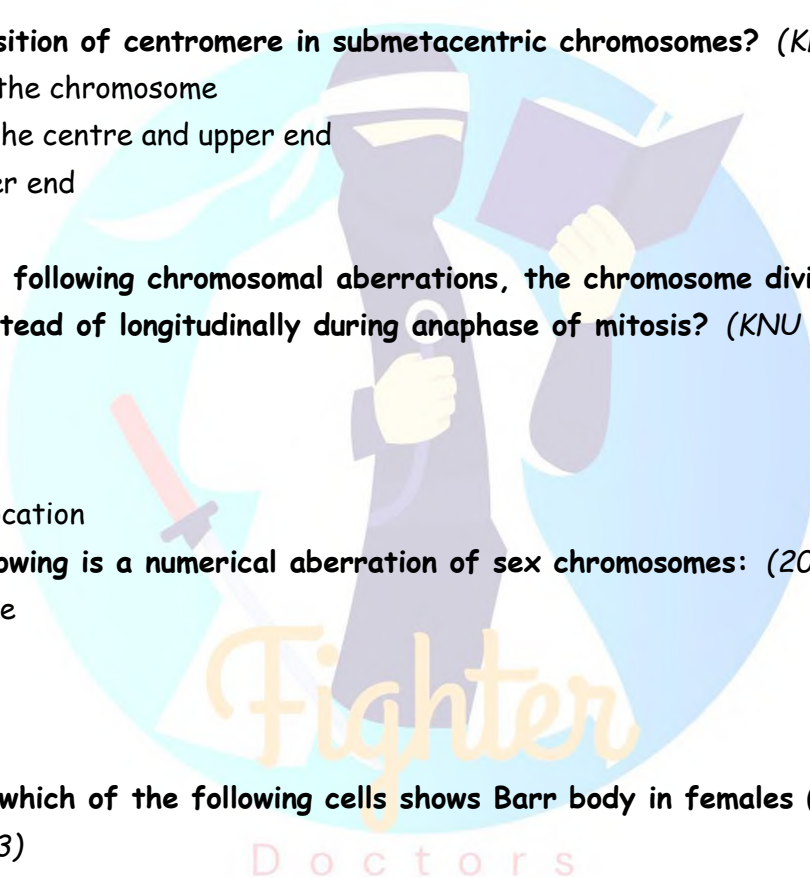
- A) Lymphocyte
- B) Neutrophil
- C) Monocyte
- D) Basophil

**26. During the cell cycle, in which stage DNA replication takes place?\* (2023)**

- A) Prophase
- B) Metaphase
- C) Anaphase
- D) Interphase

**27. In which of the following types, the chromosomes have short and long arms of unequal length with the centromere presents more towards one end?\* (2023)**

- A) Acrocentric





- B) Telocentric
- C) Submetacentric
- D) Metacentric

**28. What does the cause of Down syndrome: (2022)**

- A) due to duplication of chromosome 22.
- B) due to X chromosome monosomy.
- C) due to multiple X syndrome.
- D) caused by chromosome 21 trisomy.

**29. Which is a non-renewing cell: (2021)**

- A) Hepatocyte.
- B) Epidermal cell.
- C) Nerve cell.
- D) Cartilage cell

**30. Which is Numerical aberrations? (2021)**

- A) Ring Chromosome.
- B) Deletion
- C) Inversion.
- D) Klinefelter s syndrome.

**31. Mitosis is characterized by the following (Fighter Histology bank)**

- A) Exchange of genes
- B) Absence of crossing over
- C) Daughter cell contains n chromosome
- D) Daughter cell is not identical to mother cell

**32. Karyotyping of chromosomes are done in (Fighter Histology bank)**

- A) Prophase
- B) Anaphase
- C) Metaphase
- D) Telophase

**33. Colchicine chromatids from separating during mitosis at (Fighter Histology bank)**

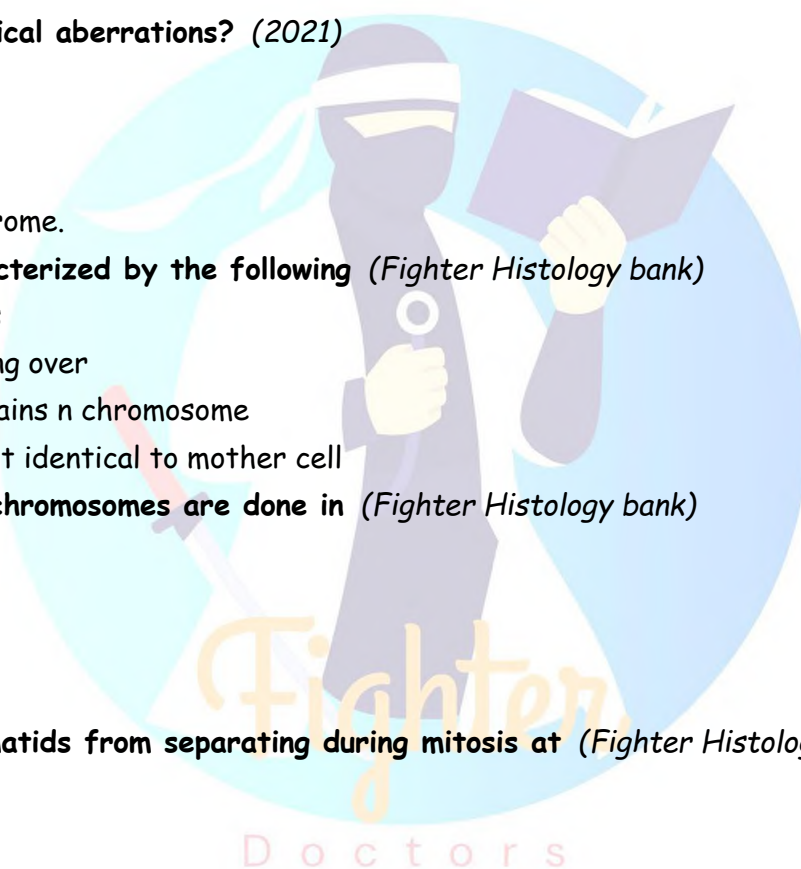
- A) Prophase
- B) Anaphase
- C) Metaphase
- D) Telophase

**34. Tubulin us synthesized in (Fighter Histology bank)**

- A) G1 phase
- B) G2 phase
- C) G0 phase
- D) S phase

**35. Philadelphia chromosome (Fighter Histology bank)**

- A) Duplication of chromosome
- B) T erminal deletion of chromosome
- C) Inversion affecting chromosome
- D) Translocation between chromosomes





## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-B  | 3-D  | 4-B  | 5-A  | 6-C  | 7-C  | 8-D  | 9-B  | 10-A |
| 11-C | 12-D | 13-C | 14-D | 15-D | 16-A | 17-B | 18-C | 19-A | 20-C |
| 21-B | 22-B | 23-B | 24-A | 25-B | 26-D | 27-C | 28-D | 29-C | 30-D |
| 31-B | 32-C | 33-C | 34-B | 35-D |      |      |      |      |      |

## Written questions

- Enumerate a- Two types of numerical chromosomal aberrations having positive sex chromatin b- Three sites of Neuroepithelium. c- Sites of Dense regular connective tissue
- Enumerate the following: B. Four events occurring during the prophase of mitosis. (2 marks)
- Mention the following: C. Two numerical chromosomal aberrations. (1 mark)
- Enumerate: A. Two types of numerical chromosomal aberrations having positive sex chromatin. (1 mark)
- Enumerate: d. Types of chromosomes according to the position of centromere. (2 marks)
- Enumerate: C. Types of chromosomal aberrations.
- Give the scientific name for the following statements: a) It is the process by which the cytoplasm and nucleus of somatic cell are divided equally into two identical daughter cells.
- Give the scientific name for the following statements: c) The period between two successive divisions during which cell grows and performs their functions.
- Give the scientific name for the following statements: h) One site for highly specialized cells leave the cycle while in G1 forever and go to G0 phase.

## 6. Epithelium

1. Pseudostratified columnar ciliated epithelium can be found in: (2026, 2024, 2023, KNU 2026)

- A) Fallopian tube
- B) Blood vessels.
- C) Respiratory passages
- D) Urinary passage

2. What type of tissue lines the urinary bladder (2022, 2019)

- A) Simple squamous epithelium
- B) Simple cuboidal epithelium
- C) Simple columnar epithelium
- D) Transitional epithelium

3. In absorptive cells, the brush border seen at Im level is: (2026, 2024, 2023, KNU 2026)

- A) Streocilia.
- B) True cilia.
- C) Microvilli.
- D) Basal infoldings.

4. Which of the following glands is considered as an apocrine gland: (2026, 2024, 2023, KNU 2026)

- A) Pancreas



B) Mammary glands

C) Salivary glands.

D) Sebaceous glands

**5. What is characteristic to stratified squamous epithelium? (2026, KNU 2026)**

A) Consists of a single layer of flattened cells

B) May be ciliated

C) Has a rich supply of capillaries running between its cell layers

D) Its protective rather than secretory or absorptive

**6. Which of the following is the only example of a unicellular exocrine gland?\* (2024, 2023)**

A) Sebaceous

B) Intestinal gland

C) Goblet cell

D) Sweat

**7. What type of epithelium lines the trachea? (2022, 2019)**

A) Simple squamous epithelium

B) Simple cuboidal epithelium

C) Simple columnar epithelium

D) Pseudostratified epithelium

**8. What type of tissue lines blood vessels? (2022, 2019)**

A) Simple squamous epithelium

B) Simple cuboidal epithelium

C) Simple columnar epithelium

D) Pseudostratified epithelium

**9. What type of epithelium has dome shaped cells on the apical surface? (2019)**

A) Epithelioid

B) Mesothelium

C) Endothelium

D) Transitional

**10. What type of gland secretes its product directly into the bloodstream? (2022, 2019)**

A) Endocrine gland

B) Multicellular gland

C) Exocrine gland

D) All of the above

**11. Which is NOT lined with simple squamous epithelium: (2022, 2019)**

A) Aorta.

B) Thyroid gland.

C) Pericardium.

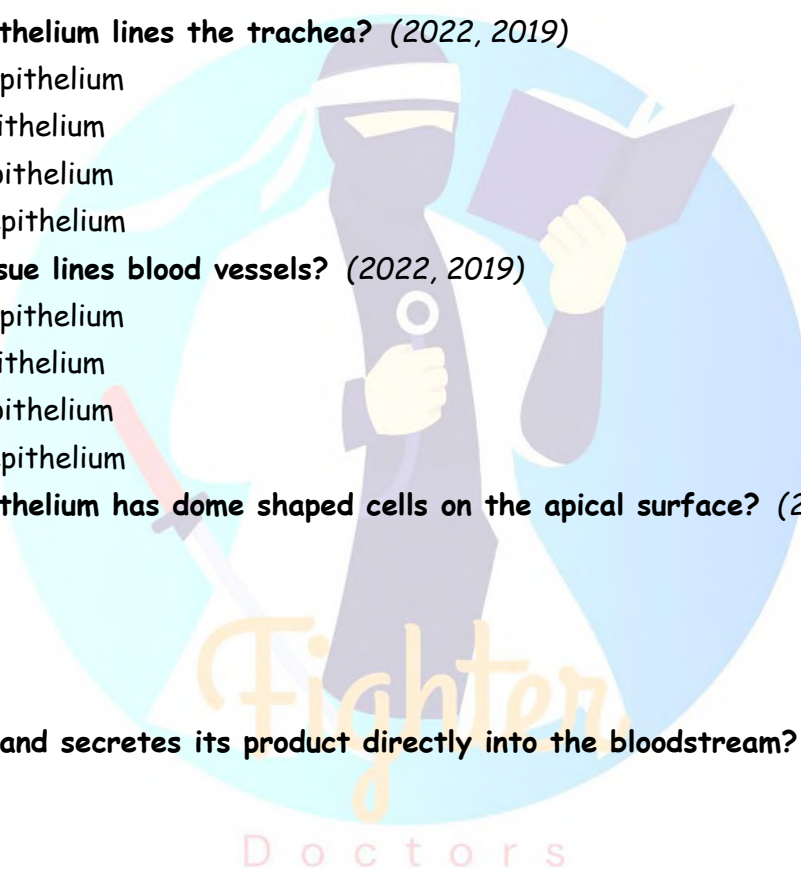
D) Alveoli of the lung

**12. Which of the following not included in the classical description of a junctional complex: (2023, KNU 2026)**

A) The fascia occludes

B) The zonula occludens

C) The zonula adherens





D) The macula adenens

**13. What is concerned with compound exocrine glands? (2026, KNU 2026)**

A) Have unbranched secretory duct

B) Secrete hormones

C) Have branched secretory ducts

D) Secrete into the blood

**14. Which of the following is not in the classical description of a junctional complex? (2026, KNU 2026)**

A) The Zonula occludens

B) The Zonula adherens

C) The macula adherens

D) Gap junction

**15. Which epithelial type is incorrectly matched with its function? (2026, KNU 2026)**

A) Simple cubical - Secretion and reabsorption

B) Simple columnar - Protection only

C) Pseudostratified - Transport of mucus

D) Transitional - Accommodates stretching

**16. Which gland type releases secretion by exocytosis without cell damage? (2026, KNU 2026)**

A) Sebaceous gland

B) Mammary gland

C) Salivary gland

D) Meibomian gland

**17. Which apical specialization increases surface area for absorption in small intestine? (2026, KNU 2026)**

A) Cilia

B) Flagella

C) Microvilli

D) Stereocilia

**18. The basement membrane is composed of which of the following? (2026, KNU 2026)**

A) Only epithelial cells

B) Basal lamina and reticular lamina

C) Only collagen type I

D) Only glycoproteins

**19. Which cell junction has Pentalaminar appearance? (2026, KNU 2026)**

A) Tight junction

B) Desmosome

C) Gap junction

D) Hemidesmosome

**20. Which of the following represents a character of transitional epithelium? (2025)**

A) Superficial cells are columnar with basal oval nuclei

B) The basal cells are flat cells with flat nuclei

C) Has no basement membrane

D) It is present in the urinary bladder



**21. Which of the following represents Tight junction? (2025)**

- A) Zonula occludens
- B) Zonula adherens
- C) Gap junction
- D) Fascia adherens

**22. Which of the following represents the typical feature of hemidesmosomes? (2025)**

- A) It is a belt-like specialization
- B) Anchors cell to the basement membrane
- C) The cytoplasmic side shows condensation of actin filaments
- D) The intercellular space between the adjacent cell membranes is 20 nm

**23. Which of the following represents the typical feature of microvilli? (2025)**

- A) It is a motile cytoplasmic projection
- B) It is present in the cells lining the fallopian tube
- C) It is formed of microtubules
- D) Its core is formed of actin filaments

**24. Which of the following represents a site for simple cubical epithelium? (2025)**

- A) Bowman's capsule of kidney
- B) Thyroid follicles
- C) Fallopian tube
- D) Vas deferens

**25. Which type of epithelium is specialized for protection? (2025)**

- A) Simple squamous epithelium
- B) Stratified squamous epithelium
- C) Simple columnar epithelium with microvilli.
- D) Pseudostratified columnar epithelium

**26. Which epithelial specialization allows passage of ions between epithelial cells? (2025)**

- A) Tight Junctions
- B) Gap junctions
- C) Desmosomes
- D) Hemidesmosomes

**27. The cilia can be characterized by which of the followings? (2025)**

- A) The basal body is formed of nine microtubules
- B) The rootlet contains twenty-seven microtubules
- C) move secretions over the tissue surface in one directions
- D) Can be found in the male genital system

**28. Steriocilia are: (2024, 2023)**

- A) Long motile cilia.
- B) True cilia.
- C) Long motile microvilli.
- D) Long microvilli.

**29. Endothelium provides lining of: (2024, 2023)**

- A) Peritoneum.
- B) Pleural cavity.



C) The heart.

D) Alveoli of the lungs.

**30. Stratified squamous epithelium: (2024, 2023)**

A) Consists of a single layer of flattened cells.

B) Is protective rather than secretory or absorptive.

C) Has a rich supply of capillaries running between its cell layers.

D) May be ciliated.

**31. Desmosomes are: (2024, 2023)**

A) Macula occludens.

B) Macula adherens.

C) Zonula occludens.

D) Zonula adherens.

**32. In zonula occludens the distance between two cell membranes is: (2024, 2023)**

A) 0 nm

B) 2 nm

C) 20 nm

D) 30 nm

**33. Compound exocrine glands: (2024, 2023)**

A) Have unbranched secretory duct

B) Have branched secretory ducts

C) Secrete hormones

D) Secrete into the blood.

**34. Pseudo-stratified columnar ciliated epithelium can be found in: (2022, 2020)**

A) Ovary

B) Blood vessels.

C) Upper respiratory passages.

D) Urinary passage

**35. What is the type of epithelium which changes its thickness based on bladder distension? (KNU 2026)**

A) Stratified squamous

B) Simple columnar

C) Transitional

D) Pseudostratified columnar

**36. What is the type of epithelium lining the small intestine and has microvilli? (KNU 2026)**

A) Simple squamous

B) Simple cuboidal

C) Simple columnar

D) Stratified squamous

**37. What is a gland that secretes directly into the bloodstream called? (KNU 2026)**

A) Exocrine

B) Mixed

C) Endocrine

D) Merocrine





**38. Mammary glands secrete by which of the following? (KNU 2026)**

- A) Merocrine secretion
- B) Apocrine secretion
- C) Holocrine secretion
- D) Phagocytosis

**39. What are microvilli characterized by? (KNU 2026)**

- A) A core of microtubules
- B) Motility
- C) Actin filament core
- D) Presence in sperm

**40. Basal folding is found in a cell associated with... (KNU 2026)**

- A) Ion transport
- B) Protein synthesis
- C) Steroid secretion
- D) Mucus secretion

**41. Which of the following junctions allows direct passage of ions and small molecules between adjacent cells and is found abundantly in cardiac muscle? (KNU 2026)**

- A) Zonula occludens
- B) Desmosome
- C) Nexus
- D) Hemidesmosome

**42. In a compound tubulo-alveolar gland like the pancreas, which of the following describes the duct system? (KNU 2026)**

- A) Unbranched duct with branched secretory portion
- B) Branched duct with flask-shaped secretory units
- C) No duct system; secretion into blood
- D) Single duct with coiled secretory portion

**43. A biopsy of an epithelium that changes from 6-8 layers when empty to 2-3 layers when distended. Which of the following is the site of epithelium? (KNU 2026)**

- A) Trachea
- B) Uterus
- C) Vas deferens
- D) Urinary bladder

**44. Which type of epithelium is specialized for absorption? (2025)**

- A) Simple squamous epithelium
- B) Stratified squamous epithelium
- C) Simple columnar epithelium with microvilli
- D) Pseudostratified columnar epithelium

**45. Which epithelial specialization anchors epithelial cells to the basement membrane? (2025)**

- A) Tight Junctions
- B) Gap junctions
- C) Desmosomes
- D) Hemidesmosomes



- 46. The cilia can be characterized by which of the followings? (2025)**  
 A) The basal body is formed of nine microtubules  
 B) The rootlet contains twenty-seven microtubules  
 C) move secretions over the tissue surface in two directions  
 D) Can be found in the female genital system
- 47. Which of the following represents a site for simple squamous epithelium? (2025)**  
 A) Fallopian tube  
 B) Thyroid follicles  
 C) Bowman's capsule of kidney  
 D) Vas deferens
- 48. Which of the following represents a typical feature of desmosomes? (2025)**  
 A) The intercellular space between adjacent cell membranes is 20-30 nm  
 B) Intermediate filaments are inserted in attachment plates in the cytoplasmic side  
 C) It is a belt-like specialization  
 D) The cytoplasmic side shows condensation of actin filaments
- 49. Which of the following represents a typical feature of stereocilia? (2025)**  
 A) It is present in the cells lining fallopian tube  
 B) It is formed of microtubules  
 C) It is a motile cytoplasmic projection  
 D) Its core is formed of actin filaments
- 50. Which of the following represents the typical feature of epithelium? (2025)**  
 A) Rich in sensory nerve endings  
 B) It has low power of regeneration  
 C) High vascular tissue  
 D) The epithelium of the skin is mesodermal in origin

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-D  | 3-C  | 4-B  | 5-D  | 6-C  | 7-D  | 8-A  | 9-D  | 10-A |
| 11-B | 12-A | 13-C | 14-D | 15-B | 16-C | 17-C | 18-B | 19-A | 20-D |
| 21-A | 22-B | 23-D | 24-B | 25-B | 26-B | 27-C | 28-D | 29-C | 30-B |
| 31-B | 32-A | 33-B | 34-C | 35-C | 36-C | 37-C | 38-B | 39-C | 40-A |
| 41-C | 42-B | 43-D | 44-C | 45-D | 46-D | 47-C | 48-B | 49-D | 50-A |

Note on Q27: Debatable: option D (male genital system) is also ciliated in efferent ductules.

## Written questions

- Enumerate: a. 3 sites of neuroepithelium. (1.5 marks) [repeated x2]
- Draw a coloured labelled diagram of Non keratinized stratified squamous epithelium [repeated x2]
- Enumerate: A. 3 sites of simple squamous epithelium (1.5 marks) [repeated x2]
- Enumerate: B. 3 types of exocrine glands according to branching of duct system (1.5 marks) [repeated x2]
- Discuss (2 marks each): A. Transitional epithelium (2 marks) [repeated x2]
- Enumerate the following: A. 3 apical specializations with the site for each one. (1.5 marks)
- Draw a colored, labeled diagram of pseudostratified columnar ciliated epithelium.



8. Enumerate: b. 3 types of exocrine glands according to mode of secretion. (1.5 marks)
9. Discuss (2 marks for each): A. Myoepithelium
10. Compare between: B. Microvilli and Cilia.
11. Enumerate: A. Types of glands according to mode of secretion with one site for each.
12. Enumerate: B. Lateral specializations with reference to intercellular diameter in each type.
13. Give the scientific name for the following statements: d) It is special type of epithelium which is modified to contract.
14. State the following:- B. one site for Pseudostratified columnar epithelium with stereocilia
15. Mention types of stratified squamous epithelium.
16. (B) Enumerate the types of columnar epithelium
17. State the histological structure of transitional epithelium.



## 7. Connective Tissue Cells

50 MCQs (top 50 of 93 available, most repeated / most recent first) | 17 written questions

**1. Which of the following would be best suited to visualize reticular fibers?** (2022, 2019, Fighter Histology bank)

- A) Hematoxylin and eosin stain
- B) Sudan stain
- C) Silver impregnation
- D) Masson's trichrome stain

**2. Concerning multilocular fat cell:** (2022, 2020, 2019)

- A) Cytochrome pigments.
- B) Supportive function for some organs like the kidney.
- C) Signet ring appearance.
- D) Flat peripheral nucleus.

**3. Which connective tissue cell type produces the ground substance in connective tissue?** (2019)

- A) Fibroblast
- B) Myofibroblast
- C) Histiocyte
- D) Plasma cell



4. Which of the following would be best suited to differentiate collagen fibers from other fibers? (2024, 2023, 2022, KNU 2026)
  - A) Hematoxylin and eosin stain
  - B) Sudan stain
  - C) Silver impregnation
  - D) Mallory trichrome stain
5. What is the shape of fat cells present in the unilocular adipose tissue? (2026, 2024, 2023, KNU 2026)
  - A) Clock face
  - B) Signet ring
  - C) Horse shoe
  - D) Kidney
6. Which of the following is NOT a fiber found in connective tissue? (2022, 2019)
  - A) Collagen fiber
  - B) Elastic fiber
  - C) Reticular fiber
  - D) All of the above are fibers found in it.
7. What type of tissue is cornea composed of? (2026, 2023, KNU 2026)
  - A) Mucous connective tissue
  - B) Mesenchyme
  - C) Dense irregular connective tissue
  - D) Dense regular connective tissue
8. What is the main ultrastructural feature (EM) of histiocyte? (2026, KNU 2026)
  - A) Smooth endoplasmic reticulum
  - B) Lysosomes
  - C) Microvilli
  - D) Rough endoplasmic reticulum
9. Which of the following forms the stroma of organs? (2025)
  - A) Elastic fibers
  - B) Collagen type III (Reticular fibers)
  - C) Collagen type I
  - D) Muscle fibers
10. Reticular fibers are primarily composed of which protein? (2025)
  - A) Collagen
  - B) Elastin
  - C) Fibrillin
  - D) Keratin
11. The main ultrastructural feature (EM) of histiocyte is the presence of: (2024, 2023)
  - A) Microvilli.
  - B) Lysosomes.
  - C) SER
  - D) RER
12. Which of the following is NOT a fiber found in connective tissue? (2022, 2019)



- A) Collagen fiber
- B) Elastic fiber
- C) Reticular fiber
- D) pigmented fibers .

**13. Which connective tissue cell is derived from B lymphocytes? (2022, 2019)**

- A) Fibroblast
- B) Myofibroblast
- C) Histiocyte
- D) Plasma cell

**14. Where are metachromatic granules present in? (2026, KNU 2026)**

- A) Reticular cells
- B) Mast cells
- C) Plasma cells
- D) Fat cells

**15. Which of the following would be best suited to differentiate collagen fibers from other fibers? (2026, KNU 2026)**

- A) Van Gieson stain
- B) Orcein stain
- C) Sudan stain
- D) Silver

**16. Fibroblasts are characterized by which of the following? (2026, KNU 2026)**

- A) Abundant rER and Golgi
- B) Signet ring appearance
- C) Metachromatic granules
- D) Cart-wheel nucleus

**17. Brown adipose tissue differs from white adipose tissue in which of the following? (2026, KNU 2026)**

- A) Stores fat as a single droplet
- B) Generates heat
- C) Is poorly vascularized
- D) Has a signet ring appearance

**18. Muroid connective tissue in adult is present in which of the following? (2026, KNU 2026)**

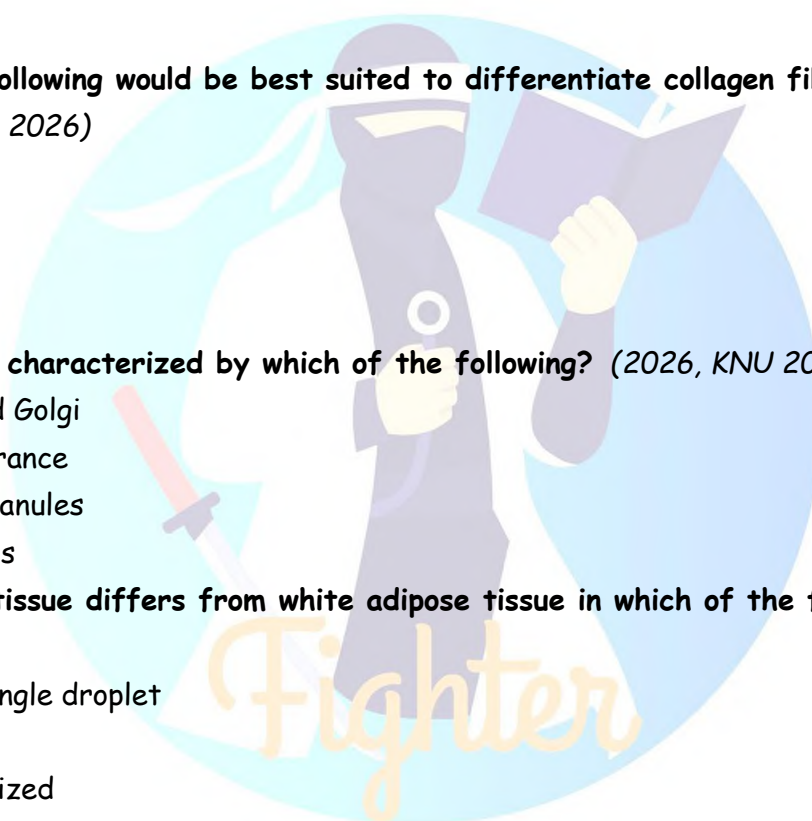
- A) Skin
- B) Umbilical cord
- C) Lymph nodes
- D) Vitreous humor of eye

**19. Which of the following represents the typical feature of connective tissue? (2025)**

- A) It has closely aggregated cells
- B) Rests on basement membrane
- C) Highly vascular tissue
- D) Ectodermal in origin

**20. Which of the following represents the typical histological structure of fibroblast? (2025)**

- A) The cytoplasm is pale basophilic





- B) Contains few rough endoplasmic reticulum
- C) The nucleus is heterochromatic
- D) Its shape is spindle-shaped with many processes

**21. Which of the following describes the white fat cell? (2025)**

- A) The cytoplasm is occupied by a single large lipid droplet
- B) The nucleus is central and rounded
- C) Contains many filamentous mitochondria
- D) Contains cytochrome pigment

**22. What is the shape of nucleus present in plasma cell? (2025)**

- A) Clock face
- B) Horse shoe
- C) Kidney
- D) Signet ring

**23. Which one is among the functions of adipose connective tissue? (2025)**

- A) Gives the skin its shape
- B) Withstands stretch
- C) Binds structures and surrounds organs & blood vessels
- D) Forms the stroma of the organs

**24. Which of the following is the site of brown adipose connective tissue? (2025)**

- A) Axilla
- B) Abdominal wall
- C) Mammary gland
- D) Mesentery

**25. The elastic fibers can be demonstrated by which stain? (2025)**

- A) H & E
- B) PAS
- C) Sudan III
- D) Orcein

**26. Plasma cells release which substances? (2025)**

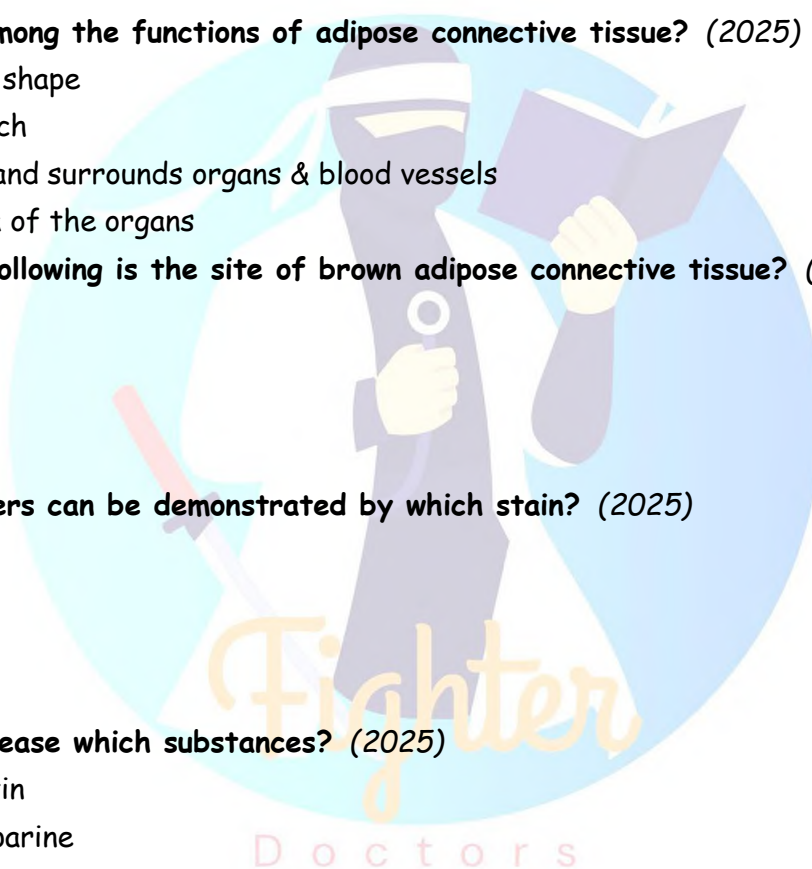
- A) Collagen and elastin
- B) Histamine and heparine
- C) Antibodies
- D) proteoglycans and fibronectin

**27. Which of the following is a characteristic of dense irregular connective tissue? (2025)**

- A) Randomly arranged collagen fibers
- B) High tensile strength in one direction
- C) Abundance of ground substance
- D) Presence of elastic fibers in all regions

**28. Plasma cells: (2024, 2023)**

- A) Have well developed smooth endoplasmic reticulum.
- B) Have phagocytic ability
- C) Are commonly found in peripheral blood
- D) Show all EM features of protein synthesizing cells.





**29. Metachromatic granules are present in: (2024, 2023)**

- A) Mast cells.
- B) Plasma cells.
- C) Fat cells.
- D) Reticular cells.

**30. Which of the following form stroma of the organs? (2024, 2023)**

- A) Loose areolar connective tissue.
- B) Reticular connective tissue.
- C) Unilocular adipose connective tissue.
- D) Muroid connective tissue.

**31. What type of tissue is a tendon composed of ? (2023, 2019)**

- A) Mucous connective tissue
- B) Mesenchyme
- C) Dense irregular connective tissue
- D) Dense regular connective tissue

**32. Concerning Collagen fibers: (2022, Fighter Histology bank)**

- A) wavy bundles formed of non-branching fibers.
- B) basophilic.
- C) one type
- D) yellow in color when fresh

**33. Concerning histiocytes have: (2019)**

- A) Darkly basophilic cytoplasm.
- B) Euchromatic nucleus.
- C) Pseudopodia.
- D) poorly developed Golgi apparatus

**34. Elastic fibers are: (2019, Fighter Histology bank)**

- A) Single, thin.
- B) Non-branching.
- C) Basophilic.
- D) Yellow with orcein stain.

**35. The network of reticular and elastic fibers is a characteristic histological feature of which cell? (KNU 2026)**

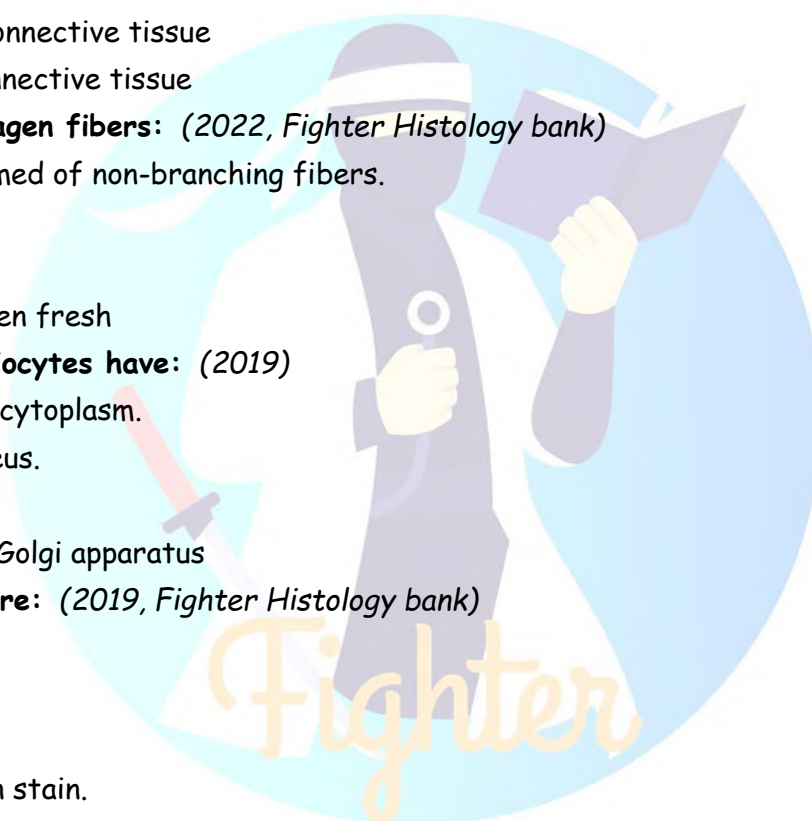
- A) Pericyte
- B) Fibroblast
- C) Plasma cell
- D) Fat cell

**36. What is the type of connective tissue forming the stroma of organs? (KNU 2026)**

- A) Reticular C.T
- B) Yellow elastic C.T
- C) White fibrous C.T
- D) Brown adipose C.T

**37. What is the characteristic feature of loose areolar connective tissue? (KNU 2026)**

- A) Contains only elastic fibers





- B) Present in the umbilical cord
- C) Contains potential cavities
- D) The least common type in the body

**38. Regular white fibrous connective tissue can be present in: (KNU 2026)**

- A) Tendons of skeletal muscles
- B) Aorta
- C) Umbilical cord
- D) Stroma of organs

**39. What is the typical feature of collagen fibers? (KNU 2026)**

- A) Run in wavy bundles
- B) Stained brown with orcein stain
- C) The individual fibers can branch
- D) Stained black with silver stain

**40. The main ultrastructural feature (EM) of histiocyte is the presence of: (KNU 2026)**

- A) Microvilli
- B) Lysosomes
- C) Smooth endoplasmic reticulum
- D) Rough endoplasmic reticulum

**41. Which of the following represents the typical histological feature of fibroblast? (KNU 2026)**

- A) The cytoplasm is acidophilic
- B) The nucleus is euchromatic
- C) Contains few rough endoplasmic reticulum
- D) Contains numerous SER

**42. What is the characteristic feature of elastic fibers? (KNU 2026)**

- A) Consist of type III collagen
- B) Stained black with silver stain
- C) Stained brown with orcein
- D) The fibers can not branch

**43. The brown color of multilocular adipose connective tissue is due to which of the followings? (KNU 2026)**

- A) Numerous rough endoplasmic reticulum
- B) Colored cytochromes inside mitochondria
- C) Poor blood supply
- D) Numerous smooth endoplasmic reticulum

**44. What is the type of connective tissue presents in the sclera of eye? (KNU 2026)**

- A) Loose areolar C.T.
- B) Regular white fibrous C.T.
- C) Reticular C.T.
- D) Irregular white fibrous C.T.

**45. Mast cells release which substances during an immune response? (2025)**

- A) Collagen and elastin
- B) Histamine and heparine



C) Antibodies and cytokines

D) proteoglycans and fibronectin

**46. Which of the following is a characteristic of dense regular connective tissue? (2025)**

A) Randomly arranged collagen fibers

B) High tensile strength in one direction

C) Abundance of ground substance

D) Presence of elastic fibers in all regions

**47. Which of the following represents the typical histological structure of fibroblasts? (2025)**

A) The nucleus is euchromatic

B) Its shape is elongated with few processes

C) The cytoplasm is pale basophilic

D) Contains few rough endoplasmic reticulum

**48. Which of the following describes the brown fat cell? (2025)**

A) The cytoplasm is occupied by a single large lipid droplet

B) The nucleus is eccentric and flattened

C) Contains few filamentous mitochondria

D) The nucleus is central and rounded

**49. Which one is among the functions of loose areolar connective tissue? (2025)**

A) Gives the skin its shape

B) Forms the stroma of the organs

C) Withstands stretch

D) Binds structures and surrounds organs & blood vessels

**50. Which of the following is the site of brown adipose connective tissue? (2025)**

A) Abdominal wall

B) Mammary gland

C) Interscapular

D) Mesentery

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-A  | 3-A  | 4-D  | 5-B  | 6-D  | 7-D  | 8-B  | 9-B  | 10-A |
| 11-B | 12-D | 13-D | 14-B | 15-A | 16-A | 17-B | 18-D | 19-C | 20-D |
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| 31-D | 32-A | 33-C | 34-A | 35-B | 36-A | 37-C | 38-A | 39-A | 40-B |
| 41-B | 42-C | 43-B | 44-D | 45-B | 46-B | 47-B | 48-D | 49-D | 50-C |

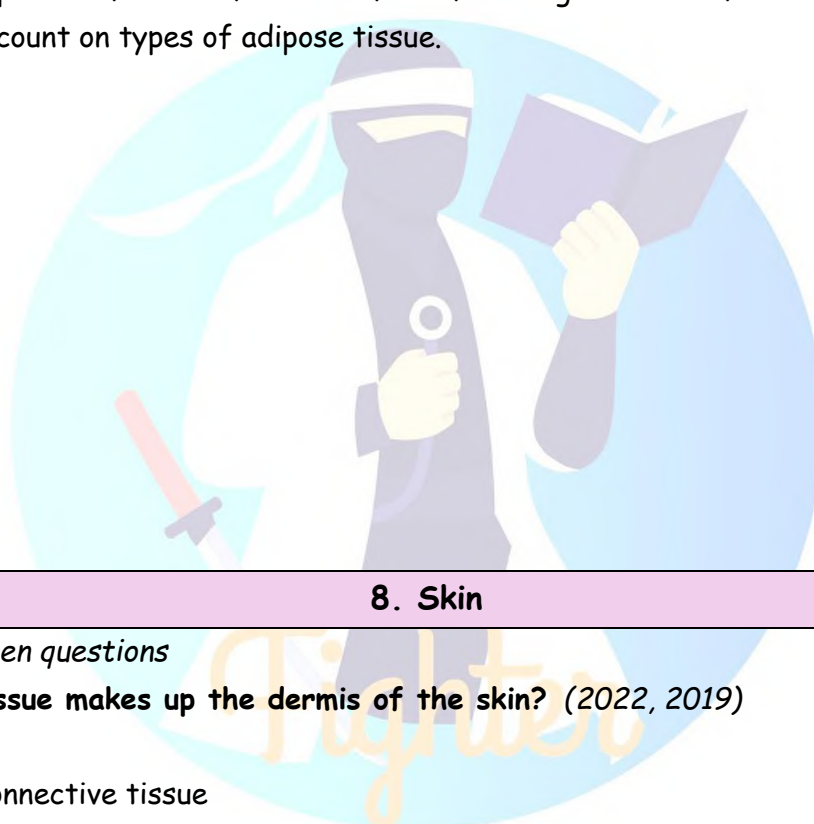
*Note on Q35: Debatable answer.*

**Written questions**

- Describe the followings A. B. Histological structure of white adipose connective tissue (3 marks) [repeated x2]
- Describe the followings E. Three connective tissue cells and mention origin and function of each one (1.5 marks) [repeated x2]
- Enumerate: C. 3 sites of Elastic connective tissue (1.5 marks) [repeated x2]
- Discuss (2 marks each): B. white Adipose Connective tissue (2 marks) [repeated x2]
- Mention the following: B. The origin of the following cells: plasma cell and macrophage. (1 mark)



6. Discuss the following: B. The histological structure of mucoid connective tissue and two sites for it. (2.5 marks)
7. Enumerate: C. Sites of Dense regular connective tissue. (2 marks)
8. Enumerate: c. 3 sites of Mucoid connective tissue. (1.5 marks)
9. Discuss (2 marks for each): B. Reticular Connective tissue
10. Compare between: A. Unilocular and multilocular adipose cell.
11. Give the scientific name for the following statements: e) Site for mucoid connective tissue in adult.
12. Compare between macrophage & plasma cell
13. Enumerate Types of Fixed connective tissue cells.
14. Discuss Reticular connective tissue (structure-function)
15. Describe the histological structure & functions of mast cells.
16. Mention one important function for each of the following: Reticular fibers
17. Give a short account on types of adipose tissue.



## 8. Skin

17 MCQs | 12 written questions

1. What type of tissue makes up the dermis of the skin? (2022, 2019)

- A) Mesenchyme
- B) Loose irregular connective tissue
- C) Dense irregular connective tissue
- D) Dense regular connective tissue

2. What is the typical feature of melanocyte in the skin? (2025)

- A) It is present between the cells in the basal cell layer
- B) Contains few lysosomes
- C) It is the sensory cells of the epidermis
- D) It is present between the cells of stratum spinosum

3. Which of the following represents the typical feature of thin skin? (2025)

- A) It contains hair follicles and sebaceous glands
- B) Contains numerous sweat glands
- C) The stratum lucidum is prominent
- D) The stratum corneum is thicker

4. Which of the following layers connected by desmosomes? (2025)



- A) Stratum basale
- B) Stratum granulosum
- C) Stratum lucidum
- D) Stratum spinosum

**5. What is the function of stratum granulosum? (2025)**

- A) Renewal of keratinocytes
- B) Resisting the effects of friction and pressure
- C) Waterproof-barrier of the skin.
- D) Receptors for fine touch sensation

**6. Which of the following best describes thick skin? (2025)**

- A) It has apparent stratum lucidum
- B) Has sebaceous glands
- C) Contains fewer sweat glands in the dermis
- D) Contains well developed hair follicles

**7. Which of the following squeezes the sebaceous gland to extrude its secretion? (KNU 2026)**

- A) Deep fascia
- B) Arrector pili muscle
- C) Hair follicle
- D) Hair root

**8. Which of the following layers contains Langer's lines? (KNU 2026)**

- A) Epidermis
- B) Dermis
- C) Superficial fascia
- D) Deep fascia

**9. In the stratum spinosum, cells are connected by numerous desmosomes. Which intermediate filament type predominates in these junctions? (KNU 2026)**

- A) Vimentin
- B) Desmin
- C) Cytokeratin
- D) Neurofilament

**10. Which of the following layers contains keratohyaline granules? (2025)**

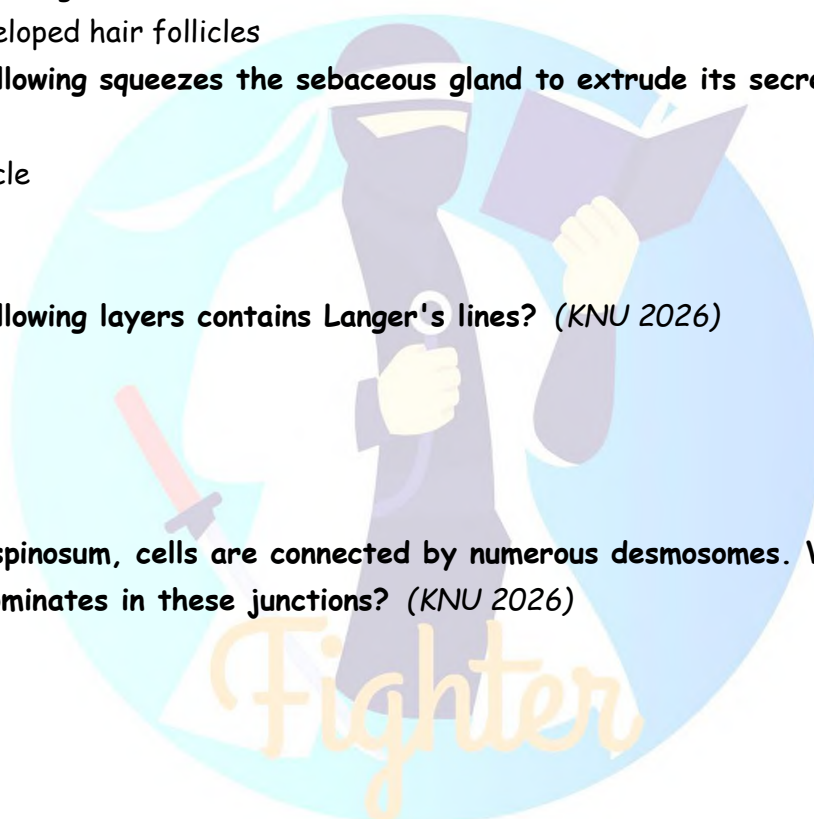
- A) Stratum basale
- B) Stratum granulosum
- C) Stratum lucidum
- D) Stratum spinosum

**11. What is the function of stratum basale? (2025)**

- A) Renewal of Keratinocytes
- B) Resisting the effects of friction and pressure
- C) Waterproof-barrier of the skin
- D) Receptors for fine touch sensation

**12. Which of the following best describes thin skin? (2025)**

- A) It has apparent stratum lucidum
- B) Contains fewer sweat glands in the dermis





C) Contains well developed papillary ridges

D) Has no sebaceous glands

**13. Which of the following is responsible for renewing the epidermis? (2025)**

A) Flattened layer of the epidermis

B) Melanocyte

C) Basal layer of epidermis

D) Elastic fibers

**14. Which of the following represents a typical feature of thick skin? (2025)**

A) Contains numerous sweat glands

B) It contains hair follicles and sebaceous glands

C) The stratum lucidum is absent

D) The stratum corneum is thinner

**15. What type of epithelium forms the epidermis? (2022)**

A) Simple cuboidal epithelium

B) Simple columnar epithelium

C) Stratified squamous epithelium

D) Pseudostratified epithelium

**16. all of the following about epidermis is correct except: (Dr Rasha)**

A) epidermis is a stratified epithelium

B) it is composed of connective tissue

C) on the palms and soles, it is extremely thick

D) it is thin on the anterior surface of the arm and forearm

**17. What type of Epithelium forms epidermis (Fighter Histology bank)**

A) Simple cuboidal epi

B) Simple columnar epi

C) Stratified squamous epi

D) Pseudostratified epi

**Answer Key**

|      |      |      |      |      |      |      |     |     |      |
|------|------|------|------|------|------|------|-----|-----|------|
| 1-C  | 2-A  | 3-A  | 4-D  | 5-C  | 6-A  | 7-B  | 8-B | 9-C | 10-B |
| 11-A | 12-B | 13-C | 14-A | 15-C | 16-B | 17-C |     |     |      |

**Written questions**

1. Mention: B. Non-keratinocytes of the skin and mention one function for each one (3 marks)  
[repeated x3]
2. Describe the followings B. Histological structure of the stratum basal in epidermis with its function (2 mark) [repeated x2]
3. Discuss the skin lines (Langer's lines).
4. Mention the following: A. Types of granules in stratum granulosum (content, type, and function). (1.5 marks)
5. Enumerate the derivatives of epidermis and describe one of them in detail
6. The cellular layer of the skin is called ..... and connective tissue layer of it is called.....
7. ....give the elasticity of the skin, ..... produce the pigment of the skin
8. Derivatives of epidermis are ....., .....and.....



9. The smooth muscle which attached to the hair follicle is called ..... and supplied by .....
10. The Sweat glands are.....and supplied by.....
11. The Sebaceous glands are.....
12. The nails are formed of .....and.....

## 9. Nervous Tissue

20 MCQs | 7 written questions

**1. Which of the following statements is incorrect regarding neuroepithelium? (2026, 2024, 2023, KNU 2026)**

- A) It is modified type of epithelium to act as receptor.
- B) Formed of sensory and supporting cells.
- C) It covers skin and forms glands.
- D) Present in taste buds.

**2. Multipolar pyramidal nerve cells can be found in which of the following? (2026, KNU 2026)**

- A) Anterior horn cells
- B) Cerebrum
- C) Cerebellum
- D) Retina

**3. Which of the following statements describe the dendrites of nerve cells? (2026, KNU 2026)**

- A) Have a constant diameter
- B) conducts nerve impulse away from the cell body
- C) Usually long
- D) Contain Nissl granules

**4. In the nervous tissue, the axons of the neuron are characterized by which one of the following? (2025)**

- A) Have uniform diameter
- B) Contain Nissl's granules
- C) Contain Golgi apparatus
- D) Convey impulses to the cell body

**5. Which of the following represents the sheath of lipoprotein complex around the axon? (2025)**

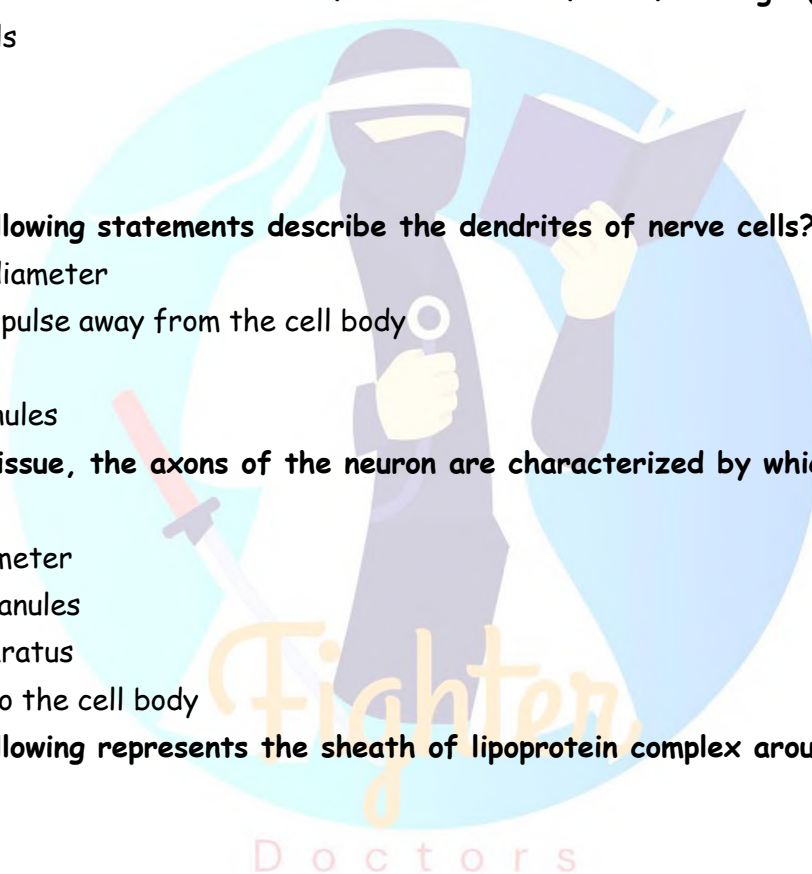
- A) Epineurium
- B) Perineurium
- C) Endoneurium
- D) Myelin

**6. The gaps between myelin sheath form which of the following structures? (2025)**

- A) Myelin sheath
- B) Axolemma
- C) Nodes of Ranvier
- D) Schmidt-Lanterman clefts

**7. Pseudounipolar neurons are present in which of the following? (2025)**

- A) Retina
- B) Spinal ganglia
- C) Sympathetic ganglia
- D) Anterior horn cells





8. What are the multiple short processes of neuron called? (2025)

- A) Nodes of Ranvier
- B) Synaptic clefts
- C) Axon terminals
- D) Dendrites

9. Which type of neuroglia forms the myelin sheath in the peripheral nervous system? (2025)

- A) Schwann cell
- B) Astrocytes
- C) Oligodendrocytes.
- D) Microglia

10. Nissl bodies in neurons correspond to which organelle at the EM level? (KNU 2026)

- A) Golgi apparatus
- B) Mitochondria
- C) Rough endoplasmic reticulum
- D) Lysosomes

11. Lipofuscin pigment accumulation in neurons is a sign of: (KNU 2026)

- A) Active phagocytosis
- B) Aging
- C) Melanin overproduction
- D) Glycogen storage

12. What are the gaps between Schwann cells in the myelin sheath called? (2025)

- A) Nodes of Ranvier
- B) Synaptic clefts
- C) Axon terminals
- D) Dendrites

13. Which type of neuroglia forms the myelin sheath in the central nervous system? (2025)

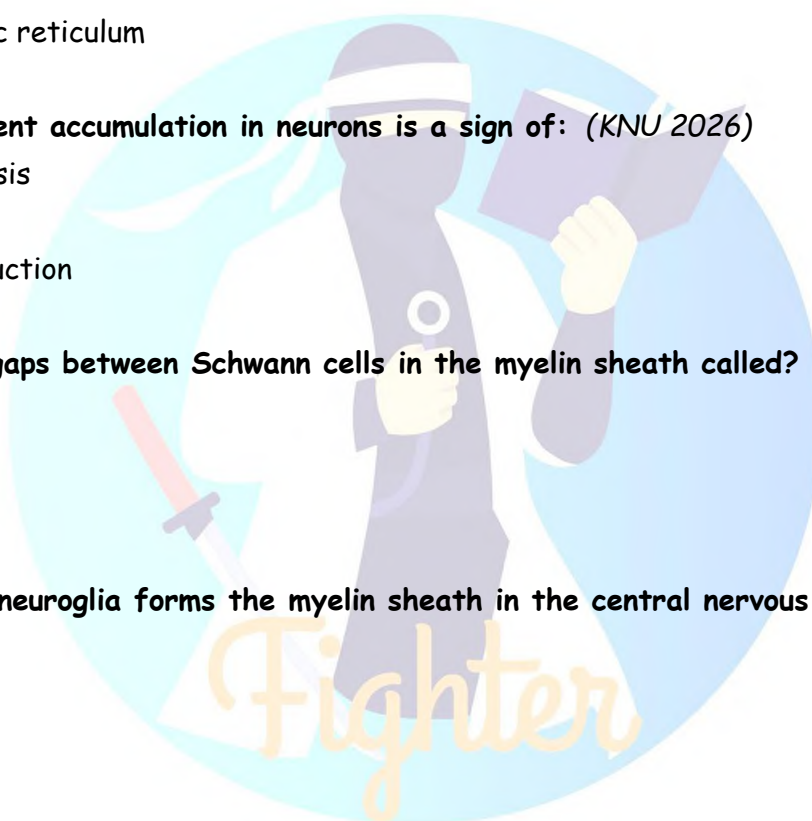
- A) Schwann cell
- B) Astrocytes
- C) Oligodendrocytes
- D) Microglia

14. The plasmalemma of Schwann cells winds and wraps around which of the following structures? (2025)

- A) Schmidt-Lanterman clefts
- B) Nodes of Ranvier
- C) Axolemma
- D) Myelin sheath

15. Which of the following represents the external sheath of dense connective tissue surrounding the nerve trunk? (2025)

- A) Endoneurium
- B) Myelin
- C) Epineurium
- D) Perineurium





**16. In the nervous tissue the dendrites of the neuron are characterized by which of the following? (2025)**

- A) Contain Golgi apparatus
- B) Have uniform diameter
- C) Convey impulse away from the cell body
- D) Contain Nissl's granules

**17. Which of the following is FALSE regarding nerve cell: (2024)**

- A) Axon doesn't contain nissel granules.
- B) Stellate nerve cell is one of bipolar nerve cells.
- C) Golgi apparatus is perinuclear.
- D) Perikaryon is devoid of centrioles.

**18. Which type of neurons is present in spinal ganglion? (2022)**

- A) Pseudo-unipolar
- B) Bipolar
- C) Multipolar
- D) Satellite

**19. Which structure carries the impulse away from the neuron cell body? (2021)**

- A) dendrite
- B) axon
- C) myelin
- D) ganglion

**20. Which is true for dendrites? (2021)**

- A) Each has a constant diameter along its entire length.
- B) They are covered by spines.
- C) Are typically myelinated.
- D) Do not branch.

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-B  | 3-D  | 4-A  | 5-D  | 6-C  | 7-B  | 8-D  | 9-A  | 10-C |
| 11-B | 12-A | 13-C | 14-D | 15-C | 16-D | 17-B | 18-A | 19-B | 20-B |

## Written questions

- Describe the followings D. State the ultrastructure of nerve cell. (1.5 marks) [repeated x2]
- Enumerate: D. Types of neurons according to number of processes (1.5 marks) [repeated x2]
- Discuss (2 marks each): C. Histological structure of perikaryon (2 marks) [repeated x2]
- Discuss (2 marks for each): C. Histological structure of Dendrites and axon
- Give the scientific name for the following statements: f) Basophilic collections of Ribosomes and rER in cytoplasm of perikaryon.
- Give the scientific name for the following statements: g) Type of neurons which has one process soon divided into two branches.
- Discuss Nerve Cell (E.M. structure)



## 1. End Module 2019 - Group A

40 MCQs

**1. following bones contain red bone marrow in the adult except:**

- A) Hip bone.
- B) shaft of tibia.
- C) sternum.
- D) upper end of femur.

**2. which of the following arteries supply the outer compact bone?**

- A) epiphyseal arteries.
- B) metaphyseal arteries.
- C) nutrient arteries.
- D) periosteal arteries.

**3. One of the following is lymphoid organ**

- A) Liver
- B) Lung
- C) Pancreas
- D) Spleen

**4. In the pulmonary circulation:**

- A) The oxygenated blood is pumped from the heart to the body
- B) The deoxygenated blood is pumped from the heart to the lung
- C) The oxygenated blood is pumped from the heart to the lung
- D) The deoxygenated blood is pumped from the heart to the body

**5. which of the following lobes of the brain contains the sensory area?**

- A) frontal lobe
- B) occipital lobe
- C) parietal lobe
- D) temporal lobe

**6. Circumpennate muscles are present in:**

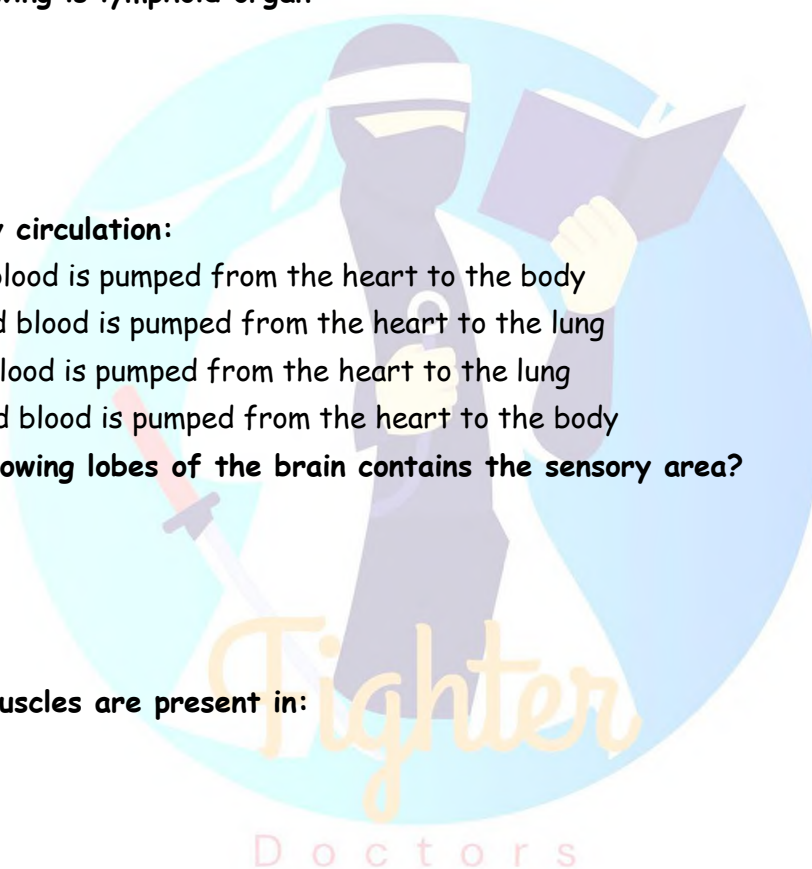
- A) Biceps muscle
- B) Deltoid muscle
- C) Rectus femoris
- D) Tibialis anterior muscle

**7. The dangerous area of the scalp is:**

- A) connective tissue layer
- B) Epicranial aponeurosis
- C) Skin layer
- D) Subaponeurotic space

**8. Which of the following muscles is present in floor of posterior triangle?**

- A) Levator scapulae muscle
- B) Omohyoid muscle
- C) Sternomastoid muscle
- D) Trapezius muscle





**9. Primary cartilaginous joints are present in:**

- A) Costochondral junction
- B) Intervertebral discs
- C) Sutures
- D) Symphysis pubis

**10. The zygomatico temporal nerve arises from:**

- A) Facial nerve
- B) Mandibular nerve
- C) Maxillary nerve
- D) Ophthalmic nerve

**11. Cells of primitive streak do not become:**

- A) Endoderm
- B) Intermediate mesoderm
- C) Paraxial mesoderm
- D) Amnioblasts

**12. The part of the sperm that contains proteolytic enzymes to digest the zona pellucida is the:**

- A) Acrosome
- B) Capacitator
- C) Corona
- D) Head

**13. The most distinctive character of a primary chorionic villus is its:**

- A) Cytotrophoblastic core
- B) Cytotrophoblastic shell
- C) Extraembryonic visceral mesodermal core
- D) Extraembryonic somatic mesodermal core

**14. Which of the following undergoes differentiation process**

- A) Primary spermatocyte
- B) Secondary spermatocyte
- C) Spermatid
- D) Spermatogonia

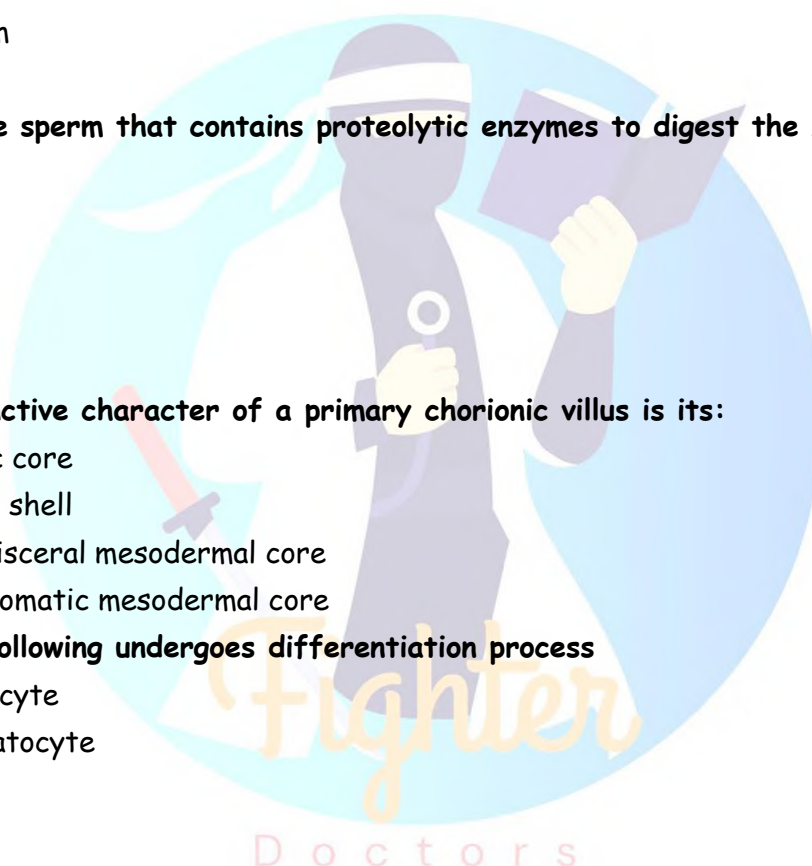
**15. Which of the following is not a component of the mature placental barrier**

- A) The basement membrane of the fetal capillaries
- B) The cytotrophoblast
- C) The endothelial lining of fetal capillaries
- D) The syncytiotrophoblast

**16. Which system is the first organ system to be visually evident in the human embryonic development**

- A) Circulatory
- B) Digestive
- C) Musculoskeletal
- D) Nervous

**17. Pregnancy tests generally are based upon the detection in maternal urine of:**





- A) Estrogen
- B) Human chorionic corticotropin
- C) Human chorionic gonadotrophin
- D) Progesterone

**18. The intraembryonic coelom is formed within:**

- A) Intermediate mesoderm
- B) Lateral plate
- C) Neural tube
- D) Somites

**19. The amniotic cavity develops:**

- A) in extraembryonic mesoderm
- B) on the ventral surface of embryonic disc
- C) within the inner cell mass near the cytotrophoblast
- D) within the outer cell mass

**20. The wall of the chorionic sac is composed of:**

- A) cytotrophoblast and syncytiotrophoblast
- B) extraembryonic splanchnic mesoderm and both layers of trophoblast
- C) trophoblast and exocoelomic membrane
- D) two layers of trophoblast lined by extraembryonic somatic mesoderm

**21. Localized cytoplasmic basophilia indicates the presence of:**

- A) Mitochondria.
- B) DNA
- C) sER.
- D) rER

**22. What type of tissue makes up the dermis of the skin?**

- A) Mesenchyme
- B) Loose irregular connective tissue
- C) Dense irregular connective tissue
- D) Dense regular connective tissue

**23. Which of the following would be best suited to visualize reticular fibers?**

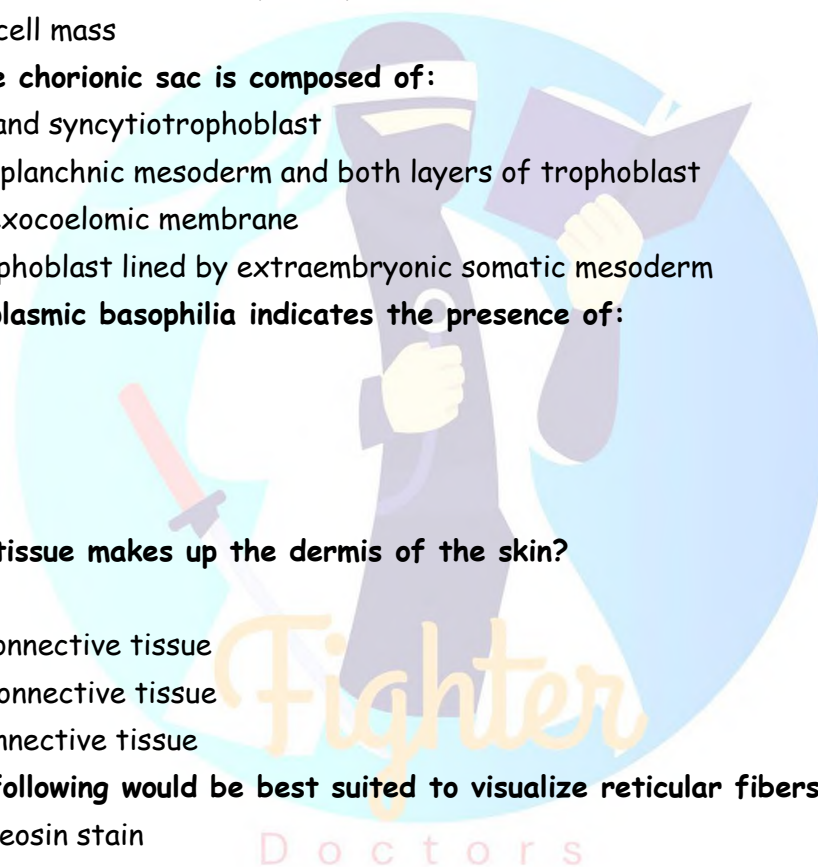
- A) Hematoxylin and eosin stain
- B) Sudan stain
- C) Silver impregnation
- D) Masson's trichrome stain

**24. Which connective tissue cell type produces the ground substance in connective tissue?**

- A) Fibroblast
- B) Myofibroblast
- C) Histiocyte
- D) Plasma cell

**25. What type of epithelium has dome shaped cells on the apical surface?**

- A) Epithelioid
- B) Mesothelium
- C) Endothelium





D) Transitional

**26. What type of gland secretes its product directly into the bloodstream?**

A) Endocrine gland

B) Exocrine gland

C) Multicellular gland

D) All of the above

**27. What type of epithelium lines the trachea?**

A) Pseudostratified epithelium

B) Simple columnar epithelium

C) Simple cuboidal epithelium

D) Simple squamous epithelium

**28. What type of tissue lines the urinary bladder?**

A) Simple squamous epithelium

B) Simple cuboidal epithelium

C) Simple columnar epithelium

D) Transitional epithelium

**29. In H&E - stained sections of plasma cell, How can Golgi apparatus appears?**

A) Apical.

B) Basal.

C) Perinuclear.

D) Unstained.

**30. Which Two organelles share in the formation of lysosomes?**

A) Golgi & cell membrane.

B) rER & Golgi.

C) rER & sER.

D) sER & Golgi.

**31. Which of the following is NOT a fiber found in connective tissue?**

A) Collagen fiber

B) Elastic fiber

C) Reticular fiber

D) All of the above are fibers found in it.

**32. What structure contributes to the cells cytoskeleton?**

A) Centrioles

B) Microfilaments

C) Peroxisome

D) Ribosomes

**33. Costal cartilage is:**

A) Hyaline cartilage.

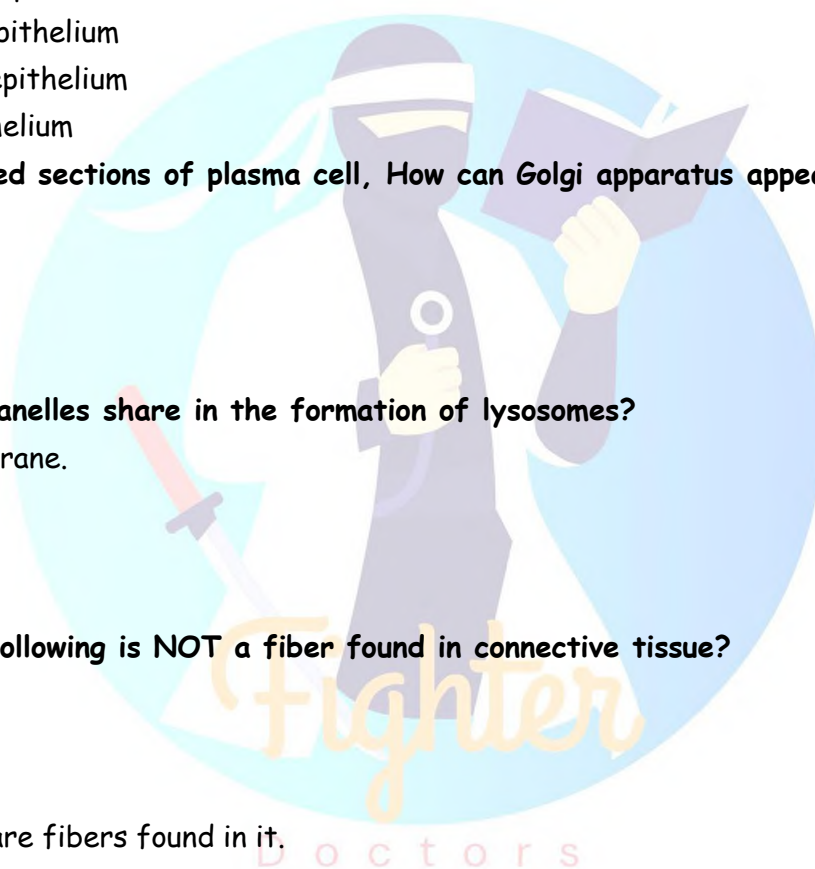
B) Yellow elastic cartilage.

C) White fibrocartilage

D) Reticular cartilage

**34. Down syndrome:**

A) caused by chromosome 21 trisomy.





B) due to duplication of chromosome 22.

C) due to multiple X syndrome.

D) due to X chromosome monosomy.

**35. Concerning multilocular fat cell:**

A) Cytochrome pigments.

B) Supportive function for some organs like kidney.

C) Signet ring appearance.

D) Flat peripheral nucleus.

**36. Concerning histiocytes have:**

A) Darkly basophilic cytoplasm.

B) Euchromatic nucleus.

C) Pseudopodia.

D) poorly developed Golgi apparatus

**37. Which of the following best suited to visualize lipids:**

A) H&E

B) Mallory

C) silver

D) sudan III

**38. Which of the following is NOT a fiber found in connective tissue?**

A) Collagen fiber.

B) Elastic fiber.

C) Reticular fiber.

D) pigmented fibers.

**39. Which is NOT lined with simple squamous epithelium:**

A) Alveoli of the lung.

B) Aorta.

C) Pericardium.

D) Thyroid gland.

**40. What type of tissue lines blood vessels?**

A) Pseudostratified epithelium

B) Simple columnar epithelium

C) Simple cuboidal epithelium

D) Simple squamous epithelium

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-B  | 2-D  | 3-D  | 4-B  | 5-C  | 6-B  | 7-D  | 8-A  | 9-A  | 10-C |
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| 31-D | 32-B | 33-A | 34-A | 35-A | 36-C | 37-D | 38-D | 39-D | 40-D |

*Note on Q14: Fighter key lists "Spermatogonia"; kept Spermatid (spermiogenesis = differentiation).*

*Note on Q15: Fighter key lists "syncytiotrophoblast"; kept Cytotrophoblast (Dr Mona's bank key: cytotrophoblast is lost in the mature barrier).*

*Note on Q18: Fighter key lists "Intermediate mesoderm"; kept Lateral plate (the intraembryonic coelom forms in lateral plate mesoderm).*



## 2. End Module 2019 - Group B

40 MCQs

1. Which of the following bones ossifies in a membrane?

- A) Radius
- B) Tibia
- C) Vault of the skull
- D) Vertebrae

2. which of the following arteries supply the bone marrow?

- A) epiphyseal arteries
- B) metaphyseal arteries
- C) nutrient arteries
- D) periosteal arteries

3. the lymph vessel is absent in following organ:

- A) Brain
- B) Liver
- C) Lung
- D) Pancreas

4. In the systemic circulation:

- A) The oxygenated blood is pumped from the heart to the lung
- B) The deoxygenated blood is pumped from the heart to the body
- C) The deoxygenated blood is pumped from the heart to the lung
- D) The oxygenated blood is pumped from the heart to the body

5. Which of the following lobes of the brain contains the visual area?

- A) Frontal lobe
- B) Occipital lobe
- C) Parietal lobe
- D) Temporal lobe

6. Bipennate muscles are present in:

- A) Extensor digitorum longus muscle
- B) Rectus abdominus muscle
- C) Rectus femoris
- D) Temporalis muscle

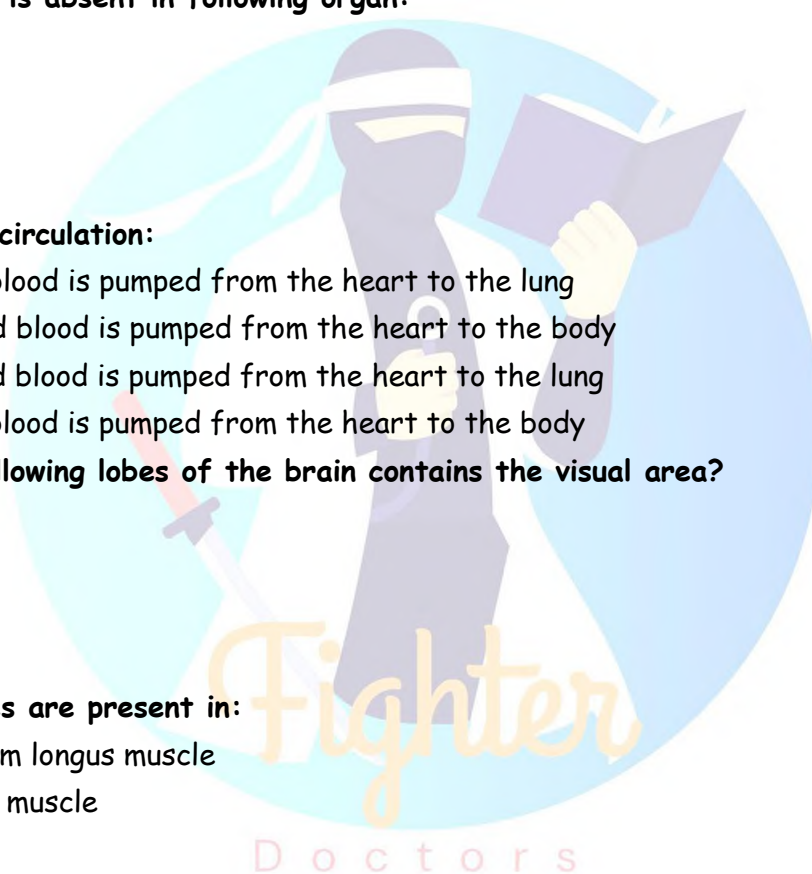
7. The occipital belly of occipitofrontalis is supplied by:

- A) Auriculotemporal nerve
- B) 3rd occipital nerve
- C) Supratrochlear nerve
- D) Posterior auricular nerve

8. Which of the following muscles is present in the floor of the posterior triangle?

- A) Levator scapulae muscle
- B) Omohyoid muscle
- C) Sternomastoid muscle
- D) Trapezius muscle

9. Pivot joints move around:





- A) Vertical axis
- B) Horizontal axis
- C) Two axis
- D) More than two axis

**10. The posterior aspect of the scalp drains into:**

- A) Mastoid lymph nodes
- B) Parotid lymph nodes
- C) Submandibular lymph nodes
- D) Submental lymph nodes

**11. Primitive streak:**

- A) is derived from the outer cells of the morula
- B) is formed during the second week in development
- C) persists as the cloacal membrane
- D) is the site of involution of epiblast cells to form mesoderm

**12. The central nervous system forms in the sequence:**

- A) Neural plate to neural crest to neural zone
- B) Neural plate to neural groove to neural tube
- C) Neural tube to neural plate to neural groove
- D) Notochord to neural plate to neural tube

**13. Which of the following structure is the postnatal remnant of the embryonic notochord?**

- A) Annulus fibrosus
- B) Nucleus pulposus
- C) Rib cage
- D) Spinal cord

**14. The time taken for a spermatogonium to develop into a mature spermatozoon is approximately:**

- A) 64 days
- B) 60 days
- C) 24 days
- D) 40 days

**15. Capacitation of the sperm:**

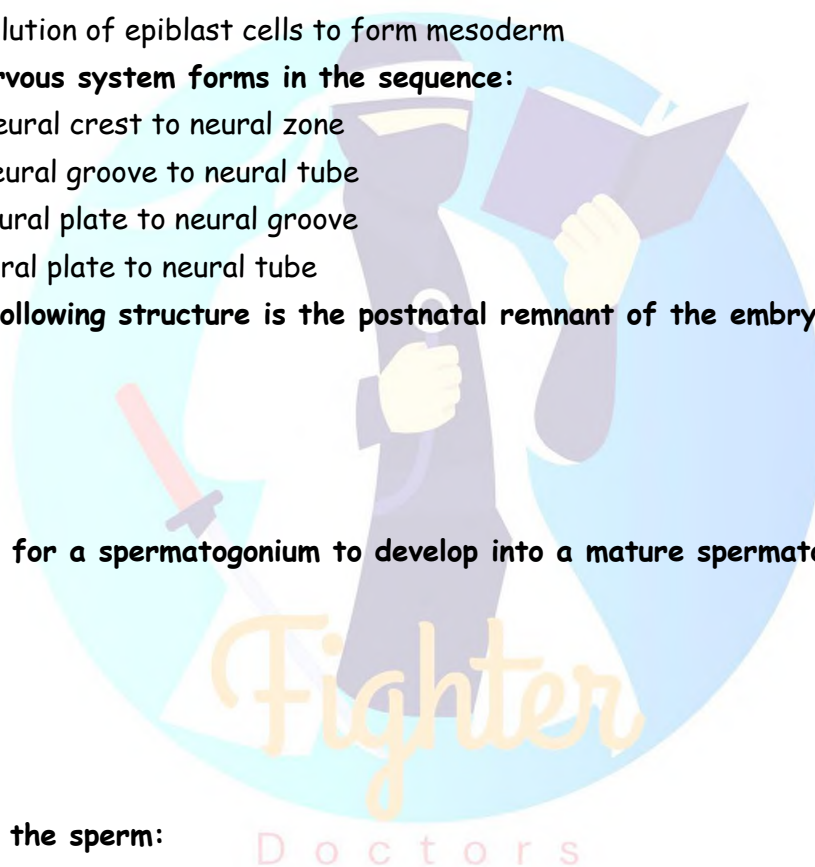
- A) is caused by the zona pellucida
- B) is essential for fertilization
- C) occurs in the male
- D) prevents polyspermy

**16. Which system is the first organ system to be visually evident in the human embryonic development?**

- A) Circulatory
- B) Digestive
- C) Musculoskeletal
- D) Nervous

**17. connecting stalk that attaches the early embryo to placenta is precursor to the:**

- A) Umbilical cord





- B) Chorionic villi
- C) Yolk sac
- D) Umbilical vein

**18. The most correct sequence of early development following fertilization is:**

- A) zygote, blastomeres, morula, blastocyst
- B) oocyte, zygote, morula, blastocyst
- C) zygote, conceptus, blastocyst
- D) polar bodies, zygote, conceptus, blastocyst

**19. The amniotic cavity develops:**

- A) in extraembryonic mesoderm
- B) on the ventral surface of embryonic disc
- C) within the inner cell mass near the cytotrophoblast
- D) within the outer cell mass

**20. Somites are developmental structures that contribute the following adult structures:**

- A) kidney, body wall connective tissue, sensory ganglia
- B) kidney, gastrointestinal tract smooth muscle, mesentery
- C) vertebra, intervertebral discs, dermis, skeletal muscle
- D) vertebra, notochord, dermis, skeletal muscle

**21. What process is nicknamed "cell drinking"?**

- A) Endocytosis
- B) Exocytosis
- C) Pinocytosis
- D) Phagocytosis

**22. Which one is an endogenous pigment?**

- A) Lead
- B) Carotenes
- C) Tattoo
- D) Hemoglobin

**23. Where are ribosomes formed?**

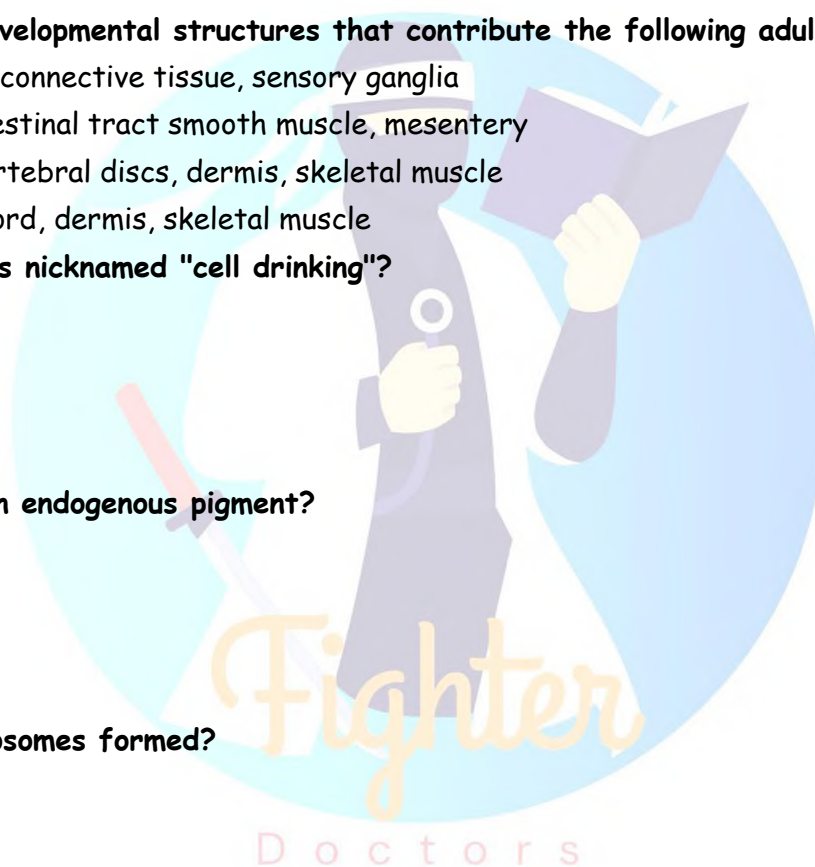
- A) Centrioles
- B) Cytoskeleton
- C) Microfilaments
- D) Nucleoli

**24. The nucleus is surrounded by which of the following?**

- A) A double membrane
- B) A single membrane
- C) No membrane around nucleus
- D) Proteins

**25. Duplication of DNA occurs in:**

- A) G1 phase
- B) G2 phase
- C) Interphase
- D) S phase





**26. What type of tissue lines blood vessels?**

- A) Simple squamous epithelium
- B) Simple cuboidal epithelium
- C) Simple columnar epithelium
- D) Stratified squamous epithelium

**27. Which organ lined by non keratinized stratified squamous epithelium?**

- A) Esophagus
- B) Stomach
- C) Small intestine
- D) Large intestine

**28. What type of epithelium has dome shaped cells on the apical surface?**

- A) Pseudostratified
- B) Mesothelium
- C) Endothelium
- D) Transitional

**29. Which of the following is a unicellular gland?**

- A) Squamous cell
- B) Goblet cell
- C) Basal cell
- D) Basket cell

**30. Which of the following can be classified as "specialized connective tissue"?**

- A) Blood
- B) Dense connective tissue
- C) Mesenchyme
- D) Mucous connective tissue

**31. What type of tissue is a tendon composed of?**

- A) Mucous connective tissue
- B) Mesenchyme
- C) Dense irregular connective tissue
- D) Dense regular connective tissue

**32. Which connective tissue cell type produces the ground substance in connective tissue?**

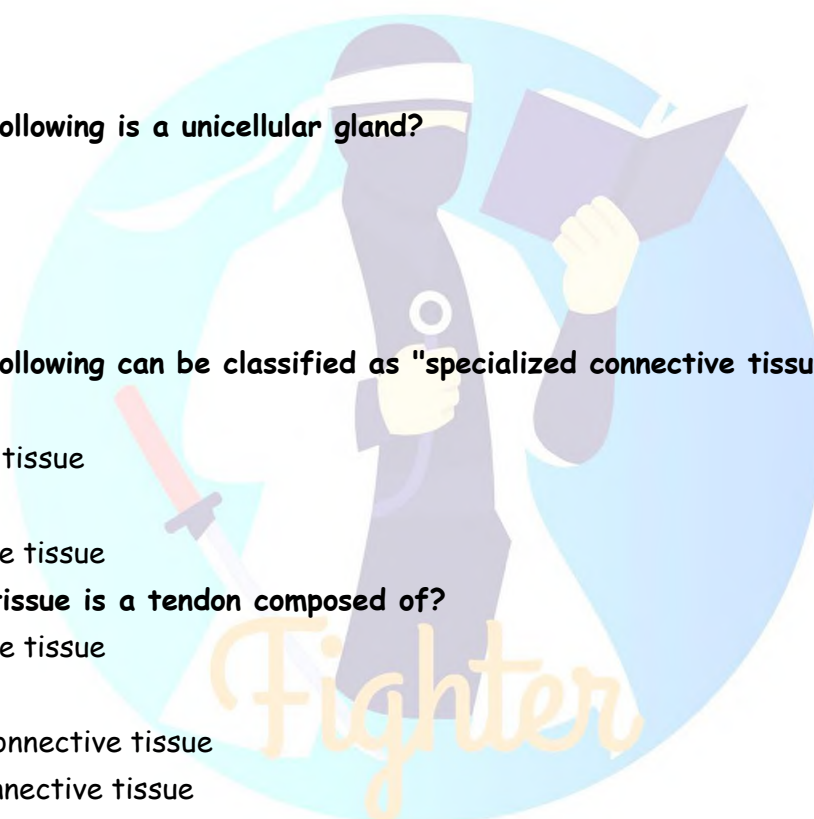
- A) Fibroblast
- B) Myofibroblast
- C) Histiocyte
- D) Plasma cell

**33. Which connective tissue cell is derived from B lymphocytes?**

- A) Fibroblast
- B) Myofibroblast
- C) Histiocyte
- D) Plasma cell

**34. Which connective tissue cell type secretes histamine?**

- A) Fibroblast
- B) Myofibroblast





C) Plasma cell

D) Mast cell

**35. The type of cartilage found in the external ear is called:**

A) Hyaline cartilage.

B) Yellow elastic cartilage.

C) White fibrocartilage.

D) Articular cartilage.

**36. Hyaline cartilage can be found in:**

A) Intervertebral disc.

B) Pubic symphysis.

C) Costal cartilage.

D) Ear pinna.

**37. Elastic fibers are:**

A) Single, thin.

B) Non-branching.

C) Basophilic.

D) Yellow with orcein stain.

**38. Costal cartilage is:**

A) Hyaline cartilage.

B) Yellow elastic cartilage.

C) White fibrocartilage.

D) Reticular cartilage.

**39. White fibrocartilage is found in the following site:**

A) Fetal skeleton.

B) Trachea.

C) Intervertebral disc

D) Eustachian tube.

**40. Concerning multilocular fat cell:**

A) Cytochrome pigments.

B) Flat peripheral nucleus.

C) Signet ring appearance.

D) Supportive function for some organs like kidney.

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
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| 31-D | 32-A | 33-D | 34-D | 35-B | 36-C | 37-A | 38-A | 39-C | 40-A |

*Note on Q3: Fighter key lists "Liver"; kept Brain (lymph vessels are absent in the brain).*

*Note on Q11: Fighter key lists option B; kept D (the primitive streak forms in the 3rd week).*

*Note on Q18: Fighter key lists option B; kept A.*

*Note on Q20: Fighter key lists "notochord" option; kept C (the notochord is not a somite derivative).*



## 3. End Semester 2019

28 MCQs

**1. What is the primitive streak ?**

- A) It is derived from the outer cells or the morula
- B) It is formed during the second week in development
- C) It is called the cloacal membrane
- D) It is the site of involution of epiblast cells to form mesoderm

**2. What is correct about the notochordal process ?**

- A) It arises from involuting endodermal Cells
- B) It extends from the prochordal plate to the primitive node
- C) It is involved in the induction of the primitive gut
- D) It becomes the appendicular skeleton

**3. When the chorionic Villi are called secondary chorionic villi ?**

- A) Contact the decidua basalis
- B) Are covered by syncytiotrophoblast
- C) Develop a mesenchymal core
- D) Give rise to branch Villi

**4. What is the region of the decidua which is essential for nourishment of the fetus ?**

- A) Basalis
- B) Capsularis
- C) Laeve
- D) Parietalis

**5. Any muscle oppose the action or prime mover is named :**

- A) Action of paradox
- B) Antagonist
- C) Fixator
- D) Synergist

**6. What is the result of the Sympathetic stimulation ?**

- A) Decrease in the heart rate
- B) Decrease in the respiratory rate
- C) Increase in the adrenaline release
- D) Increase the digestion

**7. Where is the insertion of the sternomastoid muscle ?**

- A) lateral surface of mastoid process and lateral part of superior nuchal line
- B) lateral surface of mastoid process and medial part of superior nuchal line
- C) Medial surface of mastoid process and lateral part of superior nuchal line
- D) Medial surface of mastoid process and medial part of superior nuchal line

**8. Which one of the following is the dangerous layer of the scalp ?**

- A) Connective subcutaneous tissue
- B) Loose areolar tissue
- C) Pericranium .
- D) Skin

**9. The motor nerve supply of the scalp behind the auricle is :**



- A) Great auricular nerve
- B) Greater occipital nerve
- C) Lesser occipital nerve
- D) Posterior auricular branch of facial nerve

**10. The following structure is not a content of the carotid triangle :**

- A) Anterior jugular vein
- B) Internal carotid artery
- C) Internal jugular vein
- D) Vagus nerve

**11. As regarding to the spleen which of the following statement is wrong?**

- A) It destroys worn-out RBCs
- B) It is Largest lymphoid organ
- C) It lies below right cupola of diaphragm
- D) It produces lymphocytes

**12. Which membranous organelle is prominent in cells synthesize steroid hormones?**

- A) Golgi apparatus
- B) Mitochondria
- C) rER
- D) sER

**13. Where the Simple cubical epithelium is can be found?**

- A) Gall bladder.
- B) Lung alveoli.
- C) Thyroid follicles
- D) urinary bladder

**14. What are the Ribosomes ?**

- A) Exogenous pigment
- B) Membranous organelles .
- C) Non membranous organelle
- D) Stored food

**15. What type of tissue lines blood vessels ?**

- A) Simple Squamous epithelium
- B) Simple cuboidal epithelium
- C) Simple columnar epithelium
- D) Pseudostratified epithelium

**16. Which of the following would be best suited to visualize Reticular fibers ?**

- A) Hematoxylin and eosin stain
- B) Masson's trichrome stain
- C) Silver impregnation
- D) Sudan stain

**17. What type of tissue makes up the dermis of the skin ?**

- A) Dense irregular connective tissue
- B) Dense regular connective tissue
- C) Loose irregular connective tissue

**18. What is the polysaccharide coating that is sometimes found on the cell membrane?**

- A) Cristae
- B) Glycocalyx
- C) Plasmalemma
- D) protoplasm

**19. What are ribosomes constructed ?**



- A) Cytoskeleton  
C) Nucleoli
- B) Microfilaments  
D) Peroxisome
- 20. What type Of tissue composes the kidney tubules ?**  
A) Simple squamous epithelium  
B) Simple columnar epithelium  
C) Pseudostratified epithelium  
D) Stratified squamous epithelium
- 21. Which connective tissue cell is derived from B lymphocytes ?**  
A) Fibroblast  
B) Histiocyte  
C) Myofibroblast  
D) Plasma cell
- 22. Which connective tissue cell type produces the ground substance in connective tissue?**  
A) Histiocyte  
B) Fibroblast  
C) Myofibroblast  
D) Plasma cell
- 23. What type of epithelium has dome shaped cells on the apical surface?**  
A) Epithelioid  
B) Endothelium  
C) Mesothelium  
D) Transitional
- 24. What is the effect of stimulation of greater splanchnic nerve?**  
A) Decreased blood volume.  
B) Increased blood glucose level.  
C) Increased gastrointestinal motility .  
D) Secretion of epinephrine 20% & norepinephrine 80% from adrenal medulla.
- 25. Which character of the followings as regard autonomic nervous system ?**  
A) Efferent nerves have either excitatory or inhibitory effects on the effector organs.  
B) Having one efferent neuron.  
C) Preganglionic neurons are located in the anterior horn of the spinal cord.  
D) Postganglionic neurons are located in the lateral horn of spinal cord.
- 26. What is the effect of activation of sympathetic system?**  
A) Bradycardia.  
B) Cutaneous vasodilatation.  
C) Pupillary constriction.  
D) Sweat secretion.
- 27. Which of the following symptoms appear in Horner's syndrome?**  
A) Dilatation of pupil .  
B) Dropping of the upper eye lid  
C) Increased sweating.  
D) Vasoconstriction of skin blood vessels
- 28. Which character belongs to parasympathetic nerves?**  
A) Are important in emergency conditions e.g. fight & flight reactions.  
B) Are catabolic in their actions.  
C) Generally exert opposite effects to those of the symp .Nerves  
D) Tend to have longer postgang. fibres than pregang. Fibers.

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-B  | 3-C  | 4-A  | 5-B  | 6-C  | 7-A  | 8-B  | 9-D  | 10-A |
| 11-C | 12-D | 13-C | 14-C | 15-A | 16-C | 17-A | 18-B | 19-C | 20-B |
| 21-D | 22-B | 23-D | 24-B | 25-A | 26-D | 27-B | 28-C |      |      |



## 4. End Semester 2019 دور ثاني

Source: MedZone | 27 MCQs | Q25-27 added from the Fighter file

1. Which sentence of the following does NOT true for fertilization?

- A) It is the union of a human ovum and sperm
- B) It is occurring in the isthmus of the fallopian tube.
- C) The result of this union is the production of a zygote cell
- D) It can occur 12-24 hours after ovulation

2. As regarding Anatomy of the Sperm which one of the following is WRONG?

- A) Its total length is about 50  $\mu\text{m}$
- B) It consists of: head, neck, middle piece and tail
- C) The head contains a nucleus, a pool of actin
- D) The acrosome is a vesicle contains the genetic material of the sperm

3. Acrosome Reaction is.....

- A) Alteration of the molecular structure of the glycoproteins in zona pellucida
- B) Removal of the surface-coating proteins derived from seminal fluid
- C) The digestive acrosomal enzymes at the tip of the sperm's head allow the sperm to burrow through the corona radiata
- D) The enzymatic content of acrosome, which can dissolve the glycoproteins in zona pellucida, is thus released.

4. When the chorionic villi are called secondary chorionic villi?

- A) Contact the decidua basalis
- B) Develop a mesenchymal core
- C) Give rise to branch villi
- D) Are covered by syncytiotrophoblast

5. What is region of the decidua which is essential for nourishment of fetus?

- A) Basalis
- B) Capsularis
- C) Laeve
- D) Parietalis

6. What is the action of orbital part of orbicularis oculi?

- A) Blinking
- B) Compressing the contents of orbit
- C) Direction of tears to lacrimal sac
- D) Supporting the lower eyelid against gravity

7. Which of the following lymph nodes drains the central part of the face?

- A) Buccal
- B) Submandibular
- C) Preauricular (Parotid)
- D) Submental

8. Frontal belly of occipitofrontalis muscle is supplied by which of following nerve?

- A) Auriculotemporal
- B) Temporal branch of facial
- C) Zygomatic branch of facial



D) Zygomaticofacial

**9. Which of the following sentence describe the Spiral muscle?**

- A) It is twisted muscle from the origin to its insertion
- B) Its fibers are arranged parallel to the line of pull of the muscle
- C) Its fibers converge from all sides on a central tendon
- D) Its tendon lies at the apex of the triangle

**10. The frontal lobe of the brain contains.....**

- A) Auditory area
- B) Motor area
- C) Sensory area
- D) Visual area

**11. What is the type of the joint between the tooth and its socket?**

- A) Ball and socket
- B) Gomphosis
- C) Sutural joint
- D) Syndesmosis

**12. The artery that supplies the bone marrow is.....**

- A) Periosteal
- B) Epiphyseal
- C) Metaphyseal
- D) Nutrient

**13. Localized cytoplasmic basophilia indicates the presence of.....**

- A) DNA
- B) Mitochondria
- C) RER
- D) SER

**14. What type of tissue makes up the dermis of the skin?**

- A) Dense irregular connective tissue
- B) Dense regular connective tissue
- C) Loose irregular connective tissue
- D) Mesenchyme

**15. Which of the following would be best suited to visualize reticular fibers?**

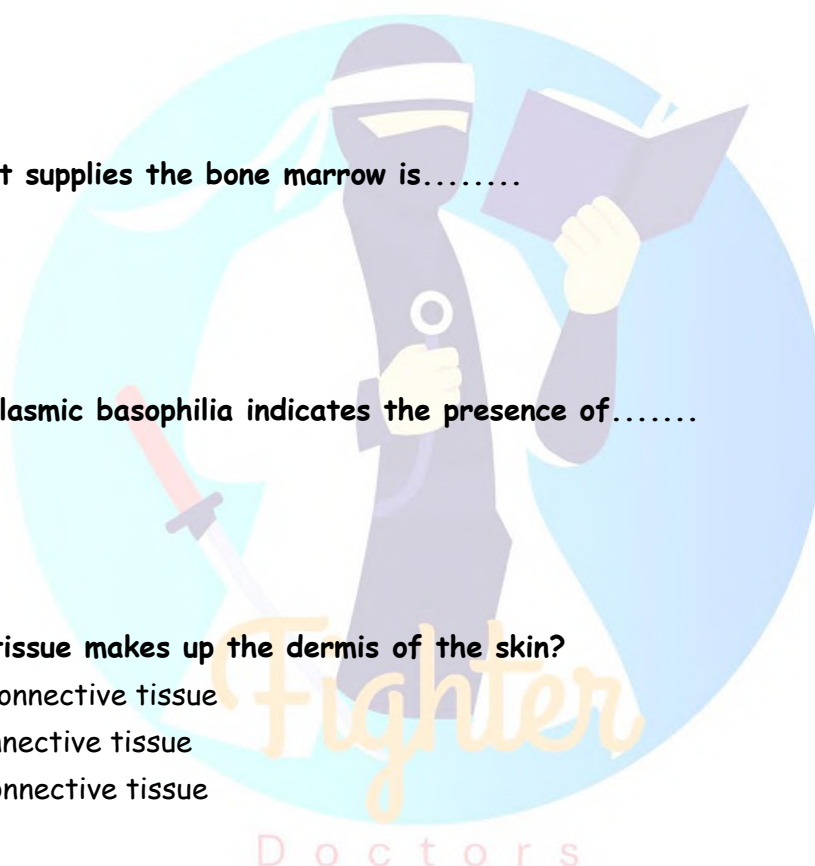
- A) Hematoxylin and eosin stain
- B) Masson's trichrome stain
- C) Silver impregnation
- D) Sudan stain

**16. Which connective tissue cell type produces ground substance in connective tissue?**

- A) Fibroblast
- B) Histiocyte
- C) Myofibroblast
- D) Plasma cell

**17. What type of epithelium has dome-shaped cells on the apical surface?**

- A) Endothelium





- B) Epithelioid
- C) Mesothelium
- D) Transitional

**18. What type of gland secretes its product directly into the bloodstream?**

- A) Endocrine
- B) Exocrine
- C) Multicellular
- D) All the above

**19. What type of epithelium lines the trachea?**

- A) Simple squamous epithelium
- B) Simple cuboidal epithelium
- C) Simple columnar epithelium
- D) Pseudostratified epithelium

**20. What type of tissue lines the urinary bladder?**

- A) Simple squamous epithelium
- B) Simple cuboidal epithelium
- C) Simple columnar epithelium
- D) Transitional epithelium

**21. In H&E stained sections of plasma cell, how can Golgi apparatus appear?**

- A) Apical
- B) Basal
- C) Perinuclear
- D) Unstained

**22. Which two organelles share in the formation of lysosomes?**

- A) Golgi & cell membrane
- B) RER & SER
- C) SER & Golgi
- D) RER & Golgi

**23. Which of the following is NOT a fiber found in connective tissue?**

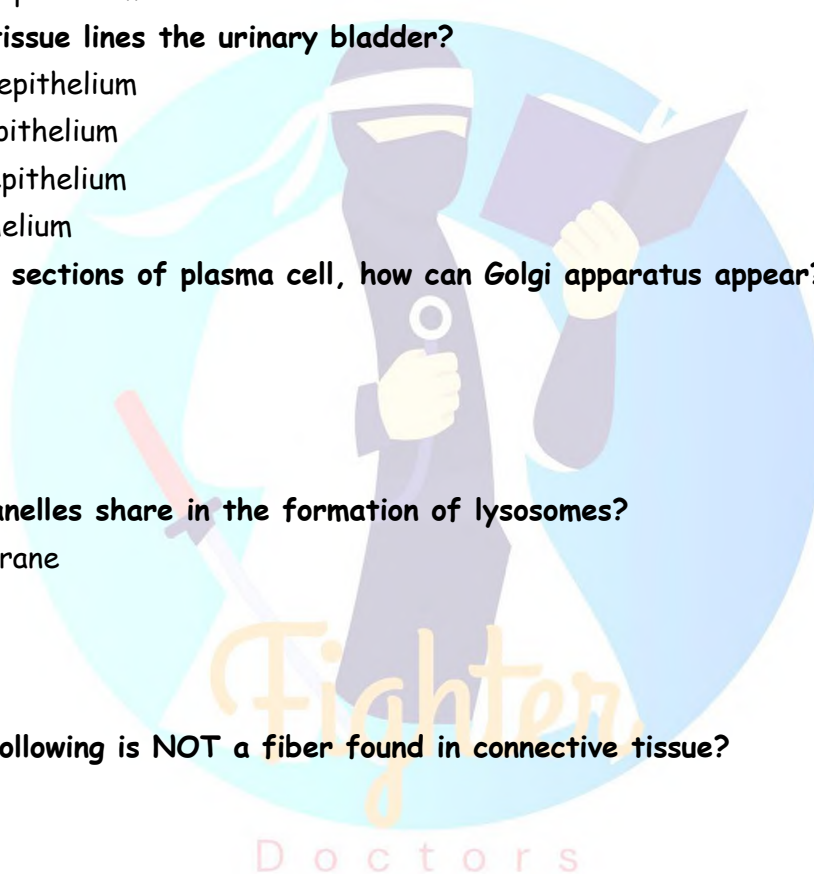
- A) Collagen fiber
- B) Reticular fiber
- C) Elastic fiber
- D) All the above are fibers found in it.

**24. What structure contributes to the cell's cytoskeleton?**

- A) Centrioles
- B) Microfilaments
- C) Peroxisome
- D) Ribosomes

**25. What is the effect of stimulation of lesser Splanchnic nerve?**

- A) Defecation.
- B) Ejaculation.
- C) Erection.
- D) Micturition.





26. Which of the following refers to autonomic nervous system?

- A) Having one efferent neuron.
- B) Efferent nerves have either excitatory or inhibitory effects on the effector organs.
- C) Postganglionic neurons are located in the lateral horn of spinal cord
- D) Preganglionic neurons are located in the anterior horn of the spinal cord.

27. What is the effect of Generalized sympathetic stimulation?

- A) Decreases blood sugar levels.
- B) Inhibition of sweating.
- C) Increase gastric motility.
- D) Rise arterial blood pressure.

### Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-B  | 2-D  | 3-D  | 4-B  | 5-A  | 6-B  | 7-B  | 8-B  | 9-A  | 10-B |
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| 21-D | 22-D | 23-D | 24-B | 25-B | 26-B | 27-D |      |      |      |

### Written questions

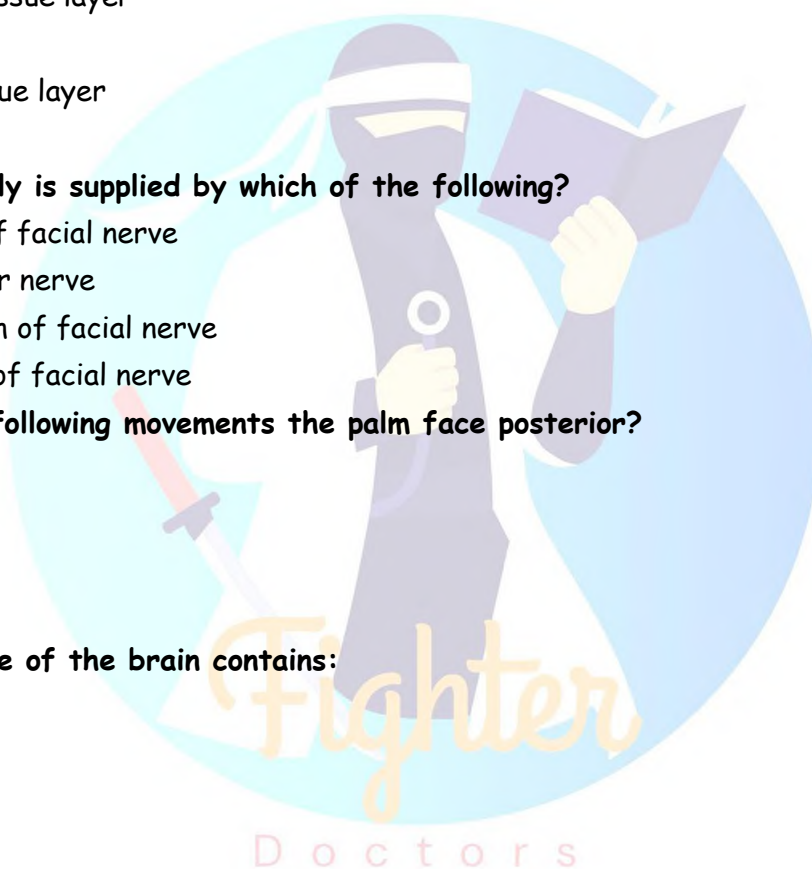
1. Draw a diagram for the venous drainage of the scalp and face.
2. Name the parts of the brain and give the function of each.
3. Enumerate the contents of the carotid triangle.
4. What is the notochord? How is this structure formed?
5. Mention one important function for each of the following:  
Lysosomes  
Cell coat  
Mast cell  
Reticular fibers
6. Mention the fate of primary lysosomes.
7. Give a short account on types of adipose tissue.
8. State the histological structure of transitional epithelium.

Doctors



## 5. End Semester 2020

1. Which of the following is an example of secondary cartilaginous joint?
  - A) Shoulder joint
  - B) Radio-ulnar joint
  - C) costo chondral joint
  - D) Symphysis pubis
2. Muscles with fibers oblique to the line of pull is present in which of the following?
  - A) Trapezius muscle
  - B) Temporalis muscle
  - C) Extensor digitorum longus muscle
  - D) Rectus femoris muscle
3. Which of the following is the dangerous area of the scalp?
  - A) The connective tissue layer
  - B) The pericranium
  - C) Loose areolar tissue layer
  - D) Skin layer
4. The occipital belly is supplied by which of the following?
  - A) Cervical branch of facial nerve
  - B) Posterior auricular nerve
  - C) Mandibular branch of facial nerve
  - D) Occipital branch of facial nerve
5. in which of the following movements the palm face posterior?
  - A) Supination
  - B) Pronation
  - C) Eversion
  - D) Inversion
6. The occipital lobe of the brain contains:
  - A) motor area
  - B) sensory area
  - C) auditory area
  - D) visual area
7. which of the following arteries supply the bone marrow?
  - A) epiphyseal arteries
  - B) metaphyseal arteries
  - C) nutrient arteries
  - D) periosteal arteries
8. which of the followings contain deoxygenated blood?
  - A) Internal carotid artery
  - B) Pulmonary artery
  - C) Pulmonary vein
  - D) External carotid artery
9. Cell coat function all except:
  - A) Cell adhesions.





- B) Cell allergy.
- C) Cell recognition.
- D) Cell transport.

**10. Which organelle produces protein for export?**

- A) SER
- B) RER
- C) Lysosome.
- D) Mitochondria
- E) Golgi apparatus

**11. The cytoplasmic side of the coated vesicle is surrounded by:**

- A) Clathrin.
- B) Tubulin.
- C) Actin.
- D) Spectrin.

**12. Which of the following can be classified as "embryonic connective tissue"?**

- A) Cartilage
- B) Mucous connective tissue
- C) Adipose tissue
- D) Bone

**13. Pseudo-stratified columnar ciliated epithelium can be found in:**

- A) Ovary
- B) Blood vessels.
- C) Upper respiratory passages.
- D) Urinary passage

**14. Plasma cell:**

- A) Originates from T-lymphocytes.
- B) Small cells
- C) Phagocytic cells
- D) Clock-face nucleus

**15. Concerning multilocular fat cell:**

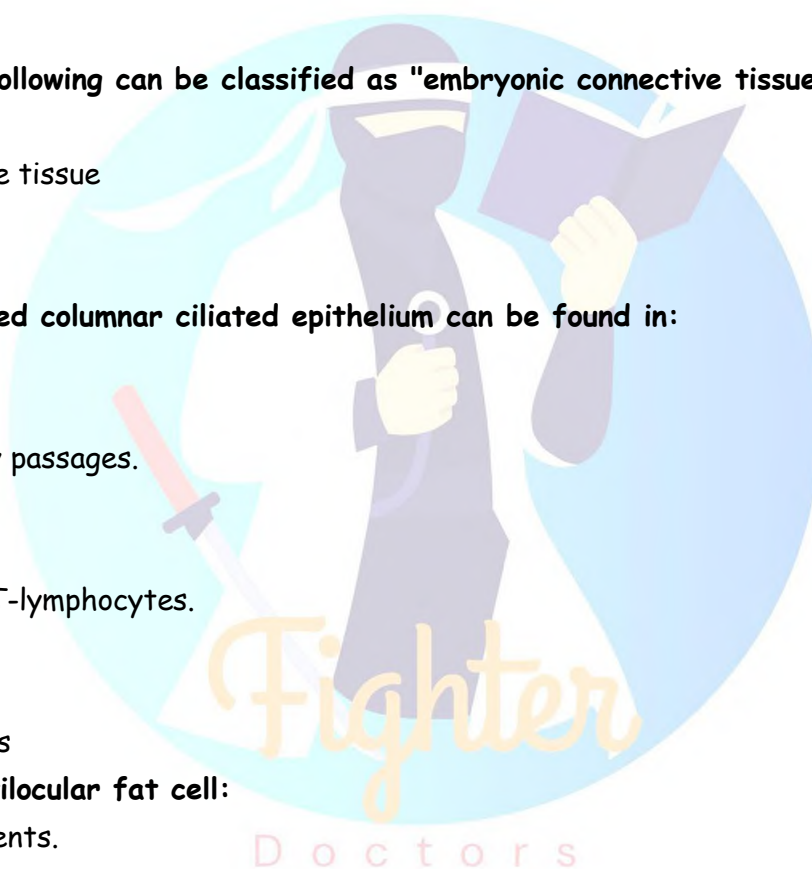
- A) Cytochrome pigments.
- B) Supportive function for some organs like the kidney.
- C) Signet ring appearance.
- D) Flat peripheral nucleus.

**16. Which of the following share in formation of cilia & flagella?**

- A) Glial cells .
- B) Keratin
- C) Centrioles .
- D) Neurofilaments.
- E) Cytoskeleton.

**17. What structure contributes to the cells cytoskeleton?**

- A) Centrioles .
- B) Peroxisome.





C) Ribosomes.

D) Microfilaments

**18. Which of the following structures is derived from neural crest cells?**

A) Dorsal root ganglion

B) Dura mater

C) Epidermis covering the back

D) Intervertebral disk

**19. What is the sequence of the development of the central nervous system?**

A) neural plate to neural crest to neural zone

B) neural plate to neural groove to neural tube

C) neural tube to neural plate to neural groove

D) notochord to neural groove to neural tube

**20. What is the incidence of abnormality the folate supplementation used to lower?**

A) abortion

B) cretinism

C) Ectopic pregnancy

D) neural tube defects

**21. which of the following plane divides the body into anterior and posterior parts?**

A) sagittal plane

B) coronal plane

C) parasagittal plane

D) horizontal plane

**22. which of the following terms means abduction of the fingers?**

A) Move the finger away from the median plane

B) Moving the finger away from the index finger

C) Moving the finger away from the middle finger

D) Moving the finger away from the little finger

**23. Which of the following bones contain red bone marrow in the adult?**

A) Shaft of femur

B) Shaft of radius

C) sternum

D) shaft of tibia

**24. which of the following arteries supply the spongy bone?**

A) epiphyseal arteries

B) metaphyseal arteries

C) nutrient arteries

D) periosteal arteries

**25. Which of the following is an example of pivot joint?**

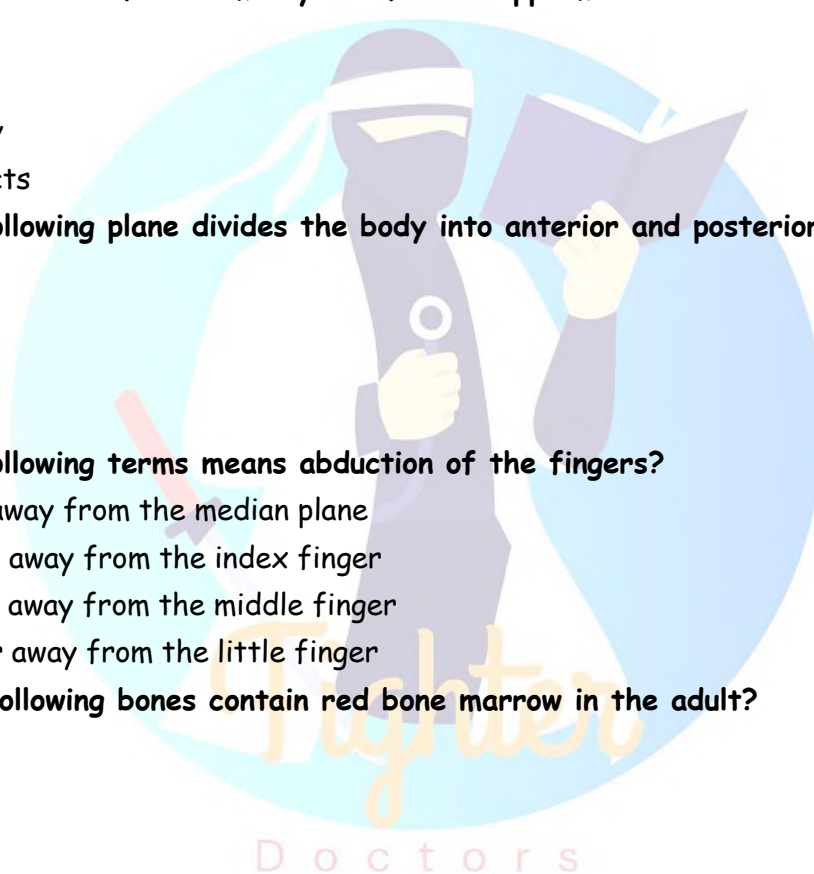
A) Shoulder joint

B) radio-ulnar joint

C) costo chondral joint

D) intervertebral discs

**26. Unipennate muscles form is present in which of the following?**





- A) trapezius muscle
- B) temporalis muscle
- C) extensor digitorum longus muscle
- D) rectus femoris muscle

**27. Which of the following is an example of the poly-axial joint?**

- A) Acromio-clavicular joint joint
- B) hip joint
- C) metacarpo-phalangeal joint
- D) inferior radio-ulnar joint

**28. The anterior facial vein is formed by which of the following?**

- A) Superficial temporal + maxillary veins
- B) Occipital vein + posterior auricular vein
- C) Supra-orbital+ supra-trochlear veins
- D) Anterior facial+ posterior facial veins.

**29. The occipital belly is supplied by which of the following?**

- A) Cervical branch of facial nerve
- B) posterior auricular nerve
- C) mandibular branch of facial nerve
- D) occipital branch of facial nerve

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-A  | 3-C  | 4-B  | 5-B  | 6-D  | 7-C  | 8-B  | 9-D  | 10-B |
| 11-A | 12-B | 13-C | 14-D | 15-A | 16-C | 17-D | 18-A | 19-B | 20-D |
| 21-B | 22-C | 23-C | 24-A | 25-B | 26-C | 27-B | 28-C | 29-B |      |

*Note on Q24: Debatable: epiphyseal vs metaphyseal arteries.*

## Written questions

1. Describe with drawing the venous drainage of the face?
2. Mention the difference between the artery, vein and lymph vessel.
3. Describe the types of filaments.
4. (A) Terminology Structure means
4. (B) Enumerate the types of columnar epithelium
5. Describe blood supply of the bones?
6. Enumerate forms of skeletal muscles and example of each type?
7. Describe the bone marrow?
8. Enumerate the results of fertilization?



## 6. End Semester 2021

24 MCQs

1. Which of the following is an example of secondary cartilaginous joint?

- A) Shoulder joint
- B) Radio-ulnar joint
- C) costo chondral joint
- D) Symphysis pubis

2. Muscles with fibers oblique to the line of pull (triangular) is present in which of the following?

- A) Trapezius muscle
- B) Temporalis muscle
- C) Extensor digitorum longus muscle
- D) Rectus femoris muscle

3. Which of the following is the dangerous area of the scalp?

- A) The connective tissue layer
- B) The pericranium
- C) Loose areolar tissue layer
- D) Skin layer

4. The occipital belly is supplied by which of the following?

- A) Cervical branch of facial nerve
- B) Posterior auricular nerve
- C) Mandibular branch of facial nerve
- D) Occipital branch of facial nerve

5. In which of the following movements the palm face posterior?

- A) Supination
- B) Pronation
- C) Eversion
- D) Inversion

6. The occipital lobe of the brain contains:

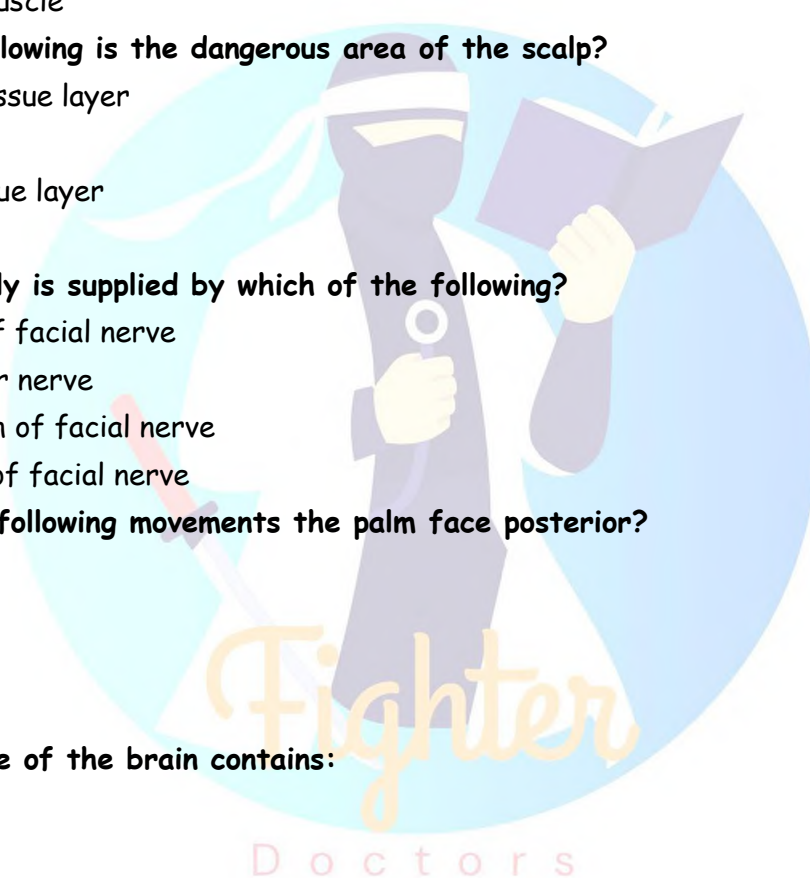
- A) Motor area
- B) Sensory area
- C) Auditory area
- D) Visual area

7. Which of the following arteries supply the bone marrow?

- A) Epiphyseal arteries
- B) Metaphyseal arteries
- C) Nutrient arteries
- D) Periosteal arteries

8. Which of the followings contain deoxygenated blood?

- A) Internal carotid artery
- B) Pulmonary artery
- C) Pulmonary vein
- D) External carotid artery





**9. When does Implantation of human embryos typically occurs?**

- A) about 1 day after fertilization
- B) about one week (i.e. 7 days) after fertilization
- C) about two weeks (i.e. 14 days) after fertilization
- D) after gastrulation

**10. What is the MOST LIKELY cause for sacroccocygeal teratoma?:**

- A) Abnormal migration and proliferation of lumbosacral neural crest.
- B) Abnormal persistence of primitive streak tissue after completion of gastrulation.
- C) Premature regression of the primitive streak during gastrulation.
- D) Abnormal closure of the posterior (caudal) neuropore of neural tube.

**11. Which of the following cavities expands and surrounds the embryo as a result of folding?**

- A) Amniotic cavity
- B) Yolk sac
- C) Chorionic cavity
- D) Blastocystic cavity

**12. Placenta previa means implantation in which of the following sites?**

- A) Abdominal cavity
- B) Uterine tube
- C) Ovary
- D) Near the cervix

**13. Which organelle is involved in lipid metabolism?**

- A) RER
- B) Golgi apparatus
- C) SER
- D) Lysosome

**14. Which is a non-renewing cell:**

- A) Hepatocyte.
- B) Epidermal cell.
- C) Nerve cell.
- D) Cartilage cell

**15. Which connective tissue cell type produces antibodies?**

- A) Fibblast
- B) Connective tissue macrophage
- C) Histiocyte
- D) Plasma cell.

**16. What is protein forming cells containing?**

- A) rER and sER
- B) rER and golgi apparatus
- C) sER and lysosomes
- D) sER and mitochondria

**17. What do you call the simple squamous epithelium that lines the serous membranes?**

- A) Epithelioid tissue
- B) Mesothelium
- C) Endothelium
- D) Transitional





**18. Which structure carries the impulse away from the neuron cell body?**

- A) dendrite
- B) axon
- C) myelin
- D) ganglion

**19. Which is Numerical aberrations?**

- A) Ring Chromosome.
- B) Deletion
- C) Inversion.
- D) Klinefelter s syndrome.

**20. What type of tissue is the sclera of the eye composed of?**

- A) Mucous connective tissue
- B) Mesenchyme
- C) Dense irregular connective tissue
- D) Dense regular connective tissue

**21. What structure contributes to the cells cytoskeleton?**

- A) Centrioles
- B) Peroxisome.
- C) Ribosomes.
- D) Microfilaments

**22. Where is Simple columnar absorptive epithelium found in?**

- A) Intestine.
- B) Fallopian tube.
- C) stomach.
- D) Trachea

**23. Which is true for dendrites?**

- A) Each has a constant diameter along its entire length.
- B) They are covered by spines.
- C) Are typically myelinated.
- D) Do not branch.

**24. Which of the following would be best suited to visualize lipid?**

- A) Hematoxylin and eosin stain
- B) Sudan stain
- C) Silver impregnation
- D) Masson's trichrome stain

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-A  | 3-C  | 4-B  | 5-B  | 6-D  | 7-C  | 8-B  | 9-B  | 10-B |
| 11-A | 12-D | 13-C | 14-C | 15-D | 16-B | 17-B | 18-B | 19-D | 20-C |
| 21-D | 22-A | 23-B | 24-B | o    | c    | t    | o    | r    | s    |

## Written questions

- Describe with drawing the venous drainage of the face?
- Mention the difference between the artery, vein and lymph vessel
- How does the embryo turn into the neurula from gastrula?
- Illustrate differentiation of intraembryonic mesoderm
- Describe EM appearance of Gogi apparatus & reference to its function.
- State the histological structure of outer & inner nuclear membrane.
- Mention types of stratified squamous epithelium.
- Describe the histological structure & functions of mast cells.



## 7. Formative 2022

20 MCQs

**1. Which organelle contains digestive enzymes?**

- A) RER
- B) Golgi apparatus
- C) SER
- D) Lysosome

**2. What structure is at the base of cilia?**

- A) Ribosomes
- B) Peroxisome
- C) Microfilaments
- D) Centrioles

**3. In H&E stained sections of plasma cell, Golgi apparatus appears:**

- A) Perinuclear.
- B) Apical.
- C) Unstained.
- D) Basal.

**4. Down syndrome:**

- A) due to duplication of chromosome 22.
- B) due to X chromosome monosomy.
- C) due to multiple X syndrome.
- D) caused by chromosome 21 trisomy.

**5. What type of tissue lines the urinary bladder**

- A) Simple squamous epithelium
- B) Simple cuboidal epithelium
- C) Simple columnar epithelium
- D) Transitional epithelium

**6. What type of epithelium forms the epidermis?**

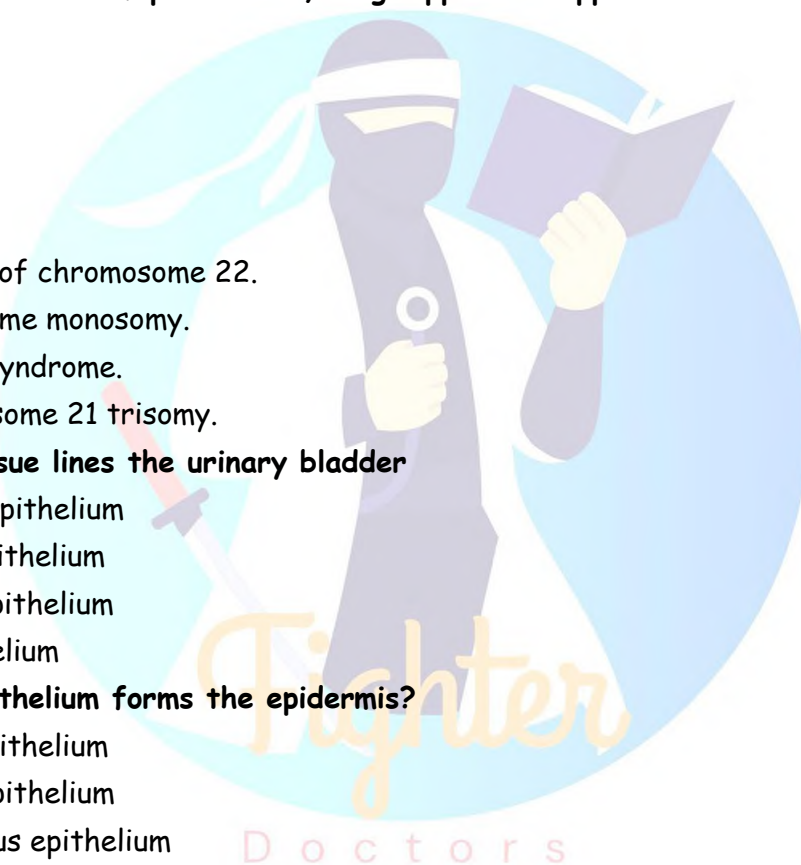
- A) Simple cuboidal epithelium
- B) Simple columnar epithelium
- C) Stratified squamous epithelium
- D) Pseudostratified epithelium

**7. Which type of neurons is present in spinal ganglion?**

- A) Pseudo-unipolar
- B) Bipolar
- C) Multipolar
- D) Satellite

**8. What type of tissue makes up the dermis of the skin?**

- A) Mesenchyme
- B) Loose irregular connective tissue
- C) Dense irregular connective tissue
- D) Dense regular connective tissue





**9. Which of the following would be best suited to differentiate collagen fibers from other fibers?**

- A) Hematoxylin and eosin stain
- B) Sudan stain
- C) Silver impregnation
- D) Mallory trichrome stain

**10. Which connective tissue cell is derived from B lymphocytes?**

- A) Fibroblast
- B) Myofibroblast
- C) Histiocyte
- D) Plasma cell

**11. When the amount of amniotic fluid exceeds two liters, the condition is called:**

- A) oligohydramnios
- B) polyhydramnios
- C) amnionitis
- D) bag of waters hydrogravidia

**12. The wall of the chorionic sac is composed of:**

- A) cytotrophoblast and syncytiotrophoblast
- B) two layers of trophoblast lined by extraembryonic somatic mesoderm
- C) trophoblast and exocoelomic membrane
- D) extraembryonic splanchnic mesoderm and both layers of trophoblast

**13. What is the most correct sequence of early development following fertilization?**

- A) zygote, blastomeres, morula, blastocyst
- B) oocyte, zygote, morula, blastocyst
- C) zygote, conceptus, blastocyst
- D) polar bodies, zygote, conceptus, blastocyst

**14. The early stages of cleavage are characterized by which of the following?**

- A) formation of a hollow ball of cells
- B) formation of the zona pellucida
- C) increase in the size of the cells in the zygote
- D) increase in the number of cells in the zygote

**15. The anterior facial vein is formed by which of the following?**

- A) Superficial temporal + maxillary veins
- B) Occipital vein + posterior auricular vein
- C) Supra-orbital+ supra-trochlear veins
- D) Anterior facial+ posterior facial veins.

**16. The frontal belly of occipitofrontalis is supplied by which of the following?**

- A) Cervical branch of facial nerve
- B) posterior auricular nerve
- C) mandibular branch of facial nerve
- D) temporal branch of facial nerve

**17. Which of the following bones is a part of axial skeleton?**

- A) Femur



- B) humerus
- C) Rib cage
- D) Metacarpal bone

**18. Which of the followings is immovable joint?**

- A) Fibrous joint
- B) Primary cartilaginous joint
- C) Secondary cartilaginous joint
- D) Synovial joint

**19. Which of the followings is the character of the cardiac muscle fiber?**

- A) It has multiple nuclei and connected by connective tissue
- B) It has central nucleus and connected by intercalated disc
- C) It is polygonal and has voluntary control
- D) It is nonstriated and has voluntary control

**20. The vessel which carries fluid from the tissue to be filtered is called**

- A) Artery
- B) Vein
- C) Lymph vessel
- D) Blood capillary

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-D  | 3-C  | 4-D  | 5-D  | 6-C  | 7-A  | 8-C  | 9-D  | 10-D |
| 11-B | 12-B | 13-A | 14-D | 15-C | 16-D | 17-C | 18-A | 19-B | 20-C |





## 8. End Module 2022

50 MCQs

1. Superficial palmer arch lies in which of the followings?

- A) Foot
- B) Forearm
- C) Hand
- D) Leg

2. Which of the followings means movement of a part of the body around its long axis?

- A) Circumduction
- B) Eversion
- C) Pronation
- D) Rotation

3. The amount of the hard bone increased at which of the followings?

- A) Lower end of the long bone
- B) Medullary cavity
- C) Middle of the shaft of the bone
- D) Upper end of the long bone

4. Which of the following joints connect between serrated margins?

- A) Gomphosis
- B) Suture
- C) Symphysis pubis
- D) Syndesmosis

5. Which of the following joint can do circumduction?

- A) Elbow joint
- B) Intercarpal joint
- C) Interphalangeal joint
- D) Wrist joint

6. Fixator muscle is..... ?

- A) Eliminate unwanted movement of the proximal joint while the prime mover act on the distal joints
- B) Form a firm base to the prime mover at the origin
- C) Has antigravity muscle
- D) Oppose the action of prime mover

7. In systemic circulation..... ?

- A) The deoxygenated blood pass from the left atrium to the left ventricle
- B) The deoxygenated blood pass from the right ventricle to the lung
- C) The oxygenated blood pass from the left ventricle to the aorta
- D) The oxygenated blood pass from the right atrium to the right ventricle

8. Which of the followings is correct about the sympathetic innervation?

- A) it increases the blood pressure and heart rate
- B) it is active at the rest
- C) it is active during the digestion
- D) it is craniosacral in origin

9. Which of the followings is the dangerous area of the scalp?



- A) Connective tissue layer
- B) Periosteum
- C) Skin layer
- D) Subaponeurotic space

**10. The common facial vein drained into which of the followings?**

- A) External jugular vein
- B) Internal jugular vein
- C) Subclavian vein
- D) Superior vena cava

**11. The sternomastoid is supplied by which of the following nerves?**

- A) Accessory nerve
- B) Facial nerve
- C) Great auricular nerve
- D) Trigeminal nerve

**12. Where does capacitation of the sperm take place?**

- A) Epididymis
- B) female genital tract
- C) prostatic gland.
- D) testis

**13. Secondary yolk sac can be seen in which of the following stages?**

- A) after folding
- B) blastocyst stage
- C) chorionic vesicle stage
- D) morula stage

**14. Which part of decidua share in formation of the placenta?**

- A) chorion leave
- B) decidua basalis
- C) decidua capsularis
- D) decidua parietalis

**15. Placental barrier at the second half of pregnancy (from 4th month onward), consists of which of the following layers?**

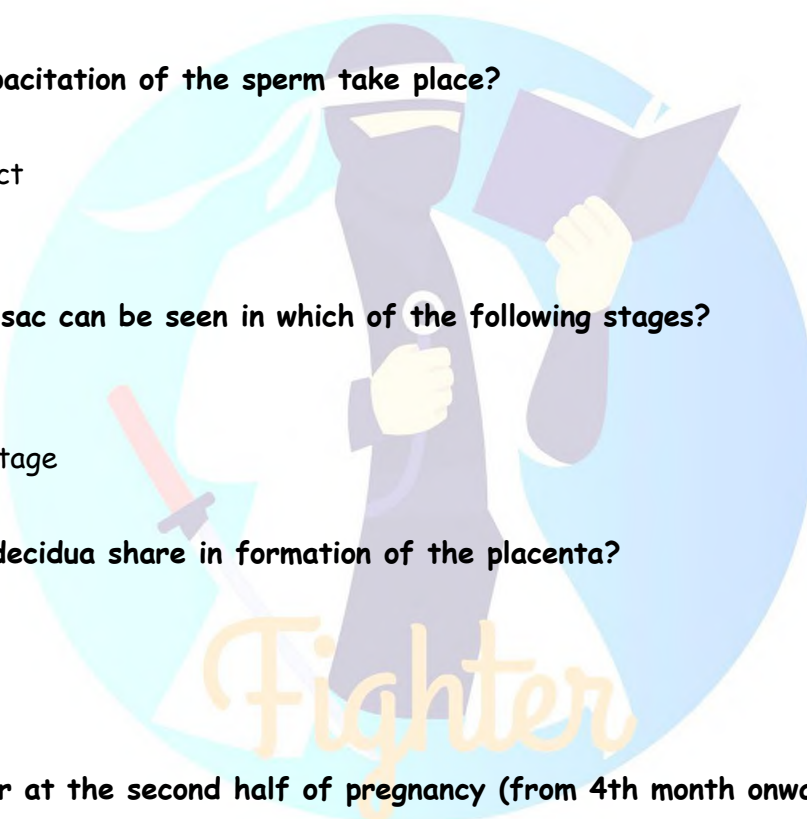
- A) The Iry mesodermal core of the villus and cytotrophoblast
- B) The Iry mesodermal core of the villus and endothelium of fetal vessels
- C) The cytotrophoblast and syncitiotrophoblast
- D) The endothelial lining of fetal vessels and syncitiotrophoblast

**16. At what stage of embryonic development does an embryo normally begin to implant in the endometrium?**

- A) Blastocyst
- B) Four-cell stage
- C) Morula
- D) Trilaminar embryo

**17. What is the site of blastocyst implantation in placenta previa?**

- A) Ovary



Doctors



- B) rectouterine pouch
- C) the tube
- D) uterine cavity

**18. Which of the followings is not formed from neural crest cells:**

- A) Central nervous system
- B) Pigment cells
- C) Adrenal gland medulla
- D) Peripheral ganglia

**19. The process by which the bilaminar disc is transformed into trilaminar disc is called:**

- A) Neurulation
- B) Gastrulation
- C) Implantation
- D) Folding

**20. The blastocyst is called late blastocyst after which of the following:**

- A) Implantation
- B) Fertilization
- C) Hatching
- D) Cleavage

**21. The cells by which the embryo nourished are which of the following:**

- A) Cytotrophoblast
- B) Decidual
- C) Amnioblast
- D) Blastomeres

**22. After the blastocytic cavity is surrounded by exocoelomic membrane it is called:**

- A) Amniotic cavity
- B) Chorionic cavity
- C) Primary yolk sac
- D) Extraembryonic mesoderm

**23. Where does fertilization of the egg occur?**

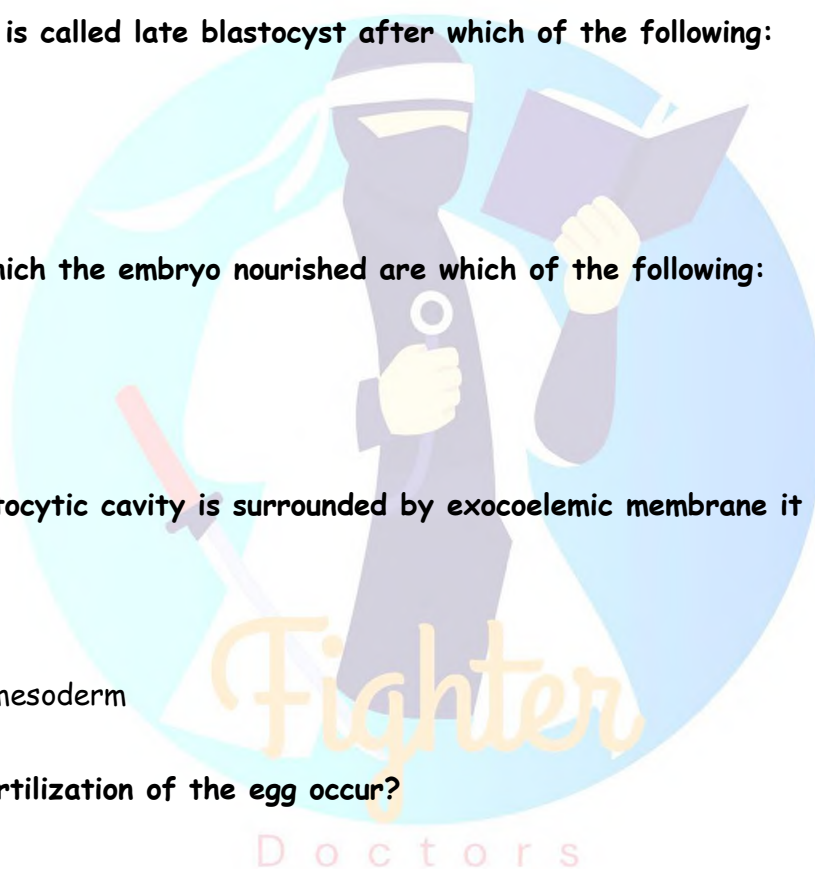
- A) Ovary
- B) Internal Os of the Uterus
- C) Fimbriae
- D) Ampulla

**24. What implants into the nutrient-rich endometrium (i.e. decidua) about 8-9 days after ovulation?**

- A) Blastocyst
- B) Morula
- C) Epiblast
- D) Hypoblast

**25. Which two embryological structures together form the bilaminar disk?**

- A) Cytotrophoblast and Hypoblast
- B) Hypoblast and Epiblast





- C) Epiblast and Cytotrophoblast  
D) Syncytiotrophoblast and Inner cell mass
- 26. In H&E - stained sections of plasma cell, How can Golgi apparatus appears?**  
A) Perinuclear.  
B) Apical.  
C) Unstained.  
D) Basal.
- 27. Which Two organelles share in the formation of lysosomes?**  
A) rER & SER.  
B) sER & Golgi.  
C) Golgi & cell membrane.  
D) rER & Golgi.
- 28. Which of the following is NOT a fiber found in connective tissue?**  
A) Collagen fiber  
B) Elastic fiber  
C) Reticular fiber  
D) All of the above are fibers found in it.
- 29. What structure contributes to the cells cytoskeleton?**  
A) Ribosomes  
B) Peroxisome  
C) Microfilaments  
D) Centrioles
- 30. What is the polysaccharide coating that is sometimes found on the cell membrane?**  
A) Plasmalemma  
B) Glycocalyx  
C) Protoplasm  
D) Cristae
- 31. What does the cause of Down syndrome:**  
A) due to duplication of chromosome 22.  
B) due to X chromosome monosomy.  
C) due to multiple X syndrome.  
D) caused by chromosome 21 trisomy.
- 32. What is the term for the process used by cells for the transport solids?**  
A) Endocytosis  
B) Exocytosis  
C) Pinocytosis  
D) Phagocytosis
- 33. What about histiocytes:**  
A) Darkly basophilic cytoplasm.  
B) Euchromatic nucleus.  
C) Pseudopodia.  
D) poorly developed Golgi apparatus
- 34. Which of the following best suited to visualize lipids:**



A) H&E

B) silver

C) Mallory

D) sudan III

**35. Which of the following is NOT a fiber found in connective tissue?**

A) Collagen fiber

B) Elastic fiber

C) Reticular fiber

D) pigmented fibers.

**36. Which is NOT lined with simple squamous epithelium:**

A) Aorta.

B) Thyroid gland.

C) Pericardium.

D) Alveoli of the lung.

**37. What type of tissue lines blood vessels?**

A) Simple squamous epithelium

B) Simple cuboidal epithelium

C) Simple columnar epithelium

D) Pseudostratified epithelium

**38. Concerning Collagen fibers:**

A) wavy bundles formed of non-branching fibers.

B) basophilic.

C) one type

D) yellow in color when fresh

**39. Concerning multilocular fat cell:**

A) Cytochrome pigments.

B) Supportive function for some organs like the kidney.

C) Signet ring appearance.

D) Flat peripheral nucleus.

**40. Cell coat function all except:**

A) Cell adhesions.

B) Cell allergy.

C) Cell recognition.

D) Cell transport

**41. The cytoplasmic side of the coated vesicle is surrounded by:**

A) Clathrin.

B) Tubulin.

C) Actin.

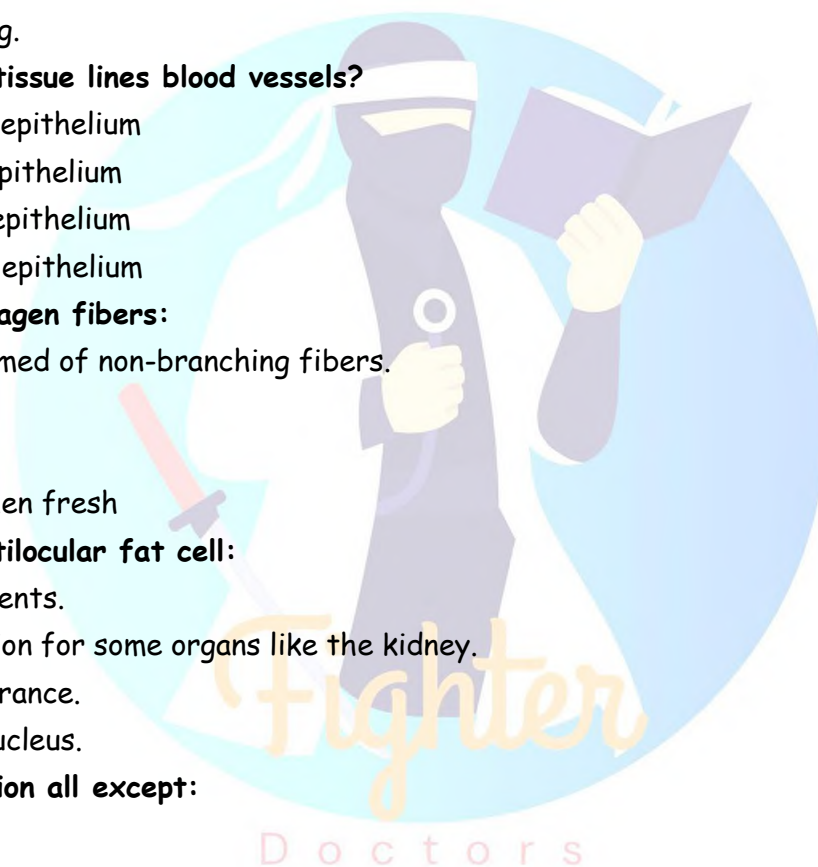
D) Spectrin.

**42. Pseudo-stratified columnar ciliated epithelium can be found in:**

A) Ovary.

B) Blood vessels.

C) Upper respiratory passages.





D) Urinary passage

**43. What is the white adipose connective tissue characterized by ?**

A) Rich in connective tissue fibers.

B) Heat insulation

C) Heart generation

D) avascular tissue.

**44. Mouth cavity is lined with:**

A) Pseudostratified columnar epithelium.

B) Stratified columnar epithelium.

C) Stratified squamous epithelium

D) Transitional epithelium.

**45. What cell type makes up the epithelium of the gall bladder?**

A) Simple squamous epithelium

B) Simple cuboidal epithelium

C) Simple columnar epithelium

D) Transitional epithelium

**46. Which cell is a connective tissue macrophage?**

A) Kupffer cells

B) Histiocyte

C) Dust cell

D) Microglia

**47. What type of tissue lines the esophagus?**

A) Simple squamous epithelium

B) Simple columnar epithelium

C) Stratified squamous epithelium

D) Pseudostratified epithelium

**48. Which of the following is NOT a fiber found in connective tissue?**

A) Collagen fiber

B) Elastic fiber

C) Reticular fiber

D) fibrocyte

**49. Which organelle is involved in lipid metabolism?**

A) rER

B) Golgi apparatus

C) sER

D) Lysosome

**50. What are the folds on the inner mitochondrial membrane called?**

A) Plasmalemma

B) Glycocalyx

C) Protoplasm

D) Cristae

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-D  | 3-C  | 4-B  | 5-D  | 6-A  | 7-C  | 8-A  | 9-D  | 10-B |
| 11-A | 12-B | 13-C | 14-B | 15-D | 16-A | 17-D | 18-A | 19-B | 20-C |
| 21-A | 22-C | 23-D | 24-A | 25-B | 26-C | 27-D | 28-D | 29-C | 30-B |
| 31-D | 32-D | 33-C | 34-D | 35-D | 36-B | 37-A | 38-A | 39-A | 40-D |
| 41-A | 42-C | 43-B | 44-C | 45-C | 46-B | 47-C | 48-D | 49-C | 50-D |



## 9. End Semester 2022

## 11 MCQs

1. The occipital belly of occipitofrontalis is supplied by which of the following?
  - A) Buccal branch of facial nerve.
  - B) posterior auricular nerve.
  - C) mandibular branch of facial nerve.
  - D) temporal branch of facial nerve
2. The lateral part of the scalp is drained into which of the following?
  - A) Parotid lymph nodes.
  - B) Mastoid lymph nodes.
  - C) Occipital lymph nodes.
  - D) Submandibular lymph nodes
3. Tendon means which of the followings?
  - A) bands of connective tissue which connect muscles with bones.
  - B) bands of connective tissue which connect bones with bones.
  - C) bands of connective tissue which connect muscles with muscles.
  - D) bands of connective tissue which connect muscles with skin.
4. The placenta is part of the chorion. The chorion consists of which of the following?
  - A) Extraembryonic mesoderm
  - B) Trophoblast
  - C) Hypoblast
  - D) A & B
5. Which of the following cells form endoderm and mesoderm?
  - A) Migrating epiblast cells
  - B) Migrating hypoblast cells
  - C) Non-migrating epiblast cells
  - D) Non-migrating hypoblast cells
6. neural tube and plate forms from:
  - A) Intermediate mesoderm
  - B) Paraxial mesoderm
  - C) Neural crest cells
  - D) Neuroectoderm
7. What type of tissue makes up the dermis of the skin?
  - A) Mesenchyme
  - B) Loose irregular connective tissue.
  - C) Dense irregular connective tissue.
  - D) Dense regular connective tissue
8. Which of the following would be best suited to visualize reticular fibers?
  - A) Hematoxylin and eosin stain
  - B) Sudan stain
  - C) Silver impregnation
  - D) Masson's trichrome stain
9. What type of gland secretes its product directly into the bloodstream?
  - A) Endocrine gland.
  - B) Multicellular gland.
  - C) Exocrine gland.
  - D) All of the above
10. What type of epithelium lines the trachea?
  - A) Simple squamous epithelium
  - B) Simple cuboidal epithelium
  - C) Simple columnar epithelium
  - D) Pseudostratified epithelium
11. What type of tissue lines the urinary bladder:
  - A) Simple squamous epithelium.
  - B) Simple cuboidal epithelium.
  - C) columnar epithelium.
  - D) Transitional epithelium.

## Answer Key

|      |     |     |     |     |     |     |     |     |      |
|------|-----|-----|-----|-----|-----|-----|-----|-----|------|
| 1-B  | 2-A | 3-A | 4-D | 5-A | 6-D | 7-C | 8-C | 9-A | 10-D |
| 11-D |     |     |     |     |     |     |     |     |      |



## 10. Formative (1) 2023

11 MCQs

1. When the protein secreting cells stained with Hematoxylin and Eosin, There is a clear unstained area near the nucleus. This pale area is called:\*

- A) Brown network or granules
- B) Cisternae
- C) Negative Golgi image
- D) Cristae

2. Transitional epithelium is a type of stratified epithelium, it lines which of the following tracts?\*

- A) Gastrointestinal
- B) Urinary
- C) Female genital
- D) Respiratory

3. The nucleus of which of the following cells shows Barr body in females (drum stick appearance):\*

- A) Lymphocyte
- B) Neutrophil
- C) Monocyte
- D) Basophil

4. What is the most common type of junction between two successive cardiac muscle fibers?\*

- A) Gap junction
- B) Desmosome
- C) M line
- D) Z line

5. During the cell cycle, in which stage DNA replication takes place?\*

- A) Prophase
- B) Metaphase
- C) Anaphase
- D) Interphase

6. Which of the following is the only example of a unicellular exocrine gland?\*

- A) Sebaceous
- B) Intestinal gland
- C) Goblet cell
- D) Sweat

7. In which of the following types, the chromosomes have short and long arms of unequal length with the centromere presents more towards one end?\*

- A) Acrocentric
- B) Telocentric
- C) Submetacentric
- D) Metacentric

8. Protein synthesizing cells are rich in which of the following structure?

- A) SER
- B) Lipid droplets
- C) Mucus granules
- D) rER

9. Actin filaments are responsible for formation of which of the following?

- A) Centriole
- B) Contractile ring
- C) Mitotic spindle
- D) Cilia

10. Which of the following is concerned with formation of ribosomes?\*

- A) Golgi
- B) Centriole
- C) RER
- D) Nucleolus

11. In the molecular structure of the cell membrane, which of the following structures fill the gap between the fatty acid tails in a ratio one by one?

- A) Cholesterol
- B) Lipid bilayer
- C) Phospholipid
- D) Integral protein

**Answer Key**

|     |     |     |     |     |     |     |     |     |      |      |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| 1-C | 2-B | 3-B | 4-A | 5-D | 6-C | 7-C | 8-D | 9-B | 10-D | 11-A |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|



## 11. Formative (2) 2023

18 MCQs

1. the secretory phase of the uterus is characterized by which of the following

- A) it is under the influence of progesterone
- B) it starts from the 5th day till 14th day of the cycle
- C) initiate the formation of a new layer of endometrium in the uterus
- D) Thickness of endometrium in this stage is 0.5 mm

2. what is the site of blastocyst implantation in placenta previa

- A) the tube
- B) uterine cavity
- C) ovary
- D) rectouterine pouch

3. Capacitation means which of the followings

- A) adaptation of sperms in female genital tract
- B) Acrosome release proteolytic enzymes such as trypsin like substances.
- C) It is induced by zona pellucida proteins
- D) release of lysosomal enzymes from cortical granules

4. In anatomical position the palm of the hand faces

- A) Anteriorly
- B) Posteriorly
- C) Superiorly
- D) Inferiorly

5. which of the followings is responsible for the growth of the length of the bone

- A) periosteum
- B) growing plate
- C) arterial supply
- D) nerve supply

6. the estrogen is secreted from

- A) granulosa cell
- B) corpus luteum
- C) cells of anterior lobe of pituitary gland
- D) endometrial cells

7. which of the followings is true about the fertilization

- A) it occurs usually in the infundibulum of the fallopian tube
- B) chromosomal sex of the embryo is determined at fertilization.
- C) the oocyte remains viable for 5 days
- D) all the sperms can penetrate the zona pellucida

8. Medial rotation means

- A) Moving the limb externally around the long axis
- B) Moving the limb internally around the long axis
- C) Moving the limb upward around horizontal axis
- D) Moving the limb downward around horizontal axis

9. Coronal plane divide the body into

- A) Upper and lower halves
- B) Anterior and posterior halves
- C) Right and left halves
- D) Superior and inferior halves

10. Bone marrow is supplied by

- A) Periosteal artery
- B) Epiphyseal artery
- C) Nutrient artery



**11. Which of the following is axial skeleton**

- A) Femur
- B) Humerus
- C) Skull
- D) Hip

**12. Planter aspect of the foot means**

- A) Sole
- B) Palm
- C) Dorsum

**13. Carpal bones are**

- A) Long bone
- B) Short bones
- C) Irregular bones
- D) Pneumatic bone

**14. which of the followings is the character of oogenesis**

- A) starts at puberty and extend till old age
- B) it takes about 64 days
- C) Primary oocyte complete its first meiotic division at puberty
- D) Secondary oocyte completes its 2nd meiotic division at the time of ovulation

**15. compaction means which of the followings**

- A) the blastomere forming a compact ball of cells held together by tight junctions
- B) changes in the zona reaction to prevent further binding and penetration of the sperm
- C) release of the acrosomal enzymes helps penetration of the sperm to the zona pellucida
- D) the ability of the sperm to fertilize the ovum

**16. morphological changes of spherical spermatids to form elongated sperm is called**

- A) spermatogenesis
- B) Spermiogenesis
- C) acrosomal reaction

**17. the spermlysin lies in which part of the sperm**

- A) Head
- B) Neck
- C) Middle piece
- D) Tail

**18. which of the following is the character of the spongy bone**

- A) it forms the out part of any bone
- B) it is consists of number of cylindrical units called Haversian system
- C) It is formed of trabeculae, the lamellae are branching and anastomose with

**Answer Key**

|      |      |      |      |      |      |      |      |     |      |
|------|------|------|------|------|------|------|------|-----|------|
| 1-A  | 2-B  | 3-A  | 4-A  | 5-B  | 6-A  | 7-B  | 8-B  | 9-B | 10-C |
| 11-C | 12-A | 13-B | 14-C | 15-A | 16-B | 17-A | 18-C |     |      |



## 12. End Module 2023

60 MCQs

1. The function of rough endoplasmic reticulum is to synthesize:

- A) Lipid.
- B) Protein that will be secreted by the cell.
- C) Cytoplasmic protein necessary for the cells' own existence.
- D) Carbohydrates.

2. Smooth endoplasmic reticulum:

- A) Sometimes has ribosomes attached to its membrane
- B) Is rarely found in skeletal muscle cells
- C) Areas of localized basophilic indicate its presence
- D) Is present in cells where drug detoxification is taking place

3. The Golgi apparatus is:

- A) The site of packaging of secretory products of the cell.
- B) The site of protein synthesis.
- C) Continuous with smooth endoplasmic reticulum.
- D) Capable of providing energy.

4. Multivesicular bodies are:

- A) Primary lysosomes.
- B) Formed as a result of digestion of solid material.
- C) Formed as a result of digestion of fluid material.
- D) Autophagic vacuoles.

5. Microtubules have important role in:

- A) Carbohydrate metabolism.
- B) Oxidative phosphorylation.
- C) Water balance.
- D) Formation of the mitotic spindle during mitosis.

6. DNA could be found in:

- A) Ribosomes.
- B) Golgi apparatus.
- C) Mitochondria.
- D) Lysosomes.

7. Heterochromatin is:

- A) Present in the nuclei of metabolically active cells.
- B) Special type of stain.
- C) Present in the nuclei of metabolically inactive cells
- D) Type of cell inclusions.

8. The nucleolus is formed of:

- A) Heterochromatin and euchromatin.
- B) Nucleolonema and pars amorpha.
- C) Nucleolus associated and peripheral chromatin.
- D) Condensed chromatin.

9. The nuclear pore:





- A) Is at regular intervals.
- B) Is bridged by a unit membrane.
- C) Is a transient structure.
- D) Allows for communication between the nucleus and the cytoplasm

## 10. Glycocalyx is:

- A) The protein component of the cell membrane.
- B) The hydrophilic end of lipid.
- C) The carbohydrate component of the cell membrane.
- D) The hydrophobic end of lipid.

## 11. Steriocilia are:

- A) Long motile cilia.
- B) True cilia.
- C) Long motile microvilli.
- D) Long microvilli.

## 12. Endothelium provides lining of:

- A) Peritoneum.
- B) Pleural cavity.
- C) The heart.
- D) Alveoli of the lungs.

## 13. In absorptive cells, the brush border seen at Im level is:

- A) Streocilia.
- B) True cilia.
- C) Microvilli.
- D) Basal infoldings.

## 14. Stratified squamous epithelium:

- A) Consists of a single layer of flattened cells.
- B) Is protective rather than secretory or absorptive.
- C) Has a rich supply of capillaries running between its cell layers.
- D) May be ciliated.

## 15. Desmosomes are:

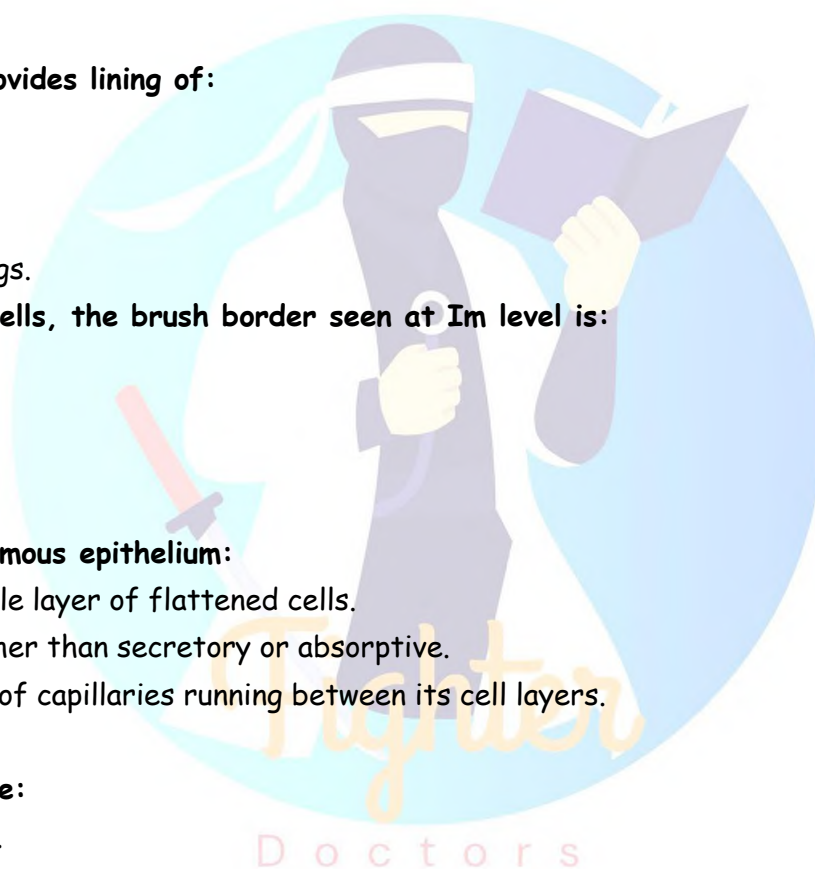
- A) Macula occludens.
- B) Macula adherens.
- C) Zonula occludens.
- D) Zonula adherens.

## 16. In zonula occludens the distance between two cell membranes is:

- A) 0 nm
- B) 2 nm
- C) 20 nm
- D) 30 nm

## 17. Compound exocrine glands:

- A) Have unbranched secretory duct
- B) Have branched secretory ducts
- C) Secrete hormones





D) Secrete into the blood.

**18. Which of the following not included in the classical description of a junctional complex:**

- A) The fascia occludes
- B) The zonula occludens
- C) The zonula adherens
- D) The macula adenens

**19. The main ultrastructural feature (EM) of histiocyte is the presence of:**

- A) Microvilli.
- B) Lysosomes.
- C) SER
- D) RER

**20. Plasma cells:**

- A) Have well developed smooth endoplasmic reticulum.
- B) Have phagocytic ability
- C) Are commonly found in peripheral blood
- D) Show all EM features of protein synthesizing cells.

**21. Metachromatic granules are present in:**

- A) Mast cells.
- B) Plasma cells.
- C) Fat cells.
- D) Reticular cells.

**22. Which of the following would be best suited to differentiate collagen fibers from other fibers?**

- A) Hematoxylin and eosin stain
- B) Sudan stain
- C) Silver
- D) Mallory trichrome stain

**23. What type of tissue is cornea composed of?**

- A) Mucous connective tissue
- B) Mesenchyme
- C) Dense irregular connective tissue
- D) Dense regular connective tissue

**24. What is the shape of fat cells present in the unilocular adipose tissue?**

- A) Clock face
- B) Signet ring
- C) Horse shoe
- D) Kidney

**25. Which of the following is the only example of a unicellular exocrine gland?**

- A) Sweat gland
- B) Intestinal gland
- C) Goblet cell.
- D) Sebaceous glands

**26. Which of the following statements is incorrect?**



- A) Microtubules are involved in ciliary movement and intracellular transport.
- B) The cis (forming) face of Golgi apparatus is where secretory granules are formed.
- C) The microtubules of the centrioles and in the basal body of cilia have a 9x3 arrangement.
- D) Ribosomes, made of rRNA and protein, are the site of protein synthesis.

**27. Pseudostratified columnar ciliated epithelium can be found in:**

- A) Fallopian tube
- B) Blood vessels.
- C) Respiratory passages
- D) Urinary passage

**28. Which of the following glands is considered as an apocrine gland:**

- A) Pancreas
- B) Mammary glands
- C) Salivary glands.
- D) Sebaceous glands

**29. Which of the following statements is incorrect regarding neuroepithelium?**

- A) It is modified type of epithelium to act as receptor.
- B) Formed of sensory and supporting cells.
- C) It covers skin and forms glands.
- D) Present in taste buds.

**30. Which of the following form stroma of the organs?**

- A) Loose areolar connective tissue.
- B) Reticular connective tissue.
- C) Unilocular adipose connective tissue.
- D) Muroid connective tissue.

**31. Which of the following is one of the characteristics of anatomical position?**

- A) The arms are abducted
- B) The palm face anterior
- C) The lower limbs are abducted
- D) The toes directed to the left.

**32. Which of the following anatomical planes divided the body into anterior and posterior parts?**

- A) Horizontal plane.
- B) Coronal plane
- C) Midsagittal plane
- D) Parasagittal plane

**33. Where does the superficial palmar arch lie?**

- A) In the anterior surface of the hand
- B) In the posterior surface of the hand
- C) In the superior surface of the foot
- D) In the inferior surface of the foot

**34. Which of the following movement of the limb upward?**

- A) Flexion
- B) Elevation
- C) Extension



D) Depression

**35. Which of the following cartilage disappears in the adult?**

A) Growing plate

B) Articular cartilage

C) Costal cartilage

D) Cartilage of the auricle

**36. There are two forms of bone, which of the following is the character of spongy bone?**

A) It consists of number of cylindrical units called haversian system

B) It is formed of trabeculae

C) It forms the surface layer of all bones

D) It provides strength for weight bearing

**37. Which of the following arteries supplies the bone marrow?**

A) Periosteal artery

B) The articular artery

C) Epiphyseal artery

D) Nutrient artery

**38. Which of the following is fibrous joint?**

A) Elbow joint

B) Shoulder joint

C) Inferior tibiofibular joint

D) Intervertebral disc

**39. What is the type of costochondral junction?**

A) Fibrous joint

B) Primary cartilaginous joint

C) Secondary cartilaginous joint

D) Synovial joint

**40. Presence of joint cavity is one of the characters of which of the following joint?**

A) Fibrous joint

B) Primary cartilaginous joint

C) Secondary cartilaginous joint

D) Synovial joint

**41. Which of the following is the character of the smooth muscle?**

A) it is voluntary non striated muscle

B) it is attached to the skeleton.

C) it is non striated involuntary muscle

D) it is connected by intercalated disc

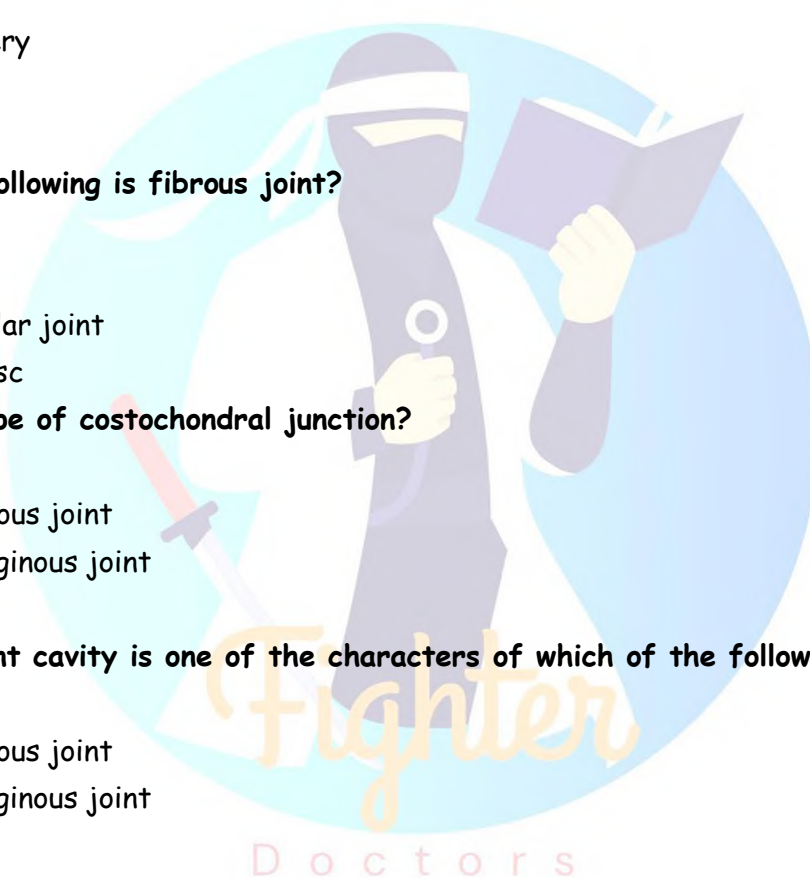
**42. When the muscle fibers are attached to one side of the tendon, it is called which of the following?**

A) triangular

B) unipennate

C) bipennate

D) spiral





**43. When the muscle removes unwanted movement in the proximal joint and the prime mover act on the distal, it is called \_\_\_\_\_:**

- A) antagonist
- B) prime mover
- C) synergist
- D) Fixator

**44. In portal circulation, which of the following ends in the inferior vena cava?**

- A) portal vein
- B) hepatic vein
- C) left gastric vein
- D) superior pancreatico duodenal vein

**45. Which of the following vessel is fenestrated?**

- A) large artery
- B) small artery
- C) vein
- D) blood capillary

**46. Which structure does not take place in the late blastocyst?**

- A) Zona pellucida
- B) Trophoblast
- C) Embryoblast
- D) Blastocoele

**47. Which statement about the neural crest is NOT true?**

- A) Pia mater can develop from the neural crest
- B) Neural crest gives rise to the cortex of the suprarenal gland
- C) Melanocytes originate from the neural crest
- D) Ganglionic cells originate from neural crest cells

**48. Which statement is true about the trophoblast?**

- A) It differentiates into outer cytotrophoblast and inner syncytiotrophoblast
- B) It differentiates into inner cytotrophoblast and outer syncytiotrophoblast
- C) Structure called "trophoblast" does not exist during the development of human
- D) It does not differentiate further

**49. Where originates the extraembryonic mesoderm?**

- A) It is a result of the migration of cells from the neuroectoderm
- B) In a site between trophoblast and embryoblast
- C) In a site between hypoblast and epiblast
- D) In a site between endoderm and ectoderm

**50. Where the fertilization occurs usually?**

- A) In the infundibular part of the Fallopian tube
- B) In the ampullary part of the Fallopian tube
- C) Horn of the uterus (cornua uteri)
- D) Cavity of the uterus

**51. What is a capacitation?**

- A) Capacitation is a process in which the glycoprotein layer over the acrosome is dissolved
- B) Capacitation is a process in which sperm penetrates zona pellucida
- C) Capacitation is a process in which the tail of the sperm is inactivated
- D) Capacitation is a process in which the tail of the sperm is activated

**52. Primitive streak originates:**

- A) In lateral parts of hypoblast
- B) In lateral parts of epiblast
- C) In the medial line of hypoblast
- D) In the medial line of epiblast



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- A) Immediately after fertilization
- B) 3rd day after fertilization
- C) 6th day after fertilization
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- A) Cells from the primitive node migrate between ectoderm and endoderm
- B) Cells from the wall of the amnionic sack migrate between ectoderm and endoderm
- C) Cells from the wall of the yolk sack migrate between ectoderm and endoderm
- D) Cells from primitive streak migrate between ectoderm and endoderm

**55. Which statement about 'formation of somites is correct?**

- A) Formation of somites is gradual: Cranial somites start, caudal somites continue later on
- B) Formation of somites is gradual: Caudal somites start, cranial somites continue later on
- C) Timing of formation of somites is less or more chaotic
- D) All somites are formed at the same time

**56. The most common site for implantation in ectopic pregnancy is:**

- A) internal os of the uterus
- B) mesentery
- C) ovary
- D) uterine tube

**57. The first week of human development is characterized by formation of the:**

- A) inner cell mass
- B) trophoblast
- C) blastocyst
- D) all of the above

**58. The ovulated mammalian oocyte is arrested at:**

- A) prophase of meiosis I
- B) metaphase of meiosis I
- C) prophase of meiosis II
- D) metaphase of meiosis II

**59. All but one of following statements about the cleavage are true. EXCEPT?**

- A) Cleavage occurs when the embryo travels through the Fallopian tube
- B) During cleavage, the amount of DNA inside the embryo increases
- C) Embryo is closed inside zona pellucida during cleavage
- D) Cleavage continues after implantation

**60. Which embryonic structure gives rise to cells in our bodies?**

- A) epiblast and hypoblast
- B) Hypoblast
- C) Epiblast
- D) Extraembryonic mesoderm

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
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| 51-A | 52-D | 53-C | 54-D | 55-A | 56-D | 57-D | 58-D | 59-D | 60-C |



## 13. End Semester 2023

16 MCQs

1. One of the functions of smooth endoplasmic reticulum is?

- A) Protein synthesis
- B) Regulation of intracellular calcium distribution
- C) Excretion
- D) Maintain the skeleton of the cell

2. Which of the following is the only example of a unicellular exocrine gland?

- A) Goblet cell
- B) Intestinal gland.
- C) Sebaceous gland.
- D) Sweat gland

3. Actine filaments are responsible for formation of which of the following?

- A) Centriole
- B) Cilia
- C) Microvilli
- D) Mitotic spindle

4. Which of the following is concerned with formation of ribosomes?

- A) Centriole
- B) Golgi
- C) Nucleolus
- D) RER

5. stratified cuboidal epithelium is found in?

- A) Large blood vessels
- B) Acini of salivary glands
- C) Gall bladder
- D) Urinary bladder

6. What type of tissue is a tendon composed of ?

- A) Mucous connective tissue
- B) Mesenchyme
- C) Dense irregular connective tissue
- D) Dense regular connective tissue

7. Which is the main organelle that plays role in muscle contraction?

- A) Free ribosomes
- B) Lysosomes
- C) RER
- D) SER

8. What is the shape of the fat cells in unilocular adipose tissue?

- A) Clock-face
- B) Flattened
- C) Horse shoe
- D) Signet ring

9. Which of the followings is true about the neural tube formation?

- A) It appears in the 4th week of development.
- B) It is endodermal in origin.
- C) It forms the central nervous system.
- D) Its caudal end divided into three vesicles

10. Somites appear in the third week, they form which of the followins ?

- A) Muscles of chest wall
- B) Pleural cavity
- C) Kidney
- D) Bone and cartilage

11. The anterior facial vein is formed by which of the following ?

- A) Superficial temporal + maxillary veins.
- B) Occipital vein + posterior auricular vein
- C) Supra-orbital supra-trochlear veins
- D) Anterior facial+ maxillary facial veins



12. Which of following is an example of secondary cartilaginous joint?

- A) Shoulder joint
- B) radio-ulnar joint
- C) costo-chondral
- D) intervertebral disc

13. Enzymes that catalyze removal or addition of hydrogen from the molecule are known as which of the following?

- A) zygote
- B) morula
- C) blastocyst
- D) embryonic disc

14. When amount of amniotic fluid is less than half liter, condition is called?

- A) oligohydramnios
- B) polyhydramnios or hydramnios
- C) amnionitis
- D) bag of water

15. middle phalanx of the little finger is example to which type of bone ?

- A) Long
- B) Short
- C) Irregular
- D) Flat

16. All the followings are branches of facial nerve except:

- A) Mental
- B) Cervical
- C) Buccal
- D) Temporal

## Answer Key

|      |      |      |      |      |      |     |     |     |      |
|------|------|------|------|------|------|-----|-----|-----|------|
| 1-B  | 2-A  | 3-C  | 4-C  | 5-B  | 6-D  | 7-D | 8-D | 9-C | 10-D |
| 11-C | 12-D | 13-- | 14-A | 15-A | 16-A |     |     |     |      |

Note on Q5: Debatable answer.

Note on Q10: Debatable: both Muscles of chest wall (myotome) and Bone and cartilage (sclerotome) are somite derivatives.

Note on Q13: Source question and options do not match (options are embryology terms) - no valid answer.

## Written questions

1) Give a comment on :-

- A) Cell inclusions :
- B) Centriole :

2) State the following:-

- A) Sites of heterochromatin in nucleus
- B) one site for Pseudostratified columnar epithelium with stereocilia

3) Compare between macrophage & plasma cell

4) Enumerate Types of Fixed connective tissue cells.

5) Discuss Reticular connective tissue (structure-function)

6) Discuss Nerve Cell (E.M. structure)

7) Describe the skeletal muscles forms with example of each type

8) Compare between sympathetic and parasympathetic fibers?

9) Describe the dangerous area of the face and explain why it is dangerous?

10) Mention normal and abnormal sites of implantation?

11) Give short note on umbilical cord development and its anomalies?

12) The second week is a period of two. Describe the cavities which appear in the second week of development?



## 14. End Module 2024

60 MCQs

1. The function of rough endoplasmic reticulum is to synthesize:

- A) Lipid
- B) Protein that will be secreted by the cell
- C) Cytoplasmic protein necessary for the cells' own existence
- D) Carbohydrates

2. Smooth endoplasmic reticulum:

- A) Sometimes has ribosomes attached to its membrane
- B) Is rarely found in skeletal muscle cells
- C) Areas of localized basophilic indicate its presence
- D) Is present in cells where drug detoxification is taking place

3. The Golgi apparatus is:

- A) The site of packaging of secretory products of the cell.
- B) The site of protein synthesis.
- C) Continuous with smooth endoplasmic reticulum.
- D) Capable of providing energy.

4. Multivesicular bodies are:

- A) Primary lysosomes
- B) Formed as a result of digestion of solid material.
- C) Formed as a result of digestion of fluid material
- D) Autophagic vacuoles.

5. Microtubules have important role in:

- A) Carbohydrate metabolism
- B) Oxidative phosphorylation.
- C) Water balance
- D) Formation of mitotic spindle during mitosis.

6. DNA could be found in:

- A) Ribosomes
- B) Golgi apparatus
- C) Mitochondria
- D) Lysosomes

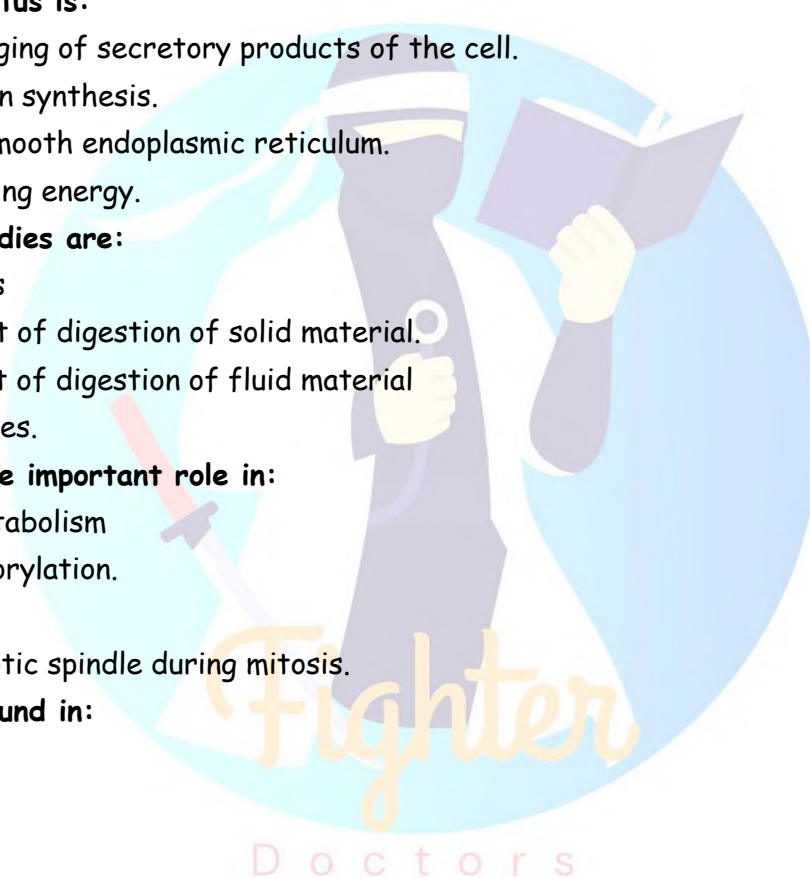
7. Heterochromatin is:

- A) Present in the nuclei of metabolically active cells.
- B) Special type of stain
- C) Present in the nuclei of metabolically inactive cells
- D) Type of cell inclusions.

8. The nucleolus is formed of:

- A) Heterochromatin and euchromatin
- B) Nucleolonema and pars amorpha
- C) Nucleolus associated and peripheral chromatin.
- D) Condensed chromatin

9. The nuclear pore.....





- A) Is at regular intervals.
- B) Is bridged by a unit membrane
- C) Is a transient structure
- D) Allows for communication between the nucleus and the cytoplasm

## 10. Glycocalyx is:

- A) The protein component of the cell membrane
- B) The hydrophilic end of lipid
- C) The carbohydrate component of the cell membrane
- D) The hydrophobic end of lipid

## 11. Steriocilia are:

- A) Long motile cilia
- B) True cilia
- C) Long motile microvilli
- D) Long microvilli

## 12. Endothelium provides lining of:

- A) Peritoneum
- B) Pleural cavity
- C) The heart
- D) Alveoli of lungs

## 13. In absorptive cells, the brush border seen at Im level is:

- A) Streocilia
- B) True cilia
- C) Microvilli
- D) Basal infoldings

## 14. Stratified squamous epithelium:

- A) Consists of a single layer of flattened cells.
- B) Is protective rather than secretory or absorptive.
- C) Has a rich supply of capillaries running between its cell layers
- D) May be ciliated

## 15. Desmosomes are:.....

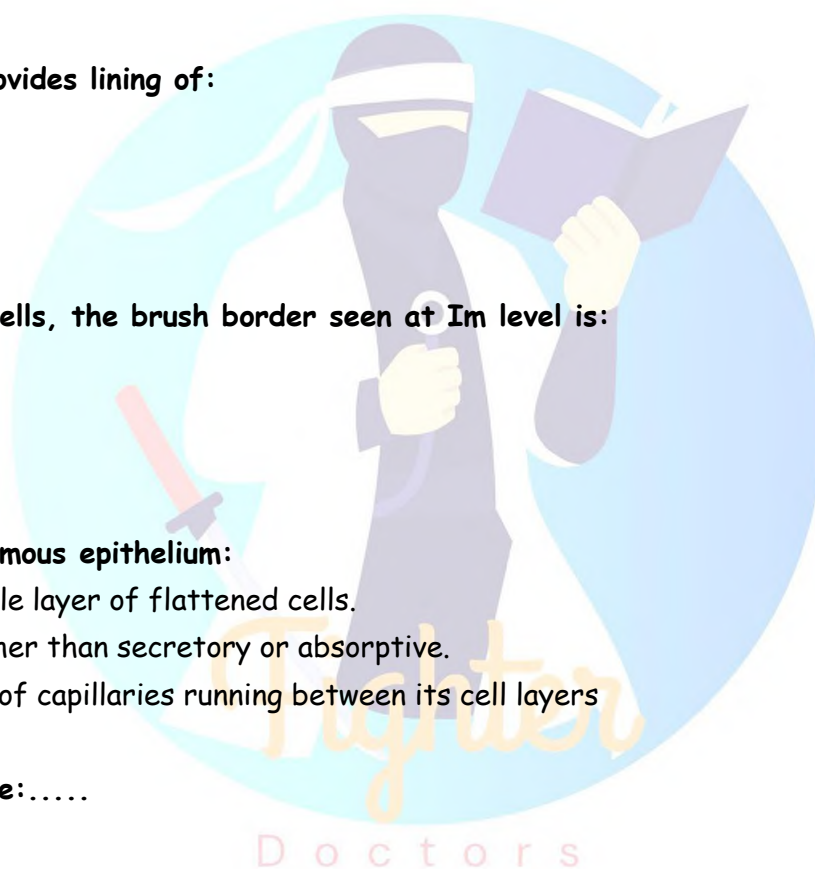
- A) Macula occludens
- B) Macula adherens
- C) Zonula closing
- D) Zonula adherens

## 16. In zonula occludens the distance between two cell membranes is:

- A) 0 nm
- B) 2 nm
- C) 20 nm
- D) 30 nm

## 17. Compound exocrine glands:

- A) Have unbranched secretory duct
- B) Have branched secretory ducts
- C) Secrete hormones





D) Secrete into the blood

**18. Which of following not included in classical description of a junctional complex :**

- A) fascia occludes
- B) zonula occludens
- C) zonula adherens
- D) macula adherens

**19. The main ultrastructural feature (EM) of histiocyte is the presence of:**

- A) Microvilli
- B) Lysosomes
- C) SER
- D) RER

**20. Plasma cells:**

- A) Have well developed smooth endoplasmic reticulum
- B) Have phagocytic ability
- C) Are commonly found in peripheral blood
- D) Show all EM features of protein synthesizing cells

**21. Metachromatic granules are present in:**

- A) Mast cells
- B) Plasma cells
- C) Fat cells
- D) Reticular cells

**22. Which of the following would be best suited to differentiate collagen fibers from other fibers?**

- A) H and E stain
- B) Sudan stain
- C) Silver.
- D) Mallory trichrome stain

**23. What type of tissue is cornea composed of?**

- A) Mucous C.T
- B) Mesenchyme
- C) Dense irregular C.T.
- D) Dense regular C.T

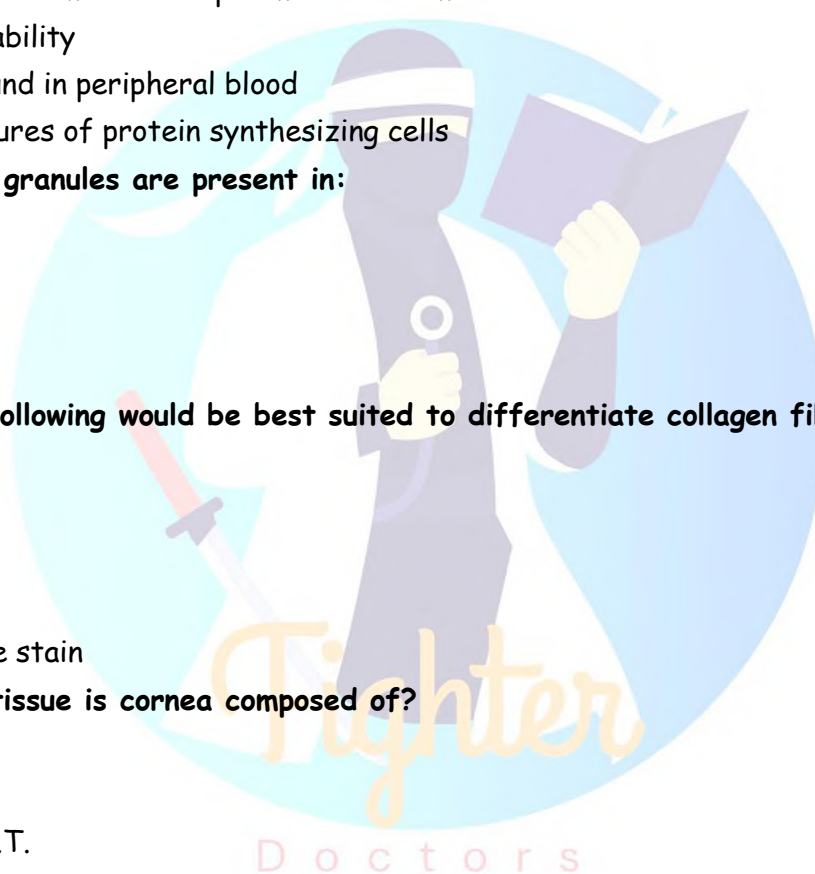
**24. What is shape of fat cells present in the unilocular adipose tissue?**

- A) Clock face
- B) Signet ring
- C) Horse shoe
- D) Kidney

**25. Which of following is only example of a unicellular exocrine gland?**

- A) Sweat gland
- B) Intestinal gland
- C) Goblet cell
- D) Sebaceous glands

**26. Which of the following statements is incorrect?**





- A) Microtubules are involved in ciliary movement and intracellular transport.
- B) cis (forming) face of Golgi apparatus is where secretory granules are formed.
- C) microtubules of the centrioles and in the basal body of cilia have a 9\times3 arrangement.
- D) Ribosomes, made of rRNA and protein, are site of protein synthesis.

**27. Pseudostratified columnar ciliated epithelium can be found in:**

- A) Fallopian tube
- B) Blood vessels.
- C) Respiratory passages.
- D) Urinary passage

**28. Which of the following glands is considered as an apocrine gland:**

- A) Pancreas
- B) Mammary glands
- C) Salivary glands.
- D) Sebaceous glands

**29. Which of the following statements is incorrect regarding neuroepithelium?**

- A) It is modified type of epithelium to act receptor
- B) Formed of sensory and supporting cells
- C) It covers skin and forms glands
- D) Present in taste buds

**30. Which of the following form stroma of the organs?**

- A) Loose areolar connective tissue.
- B) Reticular connective tissue.
- C) Unilocular adipose connective tissue.
- D) Muroid connective tissue.

**31. Which of the following is one of characteristics of anatomical position?**

- A) arms are abducted
- B) palm face anterior
- C) lower limbs are abducted
- D) toes directed to left.

**32. Which of the following anatomical planes divided the body into anterior and posterior parts?**

- A) Horizontal plane
- B) Midsagittal plane
- C) Coronal plane.
- D) Parasagittal plane

**33. Where does the superficial palmar arch lie?**

- A) In the anterior surface of the hand
- B) In the posterior surface of the hand
- C) In the superior surface of the foot
- D) In the inferior surface of the foot

**34. Which of the following movement of the limb upward?**

- A) Flexion
- B) Elevation
- C) Extension



D) Depression

**35. Which of the following cartilage disappears in the adult?**

A) Growing plate

B) Articular cartilage

C) Costal cartilage

D) Cartilage of the auricle

**36. There are two forms of bone, which of the following is the character of spongy bone?**

A) It consists of number of cylindrical units called haversian system

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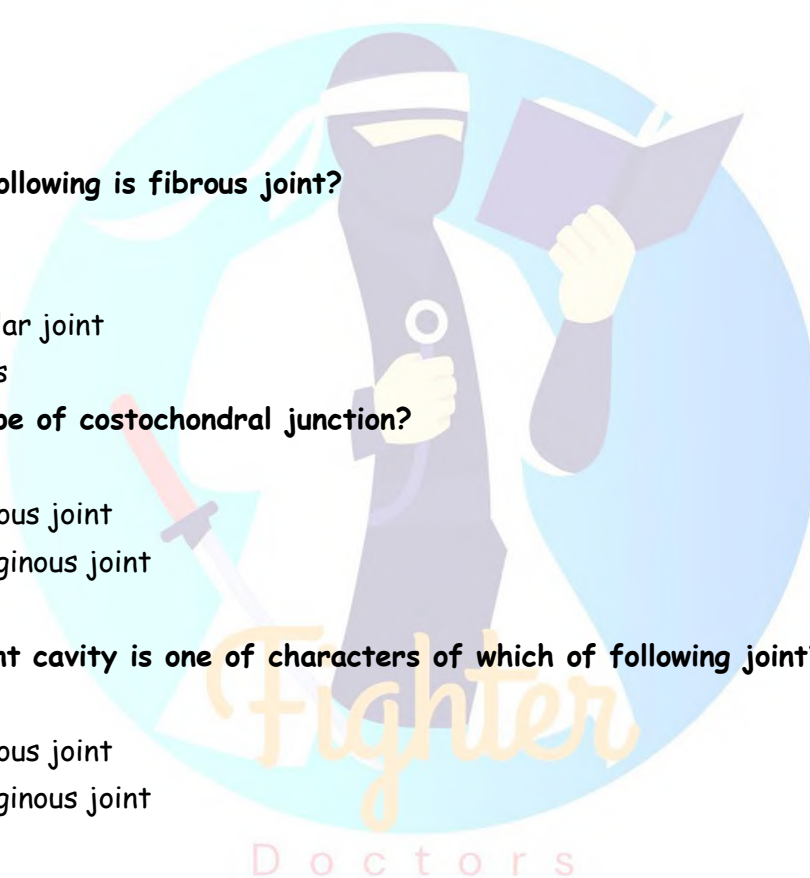
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| 41-C | 42-B | 43-D | 44-B | 45-D | 46-A | 47-B | 48-B | 49-B | 50-B |
| 51-A | 52-D | 53-C | 54-D | 55-A | 56-D | 57-D | 58-D | 59-D | 60-C |



## 15. End Semester 2024

Source: MedZone | 38 MCQs

1. All of the following are characters of Golgi apparatus except:

- A) Shows -Ve image in deeply basophilic cytoplasm.
- B) Stained with silver.
- C) Supranuclear in nerve cells.
- D) Has mature trans- face & immature cis- face.

2. The basal body of a cilium has a microtubular arrangement which is identical to the pattern found in:

- A) microvilli
- B) centrioles
- C) spindle fibers
- D) stereocilia.

3. The organelle most prominent in steroid secreting cells is:

- A) RER
- B) SER
- C) Lysosomes
- D) Cilia

4. The main ultrastructural feature (EM) of histiocyte is the presence of:

- A) Microvilli.
- B) Lysosomes.
- C) SER
- D) RER

5. Choose the unrelated structures:

- A) Cilia / Centriole
- B) Down syndrome / Numerical chromosomal aberration
- C) Nerve cell axon / Nissel granules.
- D) Mast cells / Metachromatic stain.

6. All of the following are true except:

- A) Intercellular space in zonula occludens is zero.
- B) Tonofilaments play an important role in zonula occludens.
- C) Intercellular space in desmosome is 30 nm.
- D) Gap junction is characteristic to cardiac muscle.

7. Which of the following cells plays an important role in healing in case of capillary injury:

- A) Pericyte
- B) Plasma cell
- C) Histiocyte
- D) Nerve cell

8. Choose the FALSE sentence:

- A) Origin of plasma cell is B. lymphocyte.
- B) Origin of macrophage is monocyte.
- C) Origin of pericyte is UMC.
- D) Origin of mast cell is fibroblast.



**9. Which of the following is FALSE regarding nerve cell:**

- A) Axon doesn't contain nissel granules.
- B) Stellate nerve cell is one of bipolar nerve cells.
- C) Golgi apparatus is perinuclear.
- D) Perikaryon is devoid of centrioles.

**10. Find the mis-matching sentence:**

- A) Dense regular CT / tendon.
- B) Loose areolar CT / abundant in brain.
- C) Reticular CT / stroma of spleen.
- D) Muroid CT / Wharton's jelly.

**11. One of the following is a numerical aberration of sex chromosomes:**

- A) Turner's syndrome
- B) Down's syndrome
- C) Translocation
- D) Isochromosome

**12. As regards ribosomes:**

- A) Their subunits are formed in the nucleolus.
- B) Cause cytoplasmic acidophilia.
- C) Are formed of DNA associated with proteins.
- D) Are formed of two subunits of equal size.

**13. During light microscopic examination of a histological section, you observed a stratified epithelium. The superficial layer is composed of large cuboidal dome shaped cells and some cells contain 2 nuclei. What is the type of this epithelium?**

- A) Stratified squamous.
- B) Transitional.
- C) Stratified columnar.
- D) Pseudo stratified columnar ciliated

**14. Which of the following characters are true regarding stereocilia?**

- A) Motile, contain filaments, increase surface area, in epididymis.
- B) Non motile, contain microtubules, increase surface area, in epididymis.
- C) Non motile, contain filaments, increase surface area, in epididymis.
- D) Motile, contain microtubules, increase surface area, in epididymis.

**15. Duplication of DNA occurs in:**

- A) G1 phase
- B) G2 phase
- C) S phase
- D) Mitosis

**16. In absorbing cubical cells, the brush border seen by LM is:**

- A) Cilia
- B) flagella
- C) microvilli
- D) stereocilia

**17. One of the following is an endogenous pigment:**



- A) Lead
- B) Carotene
- C) Tattoo
- D) Hemoglobin

**18. Which of the following statements is incorrect?**

- A) Heterochromatin, as seen in EM, is lighter staining DNA.
- B) The nucleolus is the region within the nucleus where rRNA is formed.
- C) The function of SER includes lipid synthesis.
- D) RER and SER can be continuous

**19. Enzymes associated with drug detoxification are associated with:**

- A) SER
- B) RER
- C) Golgi
- D) Mitochondria

**20. Which of the following structures is avascular?**

- A) Articular cartilage
- B) intercalated disc
- C) ribs
- D) muscle

**21. A 13 years old male patient came to the hospital with bleeding in the scalp region on examination the wound was without gapping, this wound may be in which of the following layers of scalp?**

- A) first and second
- B) third
- C) fourth
- D) Fifth

**22. The external jugular vein is one of veins of the face and scalp, how is it formed?**

- A) by the union of superficial temporal and maxillary veins
- B) by the union of anterior division of retromandibular vein with anterior facial vein
- C) by the union of posterior division of retromandibular vein with posterior auricular vein
- D) union of posterior auricular with the occipital vein

**23. The dangerous area of the face is triangular in shape include which of the following?**

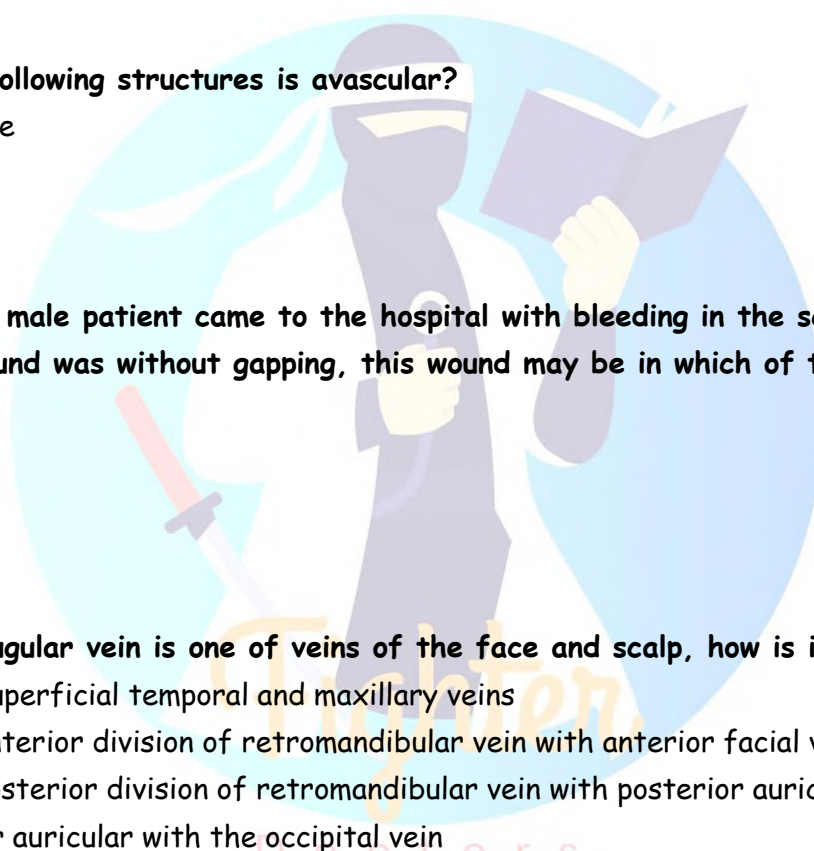
- A) nose and upper lip
- B) cheek
- C) lower lip
- D) zygomatic region

**24. Which of the following fascia Form the floor of posterior triangle?**

- A) Investing layer
- B) Prevertebral Fascia
- C) Pretracheal Fascia
- D) The carotid sheath

**25. Which of the following is the character of the artery?**

- A) It Carry blood under high pressure from the heart





- B) It has valves  
C) Its wall is semi-permeable  
D) Its wall is thin
- 26. Which of the following is part of peripheral nervous system?**  
A) Medulla  
B) Mid brain  
C) Cerebellum  
D) Somatic nerves
- 27. What is the type of symphysis pubis?**  
A) Fibrous joint  
B) Primary cartilaginous joint  
C) Secondary cartilaginous joint  
D) Synovial joint
- 28. Which of the following muscles is voluntary non striated?**  
A) Muscles of the upper limb  
B) Muscles of the trunk  
C) Muscles of the heart  
D) Muscle of the wall of blood vessel
- 29. Which of the following nerves carry sensory innervation from the lower third of the face?**  
A) Maxillary nerve  
B) Ophthalmic nerve  
C) Mandibular nerve  
D) Facial nerve
- 30. The secondary oocyte undergoes completion of the second meiotic division:**  
A) Before ovulation  
B) At fertilization  
C) Prenatal  
D) At puberty
- 31. The embryonic epiblast gives rise to the following:**  
A) Ectoderm  
B) Mesoderm  
C) Endoderm  
D) All of the above
- 32. By the 4th month the placental barrier is formed of which of the following:**  
A) Syncytiotrophoblast and endothelium  
B) Cytotrophoblast and syncytiotrophoblast  
C) Cytotrophoblast and endothelium  
D) Endothelium and extraembryonic mesoderm
- 33. When chorionic villi become vascularized they are called \_\_\_\_\_ villi:**  
A) branch  
B) stem  
C) tertiary  
D) anchoring



**34. The wall of the chorionic sac is composed of:**

- A) cytotrophoblast and syncytiotrophoblast
- B) two layers of trophoblast lined by extraembryonic somatic mesoderm
- C) trophoblast and exocoelomic membrane
- D) extraembryonic splanchnic mesoderm and both layers of trophoblast

**35. At the time of ovulation the endometrium is in the:**

- A) Menstrual phase
- B) Early proliferative phase
- C) Late proliferative phase
- D) Late secretory phase

**36. Which of the following is derived from neural crest cells:**

- A) Dura mater
- B) Stomach
- C) Urinary bladder.
- D) Adrenal medulla

**37. On examination, you noticed that the amount of amniotic fluid in a pregnant female is more than 2 litres. What is the possible cause of this case?**

- A) Oesophageal atresia
- B) Congenital anomalies in the heart
- C) Placental insufficiency
- D) Changes in trophoblast

**38. The germ layer that gives origin to the central nervous system is:**

- A) Ectoderm
- B) Mesoderm
- C) Endoderm
- D) Neural crest

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-B  | 3-B  | 4-B  | 5-C  | 6-B  | 7-A  | 8-D  | 9-B  | 10-B |
| 11-A | 12-A | 13-B | 14-C | 15-C | 16-C | 17-D | 18-A | 19-A | 20-A |
| 21-A | 22-C | 23-A | 24-B | 25-A | 26-D | 27-C | 28-D | 29-C | 30-B |
| 31-D | 32-A | 33-C | 34-B | 35-C | 36-D | 37-A | 38-A |      |      |

## Written questions

1) Compare between:

- A. Unilocular and multilocular adipose cell.
- B. Microvilli and Cilia.

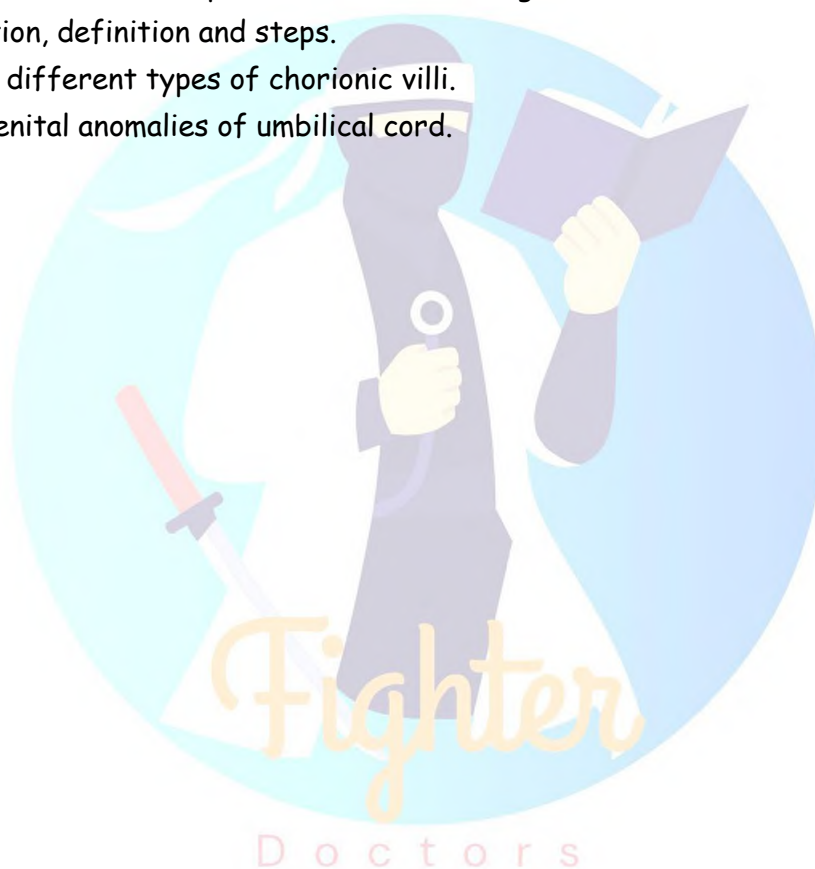
2) Enumerate:

- A. Types of glands according to mode of secretion with one site for each.
- B. Lateral specializations with reference to intercellular diameter in each type.
- C. Types of chromosomal aberrations.

3) Give the scientific name for the following statements:



- a) It is the process by which the cytoplasm and nucleus of somatic cell are divided equally into two identical daughter cells.
- b) They have the same structure as cilia but are much longer. The only example in human is sperm.
- c) The period between two successive divisions during which cell grows and performs their functions.
- d) It is special type of epithelium which is modified to contract.
- e) Site for mucoid connective tissue in adult.
- f) Basophilic collections of Ribosomes and rER in cytoplasm of perikaryon.
- g) Type of neurons which has one process soon divided into two branches.
- h) One site for highly specialized cells leave the cycle while in G1 forever and go to G0 phase.
- 4) Describe with drawing the Characters of synovial joints.
- 5) What do you know about the portal and pulmonary circulation?
- 6) Enumerate the layers of the scalp and describe the dangerous area of the scalp.
- 7) Describe neurulation, definition and steps.
- 8) Compare between different types of chorionic villi.
- 9) Mention the congenital anomalies of umbilical cord.





## 16. End Module 2025 (1st exam)

60 MCQs

1. Which of the following best describes the anatomical position of the lower limbs and toes?

- A) Knees flexed, the toes turned inward
- B) Legs abducted, feet rotated outward
- C) Legs extended and parallel, feet flat with toes directed forward
- D) Knees flexed, feet together with toes pointing sideways

2. Which plane divides the body into anterior and posterior parts?

- A) Midsagittal plane
- B) Parasagittal plane
- C) Coronal plane.
- D) Horizontal plane

3. What is the meaning of superficial dissection of the dorsal aspect of the neck?

- A) Dissection of the skin and front of the neck
- B) Dissection of the skin and back of the neck
- C) Dissection of the muscles and vessels of the back of the neck
- D) Dissection of the muscles and vessels of the front of the neck

4. Moving the distal segment away from the proximal segment is called which of the following?

- A) Adduction
- B) Abduction
- C) Flexion
- D) Extension

5. Which of the following is the character of the compact bone?

- A) Contains large spaces filled with bone marrow
- B) Composed of loosely arranged trabeculae
- C) Dense and organized into a number of cylindrical units
- D) Found only in the epiphysis of long bones

6. Which of the following arteries can supply the bone marrow?

- A) Periosteal
- B) Epiphyseal
- C) Metaphyseal
- D) Nutrient

7. Which of the following is a secondary cartilaginous joint?

- A) Intervertebral disc
- B) Suture
- C) Elbow.
- D) Sternocostal junction

8. Which of the following is considered a short bone?

- A) Carpal bone
- B) Phalanges
- C) Patella
- D) Clavicle

9. During elbow flexion, which muscle acts as the prime mover and which as the antagonist?



- A) Biceps brachii as antagonist, triceps brachii as prime mover
- B) Biceps brachii as prime mover, triceps brachii as antagonist
- C) Deltoid as prime mover, biceps brachii as antagonist
- D) Triceps brachii as prime mover, deltoid as antagonist

**10. What is the role of a synergist muscle during movement?**

- A) It opposes the action of the prime mover
- B) It initiates the movement as the main force
- C) It stabilizes the origin of the prime mover
- D) It assists the prime mover by reducing unnecessary movement

**11. During the portal circulation, the blood from the digestive system passes to which of the following?**

- A) From portal vein to the heart
- B) From the portal vein to the liver
- C) From inferior vena cava to the liver
- D) From the portal vein to the inferior vena cava

**12. The blood vessels that carry blood to the heart and have thin walls are called:**

- A) Arteries
- B) Veins
- C) Capillaries
- D) Lymph vessels

**13. The opposed bony surfaces connected by an interosseous ligament are found in which of the following?**

- A) Syndesmosis
- B) Suture
- C) Hinge joint.
- D) Plane synovial joint

**14. Primary cartilaginous joints may be temporary, like which of the following?**

- A) Costochondral junction
- B) Growing plate
- C) Intervertebral disc
- D) Symphysis pubis

**15. Which of the following contains the visual area in the brain?**

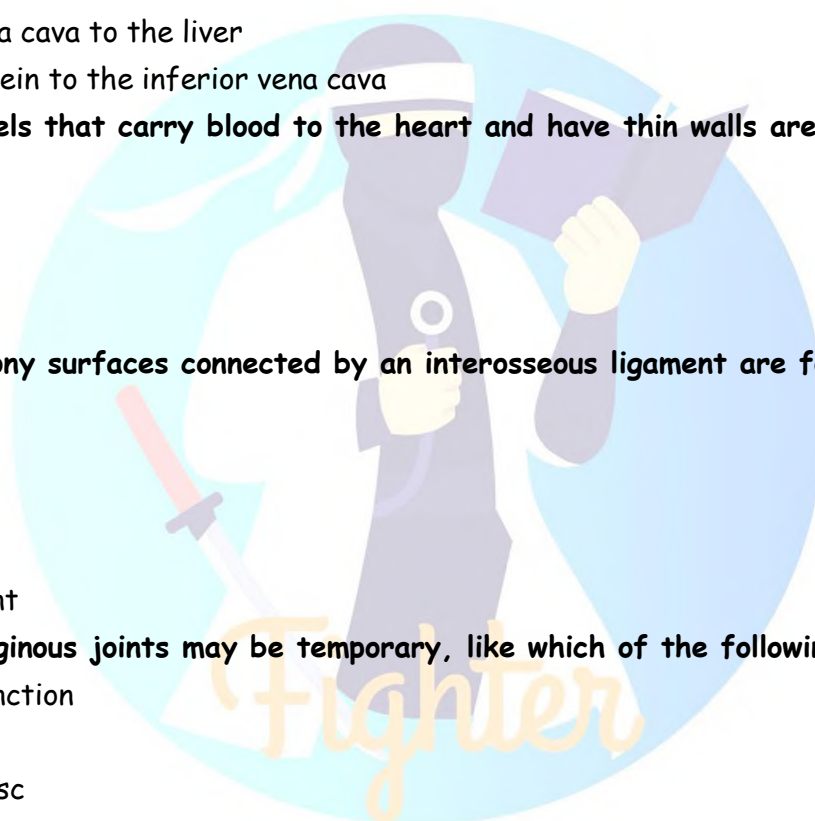
- A) Frontal lobe.
- B) Occipital lobe
- C) Temporal lobe.
- D) Parietal lobe

**16. If Sara's cycle is 38 days, on which day is she most likely to ovulate?**

- A) Day 14
- B) Day 24
- C) Day 28
- D) Day 35

**17. Which hormone surges right before ovulation, triggering the release of an egg?**

- A) Progesterone





- B) Estrogen
- C) Luteinizing hormone (LH)
- D) Follicle-stimulating hormone (FSH)

**18. If Sarah doesn't conceive, what event marks the end of the luteal phase?**

- A) Ovulation
- B) Menstruation
- C) LH surge
- D) Implantation

**19. Which hormone is released after implantation and is used to confirm pregnancy?**

- A) LH (Luteinizing hormone)
- B) FSH (Follicle-stimulating hormone)
- C) HCG (Human chorionic Gonadotropin)
- D) Estrogen

**20. At what stage does the fertilized egg typically implant in the uterus?**

- A) Zygote
- B) Morula
- C) Blastocyst
- D) Embryo

**21. Which term describes complete absence of sperm in semen analysis?**

- A) Azoospermia
- B) Oligospermia
- C) Teratozoospermia
- D) Asthenozoospermia

**22. What major event typically occurs at the start of the second week of development?**

- A) Gastrulation
- B) Fertilization
- C) Implantation of the blastocyst
- D) Formation of the neural tube

**23. What cavity begins to develop during the end of the second week?**

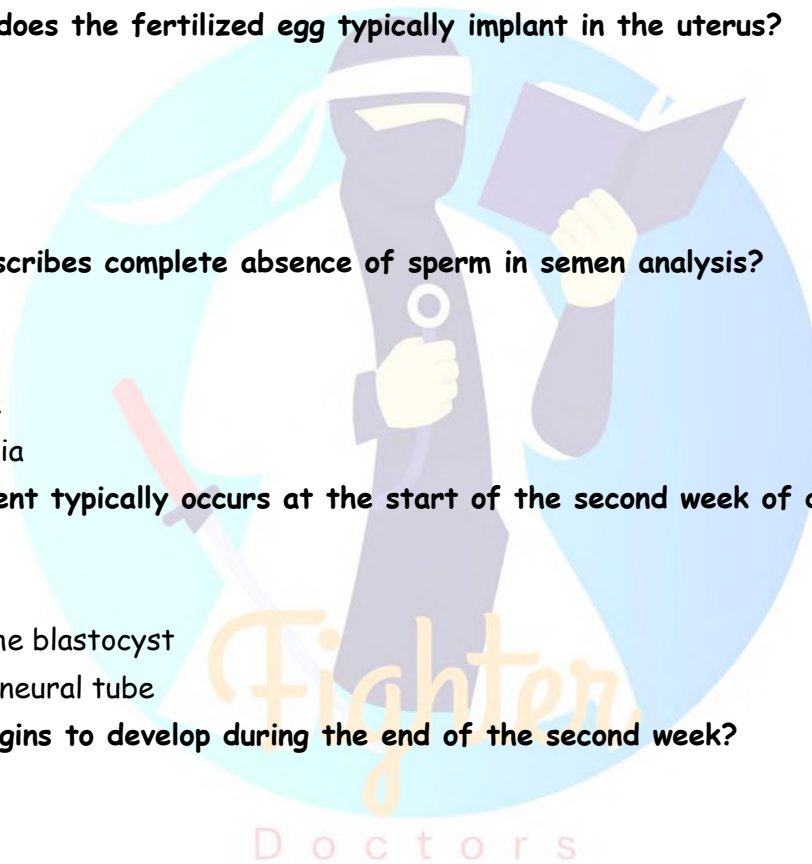
- A) Chorionic cavity
- B) Amniotic cavity
- C) Yolk sac.
- D) Blastocoele

**24. Which part of the blastocyst directly interacts with the endometrial lining during implantation?**

- A) Inner cell mass (Embryoblast)
- B) Syncytiotrophoblast
- C) Amniotic cavity
- D) Cytotrophoblast

**25. What hormone prepares the endometrium for implantation?**

- A) Estrogen
- B) Follicle-stimulating hormone (FSH)
- C) Progesterone





D) Luteinizing hormone (LH)

**26. Which layer of the bilaminar disc contributes to the formation of the primitive yolk sac?**

A) Hypoblast

B) Epiblast

C) Cytotrophoblast.

D) Syncytiotrophoblast

**27. Which event forms the definitive yolk sac?**

A) Gastrulation

B) Neurulation

C) Folding

D) Fertilization

**28. Which cells form the inner layer of the chorion?**

A) Epiblast cells

B) Syncytiotrophoblast

C) Hypoblast cells

D) Primary mesoderm

**29. During which week of development does the chorionic cavity form?**

A) First week

B) Second week

C) Third week

D) Fourth week

**30. Which structure is responsible for the establishment of uteroplacental circulation?**

A) Blastocyst

B) Amnion

C) Yolk sac

D) Syncytiotrophoblast

**31. Which of the following represents a character of transitional epithelium?**

A) Superficial cells are columnar with basal oval nuclei

B) The basal cells are flat cells with flat nuclei

C) Has no basement membrane

D) It is present in the urinary bladder

**32. Which of the following represents the typical feature of connective tissue?**

A) It has closely aggregated cells

B) Rests on basement membrane

C) Highly vascular tissue

D) Ectodermal in origin

**33. Which of the following represents Tight junction?**

A) Zonula occludens

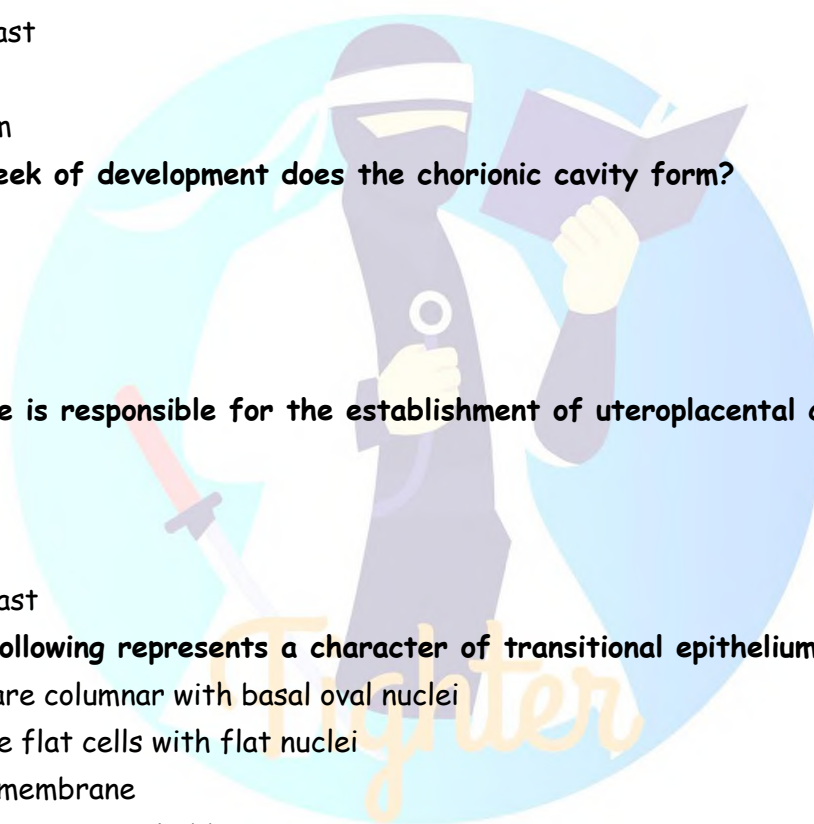
B) Zonula adherens

C) Gap junction

D) Fascia adherens

**34. Which of the following represents the typical feature of hemidesmosomes?**

A) It is a belt-like specialization





- B) Anchors cell to the basement membrane
- C) The cytoplasmic side shows condensation of actin filaments
- D) The intercellular space between the adjacent cell membranes is 20 nm

**35. Which of the following represents the typical histological structure of fibroblast?**

- A) The cytoplasm is pale basophilic
- B) Contains few rough endoplasmic reticulum
- C) The nucleus is heterochromatic
- D) Its shape is spindle-shaped with many processes

**36. Which of the following describes the white fat cell?**

- A) The cytoplasm is occupied by a single large lipid droplet
- B) The nucleus is central and rounded
- C) Contains many filamentous mitochondria
- D) Contains cytochrome pigment

**37. What is the typical feature of melanocyte in the skin?**

- A) It is present between the cells in the basal cell layer
- B) Contains few lysosomes
- C) It is the sensory cells of the epidermis
- D) It is present between the cells of stratum spinosum

**38. Which of the following represents the typical feature of thin skin?**

- A) It contains hair follicles and sebaceous glands
- B) Contains numerous sweat glands
- C) The stratum lucidum is prominent
- D) The stratum corneum is thicker

**39. Which of the following represents the typical feature of microvilli?**

- A) It is a motile cytoplasmic projection
- B) It is present in the cells lining the fallopian tube
- C) It is formed of microtubules
- D) Its core is formed of actin filaments

**40. Which of the following represents a site for simple cubical epithelium?**

- A) Bowman's capsule of kidney
- B) Thyroid follicles
- C) Fallopian tube
- D) Vas deferens

**41. Which of the following is true as regard Ribosomes?**

- A) Appears basophilic when stained with H & E.
- B) Stains red when stained with silver.
- C) It is found supranuclear in nerve cells.
- D) Appears as unstained area in deeply basophilic cytoplasm.

**42. Which of the following is absent from matrix of mitochondria?**

- A) Circular DNA & RNA.
- B)  $\text{Ca}^{2+}$  acting as catalysts condensed within dense granules.
- C) Oxidative enzymes of Krebs's cycle.
- D) Hydrolytic digestive enzymes.



**43. Which of the following is truly matching as regard each organelle?**

- A) Inner mitochondrial membrane → Cristae.
- B) Multivesicular bodies → Engulfed solid foreign bodies.
- C) Protein synthesis → SER.
- D) Euchromatin → Nucleolus associated.

**44. Which of the following is mis-matching as regard each organelle?**

- A) Attached ribosomes → Synthesis of cytosol proteins.
- B) Primary lysosomes → Originate from trans-face of Golgi apparatus.
- C) Initial protein modifications → RER.
- D) Transfer vesicles → Transport proteins from RER to cis-face.

**45. Which of the following is true regarding stains?**

- A) Collagen fibers → Pink with H & E.
- B) Mast cells → Blue with Toluidine blue.
- C) Mitochondria → Blue with Janus green.
- D) Reticular fibers → Red with silver.

**46. Which of the following forms the stroma of organs?**

- A) Elastic fibers
- B) Collagen type III (Reticular fibers)
- C) Collagen type I
- D) Muscle fibers

**47. What is the special stain of Golgi apparatus?**

- A) Silver
- B) Best's Carmine
- C) Sudan III
- D) Iron HX

**48. Which of the following best describes peripheral chromatin?**

- A) Connected to inner border of nuclear envelope
- B) Connected to nucleoli
- C) Clumped and scattered in the nuclear sap
- D) Fine granules and scattered in the nuclear sap

**49. Which of the following statements describes active cells?**

- A) Dark nucleus
- B) Heterochromatic nucleus
- C) Euchromatic nucleus
- D) Few mitochondria

**50. What about heterochromatin?**

- A) Makes the nucleus pale by LM.
- B) Appears as coarse granules in the nucleus.
- C) Contains active genes.
- D) Represents extended parts of chromatin.

**51. What is the shape of nucleus present in plasma cell?**

- A) Clock face
- B) Horse shoe



C) Kidney

D) Signet ring

**52. In the nervous tissue, the axons of the neuron are characterized by which one of the following?**

A) Have uniform diameter

B) Contain Nissl's granules

C) Contain Golgi apparatus

D) Convey impulses to the cell body

**53. Which of the following represents the sheath of lipoprotein complex around the axon?**

A) Epineurium

B) Perineurium

C) Endoneurium

D) Myelin

**54. The gaps between myelin sheath form which of the following structures?**

A) Myelin sheath

B) Axolemma

C) Nodes of Ranvier

D) Schmidt-Lanterman clefts

**55. Which one is among the functions of adipose connective tissue?**

A) Gives the skin its shape

B) Withstands stretch

C) Binds structures and surrounds organs & blood vessels

D) Forms the stroma of the organs

**56. Which of the following is the site of brown adipose connective tissue?**

A) Axilla

B) Abdominal wall

C) Mammary gland

D) Mesentery

**57. The elastic fibers can be demonstrated by which stain?**

A) H & E

B) PAS

C) Sudan III

D) Orcein

**58. Pinocytosis is the process by which the cell membrane can engulf which of the following?**

A) Fluid droplets.

B) Solid particles

C) Bacteria

D) Foreign bodies

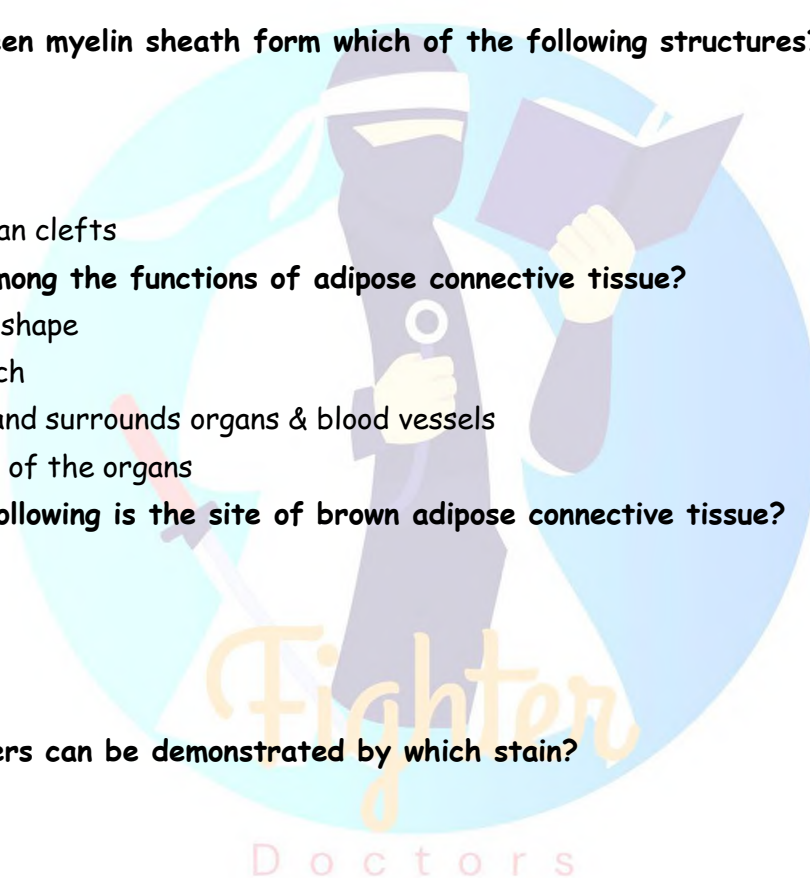
**59. Which one is regarded as an exogenous pigment?**

A) Haemoglobin

B) Melanin

C) Carotene

D) Lipofuscin





60. Pseudounipolar neurons are present in which of the following?

- A) Retina
- B) Spinal ganglia
- C) Sympathetic ganglia
- D) Anterior horn cells

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-C  | 3-B  | 4-B  | 5-C  | 6-D  | 7-A  | 8-A  | 9-B  | 10-D |
| 11-B | 12-B | 13-A | 14-B | 15-B | 16-B | 17-C | 18-B | 19-C | 20-C |
| 21-A | 22-C | 23-A | 24-B | 25-C | 26-A | 27-C | 28-D | 29-B | 30-D |
| 31-D | 32-C | 33-A | 34-B | 35-D | 36-A | 37-A | 38-A | 39-D | 40-B |
| 41-A | 42-D | 43-A | 44-A | 45-A | 46-B | 47-A | 48-A | 49-C | 50-B |
| 51-A | 52-A | 53-D | 54-C | 55-A | 56-A | 57-D | 58-A | 59-C | 60-B |

*Note on Q45: Debatable: Janus green (mitochondria) also true.*





## 17. End Semester 2025 (1st exam)

38 MCQs

1. What are the multiple short processes of neuron called?
  - A) Nodes of Ranvier
  - B) Synaptic clefts
  - C) Axon terminals
  - D) Dendrites
2. Which type of neuroglia forms the myelin sheath in the peripheral nervous system?
  - A) Schwann cell
  - B) Astrocytes
  - C) Oligodendrocytes.
  - D) Microglia
3. Reticular fibers are primarily composed of which protein?
  - A) Collagen
  - B) Elastin
  - C) Fibrillin
  - D) Keratin
4. Plasma cells release which substances?
  - A) Collagen and elastin
  - B) Histamine and heparine
  - C) Antibodies
  - D) proteoglycans and fibronectin
5. Which of the following is a characteristic of dense irregular connective tissue?
  - A) Randomly arranged collagen fibers
  - B) High tensile strength in one direction
  - C) Abundance of ground substance
  - D) Presence of elastic fibers in all regions
6. Which type of epithelium is specialized for protection?
  - A) Simple squamous epithelium
  - B) Stratified squamous epithelium
  - C) Simple columnar epithelium with microvilli.
  - D) Pseudostratified columnar epithelium
7. Which epithelial specialization allows passage of ions between epithelial cells?
  - A) Tight Junctions
  - B) Gap junctions
  - C) Desmosomes
  - D) Hemidesmosomes
8. Which organelle is responsible for cell digestion?
  - A) rER
  - B) Golgi apparatus
  - C) sER
  - D) Lysosome
9. What is the term for the process by which the cells transport solids?



- A) Endocytosis
- B) Exocytosis
- C) Pinocytosis
- D) Phagocytosis

**10. Which of the following layers connected by desmosomes?**

- A) Stratum basale
- B) Stratum granulosum
- C) Stratum lucidum
- D) Stratum spinosum

**11. What is the function of stratum granulosum?**

- A) Renewal of keratinocytes
- B) Resisting the effects of friction and pressure
- C) Waterproof-barrier of the skin.
- D) Receptors for fine touch sensation

**12. Which of the following best describes the phospholipid molecules of cell membrane?**

- A) The head is non-charged and hydrophobic
- B) The head is directed inwards away from the aqueous solutions
- C) The fatty acid tail is charged and hydrophilic
- D) The head is directed towards the aqueous solutions

**13. Regarding rER, which of the following describes it?**

- A) Form proteins secreted by the cells
- B) Form proteins used within the cytosol
- C) Linked together by tRNA forming polysomes.
- D) Bind to the outer surface of SER

**14. Which feature of the followings describes centrioles?**

- A) They are a membranous organelle
- B) Can be demonstrated with silver stain
- C) Formed of nine triplets of microtubules
- D) Have the same number of microtubules as the shaft of cilia

**15. What is the typical feature of condensed chromatin?**

- A) Appears dark acidophilic by LM
- B) Appears as fine granules by EM
- C) Contains the inactive genes
- D) Appears pale basophilic by LM

**16. Which of the following best describes thick skin?**

- A) It has apparent stratum lucidum
- B) Has sebaceous glands
- C) Contains fewer sweat glands in the dermis
- D) Contains well developed hair follicles

**17. The cilia can be characterized by which of the followings?**

- A) The basal body is formed of nine microtubules
- B) The rootlet contains twenty-seven microtubules
- C) move secretions over the tissue surface in one directions



D) Can be found in the male genital system

**18. Which one of the following stains demonstrates lipid in the liver cells?**

A) Sudan III

B) PAS

C) Silver

D) trypan blue

**19. What is the origin of secretory vesicles of Golgi apparatus?**

A) Smooth endoplasmic reticulum

B) Rough endoplasmic reticulum

C) The trans face of Golgi apparatus

D) The cell membrane

**20. Which of the following structures is connect the fibers of cardiac muscles?**

A) connective tissue

B) intercalated disc

C) ligament

D) tendon

**21. Why is the loose areolar tissue considered dangerous?**

A) It is highly vascularized

B) It contains sweat glands

C) It allows infection to spread to the brain via emissary veins

D) It is the thickest layer of the scalp

**22. Which veins unite to form the external jugular vein?**

A) Superficial temporal and maxillary veins

B) anterior division of retromandibular vein and anterior facial vein

C) posterior division of retromandibular vein and posterior auricular vein

D) union of posterior auricular with the occipital vein

**23. A 35-year-old female complains of pain in the middle third of her face. Which of the following nerves is responsible for this pain?**

A) mandibular

B) maxillary

C) ophthalmic

D) facial

**24. Which of the following is one of the content of carotid sheath?**

A) sternomastoid muscle

B) vagus nerve

C) external jugular vein

D) facial artery

**25. What is the primary function of pulmonary circulation?**

A) Regulate blood pressure

B) Exchange gases between blood and lungs

C) Deliver nutrients to tissues

D) Transport oxygenated blood to the body

**26. Which layer of the meninges is the outermost and toughest??**

A) arachnoid mater

B) pia mater

C) dura mater.

D) subarachnoid space

A) Cortical reaction



- C) Acrosome reaction  
D) Cleavage division
- 36. Where originates the extraembryonic mesoderm?**
- A) It is a result of the migration of cells from the neuroectoderm  
B) In a site between trophoblast and embryoblast  
C) In a site between hypoblast and epiblast  
D) In a site between endoderm and ectoderm

- 37. Which embryonic structure is formed adjacent to the notochord?**
- A) Endoderm  
B) Somites  
C) Amnion  
D) Allantois

- 38. At approximately what embryonic age does an embryo typically have about 27 pairs of somites?**
- A) 21 days  
B) 29 days  
C) 30 days  
D) 40 days

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-A  | 3-A  | 4-C  | 5-A  | 6-B  | 7-B  | 8-D  | 9-D  | 10-D |
| 11-C | 12-D | 13-A | 14-C | 15-C | 16-A | 17-C | 18-A | 19-C | 20-B |
| 21-C | 22-C | 23-B | 24-B | 25-B | 26-C | 27-A | 28-C | 29-B | 30-A |
| 31-A | 32-A | 33-C | 34-B | 35-A | 36-B | 37-B | 38-B |      |      |

Note on Q17: Debatable: option D (male genital system) is also ciliated in efferent ductules.

## Written questions

Written part (36 marks)

1) Enumerate (6 marks):

- A. 3 sites of simple squamous epithelium (1.5 marks)  
B. 3 types of exocrine glands according to branching of duct system (1.5 marks)  
C. 3 sites of Elastic connective tissue (1.5 marks)  
D. Types of neurons according to number of processes (1.5 marks)

2) Mention (6 marks):

- A. 3 parts of nucleolus (1.5 marks)  
B. Non-keratinocytes of the skin and mention one function for each one (3 marks)  
C. 3 functions for sER (1.5 marks)

3) Discuss (2 marks each):

- A. Transitional epithelium (2 marks)  
B. white Adipose Connective tissue (2 marks)  
C. Histological structure of perikaryon (2 marks)

4- Describe the forms of the skeletal muscle (3 marks)

5- Describe the bone marrow (definition, function, forms according to the age) (3 marks)

6- Mention the origin, insertion, nerve supply and action of buccinator muscle (3 marks)

7- Explain the congenital anomalies of placenta (3 marks)

8- Describe the steps of gastrulation (3 marks)

9- compare between chorionic villi as regarding their structure (3 marks)



## 18. End Module 2025 (2nd exam)

60 MCQs

1. Which of the following best describes the anatomical position of the lower limbs and toes?

- A) Knees flexed, the toes turned inward.
- B) Legs abducted, feet rotated outward.
- C) Legs extended and parallel, feet flat with toes directed forward.
- D) Knees flexed, feet together with toes pointing sideways.

2. Which plane divides the body into anterior and posterior parts?

- A) midsagittal.
- B) parasagittal.
- C) coronal.
- D) horizontal.

3. What is the meaning of superficial dissection of the dorsal aspect of the neck?

- A) dissection of the skin and front of the neck.
- B) dissection of the skin and back of the neck.
- C) dissection of the muscles and vessels and back of the neck.
- D) dissection of the muscles and vessels and front of the neck.

4. Moving the distal segment away from the proximal segment is called which of the following?

- A) Adduction.
- B) abduction.
- C) flexion.
- D) extension.

5. Which of the following is the character of the compact bone?

- A) Contains large spaces filled with bone marrow
- B) Composed of loosely arranged trabeculae.
- C) Dense and organized into number of cylindrical units
- D) Found only in the epiphysis of long bones.

6. Which of the following arteries can supply the bone marrow?

- A) periosteal
- B) epiphyseal
- C) metaphyseal
- D) nutrient

7. Which of the following is secondary cartilaginous joint?

- A) Intervertebral disc
- B) suture
- C) elbow
- D) sternocostal junction.

8. Which of the following is considered short bone?

- A) Carpal bone
- B) phalanges
- C) patella
- D) clavicle

9. During elbow flexion, which muscle acts as the prime mover and which as the antagonist?



- A) Biceps brachii as antagonist, triceps brachii as prime mover.
- B) Biceps brachii as prime mover, triceps brachii as antagonist
- C) Deltoid as prime mover, biceps brachii as antagonist
- D) Triceps brachii as prime mover, deltoid as antagonist

**10. What is the role of a synergist muscle during movement?**

- A) It opposes the action of the prime mover
- B) It initiates the movement as the main force
- C) It stabilizes the origin of the prime mover
- D) It assists the prime mover by reducing unnecessary movement

**11. During the portal circulation, the blood from the digestive system pass to which of the following?**

- A) From portal vein to the heart
- B) From the portal vein to the liver
- C) From inferior Vena Cava to the liver
- D) From the portal vein to the inferior vena cava

**12. blood vessels that carry blood to heart, have thin wall is called which of the following?**

- A) Artery
- B) vein
- C) capillary
- D) lymph vessel

**13. The opposed bony surfaces are connected by interosseous ligament is found in which of the following?**

- A) Syndesmosis
- B) suture
- C) hinge joint
- D) plane synovial joint

**14. Primary cartilaginous joint may be temporary like which of the following?**

- A) costochondral junction.
- B) growing plate.
- C) intervertebral disc
- D) symphysis pubis

**15. Which of the following contains the visual area in the brain?**

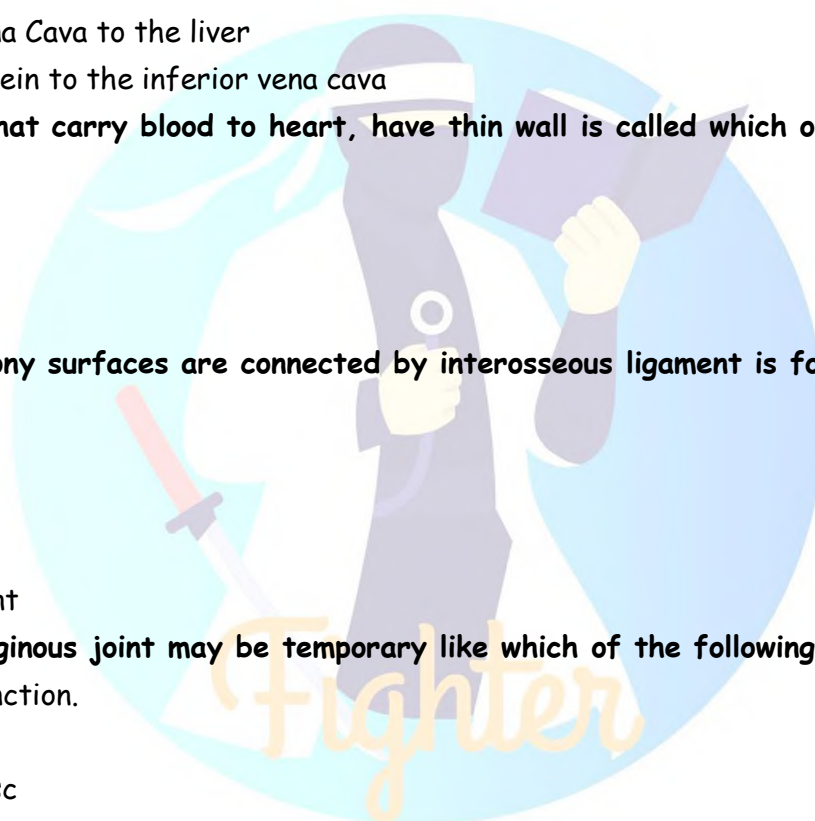
- A) Frontal lobe
- B) Occipital lobe
- C) Temporal lobe
- D) parietal lobe

**16. If Sara's cycle is 38 days, on which day is she most likely to ovulate?**

- A) Day 14
- B) Day 24
- C) Day 28
- D) Day 35

**17. Which hormone surges right before ovulation, triggering the release of an egg?**

- A) Progesterone





B) Estrogen

C) (LH)

D) (FSH)

**18. If Sarah doesn't conceive, what event marks the end of the luteal phase?**

A) Ovulation

B) Menstruation

C) LH surge

D) Implantation

**19. Which hormone is released after implantation and is used to confirm pregnancy?**

A) LH

B) FSH

C) HCG (Human chorionic Gonadotropin)

D) Estrogen

**20. At what stage does the fertilized egg typically implant in the uterus?**

A) Zygote

B) Morula

C) Blastocyst

D) Embryo

**21. Which term describes complete absence of sperm in semen analysis?**

A) Azoospermia

B) Oligospermia

C) Teratozoospermia

D) Asthenozoospermia

**22. What major event typically occurs at the start of the second week of development?**

A) Gastrulation

B) Fertilization

C) Implantation of the blastocyst

D) Formation of the neural tube

**23. What cavity begins to develop during the end of the second week?**

A) Chorionic cavity

B) Amniotic cavity

C) Yolk sac

D) Blastocoele

**24. Which part of the blastocyst directly interacts with the endometrial lining during implantation?**

A) Inner cell mass (embryoblast)

B) Syncytiotrophoblast

C) Amniotic cavity

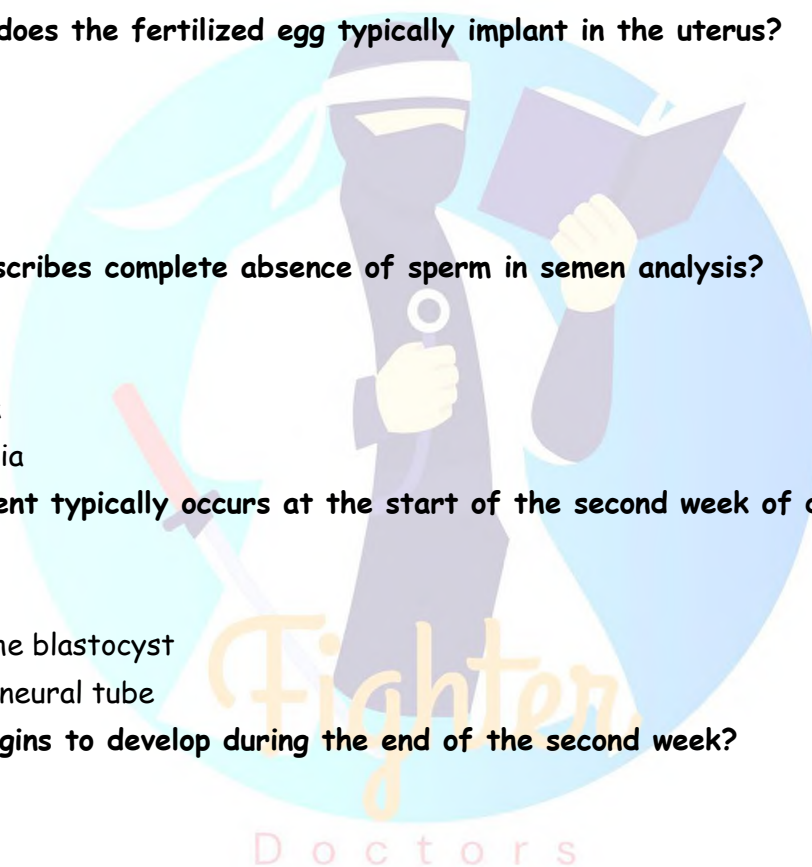
D) Cytotrophoblast

**25. What hormone prepares the endometrium for implantation?**

A) Estrogen

B) (FSH)

C) Progesterone





D) (LH)

**26. Which layer of the bilaminar disc contributes to the formation of the primitive yolk sac?**

- A) Hypoblast.
- B) Epiblast.
- C) Cytotrophoblast.
- D) Syncytiotrophoblast.

**27. Which event forms definitive yolk sac?**

- A) Gastrulation
- B) Neurulation.
- C) Folding
- D) Fertilization

**28. Which cells form the inner layer of the chorion?**

- A) Epiblast cells
- B) Syncytiotrophoblast
- C) Hypoblast cells
- D) Primary mesoderm

**29. During which week of development does the chorionic cavity form?**

- A) First week
- B) Second week
- C) Third week
- D) Fourth week

**30. Which structure is responsible for the establishment of uteroplacental circulation?**

- A) Blastocyst.
- B) Amnion.
- C) Yolk sac.
- D) Syncytiotrophoblast.

**31. Which of the following represents a character of transitional epithelium?**

- A) Superficial cells are columnar with basal oval nuclei
- B) The basal cells are flat cells with flat nuclei
- C) Has no basement membrane
- D) It is present in the urinary bladder.

**32. Which of the following represents the typical feature of connective tissue?**

- A) It has closely aggregated cells
- B) Rests on basement membrane
- C) Highly vascular tissue
- D) Ectodermal in origin

**33. Which of the following represents Tight junction?**

- A) Zonula occludens
- B) Zonula adherens
- C) Gap junction
- D) Fascia adherens

**34. Which of the following represents the typical feature of hemidesmosomes?**

- A) It is a belt-like specialization



- B) Anchors cell to the basement membrane
- C) The cytoplasmic side shows condensation of actin filaments
- D) The intercellular space between the adjacent cell membranes is 20 nm

**35. Which of the following represents the typical histological structure of Fibroblast?**

- A) The cytoplasm is pale basophilic
- B) Contains few rough endoplasmic reticulum
- C) The nucleus is heterochromatic
- D) Its shape is spindle shaped with many processes

**36. Which of the following describes the white fat cell?**

- A) The cytoplasm is occupied by a single large lipid droplet
- B) The nucleus is central and rounded
- C) Contains many filamentous mitochondria
- D) Contain cytochrome pigment

**37. What is the typical feature of melanocyte in the skin?**

- A) It is present between the cells in the basal cell layer
- B) Contains few lysosomes
- C) It is the sensory cells of the epidermis
- D) It is present between the cells of stratum spinosum

**38. Which of the following represents the typical feature of thin skin?**

- A) It contains hair follicles and sebaceous glands
- B) Contains numerous sweat glands
- C) The Stratum lucidum is prominent
- D) The Stratum corneum is thicker

**39. Which of the following represents the typical feature of Microvilli?**

- A) It is a motile cytoplasmic projection
- B) It is present in the cells lining fallopian tube
- C) It is formed of microtubules
- D) Its core is formed of actin filaments

**40. Which of the following represents a site for simple cubical epithelium?**

- A) Bowman's capsule of kidney
- B) Thyroid follicles
- C) Fallopian tube
- D) Vas deferens

**41. Which of the following is true as regard Ribosomes?**

- A) Appears basophilic when stained with H & E
- B) Stains red when stained with silver
- C) It is found supranuclear in nerve cell
- D) Appears as unstained area in deeply basophilic cytoplasm

**42. Which of the following is absent from matrix of mitochondria?**

- A) Circular DNA & RNA
- B)  $Ca^{+2}$  acting as catalysts condensed within dense granules
- C) Oxidative enzymes of Krebs' cycle
- D) Hydrolytic digestive enzymes



**43. Which of the following is truly- matching as regard each organelle?**

- A) Cristae: inner mitochondrial membrane
- B) Multivesicular bodies: Engulfed solid foreign bodies
- C) SER: protein synthesis
- D) Euchromatin: Nucleolus associated

**44. Which of the following is mis matching as regard each organelle?**

- A) Attached ribosomes: Synthesis of cytosol proteins
- B) Primary lysosomes: Originate from trans- face of Golgi apparatus
- C) RER: Initial protein modifications
- D) Transfer vesicles: Transport proteins from RER to cis-face

**45. Which of the following is true regarding stains?**

- A) Collagen fibers: pink with H & E
- B) Mast cells: Blue with Toluidine blue
- C) Mitochondria: Blue with Janus green
- D) Reticular fibers: red with silver

**46. Which of the following form the stroma of organs?**

- A) Elastic fibers.
- B) Collagen type III (Reticular fibers)
- C) Collagen type I
- D) Muscle fibers

**47. What is the special stain of Golgi apparatus?**

- A) Silver
- B) Best's Carmine
- C) Sudan III
- D) Iron HX

**48. Which of the following best describes peripheral chromatin?**

- A) Connected to inner border of nuclear envelope
- B) Connected to nucleoli
- C) Clumped and scattered in the nuclear sap
- D) Fine granules and scattered in the nuclear sap

**49. Which of the following statements describe active cells?**

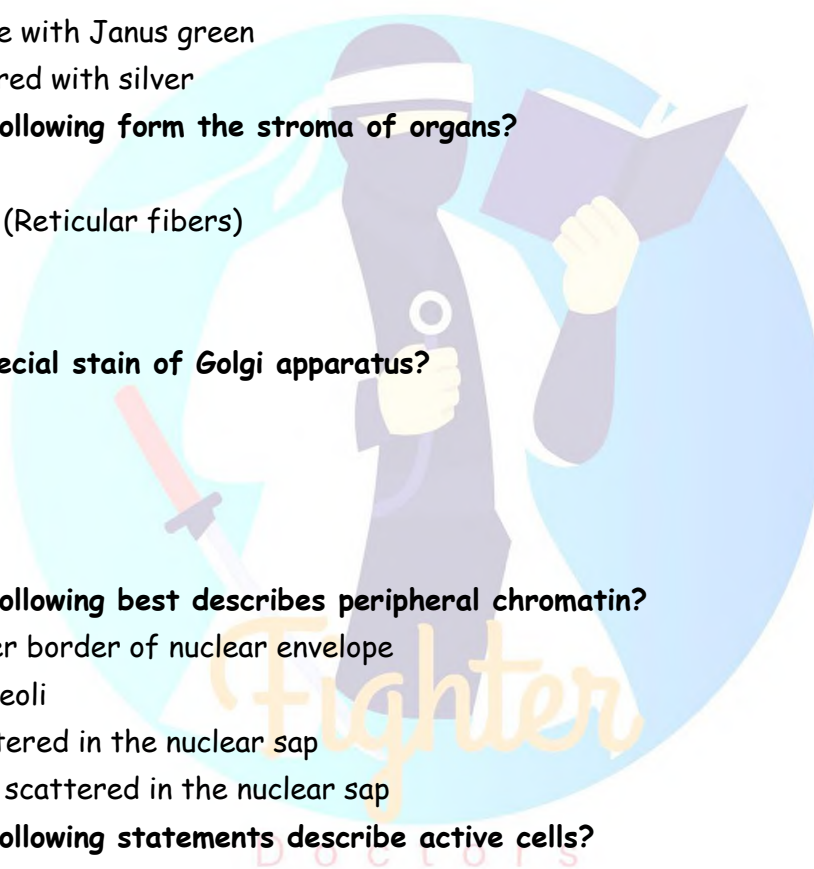
- A) Dark nucleus
- B) Heterochromatic nucleus
- C) Euchromatic nucleus
- D) Few Mitochondria.

**50. What about heterochromatin?**

- A) Makes the nucleus pale by LM
- B) Appears as coarse granules in the nucleus
- C) Contains active genes
- D) Represents extended parts of chromatin

**51. What is the shape of nucleus present in plasma cell?**

- A) Clock face
- B) Horse shoe.





C) Kidney

D) Signet ring

**52. In the nervous tissue, the axons of the neuron are characterized by which one of the following?**

A) Have uniform diameter

B) Contain Nissl's granules

C) Contain Golgi apparatus

D) Convey impulses to the cell body

**53. Which of the following represents the sheath of lipoprotein complex around the axon?**

A) Epineurium

B) Perineurium.

C) Endoneurium

D) Myelin

**54. The gaps between myelin sheath forming which of the following structures?**

A) Myelin sheath

B) Axolemma

C) Nodes of Ranvier

D) Schmidt-Lanterman clefts

**55. Which one is among the functions of adipose connective tissue?**

A) Gives the skin its shape

B) Withstands stretch

C) Binds structures and surround organs & blood vessels

D) Forms the stroma of the organs

**56. What of the following is the site of Brown adipose connective tissue?**

A) Axilla

B) Abdominal wall

C) Mammary gland

D) Mesentery

**57. The elastic fibers can be demonstrated by which stain?**

A) H & E

B) PAS

C) Sudan III

D) Orcein

**58. Pinocytosis is the process by which cell membrane can engulf which of the following?**

A) Fluid droplets

B) solid particles

C) Bacteria

D) Foreign bodies

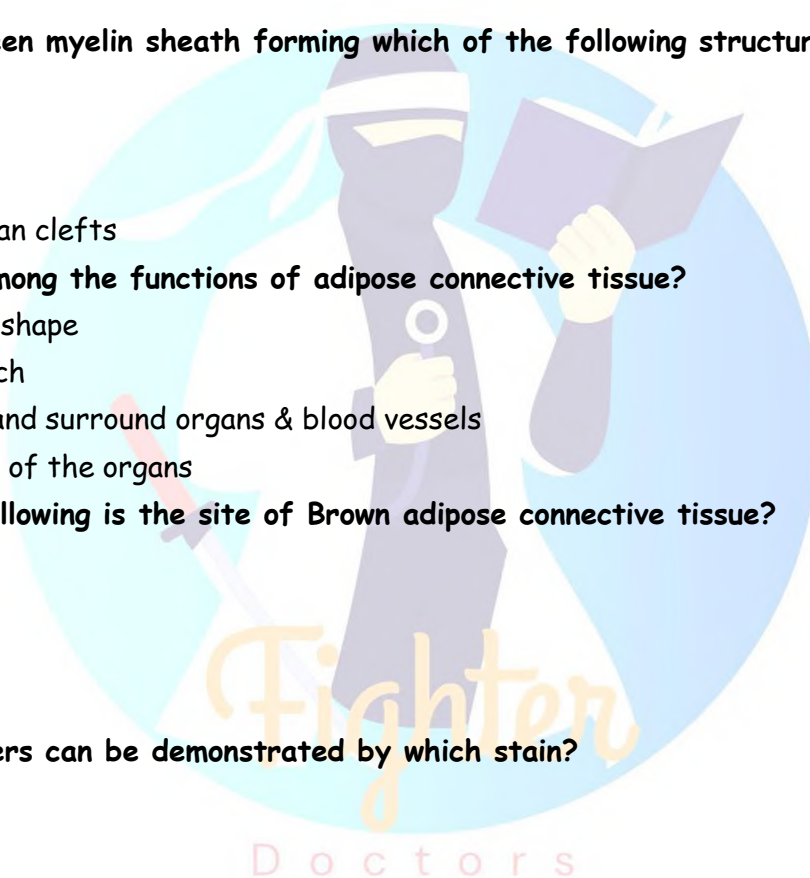
**59. Which one is regarded as exogenous pigment?**

A) Haemoglobin

B) Melanin

C) Carotene

D) Lipofuscin





60. Pseudounipolar neurons are present in which of the following?

- A) Retina
- B) Spinal ganglia
- C) Sympathetic ganglia
- D) Anterior horn cells

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-C  | 3-B  | 4-B  | 5-C  | 6-D  | 7-A  | 8-A  | 9-B  | 10-D |
| 11-B | 12-B | 13-A | 14-B | 15-B | 16-B | 17-C | 18-B | 19-C | 20-C |
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| 31-D | 32-C | 33-A | 34-B | 35-D | 36-A | 37-A | 38-A | 39-D | 40-B |
| 41-A | 42-D | 43-A | 44-A | 45-A | 46-B | 47-A | 48-A | 49-C | 50-B |
| 51-A | 52-A | 53-D | 54-C | 55-A | 56-A | 57-D | 58-A | 59-C | 60-B |

*Note on Q45: Debatable: Janus green (mitochondria) also true.*





## 19. End Semester 2025 (2nd exam)

38 MCQs

1. What are the multiple short processes of neuron called?
  - A) Nodes of Ranvier
  - B) Synaptic clefts
  - C) Axon terminals
  - D) Dendrites
2. Which type of neuroglia forms the myelin sheath in the peripheral nervous system?
  - A) Schwann cell
  - B) Astrocytes
  - C) Oligodendrocytes
  - D) Microglia
3. Reticular fibers are primarily composed of which protein?
  - A) Collagen.
  - B) Elastin.
  - C) Fibrillin.
  - D) Keratin.
4. Plasma cells release which substances?
  - A) Collagen and elastin
  - B) Histamine and heparine.
  - C) Antibodies
  - D) proteoglycans and fibronectin.
5. Which of the following is a characteristic of dense irregular connective tissue?
  - A) Randomly arranged collagen fibers
  - B) High tensile strength in one direction.
  - C) Abundance of ground substance
  - D) Presence of elastic fibers in all regions.
6. Which type of epithelium is specialized for protection?
  - A) Simple squamous epithelium
  - B) Stratified squamous epithelium
  - C) Simple columnar epithelium with microvilli
  - D) Pseudostratified columnar epithelium
7. Which epithelial specialization allows passage of ions between epithelial cells?
  - A) Tight Junctions
  - B) Gap junctions
  - C) Desmosomes
  - D) Hemidesmosomes.
8. Which organelle is responsible for cell digestion?
  - A) RER
  - B) Golgi apparatus
  - C) SER
  - D) Lysosome
9. What is the term for the process by which the cells transport solids?



- A) Endocytosis.
- B) Exocytosis
- C) Pinocytosis
- D) Phagocytosis

**10. Which of the following layers is connected by desmosomes?**

- A) Stratum basale
- B) Stratum granulosum
- C) Stratum lucidum
- D) Stratum spinosum

**11. What is the function of stratum granulosum?**

- A) Renewal of keratinocytes
- B) Resisting effects of friction and pressure
- C) Waterproof-barrier of the skin
- D) Receptors for fine touch sensation

**12. Which of the following best describes the phospholipid molecules of cell membrane?**

- A) The head is non-charged and hydrophobic
- B) The head is directed inwards away from the aqueous solutions
- C) The fatty acid tail is charged and hydrophilic
- D) The head is directed towards the aqueous solutions

**13. Regarding rER, which of the following describes it?**

- A) Form proteins secreted by the cells
- B) Form proteins used within the cytosol
- C) Linked together by tRNA forming polysomes.
- D) Bind to the outer surface of SER

**14. Which feature of the followings describes centrioles?**

- A) They are a membranous organelle.
- B) Can be demonstrated with silver stain.
- C) Formed of nine triplets of microtubules
- D) Have the same number of microtubules as the shaft of cilia

**15. What is the typical feature of condensed chromatin?**

- A) Appears dark acidophilic by LM
- B) Appears as fine granules by EM
- C) Contains the inactive genes
- D) Appears pale basophilic by LM

**16. Which of the following best describes thick skin?**

- A) It has apparent stratum lucidum
- B) Has sebaceous glands
- C) Contains fewer sweat glands in the dermis
- D) Contains well developed hair follicles

**17. The cilia can be characterized by which of the followings?**

- A) The basal body is formed of nine microtubules
- B) The rootlet contains twenty-seven microtubules
- C) move secretions over the tissue surface in one direction



D) Can be found in the male genital system

**18. Which one of the following stains demonstrates lipid in the liver cells?**

A) Sudan III

B) PAS

C) Silver

D) trypan blue

**19. What is the origin of secretory vesicles of Golgi apparatus?**

A) SER

B) RER

C) trans face of Golgi apparatus.

D) cell membrane

**20. Which of the following structures connects the fibers of cardiac muscles?**

A) connective tissue

B) intercalated disc

C) ligament

D) tendon

**21. Why is the loose areolar tissue considered dangerous?**

A) It is highly vascularized

B) It contains sweat glands

C) It allows infection to spread to the brain via emissary veins

D) It is the thickest layer of the scalp

**22. Which veins unite to form the external jugular vein?**

A) Superficial temporal and maxillary veins

B) anterior division of retromandibular vein and anterior facial vein

C) posterior division of retromandibular vein and posterior auricular vein

D) union of posterior auricular with the occipital vein

**23. A 35-year-old female complains of pain in the middle third of her face. Which of the following nerves is responsible for this pain?**

A) mandibular

B) maxillary

C) ophthalmic

D) facial

**24. Which of the following is one of the content of carotid sheath?**

A) sternomastoid muscle

B) vagus nerve

C) vagus nerve

D) facial artery

**25. What is the primary function of pulmonary circulation?**

A) Regulate blood pressure

B) Exchange gases between blood and lungs

C) Deliver nutrients to tissues

D) Transport oxygenated blood to the body

**26. Which layer of the meninges is the outermost and toughest?**

A) arachnoid mater.

B) pia mater.

C) dura mater.

D) subarachnoid space.

**27. What type of joint connects the bones of the vault of the skull?**

A) Primary cartilaginous joint

B) Fibrous joint.

C) Secondary cartilaginous joint



D) Synovial joint

**28. Which of the following muscles is involuntary striated?**

A) Muscles of the upper limb

B) Muscles of the trunk

C) Muscles of the heart

D) Muscle of the wall of blood vessel

**29. Which of the following arteries supply the outer compact bone?**

A) Nutrient artery

B) Periosteal artery

C) Epiphyseal artery

D) Metaphyseal artery

**30. By the 4th month the placental barrier is formed of which of the following?**

A) Syncytiotrophoblast and endothelium.

B) Cytotrophoblast and syncytiotrophoblast.

C) Cytotrophoblast and endothelium.

D) Endothelium and extraembryonic mesoderm.

**31. On examination, you noticed that the the placenta extended into urinary bladder. What is the possible cause of this case?**

A) failure of formation of cytotrophoblastic shell

B) Changes in trophoblast

C) Congenital anomalies in the heart

D) Placental insuffecincy.

**32. During the 4th week of embryonic development, lateral folding occurs due to rapid growth lateral body folds. Which structure is most affected by lat folding?**

A) Yolk sac

B) Heart tube

C) Septum transversum

D) Allantois

**33. What is the primary role of the notochord, which forms during gastrulation?**

A) To serve as a precursor to the heart

B) To form the body cavity

C) To induce the formation of the neural tube

D) To give rise to the digestive tract

**34. A 28-year-old woman is in her first trimester of pregnancy and is informed by her doctor that gastrulation is about to occur. What is the primary developmental process that occurs during gastrulation?**

A) Formation of the neural tube

B) Establishment of the three germ layers

C) Development of the heart

D) Formation of the placenta

**35. What is the primary mechanism that prevents polyspermy after the first sperm enters the egg?**

A) Cortical reaction



- B) Decidual reaction
- C) Acrosome reaction
- D) Cleavage division

### 36. Where originates the extraembryonic mesoderm?

- A) It is a result of the migration of cells from the neuroectoderm
- B) In a site between trophoblast and embryoblast
- C) In a site between hypoblast and epiblast
- D) In a site between endoderm and ectoderm

### 37. Which embryonic structure is formed adjacent to the notochord?

- A) Endoderm
- B) Somites
- C) Amnion
- D) Allantois

### 38. At approximately what embryonic age does an embryo typically have about 27 pairs of somites?

- A) 21 days
- B) 29 days
- C) 30 days
- D) 40 days

### Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-A  | 3-A  | 4-C  | 5-A  | 6-B  | 7-B  | 8-D  | 9-D  | 10-D |
| 11-C | 12-D | 13-A | 14-C | 15-C | 16-A | 17-C | 18-A | 19-C | 20-B |
| 21-C | 22-C | 23-B | 24-B | 25-B | 26-C | 27-B | 28-C | 29-B | 30-A |
| 31-A | 32-A | 33-C | 34-B | 35-A | 36-B | 37-B | 38-B |      |      |

### Written questions

#### Written part

#### 1. 1)Enumerate:

- a) 3 sites of simple squamous epithelium
- b) 3 types of exocrine glands according to branching of duct system
- c) 3 sites of Elastic connective tissue
- d) Types of neurons according to number of processes

#### 2. 2)Mention:

- a) 3 parts of nucleolus
- b) Non-keratinocytes of the skin and mention one function for each one
- c) 3 functions for SER

#### 3. 4)Discuss:

- a) Transitional epithelium
- b) white Adipose Connective tissue
- c) Histological structure of perikaryon

#### 4. Describe the forms of the skeletal muscle

#### 5. Describe the bone marrow (definition, function, forms according to the age)

#### 6. Mention the origin, insertion, nerve supply and action of buccinator muscle

#### 7. Explain the congenital anomalies of placenta

#### 8. Describe the steps of gastrulation

#### 9. Compare between chorionic villi as regarding their structure



## 20. End Semester 2025

38 MCQs

1. What are the gaps between Schwann cells in the myelin sheath called?

- A) Nodes of Ranvier
- B) Synaptic clefts
- C) Axon terminals
- D) Dendrites

2. Which type of neuroglia forms the myelin sheath in the central nervous system?

- A) Schwann cell
- B) Astrocytes
- C) Oligodendrocytes
- D) Microglia

3. Reticular fibers are primarily composed of which protein?

- A) Collagen
- B) Elastin
- C) Fibrillin
- D) Keratin

4. Mast cells release which substances during an immune response?

- A) Collagen and elastin
- B) Histamine and heparine
- C) Antibodies and cytokines
- D) proteoglycans and fibronectin

5. Which of the following is a characteristic of dense regular connective tissue?

- A) Randomly arranged collagen fibers
- B) High tensile strength in one direction
- C) Abundance of ground substance
- D) Presence of elastic fibers in all regions

6. Which type of epithelium is specialized for absorption?

- A) Simple squamous epithelium
- B) Stratified squamous epithelium
- C) Simple columnar epithelium with microvilli
- D) Pseudostratified columnar epithelium

7. Which epithelial specialization anchors epithelial cells to the basement membrane?

- A) Tight Junctions
- B) Gap junctions
- C) Desmosomes
- D) Hemidesmosomes

8. Which organelle is responsible for maintenance of cell health?

- A) Rough endoplasmic reticulum
- B) Golgi apparatus
- C) Smooth endoplasmic reticulum
- D) Lysosome



**9. What is the term for the process by which the cells transport fluids?**

- A) Endocytosis
- B) Exocytosis
- C) Pinocytosis
- D) Phagocytosis

**10. Which of the following layers contains keratohyaline granules?**

- A) Stratum basale
- B) Stratum granulosum
- C) Stratum lucidum
- D) Stratum spinosum

**11. What is the function of stratum basale?**

- A) Renewal of Keratinocytes
- B) Resisting the effects of friction and pressure
- C) Waterproof-barrier of the skin
- D) Receptors for fine touch sensation

**12. Which of the following best describes the phospholipid molecules of cell membrane?**

- A) The head is non-charged and hydrophobic
- B) The head is directed inwards away from the aqueous solutions
- C) The fatty acid tail is charged and hydrophilic
- D) The head is directed towards the aqueous solutions

**13. Regarding free ribosomes, which of the following describes it?**

- A) Form proteins secreted by the cells
- B) Form proteins used within the cytosol
- C) Linked together by tRNA forming polysomes
- D) Bind to the outer surface of rER

**14. Which feature of the followings describes centrioles?**

- A) They are a membranous organelle
- B) Can be demonstrated with iron hematoxylin stain
- C) Formed of eight bundles of microtubules
- D) Have the same number of microtubules as the shaft of cilia

**15. What is the typical feature of extended chromatin?**

- A) Appears pale basophilic by LM
- B) Appears as coarse granules by EM
- C) Contains the inactive genes
- D) Appears dark basophilic by LM

**16. Which of the following best describes thin skin?**

- A) It has apparent stratum lucidum
- B) Contains fewer sweat glands in the dermis
- C) Contains well developed papillary ridges
- D) Has no sebaceous glands

**17. The cilia can be characterized by which of the followings?**

- A) The basal body is formed of nine microtubules
- B) The rootlet contains twenty-seven microtubules
- C) move secretions over the tissue surface in two directions
- D) Can be found in the female genital system



18. Which one of the following stains demonstrates glycogen in the liver cells?
  - A) Sudan III
  - B) PAS
  - C) Silver
  - D) Sudan black
19. What is the origin of transfer vesicles of Golgi apparatus?
  - A) Smooth endoplasmic reticulum
  - B) Rough endoplasmic reticulum
  - C) The trans face of Golgi apparatus
  - D) The cell membrane
20. Which of the following structures connect the fibers of cardiac muscles?
  - A) connective tissue
  - B) intercalated disc
  - C) ligament
  - D) tendon
21. A 40 years old male patient came to the hospital with bleeding in the scalp region on examination the wound showing a large gap, what is the cause of this gap?
  - A) cutting of the epicranial aponeurosis
  - B) cutting of the second layer of the scalp
  - C) cutting of the pericranium
  - D) the wound is superficial
22. The common facial vein is one of veins of the face and scalp, how is formed?
  - A) by the union of superficial temporal and maxillary veins
  - B) by the union of anterior division of retromandibular vein with anterior facial vein
  - C) by the union of posterior division of retromandibular vein with posterior auricular vein
  - D) union of posterior auricular with the occipital vein
23. A 35-year-old female complains of pain in the upper third of her face. Which of the following nerves is responsible for this pain?
  - A) mandibular
  - B) maxillary
  - C) ophthalmic
  - D) facial
24. Which of the following Form the floor of submental triangle?
  - A) sternomastoid muscle
  - B) Prevertebral Fascia
  - C) mylohyoid
  - D) The carotid sheath
25. in systemic circulation the blood pass to the body, how it can reach the heart again?
  - A) Through the inferior and superior vena cava which open into the left atrium
  - B) Through the inferior and superior vena cava which open into the right atrium
  - C) Through the pulmonary veins which open into the left atrium
  - D) Through the pulmonary arteries which open into the right ventricles
26. Which of the following can Maintain posture and balance?



- A) frontal lobe
- B) Mid brain
- C) Cerebellum
- D) occipital lobe

**27. What is the type of the joint between the teeth and the mandible ?**

- A) Fibrous joint
- B) Primary cartilaginous joint
- C) Secondary cartilaginous joint
- D) Synovial joint

**28. Which of the following muscles is involuntary non striated?**

- A) Muscles of the upper limb
- B) Muscles of the trunk
- C) Muscles of the heart
- D) Muscle of the wall of blood vessel

**29. the deep fascia is well developed in some regions and thin in others, where it is thin?**

- A) Upper limb
- B) Thorax
- C) Lower limb
- D) neck

**30. by the 4th month the placental barrier is formed of which of the following?**

- A) Syncytiotrophoblast and endothelium
- B) Cytotrophoblast and syncytiotrophoblast
- C) Cytotrophoblast and endothelium
- D) Endothelium and extraembryonic mesoderm

**31. on examination, you noticed that the amount of amniotic fluid in a pregnant female is more than 2 litres What is the possible cause of this case**

- A) Oesophageal atresia
- B) Congenital anomalies in the heart
- C) Placental insufficiency
- D) Changes in trophoblast

**32. During the 4th week of embryonic development, cranial folding occurs due to rapid growth of the neural tube and somites. This folding shifts the developing heart tube into the thoracic region. Which structure is most affected by cranial folding?**

- A) Yolk sac
- B) Amniotic cavity
- C) Septum transversum
- D) Allantois

**33. What is the primary role of the notochord, which forms during gastrulation?**

- A) To serve as a precursor to the heart
- B) To induce the formation of the neural tube
- C) To give rise to the digestive tract
- D) To form the body cavity



**34. A 28-year-old woman is in her first trimester of pregnancy and is informed by her doctor that gastrulation is about to occur. What is the primary developmental process that occurs during gastrulation?**

- A) Formation of the neural tube
- B) Establishment of the three germ layers
- C) Development of the heart
- D) Formation of the placenta

**35. What is the primary mechanism that prevents polyspermy after the first sperm enters the egg?**

- A) Cortical reaction
- B) Decidual reaction
- C) Acrosome reaction
- D) Cleavage division

**36. Where originates the extraembryonic mesoderm?**

- A) It is a result of the migration of cells from the neuroectoderm
- B) In a site between trophoblast and embryoblast
- C) In a site between hypoblast and epiblast
- D) In a site between endoderm and ectoderm

**37. Which embryonic structure is formed adjacent to the notochord?**

- A) Endoderm
- B) Somites
- C) Amnion
- D) Allantois

**38. At approximately what embryonic age does an embryo typically have about 30 pairs of somites?**

- A) 21 days
- B) 27 days
- C) 30 days
- D) 40 days

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-A  | 2-C  | 3-A  | 4-B  | 5-B  | 6-C  | 7-D  | 8-D  | 9-C  | 10-B |
| 11-A | 12-D | 13-B | 14-B | 15-A | 16-B | 17-D | 18-B | 19-B | 20-B |
| 21-A | 22-B | 23-C | 24-C | 25-B | 26-C | 27-A | 28-D | 29-B | 30-A |
| 31-A | 32-C | 33-B | 34-B | 35-A | 36-B | 37-B | 38-C |      |      |

## Written questions

Written questions (36 marks)

1) Enumerate (12 marks):

- a. 3 sites of neuroepithelium. (1.5 marks)
- b. 3 types of exocrine glands according to mode of secretion. (1.5 marks)
- c. 3 sites of Mucoïd connective tissue. (1.5 marks)
- d. Types of chromosomes according to the position of centromere. (2 marks)
- e. Sites of heterochromatin. (1.5 marks)



- f. Non-keratinocytes of the skin and mention one function for each one. (3 marks)
- g. Two functions for Golgi apparatus. (1 mark)
- 2) Discuss (6 marks) (2 marks for each):
  - A. Myoepithelium
  - B. Reticular Connective tissue
  - C. Histological structure of Dendrites and axon
- 3) Enumerate the derivatives of epidermis and describe one of them in detail (3 marks)
- 4) Describe the blood supply of the bone. (3 marks)
- 5) Describe the dangerous area of the face and its clinical significance. (3 marks)
- 6) Explain the congenital anomalies of placenta. (3 marks)
- 7) Describe the steps of notochord. (3 marks)
- 8) Describe the congenital anomalies of the yolk sac. (3 marks)

*Note: Note printed in the source file: derivatives of the epidermis = 1- Sweat glands 2- Sebaceous glands 3- Nails 4- Hair. Sweat glands: long tubular glands that lie coiled in the dermis; sweat is normally odorless, but its odor may be produced by skin-surface bacteria.*





## 21. End Semester 2026 حڪومي دور اول

38 MCQs

1. Movement of the shoulder superiorly is termed which of the following?

- A) Protraction
- B) Elevation
- C) Depression
- D) Retraction

2. During the X-ray examination of a child, the bones appeared incomplete. The physician explained that the bones are still in the process of growing. Which of the following structures is responsible for increase the length of the bone?

- A) Periosteum
- B) Endosteum
- C) Epiphyseal plate (growth plate)
- D) Articular cartilage

3. The attachment of a muscle to the bone that moves when the muscle contracts is called which of the following?

- A) Origin
- B) Insertion
- C) Belly
- D) Ligament

4. Which of the following structures receives oxygenated blood through the pulmonary veins?

- A) Right atrium
- B) Left atrium
- C) Right ventricle
- D) Pulmonary trunk

5. Which of the following joints are formed when two bones are connected by fibrocartilage?

- A) Symphysis pubis
- B) Shoulder joint
- C) Skull sutures
- D) Hip joint

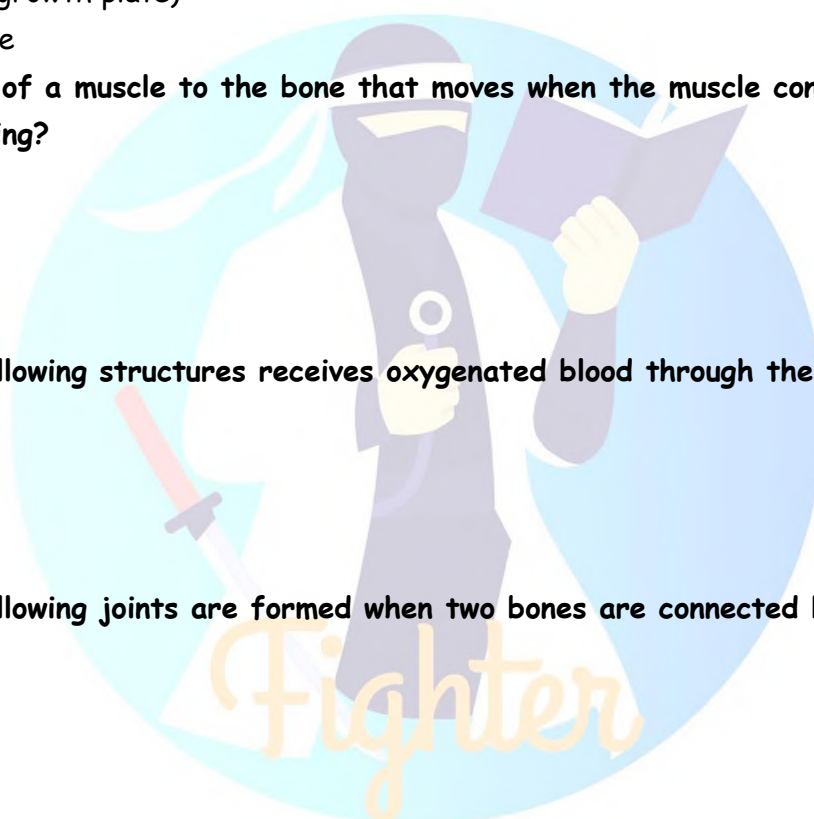
6. Which of the following provides nutrients and acts as a shock absorber in the brain?

- A) Cerebrospinal fluid (CSF)
- B) Blood plasma
- C) Synovial fluid
- D) Lymph

7. An 8-year-old child presents with fever and swelling of the scalp after a minor injury. The physician cautions that the infection may spread to the brain. Through which layer of the scalp can the infection reach the intracranial structures?

- A) Skin of the scalp
- B) Connective tissue layer (dense subcutaneous tissue)
- C) Loose areolar tissue layer
- D) Pericranium

8. The common facial vein is formed by the union of which veins?





- A) Anterior Facial vein and anterior jugular vein
- B) Anterior Facial vein and anterior division of retromandibular vein
- C) Posterior division of posterior facial vein and external jugular vein
- D) Anterior division of retromandibular vein and posterior auricular vein

**9. A 50-year-old man complains of headache and pain around the eye. Which of the following nerves supply this region?**

- A) Maxillary nerve (V2)
- B) Mandibular nerve (V3)
- C) Ophthalmic nerve (V1)
- D) Greater occipital nerve

**10. Which of the following structures is a content of the digastric triangle?**

- A) Submandibular gland
- B) Parotid gland
- C) Common carotid artery
- D) External jugular vein

**11. A newborn presents with a sacrococcygeal mass containing tissues derived from all three germ layers. The most likely embryological defect is failure of which of the following?**

- A) Regression of the primitive streak
- B) Formation of the primitive node
- C) Fusion of the neural folds
- D) Differentiation of paraxial mesoderm

**12. Absence of the notochord would primarily interfere with development of which of the following?**

- A) Epidermis
- B) Neural tube
- C) Amnion
- D) Extraembryonic mesoderm

**13. An embryo has 7 cervical, 12 thoracic, 5 lumbar, and 5 sacral somites identified. What is the most likely embryonic age?**

- A) 21 days
- B) 23 days
- C) 25 days
- D) 29 days

**14. A fetus is diagnosed on ultrasound with absence of the cranial vault and cerebral hemispheres. What is the embryological defect?**

- A) Neural plate induction
- B) Neural fold elevation and closure
- C) Notochord formation
- D) Neural crest cell differentiation

**15. An embryo shows failure of formation of the amniotic cavity. Which layer of the bilaminar disc is primarily affected?**

- A) Hypoblast
- B) Epiblast



- C) Cytotrophoblast
- D) Syncytiotrophoblast

**16. Failure of formation of extraembryonic mesoderm would prevent development of which of the following?**

- A) Amniotic cavity
- B) Chorionic cavity
- C) Bilaminar germ disc
- D) Trophoblastic shell

**17. An embryo is found attached to the chorionic plate by a thick band of tissue. This structure will later develop which of the following?**

- A) Allantois
- B) Vitelline duct
- C) Umbilical cord
- D) Amnion

**18. A placenta later shows placenta previa. The abnormality most likely originated from implantation of the blastocyst in which of the following?**

- A) Fundus of uterus
- B) Posterior uterine wall
- C) Lower uterine segment
- D) Myometrium

**19. A 15-year-old girl presents with primary amenorrhea, short stature, and webbed neck. What is her karyotype?**

- A) 46,XX
- B) 46,XY
- C) 45,X
- D) 47,XXX

**20. Regarding the prophase of the first meiotic division, crossing over occurs in which of the following stages?**

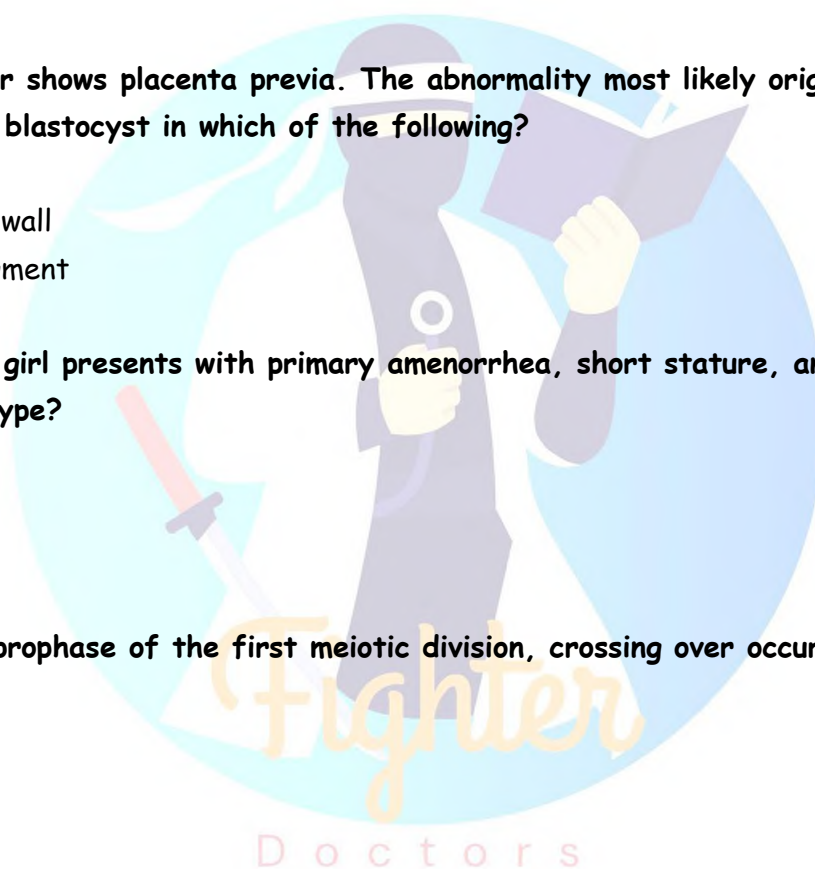
- A) Leptotene
- B) Zygotene
- C) Pachytene
- D) Diplotene

**21. Which of the following describes the chromosomal microtubules?**

- A) They are continuous
- B) Arise from MTOC surrounding the centrioles
- C) Responsible for elongation of the cell
- D) Responsible for pulling the sister chromatids

**22. Which of the following is considered as aneuploidy?**

- A) Triploid
- B) Trisomy
- C) Tetraploid
- D) Polyploidy





**23. In which of the following chromosomal aberrations two copies of the same gene can be found on the same chromosome?**

- A) Isochromosome
- B) Ring chromosome
- C) Addition
- D) Translocation

**24. Multipolar pyramidal nerve cells can be found in which of the following?**

- A) Anterior horn cells
- B) Cerebrum
- C) Cerebellum
- D) Retina

**25. Which of the following statements describe the dendrites of nerve cells?**

- A) Have a constant diameter
- B) Conducts nerve impulse away from the cell body
- C) Usually long
- D) Contain Nissl granules

**26. Regarding the centrioles, which of the following describes it?**

- A) Appear as two dark bodies with H&E
- B) Share in the formation of cilia
- C) Present as two pairs in non-dividing cells
- D) Formed by dynamic microtubules

**27. What is the typical description of glycogen granules?**

- A) Alpha glycogen forms rosette aggregation in the liver
- B) Appears red with H&E stain
- C) Can be stained with osmic acid
- D) Beta glycogen forms rosette aggregation in the liver

**28. Which of the following stains can demonstrate the plasma membrane?**

- A) PAS
- B) Osmic acid
- C) Sudan II
- D) Orcein

**29. Diffuse cytoplasmic basophilia caused by ribosome can be found in which cells?**

- A) Embryonic cells
- B) Pancreatic cells
- C) Nerve cells
- D) Glial cells

**30. Regarding nuclear chromatin, what is the typical character?**

- A) Extended type appears dark basophilic with LM
- B) Condensed type appears pale basophilic with LM
- C) Extended type contains the active genes
- D) Euchromatin appears as coarse granules with EM

**31. Fibroblasts are characterized by which of the following?**

- A) Abundant rER and Golgi



- B) Signet ring appearance
- C) Metachromatic granules
- D) Cart-wheel nucleus

**32. Which epithelial type is incorrectly matched with its function?**

- A) Simple cuboidal - Secretion and reabsorption
- B) Simple columnar - Protection only
- C) Pseudostratified - Transport of mucus
- D) Transitional - Accommodates stretching

**33. Which gland type releases secretion by exocytosis without cell damage?**

- A) Sebaceous gland
- B) Mammary gland
- C) Salivary gland
- D) Meibomian gland

**34. Which apical specialization increases surface area for absorption in small intestine?**

- A) Cilia
- B) Flagella
- C) Microvilli
- D) Stereocilia

**35. The basement membrane is composed of which of the following?**

- A) Only epithelial cells
- B) Basal lamina and reticular lamina
- C) Only collagen type I
- D) Only glycoproteins

**36. Brown adipose tissue differs from white adipose tissue in which of the following?**

- A) Stores fat as a single droplet
- B) Generates heat
- C) Is poorly vascularized
- D) Has a signet ring appearance

**37. Muroid connective tissue in adult is present in which of the following?**

- A) Skin
- B) Umbilical cord
- C) Lymph nodes
- D) Vitreous humor of eye

**38. Which cell junction has Pentalaminar appearance?**

- A) Tight junction
- B) Desmosome
- C) Gap junction
- D) Hemidesmosome

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-B  | 2-C  | 3-B  | 4-B  | 5-A  | 6-A  | 7-C  | 8-B  | 9-C  | 10-A |
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| 21-D | 22-B | 23-A | 24-B | 25-D | 26-B | 27-A | 28-B | 29-A | 30-C |
| 31-A | 32-B | 33-C | 34-C | 35-B | 36-B | 37-D | 38-A |      |      |



## Written questions

## Part II: Written Questions

1. Describe the following:
  - A. Histological structure of white adipose connective tissue. (3 marks)
  - B. Histological structure of the stratum basale in epidermis with its function. (2 marks)
  - C. Histological structure of nuclear envelope and its function. (2 marks)
  - D. State the ultrastructure of nerve cell. (1.5 marks)
  - E. Three connective tissue cells and mention origin and function of each one. (1.5 marks)
2. Enumerate:
  - A. Two types of numerical chromosomal aberrations having positive sex chromatin. (1 mark)
  - B. Three sites of Neuroepithelium. (2 marks)
  - C. Sites of Dense regular connective tissue. (2 marks)
3. Draw a coloured labelled diagram of Non keratinized stratified squamous epithelium. (3 marks)
4. Draw a coloured labelled diagram of the posterior triangle and the carotid triangle of the neck. (3 marks)
5. Describe the origin and the nerve supply of the following muscles: (3 marks) a- Buccinator b- Orbicularis oculi c- Sternomastoid
6. Name three sites where cartilage is found and three sites where red bone marrow is located in the adult.
7. A patient experiences severe postpartum hemorrhage due to placenta accreta. Discuss the normal development of chorionic villi and decidua basalis, and explain the embryological base of placenta accreta. (3 marks)
8. A fetus develops unexplained intrauterine growth restriction, and delivery reveals a true knot in the umbilical cord. Discuss the formation and growth of the umbilical cord. (3 marks)
9. Describe the types and congenital anomalies of twins. (3 marks)

Doctors



## 22. End Module 2026 حڪومي دور اول

52 MCQs

1. Which of the following glands is considered as an apocrine gland?

- A) Pancreas
- B) Mammary glands
- C) Salivary glands
- D) Sebaceous glands

2. A child presents with cranial deformities due to premature fusion of skull bones. Which joint type connected the skull bones?

- A) primary cartilaginous joint
- B) secondary cartilaginous joint
- C) fibrous joint
- D) synovial joint

3. During dissection, the brain was sectioned into anterior and posterior parts. Which anatomical plane is used for the examination?

- A) coronal plane
- B) parasagittal plane
- C) midsagittal plane
- D) horizontal plane

4. Which of muscle has central tendon and the muscle fibers attached to the both sides of the tendon?

- A) triangular muscle
- B) Spiral muscle
- C) Unipennate muscle
- D) Bipennate muscle

5. Which of the following arteries can supply the bone marrow?

- A) epiphyseal
- B) periosteal
- C) metaphyseal
- D) nutrient

6. In which anatomical location are the deep plantar vessels found?

- A) in the palm of the hand away from the skin
- B) in the sole of the foot near the skin
- C) in the palm of the hand near the skin
- D) in the sole of the foot away from the skin

7. Which of the following contains the motor area in the brain?

- A) temporal lobe
- B) frontal lobe
- C) parietal lobe
- D) occipital lobe

8. Which part of spinal nerve is mixed and form the plexuses in the body?

- A) ventral rami
- B) dorsal root
- C) ventral root
- D) dorsal rami

9. The amount of compact bone is increased in the middle of shaft of long bone. Why?

- A) To increase the length
- B) As it looks like a sponge to sustain the pressure
- C) As it makes bone narrow in the middle
- D) As it provides strength for weight



**10. During the flexion of the elbow, the biceps acts as:**

- A) Prime mover
- B) Synergist
- C) Antagonist
- D) Fixator

**11. Which of the following can withstand the blood pressure created by the heart's beating?**

- A) Large conducting arteries
- B) Small distributing arteries
- C) Large veins
- D) Blood capillaries

**12. During the pulmonary circulation, the deoxygenated blood is pumped from the right ventricle to the lungs through which of the following?**

- A) Pulmonary veins
- B) Pulmonary trunk
- C) Aorta
- D) Inferior vena cava

**13. To put the person in anatomical position, what is the correct position of the hand and the foot?**

- A) The palm and the toes face anteriorly
- B) The palm faces posterior and the toes face anterior
- C) The palm faces anterior and the toes face posterior
- D) The palm faces anterior and the toes face anterior

**14. During the flexion of the elbow, the triceps act as:**

- A) synergist
- B) prime mover
- C) fixator
- D) antagonist

**15. Which term describes the movement of the distal end of the limb in a circle of 360 degree?**

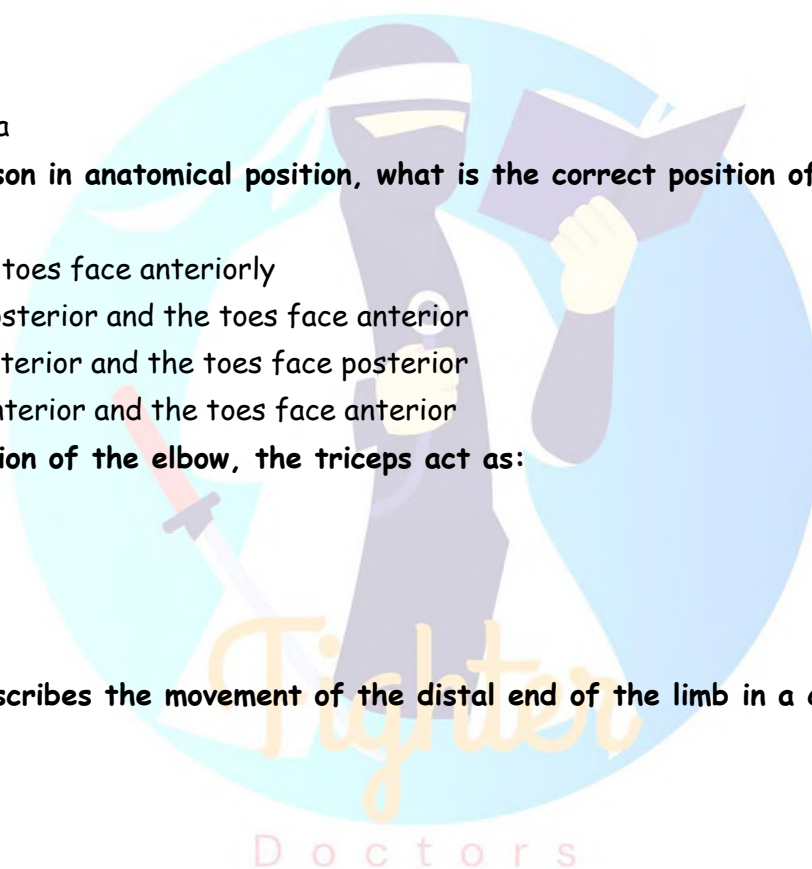
- A) circumduction
- B) adduction
- C) rotation
- D) flexion

**16. Presence of joint cavity is one of the characters of which of the following joint?**

- A) primary cartilaginous joint
- B) synovial joint
- C) fibrous joint
- D) secondary cartilaginous joint

**17. A researcher examines an embryo at day 9 post-fertilization and notes fluid accumulation between the epiblast and trophoblast. The cavity will develop into which structure?**

- A) Chorionic cavity





- B) Amniotic cavity
- C) Yolk sac
- D) Blastocyst cavity

**18. A 25-year-old male presents with infertility. Semen analysis shows azoospermia. Testicular biopsy reveals the presence of Sertoli cells but absence of germ cells. Which stage of gametogenesis is most likely affected?**

- A) Meiosis I in primary spermatocytes
- B) Mitosis of spermatogonia
- C) Spermiogenesis
- D) Migration of primordial germ cells

**19. A researcher observes an embryo with a visible primitive streak on its epiblast surface. Which of the following best describes the significance of the structure?**

- A) Initiation of heart tube formation
- B) Formation of the yolk sac
- C) Origin of the three germ layers
- D) Site of neural tube closure

**20. On day 5 post-fertilization, an embryo develops a fluid-filled cavity and differentiates into two cell types. Which gives rise to embryo proper?**

- A) Syncytiotrophoblast
- B) Cytotrophoblast
- C) Trophoblast
- D) Inner cell mass

**21. Implantation is complete by day 10-12 and the conceptus is fully embedded in the endometrium. Which layer of maternal trophoblast is directly responsible for eroding maternal capillaries to access blood?**

- A) Syncytiotrophoblast
- B) Cytotrophoblast
- C) Epiblast
- D) Extraembryonic mesoderm

**22. During gastrulation, a subset of epiblast cells moves through the primitive streak and displaces the hypoblast. These cells will become which germ layer?**

- A) Ectoderm
- B) Hypoblast
- C) Mesoderm
- D) Endoderm

**23. After fertilization, the zygote begins dividing in a process known as cleavage. 16 cell stage is?**

- A) Blastocyst
- B) Zygote
- C) Morula
- D) Embryoblast

**24. Regarding metaphase I in second meiotic division, which of the following occurs?**



- A) Crossing over occurs at this stage
- B) The d-chromosome splits at the centromere into two chromatids
- C) The 23-bivalent d-chromosomes become aligned in the equatorial plane
- D) The 23 d-chromosomes are aligned at the equatorial plate

**25. A pathology report describes an embryo with a tumor in the sacrococcygeal region composed of tissues from all three germ layers. What is the embryological origin of this tumor?**

- A) Abnormal yolk sac differentiation
- B) Persistence of primitive streak remnants
- C) Neural crest cell migration defect
- D) Notochord remnants

**26. A 30-year-old woman undergoes in vitro fertilization (IVF). Eggs are retrieved after hormonal stimulation and are found to be arrested in metaphase II. What triggers the completion of meiosis II in the oocyte?**

- A) Sperm penetration
- B) LH surge
- C) Estrogen peak
- D) FSH rise

**27. During fertilization a sperm secretes enzymes to penetrate the oocyte's outer layer. Which structure prevents polyspermy?**

- A) Zona pellucida
- B) Cell membrane
- C) Endometrium
- D) Corona radiata

**28. In an embryo at day 12 post-fertilization lacunar spaces appear within syncytiotrophoblast filled with maternal blood. What is this process critical for?**

- A) Neurulation process
- B) Gastrulation process
- C) Uteroplacental circulation
- D) Cleavage process

**29. Which of the following describes Barr body?**

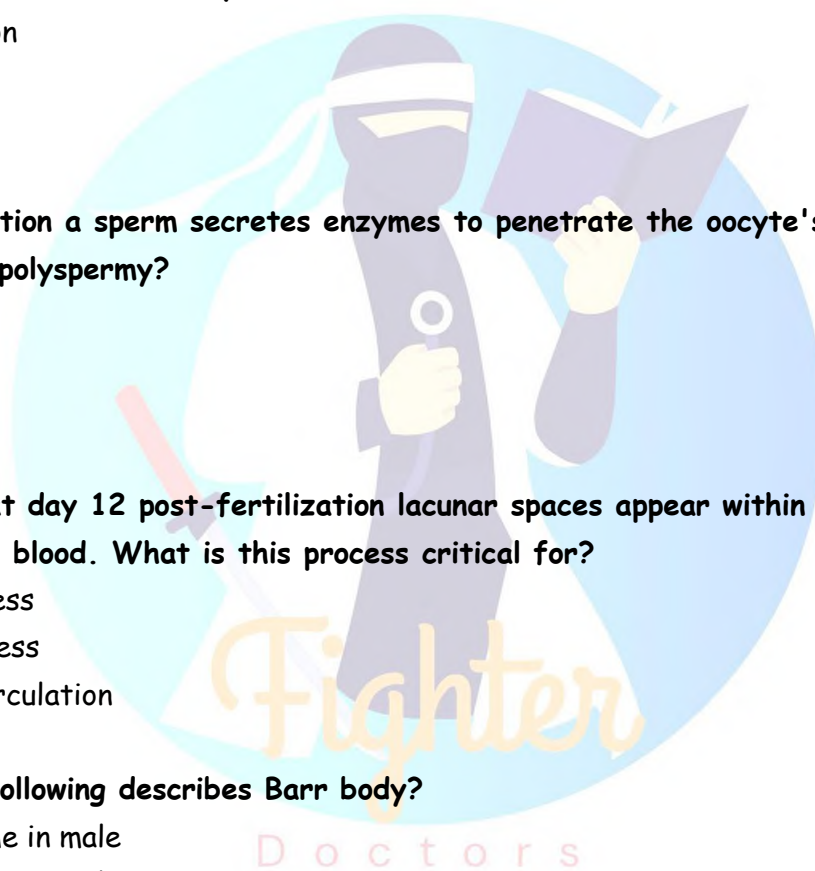
- A) The Y chromosome in male
- B) The X chromosome in males
- C) The XX chromosomes in females
- D) The inactive X chromosome in females

**30. What is concerned with Compound exocrine glands?**

- A) Have unbranched secretory duct
- B) Secrete hormones
- C) Have branched secretory ducts
- D) Secrete into the blood

**31. In absorptive cells, where the brush border is seen at LM level?**

- A) True cilia
- B) Stereocilia
- C) Microvilli





D) Basal infoldings

**32. What type of tissue is cornea composed of?**

A) Mesenchyme

B) Dense irregular connective tissue

C) Dense regular connective tissue

D) Mucous connective tissue

**33. Which of the following has valves and constricted at their sites showing beaded appearance?**

A) Capillaries

B) Veins

C) Lymph vessels

D) Arteries

**34. Which of the following is not in the classical description of a junctional complex?**

A) The Zonula occludens

B) The Zonula adherens

C) The Macula adherens

D) Gap junction

**35. Which organelle is responsible for maintenance of cell health?**

A) Lysosome

B) Golgi apparatus

C) Rough endoplasmic reticulum

D) Smooth endoplasmic reticulum

**36. Where are metachromatic granules present in?**

A) Reticular cells

B) Mast cells

C) Plasma cells

D) Fat cells

**37. Which of the following would be best suited to differentiate collagen fibers from other fibers?**

A) Van Gieson stain

B) Orcein stain

C) Sudan stain

D) Silver

**38. What is the shape of fat cells present in the unilocular adipose tissue?**

A) Signet ring

B) Clock face

C) Kidney

D) Horse shoe

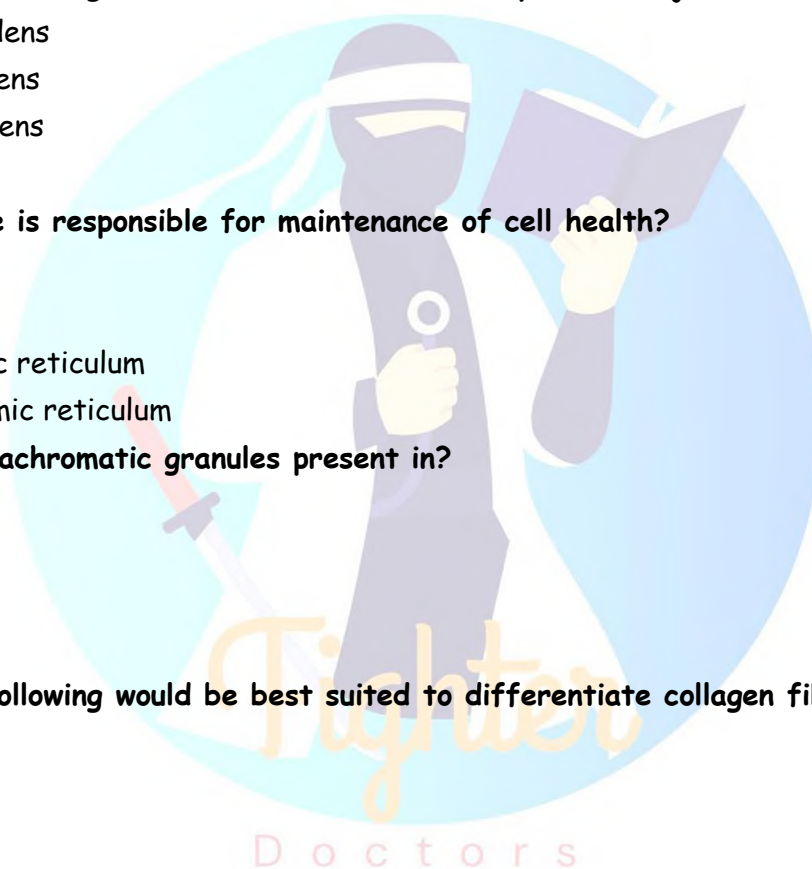
**39. Which of the following cells are continuously turnover cells?**

A) Heart muscles

B) Blood cells

C) Nerve cells

D) Liver cells





**40. Which of the following describes Multivesicular bodies?**

- A) Primary lysosomes
- B) Autophagic vacuoles
- C) Formed as a result of digestion of solid material
- D) Formed as a result of digestion of fluid material

**41. What is characteristic to stratified squamous epithelium?**

- A) Consists of a single layer of flattened cells
- B) May be ciliated
- C) Has a rich supply of capillaries running between its cell layers
- D) It is protective rather than secretory or absorptive

**42. What is concerned with Glycocalyx?**

- A) The protein component of the cell membrane
- B) The hydrophobic end of lipid
- C) The hydrophilic end of lipid
- D) The carbohydrate component of the cell membrane

**43. Where pseudostratified columnar ciliated epithelium can be found?**

- A) Respiratory passages
- B) Blood vessels
- C) Fallopian tube
- D) Urinary passage

**44. Which of points related to Heterochromatin?**

- A) Special type of stain
- B) Present in the nuclei of metabolically active cells
- C) Type of cell inclusions
- D) Present in the nuclei of metabolically inactive cells

**45. Regarding the Gap 1 phase of interphase, which of the following describes it?**

- A) Each cell contains 92 s chromosomes
- B) The nucleus of the daughter cell contains 46s chromosomes
- C) The more specialized the cell, the shorter the G1-phase
- D) Each cell contains 46 d chromosome

**46. What is the main ultrastructural feature (EM) of histiocyte?**

- A) Smooth endoplasmic reticulum
- B) Lysosomes
- C) Microvilli
- D) Rough endoplasmic reticulum

**47. What is the typical feature of submetacentric chromosome?**

- A) The centromere is terminal
- B) The centromere is at the center of the chromosome
- C) The centromere is midway between the center and upper end
- D) The centromere is very close to upper end

**48. Which of the following statements is incorrect regarding neuroepithelium?**

- A) Present in taste buds
- B) It is modified type of epithelium to act as receptor



C) Formed of sensory and supporting cells

D) It covers skin and forms glands

**49. What is true about smooth endoplasmic reticulum?**

A) Areas of localized basophilia indicate its presence

B) Is rarely found in skeletal muscle cells

C) Is present in cells where drug detoxification is taking place

D) Sometimes has ribosomes attached to its membrane

**50. What about the nuclear pore?**

A) Allows for communication between the nuclei & cytoplasm

B) Is a transient structure

C) Is at regular intervals

D) Is bridged by a unit membrane

**51. What is the function of rough endoplasmic reticulum?**

A) Cytoplasmic protein necessary for the cell's own existence

B) Carbohydrate synthesis

C) Lipid synthesis

D) Protein that will be secreted by the cell

**52. Isochromosome occurs mostly in which type of chromosomes?**

A) Acrocentric

B) Metacentric

C) Telocentric

D) Submetacentric

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-B  | 2-C  | 3-A  | 4-D  | 5-D  | 6-D  | 7-B  | 8-A  | 9-D  | 10-A |
| 11-A | 12-B | 13-D | 14-D | 15-A | 16-B | 17-B | 18-D | 19-C | 20-D |
| 21-A | 22-D | 23-C | 24-D | 25-B | 26-A | 27-A | 28-C | 29-D | 30-C |
| 31-C | 32-C | 33-C | 34-D | 35-A | 36-B | 37-A | 38-A | 39-B | 40-D |
| 41-D | 42-D | 43-A | 44-D | 45-B | 46-B | 47-C | 48-D | 49-C | 50-A |
| 51-D | 52-A |      |      |      |      |      |      |      |      |

Doctors



## 23. End Module 2026 اهليه

60 MCQs

1. To put the person in anatomical position, what is the correct position of the arm?

- A) The palm faces anterior and the arm is adducted
- B) The palm faces anterior and the arm is abducted
- C) The palm faces posterior and the arm is adducted
- D) The palm faces posterior and the arm is abducted

2. During dissection, the brain was sectioned into equal right and left parts. Which plane is used for this division?

- A) Coronal plane
- B) Horizontal plane
- C) Midsagittal plane
- D) Parasagittal plane

3. In which anatomical location are the deep palmar vessels found?

- A) In the palm of the hand near the skin
- B) In the sole of the foot near the skin
- C) In the sole of the foot deep
- D) In the palm of the hand away from the skin

4. Which term describes the movement of the distal end of the limb in a circle?

- A) Adduction
- B) Circumduction
- C) Flexion
- D) Rotation

5. The amount of compact bone is increased in the middle of shaft of long bones because:

- A) As it contains bone marrow in its meshes
- B) As it looks like a sponge to sustain the pressure
- C) As it provides strength for weight bearing
- D) To increase its length

6. Which of the following arteries can supply the outer compact bone?

- A) Epiphyseal
- B) Metaphyseal
- C) Nutrient
- D) Periosteal

7. Which form of muscle has central tendon and the muscle fibers attached to one side of the tendon?

- A) Bipennate muscle
- B) Spiral muscle
- C) Triangular muscle
- D) Unipennate muscle

8. During the flexion of the elbow, the muscles of the shoulder act as:

- A) Antagonist
- B) Fixator
- C) Prime mover

Fighter  
Doctors



D) Opposing

**9. Presence of joint cavity is seen in:**

A) Primary cartilaginous joint

B) Synovial joint

C) Secondary cartilaginous joint

D) Fibrous joint

**10. Which of the following squeezes the sebaceous gland to extrude its secretion?**

A) Deep fascia

B) Arrector pili muscle

C) Hair follicle

D) Hair root

**11. Which of the following layers contains Langer's lines?**

A) Epidermis

B) Dermis

C) Superficial fascia

D) Deep fascia

**12. Which of the following hold the tendons in place to facilitate action of muscles?**

A) Aponeurosis

B) Intermuscular septa

C) Retinacula

D) Superficial fascia

**13. The number of cervical segments of the spinal cord are:**

A) Eight segments

B) Five segments

C) Seven segments

D) Twelve segments

**14. From outward to inward, the layers of meninges are arranged as:**

A) Pia, arachnoid, dura

B) Dura, arachnoid, pia

C) Arachnoid, dura, pia

D) Pia, dura, arachnoid

**15. The two cerebral hemispheres are connected together by:**

A) Cerebellum

B) Corpus callosum

C) Brain stem

D) Spinal cord

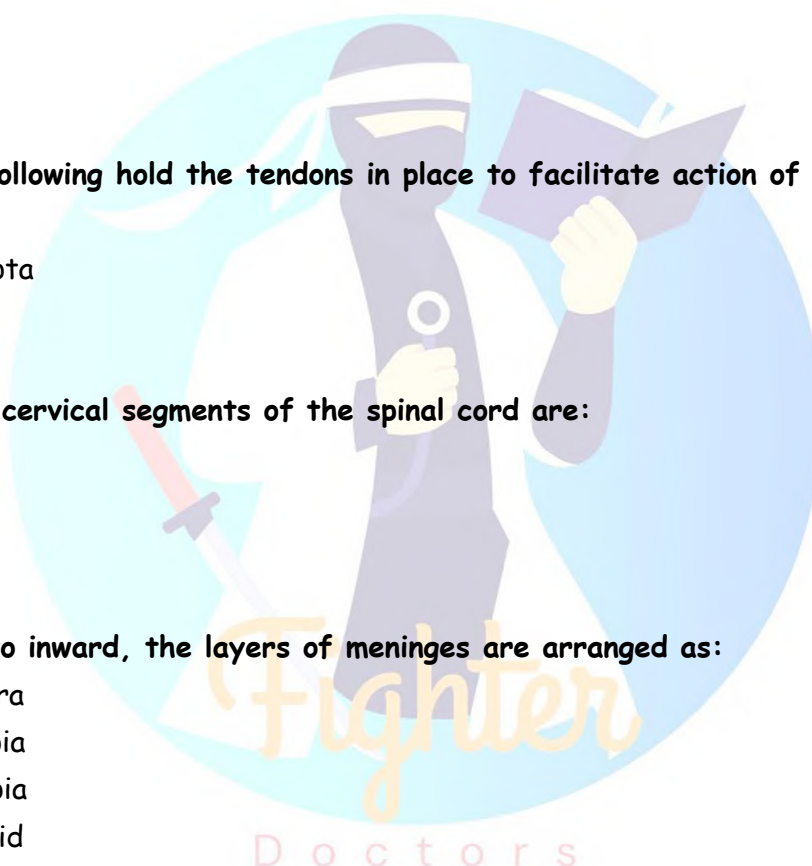
**16. A patient shows abnormal development of the yolk sac during the second week. Which embryonic layer directly contributes to the formation of the primary yolk sac?**

A) Epiblast

B) Hypoblast

C) Cytotrophoblast

D) Extraembryonic mesoderm





17. Histological examination of an implanted blastocyst shows persistence of the exocoelomic (Heuser's) membrane with inadequate hypoblast proliferation. Which structure will be affected?
- Amniotic cavity
  - Yolk sac
  - Chorionic cavity
  - Connecting stalk
18. Which event marks the transition from week 1 to week 2 of human development?
- Appearance of the primitive streak
  - Completion of implantation
  - Formation of trilaminar disc
  - Closure of neural tube
19. A teratogenic insult specifically damages the extraembryonic mesoderm during the second week. Which of the following structures is most likely to be malformed?
- Neural tube
  - Chorionic cavity
  - Epiblast
  - Primitive node
20. A researcher observes failure of epiblast cell migration through the primitive streak in an early embryo. Which of the following structures will be most immediately affected?
- Amniotic cavity
  - Trilaminar germ disc
  - Chorionic cavity
  - Secondary yolk sac
21. A mutation causes abnormal regression of the primitive streak after gastrulation. Which congenital anomaly is most likely to develop?
- Neural tube defect
  - Sacroccygeal teratoma
  - Absent connecting stalk
  - Absent chorionic cavity
22. A specimen shows a normal trilaminar disc, but absence of proper notochord formation. Which developmental event will be most severely impaired?
- Somitogenesis
  - Neural plate induction
  - Formation of lateral plate mesoderm
  - Formation of intermediate mesoderm
23. A teratogen disrupts the function of the primitive node (Hensen's node) during week 3. Which structure normally originates from this region?
- Neural crest cells
  - Notochord
  - Somites
  - Lateral plate mesoderm
24. A 28-year-old woman has regular 28-day cycles. On day 21, serum progesterone is markedly elevated. Which ovarian structure is primarily responsible for this hormone level?



- A) Primary follicle
- B) Secondary follicle
- C) Corpus luteum
- D) Corpus albicans

**25. A woman undergoing infertility evaluation shows failure of the LH surge despite adequate estrogen levels. Which phase of the reproductive cycle is most directly affected?**

- A) Menstrual phase
- B) Follicular phase
- C) Ovulation
- D) Luteal phase

**26. A genetic defect prevents cortical granule release after fertilization. Which consequence is most likely?**

- A) Failure of sperm capacitation
- B) Polyspermy
- C) Failure of the acrosome reaction
- D) Inability of the zygote to implant

**27. A mutation in the embryo prevents the differentiation and formation of the syncytiotrophoblast. Which of the following stages will be directly blocked?**

- A) Fertilization
- B) Cleavage
- C) Implantation
- D) Pronuclear formation

**28. Which of the following is a condition characterized by absent sperms?**

- A) Teratozoospermia
- B) Asthenozoospermia
- C) Oligospermia
- D) Azoospermia

**29. Endometrial biopsy taken on day 23 of the cycle (28 days cycle) shows straight glands. This finding suggests deficiency of which hormone?**

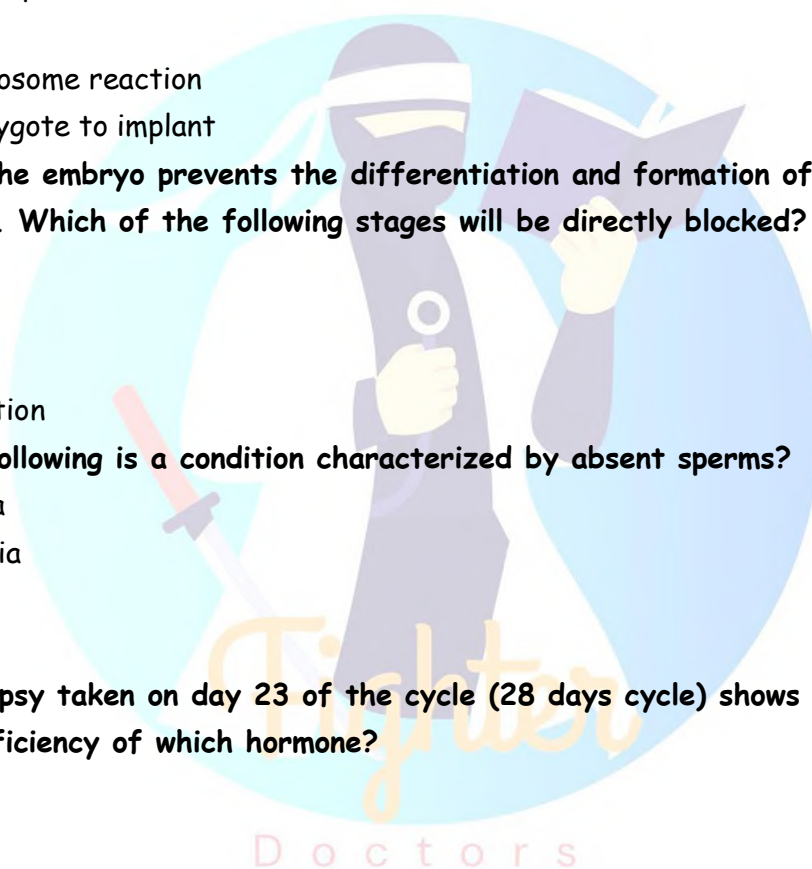
- A) Estrogen
- B) Progesterone
- C) LH
- D) FSH

**30. Which of the following processes involves morphological transformation of spermatids into mature sperm?**

- A) Spermatogenesis
- B) Spermiogenesis
- C) Spermiation
- D) Sperm capacitation

**31. The bivalent chromosomes separate from one another at the chiasma in which stage of meiosis?**

- A) Anaphase 2
- B) Metaphase 1





C) Metaphase 2

D) Anaphase 1

**32. What is the type of epithelium which changes its thickness based on bladder distension?**

A) Stratified squamous

B) Simple columnar

C) Transitional

D) Pseudostratified columnar

**33. Which of the following represents a structural chromosomal aberration?**

A) Trisomy

B) Monosomy

C) Turner's syndrome

D) Deletion

**34. What is the type of epithelium lining the small intestine and has microvilli?**

A) Simple squamous

B) Simple cuboidal

C) Simple columnar

D) Stratified squamous

**35. Pseudostratified columnar ciliated epithelium can be found in:**

A) Fallopian tube

B) Blood vessels

C) Respiratory passages

D) Urinary passage

**36. Apoptosis, which of the following describes it?**

A) It is a physiological condition

B) The cells add their organelles and nucleus

C) Occurs normally at the end of life span of the cell

D) May result from exposure to toxins and anemia

**37. Which of the following represents a potentially renewable cell?**

A) Nerve cell

B) Liver cell

C) Cardiac muscle cell

D) Blood cell

**38. What is the nuclear description of the interphase in cell cycle?**

A) It lasts for a short period

B) The daughter cell contains 92 chromosomes in the G1 phase

C) The changes are visible with the microscope

D) The daughter cell contains 92 chromosomes in the G0 phase

**39. Which of the following occurs during the prophase of mitosis?**

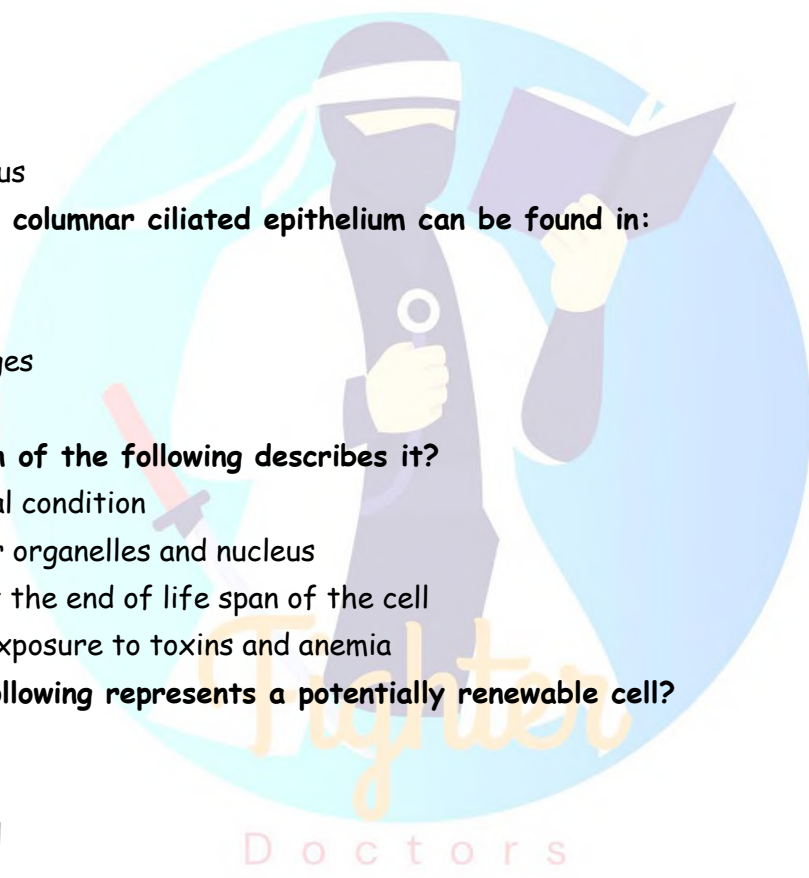
A) Disappearance of nuclear membrane

B) The chromosomes are arranged in the equatorial plate

C) The chromosomes split longitudinally at the centromere

D) The nucleolus reappears

**40. What is a gland that secretes directly into the bloodstream called?**





- A) Exocrine
- B) Mixed
- C) Endocrine
- D) Merocrine

**41. Mammary glands secrete by which of the following?**

- A) Merocrine secretion
- B) Apocrine secretion
- C) Holocrine secretion
- D) Phagocytosis

**42. The network of reticular and elastic fibers is a characteristic histological feature of which cell?**

- A) Pericyte
- B) Fibroblast
- C) Plasma cell
- D) Fat cell

**43. What is the type of connective tissue forming the stroma of organs?**

- A) Reticular C.T
- B) Yellow elastic C.T
- C) White fibrous C.T
- D) Brown adipose C.T

**44. What is the characteristic feature of loose areolar connective tissue?**

- A) Contains only elastic fibers
- B) Present in the umbilical cord
- C) Contains potential cavities
- D) The least common type in the body

**45. Regular white fibrous connective tissue can be present in:**

- A) Tendons of skeletal muscles
- B) Aorta
- C) Umbilical cord
- D) Stroma of organs

**46. What is the nuclear envelope?**

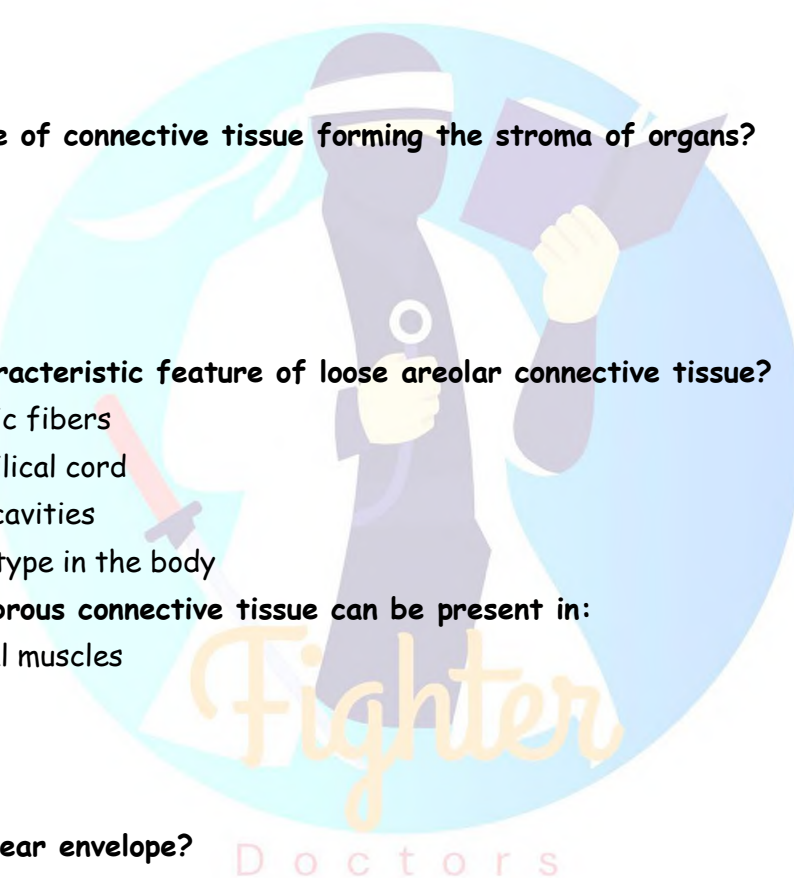
- A) A single membrane
- B) A double membrane with pores
- C) Without pores
- D) Continuous with the Golgi apparatus

**47. What is the typical feature of collagen fibers?**

- A) Run in wavy bundles
- B) Stained brown with orcein stain
- C) The individual fibers can branch
- D) Stained black with silver stain

**48. The cell coat (glycocalyx) is formed of which of the following?**

- A) Integral protein
- B) Peripheral proteins





C) Glycosaminoglycans

D) Glycoproteins and glycolipids

**49. What are microvilli characterized by?**

A) A core of microtubules

B) Motility

C) Actin filament core

D) Presence in sperm

**50. Euchromatin is characterized by which of the following?**

A) Dark staining and coiled DNA

B) Light staining and uncoiled DNA

C) Being transcriptionally inactive

D) Located only at the nuclear periphery

**51. The nucleolus is involved in which of the following?**

A) DNA replication

B) rRNA synthesis and ribosomal subunit assembly

C) Protein degradation

D) Lipid metabolism

**52. Detoxification of drugs in liver cells is the function of which organelle?**

A) rER

B) SER

C) Golgi apparatus

D) Lysosome

**53. Which of the following represents a special stain for mitochondria?**

A) PAS stain

B) Janus green

C) Sudan black

D) Hematoxylin

**54. Which of the following would be best suited to differentiate collagen fibers from other fibers?**

A) Hematoxylin and eosin stain

B) Sudan stain

C) Silver

D) Mallory trichrome stain

**55. Type of intermediate filaments found in the epithelium:**

A) Cytokeratin

B) Vimentin

C) Neuro filaments

D) Desmin

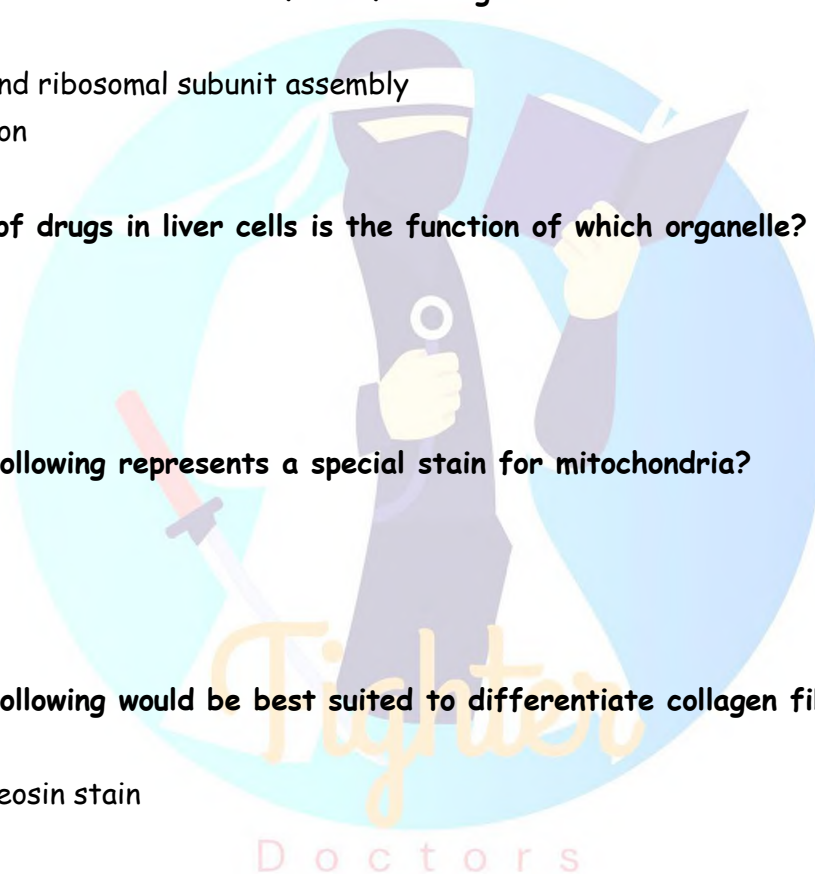
**56. Basal folding is found in a cell associated with...**

A) Ion transport

B) Protein synthesis

C) Steroid secretion

D) Mucus secretion





**57. Which of the following statements is incorrect?**

- A) Microtubules are involved in ciliary movement and intracellular transport.
- B) The cis (forming) face of Golgi apparatus is where secretory granules are formed.
- C) The microtubules of the centrioles and in the basal body of cilia have a 9x3 arrangement.
- D) Ribosomes, made of rRNA and protein, are the site of protein synthesis.

**58. Which of the following is not included in the classical description of a junctional complex?**

- A) The fascia occludens
- B) The zonula occludens
- C) The zonula adherens
- D) The macula adherens

**59. The main ultrastructural feature (EM) of histiocyte is the presence of:**

- A) Microvilli
- B) Lysosomes
- C) Smooth endoplasmic reticulum
- D) Rough endoplasmic reticulum

**60. Which of the following statements is incorrect regarding neuroepithelium?**

- A) It is modified type of epithelium to act as receptor.
- B) Formed of sensory and supporting cells.
- C) It covers skin and forms glands.
- D) Present in taste buds.

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-A  | 2-C  | 3-D  | 4-B  | 5-C  | 6-D  | 7-D  | 8-B  | 9-B  | 10-B |
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| 21-B | 22-B | 23-B | 24-C | 25-C | 26-B | 27-C | 28-D | 29-B | 30-B |
| 31-D | 32-C | 33-D | 34-C | 35-C | 36-A | 37-B | 38-C | 39-A | 40-C |
| 41-B | 42-B | 43-A | 44-C | 45-A | 46-B | 47-A | 48-D | 49-C | 50-B |
| 51-B | 52-B | 53-B | 54-D | 55-A | 56-A | 57-B | 58-A | 59-B | 60-C |

*Note on Q36: Debatable answer.*

*Note on Q38: Debatable answer (statement wording is unclear in the source).*

*Note on Q42: Debatable answer.*



## 24. End Semester 2026 اهليه

36 MCQs

**1. Which of the following closes the eye tightly?**

- A) Frontal belly of occipitofrontalis
- B) Lacrimal part of orbicularis oculi
- C) Orbital part of orbicularis oculi
- D) Palpebral part of orbicularis oculi

**2. The occipital belly of occipitofrontalis is supplied by which of the following?**

- A) Greater occipital nerve
- B) Posterior auricular branch of facial nerve
- C) Supratrochlear nerve
- D) Temporal branch of facial nerve

**3. The retromandibular vein is formed by union of which of the following?**

- A) Anterior and posterior facial veins
- B) Common facial and occipital veins
- C) Maxillary and superficial temporal veins
- D) Supraorbital and supratrochlear veins

**4. The posterior triangle is divided into two triangles by which of the following?**

- A) Inferior belly of omohyoid
- B) Levator scapulae
- C) Sternomastoid muscle
- D) Trapezius muscle

**5. Which of the following is the dangerous area of the scalp?**

- A) Subaponeurotic space
- B) Connective tissue layer
- C) Aponeurotic layer
- D) Periosteal layer

**6. Which of the following bones are ossified by membranous ossification?**

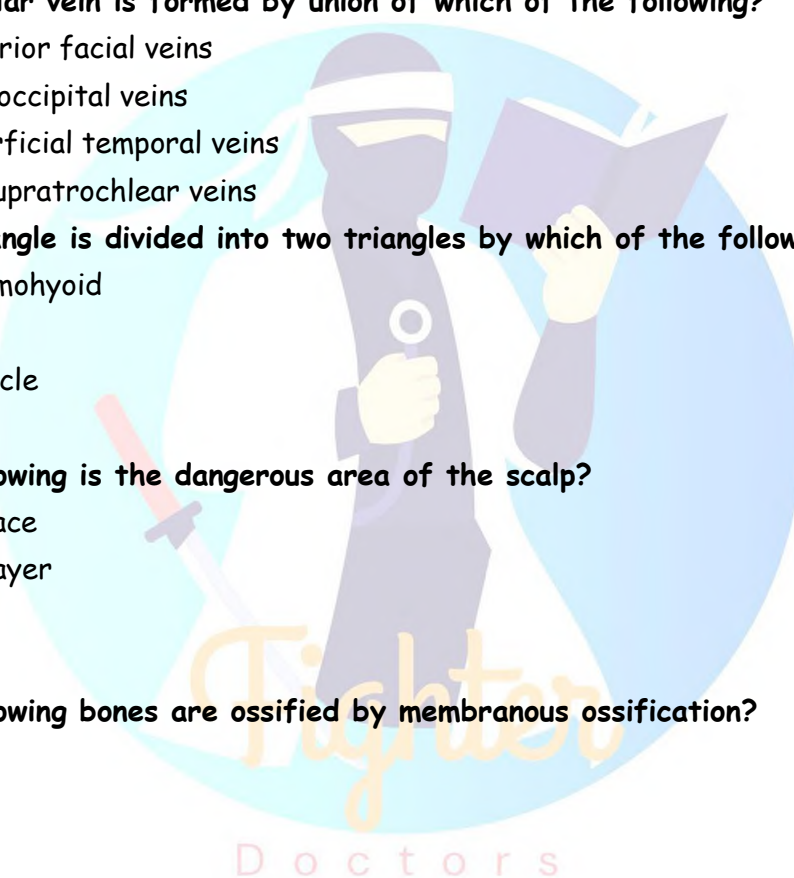
- A) Vault of the skull
- B) Humerus
- C) Femur
- D) Radius

**7. The type of joint between the teeth and the jaw is:**

- A) Synovial joint
- B) Suture
- C) Gomphosis
- D) Syndesmosis

**8. In which movement does the palm face posterior?**

- A) Flexion
- B) Supination
- C) Pronation
- D) Protraction





9. A firm grip requires the wrist to be held in extension. In this position, the wrist extensors are acting as:

- A) Synergist
- B) Antagonist
- C) Prime mover
- D) Fixator

10. A 22-year-old pregnant woman comes for her first-trimester ultrasound at 6 weeks of gestation. The scan shows a gestational sac with an embryo and a well-defined yolk sac. The yolk sac is connected to the midgut of the embryo by which structure?

- A) Allantois
- B) Vitello-intestinal duct
- C) Umbilical vein
- D) Chorionic villi

11. A 30-year-old pregnant woman at 28 weeks of gestation complains of rapid abdominal enlargement and discomfort. On ultrasound, the fetus is seen floating freely with an excessive amount of amniotic fluid. Which fetal abnormality is commonly associated with this condition?

- A) Renal agenesis
- B) Esophageal atresia
- C) Potter sequence
- D) Pulmonary hypoplasia

12. As regarding the urinary bladder development, the apex of the bladder is formed by:

- A) Mesonephric duct
- B) Metanephric cap
- C) Allantois
- D) Primitive urogenital sinus

13. As regarding to Sex chromosome abnormalities, monosomy means:

- A) Klinefelter's syndrome
- B) Super female syndrome
- C) Turner's syndrome
- D) Down's syndrome

14. A 32-year-old woman with a history of two previous cesarean sections presents with severe postpartum hemorrhage. Histology shows absence of decidua basalis with direct attachment of chorionic villi to the myometrium. The most likely diagnosis is:

- A) Placenta previa
- B) Placenta accreta
- C) Placenta increta
- D) Placenta percreta

15. Which component of the placenta is derived from extra-embryonic mesoderm?

- A) Syncytiotrophoblast
- B) Cytotrophoblast
- C) Chorionic villous core
- D) Decidua basalis



16. Post-delivery bleeding occurs despite apparent complete placental expulsion. Ultrasound reveals retained placental tissue. This complication is most commonly associated with:
- Placenta circumvallata
  - Placenta succenturiata
  - Placenta membranacea
  - Placenta fenestrata
17. Which layer is lost by late pregnancy, significantly thinning the placental barrier?
- Endothelium of fetal capillaries
  - Syncytiotrophoblast
  - Cytotrophoblast
  - Basement membrane
18. A severe embryonic malformation shows failure of neural tube closure with abnormal ventral body wall formation. The most likely underlying issue is disruption of:
- Ectoderm differentiation only
  - Mesoderm differentiation only
  - Coordinated neurulation and embryonic folding
  - Endodermal differentiation
19. Nissl bodies in neurons correspond to which organelle at the EM level?
- Golgi apparatus
  - Mitochondria
  - Rough endoplasmic reticulum
  - Lysosomes
20. In a patient with a genetic defect in tubulin, which of the following cellular structures would be most directly affected?
- Formation of the nuclear envelope
  - Assembly of the mitotic spindle
  - Organization of microfilaments
  - Synthesis of ribosomal RNA
21. Which of the following junctions allows direct passage of ions and small molecules between adjacent cells and is found abundantly in cardiac muscle?
- Zonula occludens
  - Desmosome
  - Nexus
  - Hemidesmosome
22. Lipofuscin pigment accumulation in neurons is a sign of:
- Active phagocytosis
  - Aging
  - Melanin overproduction
  - Glycogen storage
23. The perinuclear space is continuous with which of the following?
- Golgi cisternae
  - Rough endoplasmic reticulum
  - Mitochondrial intermembrane space
  - Lysosomal lumen
24. In a compound tubulo-alveolar gland like the pancreas, which of the following describes the duct system?
- Unbranched duct with branched secretory portion
  - Branched duct with flask-shaped secretory units
  - No duct system; secretion into blood
  - Single duct with coiled secretory portion
25. Which of the following is true regarding heterochromatin?



- A) It is transcriptionally active  
 B) It is predominant in protein-synthesizing cells  
 C) It is concentrated at the nuclear periphery  
 D) It stains lightly with basic dyes
- 26. In the stratum spinosum, cells are connected by numerous desmosomes. Which intermediate filament type predominates in these junctions?**  
 A) Vimentin  
 B) Desmin  
 C) Cytokeratin  
 D) Neurofilament
- 27. A biopsy of an epithelium that changes from 6-8 layers when empty to 2-3 layers when distended. Which of the following is the site of epithelium?**  
 A) Trachea  
 B) Uterus  
 C) Vas deferens  
 D) Urinary bladder
- 28. Which of the following represents the typical histological feature of fibroblast?**  
 A) The cytoplasm is acidophilic  
 B) The nucleus is euchromatic  
 C) Contains few rough endoplasmic reticulum  
 D) Contains numerous SER
- 29. What is the characteristic feature of elastic fibers?**  
 A) Consist of type III collagen  
 B) Stained black with silver stain  
 C) Stained brown with orcein  
 D) The fibers can not branch
- 30. The brown color of multilocular adipose connective tissue is due to which of the followings?**  
 A) Numerous rough endoplasmic reticulum  
 B) Colored cytochromes inside mitochondria  
 C) Poor blood supply  
 D) Numerous smooth endoplasmic reticulum
- 31. What is the type of connective tissue presents in the sclera of eye?**  
 A) Loose areolar C.T.  
 B) Regular white fibrous C.T.  
 C) Reticular C.T.  
 D) Irregular white fibrous C.T.
- 32. The cleavage furrow develops at which of the following stages of mitosis?**  
 A) Anaphase  
 B) Telophase  
 C) Prophase  
 D) Metaphase
- 33. What is the position of centromere in submetacentric chromosomes?**  
 A) At the center of the chromosome  
 B) Midway between the centre and upper end  
 C) Very close to upper end  
 D) Terminal
- 34. In which of the following chromosomal aberrations, the chromosome divides transversely at the centromeres instead of longitudinally during anaphase of mitosis?**  
 A) Inversion  
 B) Isochromosomes



- C) Ring Chromosome D) Reciprocal translocation
35. The membrane of cytoplasmic vesicles can be added to the cell membrane in which of the following processes?
- A) Phagocytosis B) Endocytosis  
C) Pinocytosis D) Exocytosis
36. Which of the following describes the histological structure of Golgi apparatus?
- A) Produces cytoplasmic basophilia with H&E  
B) Appears as a network of brown granules with silver stain  
C) Appears as a basophilic area around the nucleus in plasma cell  
D) The Cis face represents the mature surface

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-B  | 3-C  | 4-A  | 5-A  | 6-A  | 7-C  | 8-C  | 9-D  | 10-B |
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| 31-D | 32-B | 33-B | 34-B | 35-D | 36-B |      |      |      |      |

## Written questions

### Part II: Written Questions (36 marks)

- Describe the blood supply of the face. (3 marks)
- Discuss the skin lines (Langer's lines). (3 marks)
- Give a definition for each of the following: Portal vein, Inferior vena cava, Lymphatic capillaries, Pulmonary vein, Right ventricle, Aorta. (3 marks)
- A 26-year-old primigravida at 20 weeks of gestation undergoes a routine antenatal ultrasound. The scan reveals a single umbilical artery and mild intrauterine growth restriction. Give a short note about the normal development of the umbilical cord, its contents, and associated congenital anomalies. (3 marks)
- Enumerate the results of embryonic folding. (3 marks)
- Enumerate the steps of notochord formation. (3 marks)
- Enumerate the following:
  - 3 apical specializations with the site for each one. (1.5 marks)
  - Four events occurring during the prophase of mitosis. (2 marks)
  - Types of Nuclear Chromatin. (1.5 marks)
- Mention the following:
  - Types of granules in stratum granulosum (content, type, and function). (1.5 marks)
  - The origin of the following cells: plasma cell and macrophage. (1 mark)
  - Two numerical chromosomal aberrations. (1 mark)
- Discuss the following:
  - Nucleolus (EM structure, function, and staining). (3 marks)
  - The histological structure of mucoid connective tissue and two sites for it. (2.5 marks)
  - The electron microscopic picture of the types of Endoplasmic Reticulum and their functions. (2 marks)
- Draw a colored, labeled diagram of pseudostratified columnar ciliated epithelium. (2 marks)



## 25. End Semester 2026 حڪومي دور تاني

Answer the following questions (All questions to be answered): (19 marks)

1. Movement of the shoulder inferior is termed which of the following?
  - a- Protraction
  - b- Elevation
  - c- Depression
  - d- Retraction
2. During the X-ray examination of a child, the bones appeared incomplete. The physician explained that the bones are still in the process of growing. Which of the following structures is responsible for increase the thickness of the bone?
  - a- Periosteum
  - b- Endosteum
  - c- Epiphyseal plate (growth plate)
  - d- Articular cartilage
3. The central, fleshy portion of a muscle that lies between its attachments is termed:
  - a- Origin
  - b- Insertion
  - c- Belly
  - d- Ligament
4. Which of the following structures receives deoxygenated blood through the superior and inferior vena cava?
  - a- Right atrium
  - b- Left atrium
  - c- Right ventricle
  - d- Pulmonary trunk
5. Which of the following joints are formed when two bones are connected by fibrous tissue?
  - a- Symphysis pubis
  - b- Shoulder joint
  - c- Skull sutures
  - d- Hip joint
6. Which of the following fluids primarily functions to lubricate joints and reduce friction between articular surfaces?
  - a- Cerebrospinal fluid (CSF)
  - b- Blood plasma
  - c- Synovial fluid
  - d- Lymph
7. In surgical procedures or trauma, which layer of the scalp is referred to as the danger area?
  - a- Skin of the scalp
  - b- Connective tissue layer (dense subcutaneous tissue)
  - c- Loose areolar tissue layer
  - d- Pericranium



**8. The external jugular vein is formed by the union of which veins?**

- a. Anterior Facial vein and anterior jugular vein
- b. Anterior Facial vein and anterior division of retromandibular vein
- c. Posterior division of posterior Facial vein and external jugular vein
- d. posterior division of retromandibular vein and posterior auricular vein

**9. A 60-year-old man presents with pain and numbness over the lower jaw and temple region.**

**Which of the following nerves supplies this area?**

- a- Maxillary nerve (V2)
- b- Mandibular nerve (V3)
- c- Ophthalmic nerve (V1)
- d- Greater occipital nerve

**10. Which of the following structures is a content of the carotid triangle?**

- a- Submandibular gland
- b- Parotid gland
- c- Internal jugular vein
- d- External jugular vein

**11. A newborn presents with a sacrococcygeal mass containing tissues derived from all three germ layers. The most likely embryological defect is failure of which of the following?**

- A. Regression of the primitive streak
- B. Formation of the primitive node
- C. Fusion of the neural folds
- D. Differentiation of paraxial mesoderm

**12. Absence of the notochord would primarily interfere with development of which of the following?**

- A. Epidermis
- B. Neural tube
- C. Amnion
- D. Extraembryonic mesoderm

**13. An embryo has 30 somites identified. What is the most likely embryonic age?**

- A. 21 days
- B. 23 days
- C. 25 days
- D. 30 days

**14. A fetus is diagnosed on ultrasound with defects in meningeal formation. What is the embryological defect?**

- A. Neural plate induction
- B. Neural fold elevation and closure
- C. Notochord formation
- D. Neural crest cell differentiation

**15. An embryo shows failure of formation of primary yolk sac cavity. Which layer of the bilaminar disc is primarily affected?**

- A. Hypoblast
- B. Epiblast





C. Cytotrophoblast

D. Syncytiotrophoblast

**16. Failure of gastrulation would prevent development of which of the followings?**

A. Intraembryonic mesoderm

B. Chorionic cavity

C. Bilaminar germ disc

D. Trophoblastic shell

**17. An embryo is found attached to the chorionic plate by a thick band of tissue. This structure will later develop which of the followings?**

A. Allantois

B. Vitelline duct

C. Umbilical cord

D. Amnion

**18. A placenta later shows placenta previa. The abnormality most likely originated from implantation of the blastocyst in which of the followings?**

A. Fundus of uterus

B. Posterior uterine wall

C. Lower uterine segment

D. Myometrium

**19. A 15-year-old girl presents with Down syndrome. What is Her karyotype?**

A. 46,XX

B. 46,XY

C. 45,X

D. 47,XX

**20. Which type of neuroglia forms the myelin sheath in the peripheral nervous system?**

A. Schwann cell

B. Astrocytes

C. Oligodendrocytes

D. Microglia

**21. Plasma cells release which substances?**

A. Collagen and elastin

B. Histamine and heparine

C. Antibodies

D. proteoglycans and fibronectin

**22. Which of the following is considered as aneuploidy?**

a- Triploid

b- Trisomy

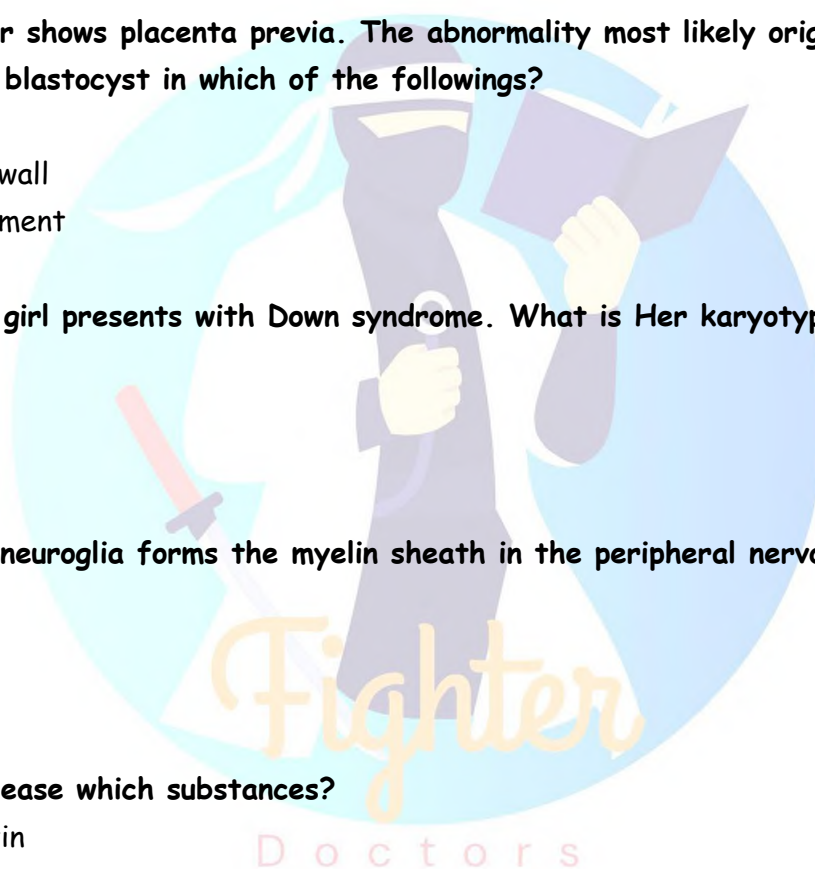
c- Tetraploid

d- Polyploidy

**23. Reticular fibers are primarily composed of which protein?**

A. Collagen

B. Elastin





C. Fibrillin

D. Keratin

**24. Multipolar pyramidal nerve cells can be found in which of the following?**

a- Anterior horn cells

b- Cerebrum

c- Cerebellum

d- Retina

**25. Which of the following statements describe the dendrites of nerve cells?**

a- Have a constant diameter

b- conducts nerve impulse away from the cell body

c- Usually long

d- Contain Nissl granules

**26. Which of the following describes the centrioles?**

A. Appear as two dark bodies with H&E

B. Share in the formation of cilia

C. Present as two pairs in non-dividing cells

D. Formed by dynamic microtubules

**27. Which of the following matching with rER ?**

A. Form proteins secreted by the cells

B. Form proteins used within the cytosol

C. Linked together by tRNA forming polysomes

D. Bind to the outer surface of sER

**28. Which one of the following stains demonstrates lipid in the liver cells?**

A. Sudan III

B. PAS

C. Silver

D. trypan blue

**29. Diffuse cytoplasmic basophilia caused by ribosome can be found in which cells?**

a- Embryonic cells

b- Pancreatic cells

c- Nerve cells

d- Glial cells

**30. What is the term for the process by which the cells transport solids?**

A. Endocytosis

B. Exocytosis

C. Pinocytosis

D. Phagocytosis

**31. Fibroblasts are characterized by which of the following?**

a- Abundant rER and Golgi

b- Signet ring appearance

c- Metachromatic granules

d- Cart-wheel nucleus

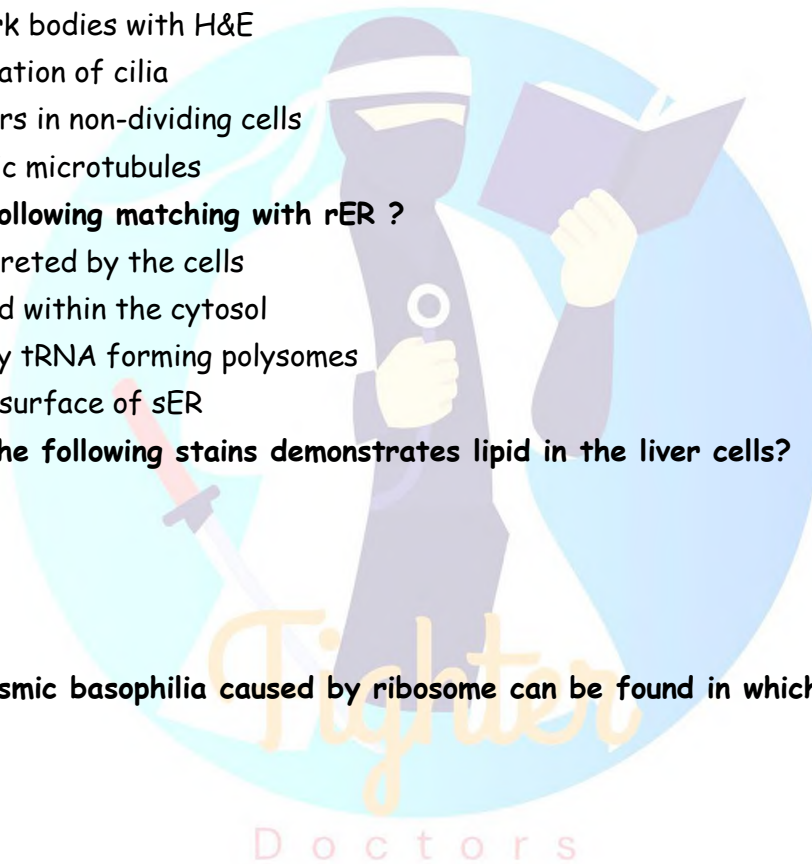
**32. Which type of epithelium is specialized for protection?**

A. Simple squamous epithelium

B. Stratified squamous epithelium

C. Simple columnar epithelium with microvilli

D. Pseudostratified columnar epithelium



a- Sebaceous gland  
c- Salivary gland

b- Mammary gland  
d- Meibomian gland

- A. Randomly arranged collagen fibers
- B. High tensile strength in one direction
- C. Abundance of ground substance
- D. Presence of elastic fibers in all regions

A) Only epithelial cells  
B) Basal lamina and reticular lamina  
C) Only collagen type I  
D) Only glycoproteins

A) Stores fat as a single droplet  
B) Generates heat  
C) Is poorly vascularized  
D) Has a signet ring appearance

A) Skin  
B) Umbilical cord  
C) Lymph nodes  
D) Vitreous humor of eye

A. Smooth endoplasmic reticulum  
B. Rough endoplasmic reticulum  
C. The trans face of Golgi apparatus  
D. The cell membrane

14. Mention the four component of the nucleus. (2 mark)



## 26. End Module 2026 دور ثاني حكومي

1. A 40-year-old man is examined after a road traffic accident. The physician asks him to stand in the standard anatomical position. Which of the following correctly describes the orientation of the body?
  - A. Head faces posterior
  - B. Eyes look forward and palms face anterior
  - C. Arms are crossed over the chest
  - D. Toes point laterally
2. During dissection, the brain was sectioned into an equal right and left parts. Which anatomical plane is used for this division?"
  - A. midsagittal plane
  - B. parasagittal plane
  - C. coronal plane
  - D. horizontal plane
3. In which anatomical location are the superficial palmar vessels found?
  - A. In the palm of the hand near the skin
  - B. In the sole of the foot away from the skin
  - C. In the sole of the foot near the skin
  - D. In the palm of the hand away from the skin
4. A physiotherapist asks a patient to move the distal end of the arm so that it traces a complete circular path, combining flexion, extension, abduction, and adduction. Which term best describes this movement?
  - A. Adduction
  - B. Circumduction
  - C. Flexion
  - D. Rotation
5. A 55-year-old man undergoes imaging of his femur. The radiologist notes abundant spongy bone at the epiphysis. What is the main character of spongy bone?
  - A. To increase its length
  - B. As it looks like a sponge to sustain the pressure
  - C. it cover the out surface of the bone
  - D. As it provides strength for weight bearing
6. Which of the following arteries can supply the outer compact bone?
  - A. periosteal
  - B. epiphyseal
  - C. metaphyseal
  - D. nutrient
7. Which of the following the muscle fibers attached to the one side of the tendon?
  - A. Triangular muscle
  - B. Unipennate muscle
  - C. Bipennate muscle
  - D. Spiral muscle
8. During the flexion of the elbow, the muscles of the shoulder act as.....
  - A. prime mover
  - B. fixator
  - C. synergist
  - D. antagonist
9. During systemic circulation, the oxygenated blood is pumped from the left ventricle to the body through which of the following?
  - A. Inferior vena cava
  - B. Pulmonary trunk
  - C. Pulmonary veins
  - D. Aorta



**10. A medical student is asked why veins are structurally different from arteries. Which of the following features allows veins to return blood efficiently to the heart despite lower pressure?**

- A. Thick muscular walls to withstand high pressure
- B. Valves that prevent backflow of blood
- C. Elastic lamina for pulsatile expansion
- D. Continuous contraction to push blood forward

**11. Which of the following has valves and constricted at their sites showing beaded appearance?**

- A. veins
- B. lymph vessel
- C. arteries
- D. capillaries

**12. absence of movement is one of the characters of which of the following joint?**

- A. Primary cartilaginous joint
- B. Synovial joint
- C. Secondary cartilaginous joint
- D. Fibrous joint

**13. intervertebral disc is which of the following joint?**

- A. Primary cartilaginous joint
- B. Synovial joint
- C. Secondary cartilaginous joint
- D. Fibrous joint

**14. Which of the following contains the visual area in the brain?**

- A- Frontal lobe
- B-Occipital lobe.
- C-Temporal lobe
- D- parietal lobe

**15. Which part of spinal nerve is only sensory ?**

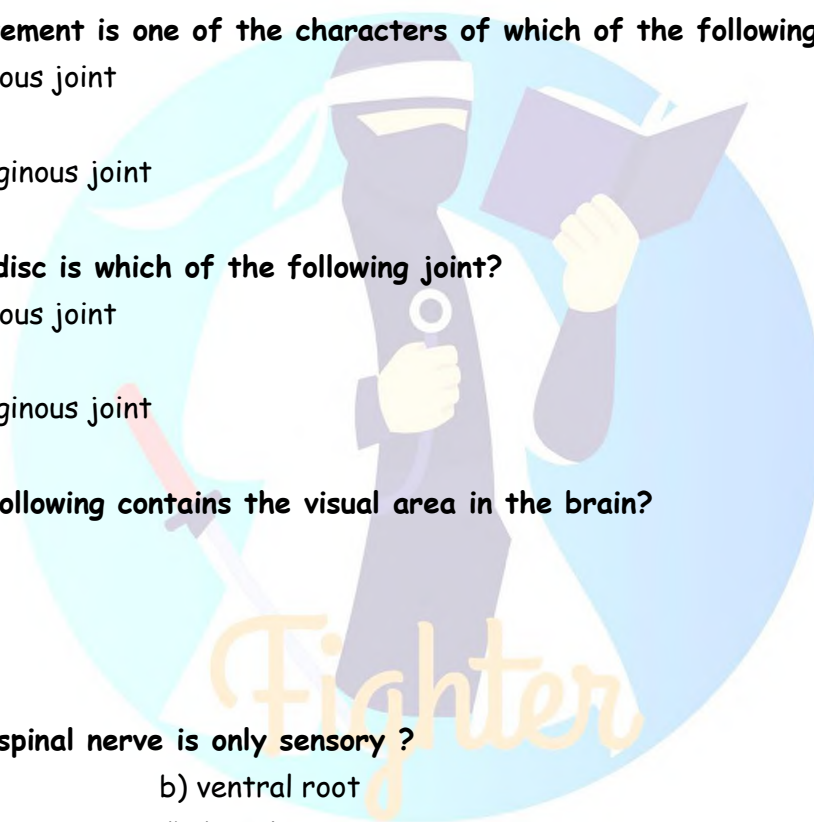
- a) dorsal root
- b) ventral root
- c) ventral rami
- d) dorsal rami

**16. A 25-year-old male presents with infertility. Semen analysis shows many abnormal sperms while testicular biopsy shows normal spermatids. Which stage of gametogenesis is most likely affected?**

- A. Mitosis of spermatogonia
- B. Meiosis I in primary spermatocytes
- C. Spermiogenesis
- D. Migration of primordial germ cells

**17. A 30-year-old woman undergoes in vitro fertilization (IVF). Eggs are retrieved after hormonal stimulation and are found to be arrested in metaphase of meiosis II. What triggers the completion of meiosis II in the oocyte?**

- A. LH surge
- B. FSH rise
- C. Sperm penetration
- D. Estrogen peak





**18. A researcher observes that in a healthy adult male, germ cells are continuously produced from puberty onward. Which cell type serves as the stem cell reservoir for lifelong spermatogenesis?**

- A. Primary spermatocyte
- B. Spermatid
- C. Spermatogonium
- D. Leydig cell

**19. After fertilization, the zygote begins dividing as it travels through the fallopian tube. On week post-fertilization, it typically consists of inner cell mass with blastocoele What is this stage called?**

- A. Morula
- B. Blastocyst
- C. Zygote
- D. Embryoblast

**20. During fertilization, a sperm releases enzymes to penetrate the oocyte's outer layers. Which structure prevents polyspermy by hardening after the first sperm entry?**

- A. Corona radiata
- B. Zona pellucida
- C. cell membrane
- D. Perivitelline space

**21. On day 5 post-fertilization, an embryo develops a fluid-filled cavity and differentiates into two cell lineages. Which lineage gives rise to the placenta?**

- A. Trophoblast
- B. Syncytiotrophoblast
- C. Cytotrophoblast
- D. Inner cell mass

**22. A 28-year-old woman undergoes a transvaginal ultrasound 10 days after her last menstrual period (LMP). The clinician identifies a small gestational sac within the endometrium but no embryo. Which structure is primarily responsible for producing the hormone detected in her positive pregnancy test?**

- A. Syncytiotrophoblast
- B. Cytotrophoblast
- C. Epiblast
- D. Hypoblast

**23. A researcher examines an embryo at day 9 post-fertilization and notes a new space between hypoblast and exocoelomic membrane. This cavity will develop into which structure?**

- A. Yolk sac
- B. Amniotic cavity
- C. Chorionic cavity
- D. Blastocyst cavity

**24. In an embryo at day 12 post-fertilization, lacunae (spaces) appear within the syncytiotrophoblast and fill with maternal blood. What is this process critical for?**

- A. Gastrulation process
- B. Neurulation process
- C. Uteroplacental circulation
- D. Cleavage process



25. Implantation is complete by day 10-12, and the conceptus is fully embedded in the endometrium. Which layer of trophoblast is directly responsible for eroding maternal capillaries to access blood?
- Cytotrophoblast
  - Extraembryonic mesoderm
  - Syncytiotrophoblast
  - Epiblast
26. A researcher observes an embryo with a visible primitive streak on its epiblast surface. Which of the following best describes the significance of this structure?
- Site of neural tube closure
  - Origin of the three germ layers
  - Formation of the yolk sac
  - Initiation of heart tube formation
27. During embryonic development, a group of cells migrates cranially from the primitive node along the midline and forms a solid cord. What is this structure called?
- Primitive streak
  - Notochord
  - Neural crest
  - Somite
28. In an experimental mouse model, the notochord fails to form. Which of the following developmental defects is most likely to occur?
- Absence of intraembryonic mesoderm
  - Failure of neural tube induction
  - Lack of yolk sac formation
  - Defective placental development
29. A pathology report describes an embryo with a tumor in the sacrococcygeal region composed of tissues from all three germ layers (e.g., skin, cartilage, gut epithelium). What is the name of this tumor?
- Teratoma
  - Notochord remnants
  - Neural crest cell migration defect
  - Abnormal yolk sac differentiation
30. During gastrulation, a subset of epiblast cells moves through the primitive streak and lies in-between ectoderm and endoderm. These cells will primarily give rise to which germ layer?
- Ectoderm
  - Mesoderm
  - Endoderm
  - Trophoblasts
31. What is the typical feature of submetacentric chromosome?
- the centromere is midway between the center and upper end
  - the centromere is at the center of the chromosome
  - the centromere is very close to upper end
  - the centromere is terminal



**32. Which one of these cells is fixed cell type found in connective tissue?**

- a. Fibroblast
- b. mast cell
- c. leukocytes
- d. Plasma cell

**33. What do you call the simple squamous epithelium that lines the blood vessels?**

- a. Epithelioid tissue
- b. Mesothelium
- c. Endothelium
- d. Transitional

**34. Which of the following describes Barr body ?**

- A. The inactive X chromosome in females.
- B. The X chromosome in males.
- C. The Y chromosome in males.
- D. The XX chromosomes in females.

**35. What is the typical feature of submetacentric chromosome?**

- A. the centromere is midway between the center and upper end
- B. the centromere is at the center of the chromosome
- C. the centromere is very close to upper end
- D. the centromere is terminal

**36. which of the following cells are Permanent cells?**

- a. Nerve cells
- b. Liver cells
- c. Blood cells
- d. Skeletal muscles

**37. Which of points related to Euchromatin?**

- A. present in the nuclei of metabolically active cells.
- B. special type of stain.
- C. present in the nuclei of metabolically inactive cells.
- D. type of cell inclusions.

**38. What is concerned with tail of phospholipid granules ?**

- A. the protein component of the cell membrane.
- B. the carbohydrate component of the cell membrane.
- C. the hydrophilic end of lipid.
- D. the hydrophobic end of lipid.

**39. What is the function of rough endoplasmic reticulum ?**

- A. lipid.
- B. carbohydrate.
- C. protein that will be secreted by the cell.
- D. cytoplasmic protein necessary for the cells own existence.

**40. What is about smooth endoplasmic reticulum?**

- A. Sometimes has ribosomes attached to its membrane
- B. Is rarely found in skeletal muscle cells
- C. Areas of localized basophilic indicate its presence
- D. Is present in cells where drug detoxification is taking place



**41. What is the feature of Golgi apparatus ?**

- A. the site of packaging of secretory products of the cell.
- B. the site of protein synthesis.
- C. continuous with smooth endoplasmic reticulum.
- D. capable of providing energy.

**42. Which of the following describes Multivesicular bodies ?**

- A. Primary lysosomes.
- B. Formed as a result of digestion of solid material.
- C. Formed as a result of digestion of fluid material.
- D. Autophagic vacuoles.

**43. Which is the role of Microtubules ?**

- A. Carbohydrate metabolism.
- B. Oxidative phosphorylation.

**43. Which is the role of Microtubules ?**

- C. Water balance.
- D. Formation of the cell spindle during mitosis.

**44. What about the nuclear pore?**

- A. Is at regular intervals
- B. Is bridged by a unit membrane
- C. Is a transient structure
- D. Allows for communication between the nucleus and the cytoplasm

**45. At which Pseudo-stratified columnar ciliated epithelium found ?**

- a) The ovary.
- b) The blood vessels.
- c) Upper respiratory passages.
- d) Urinary passage.

**46. What type of tissue forms the alveoli in the lung?**

- a. Simple squamous epithelium
- b. Simple cuboidal epithelium
- c. Simple columnar epithelium
- d. Stratified squamous epithelium

**47. What about Stereocilia is?**

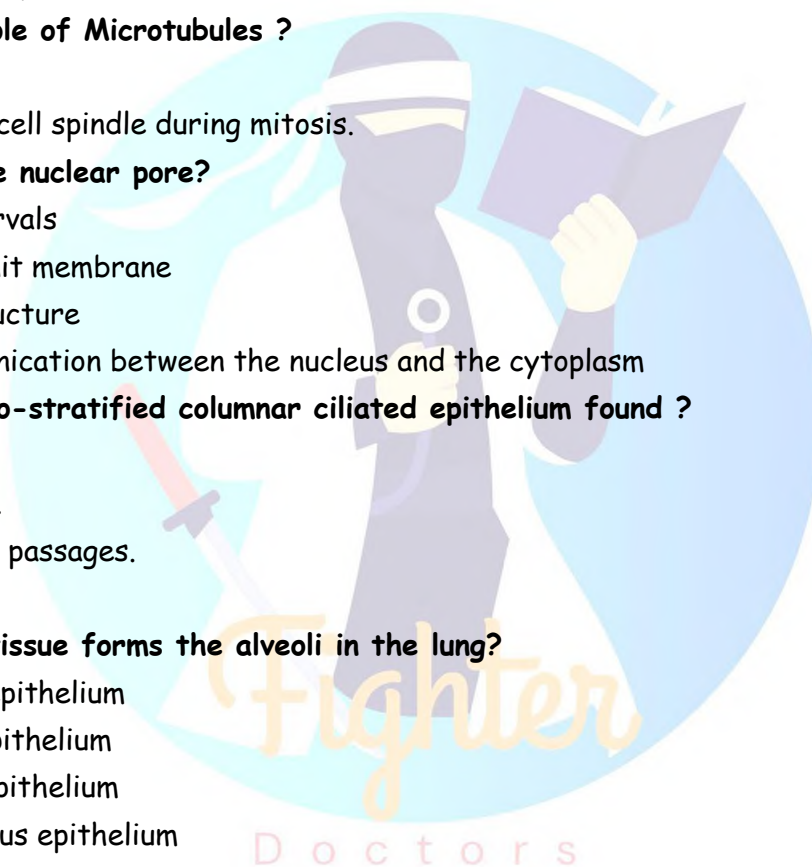
- a) Formed of intermediate keratin filaments.
- b) Motile true cilia.
- c) Long microvilli.
- d) Present in respiratory epithelium

**48. How can stratified epithelium classified according the shape of ?**

- a) Superficial cell layers.
- b) Intermediate cell layers.
- c) Basal cell layer.
- D) middle layer

**49. Which epithelium is modified to act as receptor ?**

- a) Myoepithelium.
- b) Glandular epithelium.
- c) Neuroepithelium.
- d) Transitional epithelium





**50. What is the main ultrastructural feature (EM) of histiocyte ?**

- A. Microvilli.
- B. Lysosomes.
- C. Smooth endoplasmic reticulum.
- D. Rough endoplasmic reticulum.

**51. What is related to Plasma cells?**

- A. Have well developed smooth endoplasmic reticulum
- B. Have phagocytic ability
- C. Are commonly found in peripheral blood
- D. Show all EM features of protein synthesizing cells.

**52. Where are metachromatic granules present in?**

- A. Mast cells.
- B. Plasma cells.
- C. Fat cells.
- D. Reticular cells.

**53. Which one of the following glands is considered as apocrine gland?**

- a) Sebaceous gland.
- b) Mammary gland.
- c) Salivary gland.
- d) Pancreas.

**54. What type of tissue is cornea composed of?**

- A. Mucous connective tissue
- B. Mesenchyme
- C. Dense irregular connective tissue
- D. Dense regular connective tissue

**55. What is the shape of fat cells present in the unilocular adipose tissue?**

- A. Clock face
- B. Signet ring
- C. Horse shoe
- D. Kidney

**56. Which is true about pseudo stratified columnar ciliated epithelium ?**

- A. May be keratinized.
- B. Is highly vascular.
- C. All cells resting on basement membrane.
- D. It is a type of stratified epithelium

**57. What type of epithelium lines the trachea?**

- a. Pseudostratified epithelium
- b. Simple cuboidal epithelium
- c. Simple columnar epithelium
- d. Stratified squamous epithelium

**58. Which of the following is a unicellular gland?**

- a. Squamous cell
- b. Goblet cell
- c. Basal cell
- d. Basket cell

**59. What is the surface modification seen on the cells of the epididymis?**

- a. Microvilli
- b. Stereocilia
- c. Cilia
- d. Keratinization

**60. What type of epithelium has dome shaped cells on the apical surface?**

- a. Epithelioid tissue
- b. Mesothelium
- c. Endothelium
- d. Transitional



Source: *بنك درشا* (Dr Rasha Abdel Kader Elsisy) | short notes, definitions, fill-in questions, MCQs and Nursing exams

### Fill in the spaces:

- ..... refers to the structure that is nearer to the upper end of body, while ..... refers to a structure that is nearer to the lower end of body.
- Medial side of the forearm is called ..... while the medial side of the leg is called .....
- ..... refers to the structure that is nearer to the median plane, while ..... refers to the structure that is away from the median plane.
- The term superficial means ..... while the term deep means .....
- Anterior surface of the hand is called ..... while the other surface of hand is called .....
- Flexion means ..... while extension means .....
- ..... means moving away from the median plane, while ..... means moving towards the median plane.
- Pronation means ..... while supination means .....
- ..... means to move apart superiorly, while ..... means to move apart inferiorly.
- The cellular layer of the skin is called ..... and connective tissue layer of it is called .....
- ..... give the elasticity of the skin, ..... produce the pigment of the skin
- Derivatives of epidermis are ..... and .....
- The smooth muscle which attached to the hair follicle is called ..... and supplied by .....
- The Sweat glands are ..... and supplied by .....
- The Sebaceous glands are .....
- The nails are formed of ..... and .....
- The fascia is divided into ..... and .....
- Superficial fascia is ..... While deep fascia is .....
- The deep fascia is well developed in ..... and ..... while it is thin in ..... and .....
- the attachment of the muscle to the bone that remains stationary is the ..... while the attachment of the muscle to the bone that moves is the .....
- The fleshy part of the muscle between the tendons of origin and/or insertion is called .....
- The cartilage remain in certain sites of the body such as ..... and .....
- Types of ossification of bone are ..... and .....
- ..... and ..... are flat bones while ..... is a sesamoid bone.
- In the adult, the red bone marrow is present in ..... and .....
- The skeleton is divided into ..... and .....
- The number of cervical vertebrae is ..... while the number of thoracic vertebrae is .....
- The sutures are ..... joints
- Intra articular structures can be seen in different joints as ..... and .....
- The elbow joint is ..... while the hip joint is .....
- Hilton `s law is .....



- The trachea is divided into.....and.....
- The urinary system is consists of ....., ....., ..... and.....
- The testis lies in the ..... as it needs .....
- The epididymis is formed of .....,..... and.....
- The uterus is divided into .....,..... and.....
- Female genital system is formed of .....,.....,..... and .....
- The main artery in the arm is ..... While in the thigh is.....
- Superficial veins of the upper limb includes.....and.....
- The parts of the skeletal system are..... And.....
- The axial skeleton is formed of the following bone ....., ....., ..... and.....
- The types of bone according to the shape is ..... example..... example.....  
..... example.....
- The skeletal muscle is supplied by..... while the smooth muscle is supplied by.....
- The parts of the nervoys system are ..... and.....
- The parts of the central nervous system are .....and .....
- The parts of the brain are ....., ..... And.....
- The parts of the brain stem are ....., ..... And.....
- The types of the peripheral nerves are ..... And.....
- The types of the meninges are ....., ..... And.....

## Choose the correct answer (Part 1)

1. The anatomical position of the human body include the following except :

- A) The person stands in erect position
- B) The palm faces anterior
- C) The arms are abducted
- D) The lower limbs are close together

2. Planter surface means :

- A) The sole of the foot
- B) The palm of the hand
- C) The superior surface of the foot
- D) The posterior surface of the hand

3. all of the following about epidermis is correct except:

- A) epidermis is a stratified epithelium
- B) it is composed of connective tissue
- C) on the palms and soles, it is extremely thick
- D) it is thin on the anterior surface of the arm and forearm

4. deep fascia is absent in:

- A) abdomen
- B) palms
- C) leg
- D) forearm

5. regarding the smooth muscle, all of the following is true except:

- A) consists of long spindle shaped cells



- B) controlled by autonomic nervous system
- C) have the property of spontaneous and rhythmic contraction
- D) found in the wall of the viscera and blood vessels

**6. Any muscle oppose the action of prime mover is named:**

- A) antagonist
- B) action of paradox
- C) synergist
- D) fixator

**7. flat tendon is named:**

- A) aponeurosis
- B) muscle belly
- C) endomysium
- D) all of the above

**8. The right lung has :**

- A) 8 broncho-pulmonary segments
- B) 2 lobes
- C) 10 broncho-pulmonary segments
- D) One fissure

**9. Symphysis pubis is:**

- A) Secondary cartilaginous joint
- B) Primary cartilaginous joint
- C) Plane synovial joint
- D) Fibrous joint

**10. Carpal bones are:**

- A) Small long bones
- B) Irregular bones
- C) Short bones
- D) Sesamoid bones

**11. One of the following is poly axial joint:**

- A) Shoulder joint
- B) Knee joint
- C) Elbow joint
- D) Wrist joint

**12. All the following bones are appendicular skeleton except:**

- A) Scapula
- B) Tibia
- C) Radius
- D) Hyoid bone

**13. Regarding the prostate gland one is wrong:**

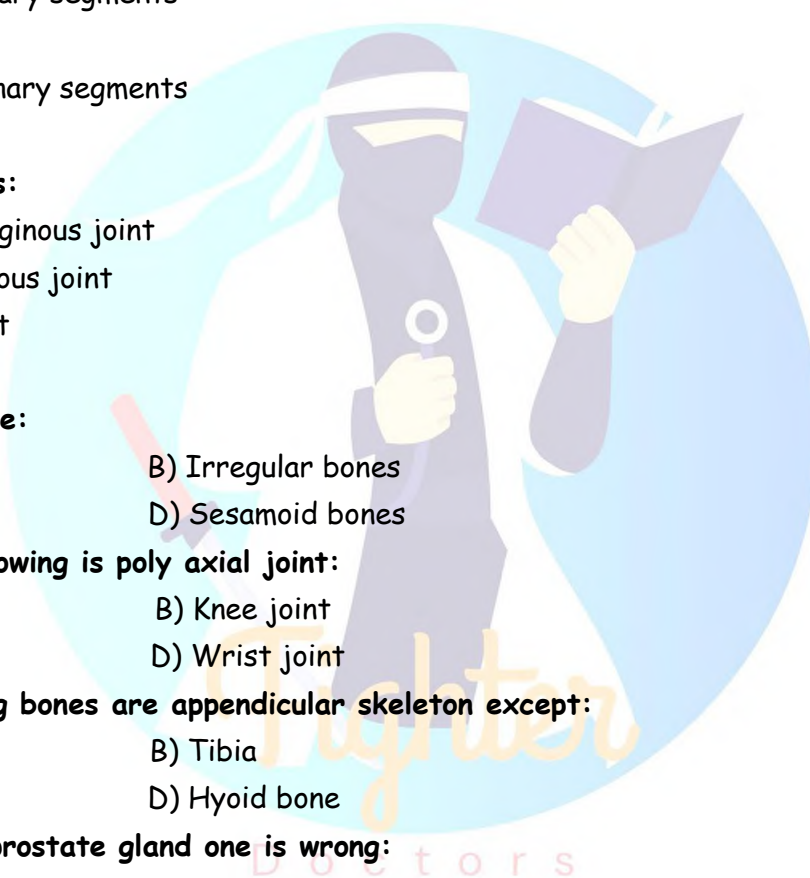
- A) It surrounds the proximal part of the urethra
- B) It secretes an alkaline fluid
- C) It lies in the scrotum
- D) The ejaculatory duct traverse it to open in the urethra

**14. Regarding medium veins one is wrong:**

- A) They drain venous plexus
- B) Most of them have valves
- C) They accompany medium arteries
- D) Superior and inferior vena cava are medium veins

**15. One of the following is lymphoid organ**

- A) Liver





B) Pancrease

C) Spleen

D) Lung

**16. The main lymphatic vessels in the body:**

A) Thoracic duct

B) Right lymphatic duct

C) Cystic duct

D) Both (a) and (b )

**17. The organ which produce male sex hormones:**

A) Testis

B) Seminal vesicle

C) Prostate

D) Bulbourethral gland

**18. One of the following arteries supplies the head and neck:**

A) Arch of aorta

B) Brachial artery

C) Common carotid artery

D) Popliteal artery

**19. With the heart in its normal position in the body:**

A) The left ventricle forms most of the anterior surface.

B) The right border is formed by right ventricle.

C) No part of the left ventricle can be seen on the diaphragmatic surface.

D) The apex lies in the 5th left intercostals space at midclavicular line

**20. The largest portion of the sternocostal surface of the heart is formed by:**

A) Left atrium

B) Right atrium

C) Right ventricle

D) Left ventricle

**21. Which group of structures empties directly into the right atrium?**

A) Superior vena cava, coronary sinus and hemiazygos vein

B) Coronary sinus and pulmonary vein

C) Pulmonary and bronchial veins

D) Superior and inferior venae cavae and coronary sinus

**22. The anterior interventricular artery is accompanied by:**

A) Anterior cardiac vein.

B) Great cardiac vein

C) Coronary sinus

D) Middle cardiac vein

**23. Which is the largest branch of the right coronary artery?**

A) The anterior interventricular artery

B) The nodal artery

C) The posterior interventricular artery.

D) The right marginal artery

**24. The vein that accompanies circumflex artery of the heart and ends in right atrium is:**

A) Great cardiac vein.

B) Middle cardiac vein.



C) Coronary sinus.

D) Anterior cardiac vein

**25. The component of conductive system that is called the pacemaker of the heart**

A) sino-atrial node .

B) Atrioventricular node.

C) Atrioventricular bundle.

D) Purkinje fibers

**26. Regarding the divisions of the brain, one is wrong:**

A) Cerebrum: formed of two cerebral hemispheres

B) Diencephalon: formed of thalamus, hypothalamus epithalamus and subthalamus

C) Cerebellum: responsible for the muscle coordination and equilibrium

D) Brain stem: formed of medulla oblongata and spinal cord

**27. Chamber of the heart that forms the apex and most of the left border is :**

A) Right ventricle.

B) Right atrium.

C) Left ventricle.

D) Left atrium .

**28. Chamber of the heart that forms the right border of the heart**

A) Right ventricle.

B) Right atrium.

C) Left ventricle.

D) Left atrium

**29. As regard the right coronary artery, one is false:**

A) Is branch of the ascending aorta

B) Is one of blood supply of the heart

C) Carry deoxygenated blood

**30. Regarding the sympathetic system, select one correct statement:**

A) It has cranio-sacral outflow.

B) It has no distribution to the limbs.

C) It causes vasoconstriction of blood vessels.

D) It causes slowness of the heart.

**33. the axial skeleton is formed of the following bone except**

A) Vertebral column

B) Femur

C) Skull

D) Sternum

**34. regarding the vertebral column. All of the followings are true except**

A) Thoracic vertebrae are 7 in number

B) It belongs to the axial skeleton

C) Sacral vertebrae are fused

D) Cervical vertebrae are 7 in number

**35. the following is an example of pneumatic bone**

A) Humerus

B) skull

C) Tibia

D) sternum

**36. concerning the bones, all the following are true except**

A) Long bones of the limb are developed be membranous ossification



- B) A sesamoid bone is a small nodule inside the tendon  
 C) Growth which takes place in the epiphyseal plate is responsible for the increase in the length of the bone
- 37. the following is a fibrous joint in which the margins are serrated**  
 A) Syndesmosis  
 B) Sutures  
 C) secondary cartilaginous joint  
 D) gomphosis
- 38. the intervertebral disc is**  
 A) Primary cartilaginous  
 B) Secondary cartilaginous  
 C) Fibrous joint  
 D) Synovial joint
- 39. the followings are synovial joint except**  
 A) Hip joint  
 B) Inferior tibiofibular joint  
 C) Wrist joint  
 D) Elbow joint  
 E) Shoulder joint
- 40. the followings are true about the joint except**  
 A) Synovial joint contains fluid  
 B) Sutures between skull are examples of fibrous joint  
 C) Secondary cartilaginous joint disappears by age  
 D) Syndesmosis is present between the lower ends of tibia and fibula
- 41. all of the following about the muscles are true except**  
 A) The smooth muscles are present in the wall of blood vessels  
 B) The skeletal muscles are attached to the skeleton  
 C) The cardiac muscle is present only in the heart  
 D) The cardiac muscle is supplied by somatic nerves
- 42. the part of the brain responsible for equilibrium**  
 A) Cerebrum  
 B) Brain stem  
 C) Cerebellum
- 43. the brain stem is formed of**  
 A) Medulla oblongata  
 B) Midbrain, medulla oblongata and pons  
 C) Cerebellum  
 D) Pons and medulla

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-A  | 3-B  | 4-A  | 5-C  | 6-A  | 7-A  | 8-C  | 9-A  | 10-C |
| 11-A | 12-D | 13-C | 14-D | 15-C | 16-D | 17-A | 18-C | 19-D | 20-C |
| 21-D | 22-B | 23-D | 24-C | 25-A | 26-D | 27-C | 28-B | 29-C | 30-C |



|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 33-B | 34-A | 35-B | 36-A | 37-B | 38-B | 39-B | 40-C | 41-D | 42-C |
| 43-B |      |      |      |      |      |      |      |      |      |

Cross match the following:

| Structure   | Type           |
|-------------|----------------|
| a-scapula   | Long bone      |
| b-humerus   | Short bone     |
| c-skull     | Irregular bone |
| d-patella   | Flat bone      |
| e-phalanx   | Pneumatic bone |
| f-vertebrae | Sesamoid bone  |

2-About the heart:

| Column "A"        | Column "B"                                |
|-------------------|-------------------------------------------|
| 1.Right atrium    | A- 1/3 diaphragmatic surface of the heart |
| 2.Right ventricle | B-All the right border of the heart       |
| 3.Left atrium     | C- Apex of the heart                      |
| 4.Left ventricle  | D- Most of the base of the heart          |
|                   | E- All the inferior border of the heart   |

3-Joint / Type:

| Joint                   | Type                    |
|-------------------------|-------------------------|
| a-intervertebral disc   | Polyaxial synovial      |
| b-shoulder joint        | Hinge synovial          |
| c-interosseous membrane | Pivot synovial          |
| d-intercarpal joint     | Fibrous joint           |
| e-atlanto axial joint   | Plane synovial          |
| f-Epiphyseal plate      | Secondary cartilagenous |
|                         | Primary cartilagenous   |



## Anatomy Final Revision - First Year Dentistry (MCQ)

**1. The ulna shares in the following joints, EXCEPT:**

- A) The elbow joint.
- B) The superior radio-ulnar joint.
- C) The wrist joint.
- D) The inferior radio-ulnar joint.

**2. The type of joint between the bodies of the vertebrae is:**

- A) Synovial joint.
- B) Fibrous joint.
- C) Primary cartilaginous joint.
- D) Secondary cartilaginous joint.

**3. The artery that supplies the bone marrow is:**

- A) Periosteal artery.
- B) Epiphyseal artery.
- C) Metaphyseal artery.
- D) Nutrient artery.

**4. Pneumatic bones are present in the:**

- A) Upper limb.
- B) Skull.
- C) Lower limb.
- D) Thorax.

**5. Vertebrae are considered as:**

- A) Long bones.
- B) Flat bones.
- C) Short bones.
- D) Irregular bones.

**6. The joints that move freely are:**

- A) Fibrous joints.
- B) Primary cartilaginous joints.
- C) Synovial joints.
- D) Secondary cartilaginous joints.

**7. The skeletal muscle action includes all of the following, EXCEPT:**

- A) Prime mover.
- B) Peristalsis.
- C) Fixator.
- D) Antagonistic.

**8. The following are long bones, EXCEPT:**

- A) Humerus.
- B) Scapula.
- C) Femur.
- D) Tibia.

**9. The following are long bones, EXCEPT:**

- A) Metacarpal bones.



Doctors



B) Phalanges.

C) Vertebrae.

D) Metatarsal bones.

**10. The axial skeleton includes all of the following, EXCEPT:**

A) The skull.

B) The clavicle.

C) The sternum.

D) The vertebra.

**11. Movement of a limb away from the midline in the coronal plane is:**

A) Adduction.

B) Abduction.

C) Flexion.

D) Extension.

**12. Regarding skeletal muscle, the following statements are correct, EXCEPT:**

A) They are voluntary muscles.

B) They have two or more attachments.

C) A prime mover is the muscle responsible for a particular movement.

D) A fixator muscle is the muscle that opposes the action of a flexor muscle.

**13. Regarding the long bones, the following statements are correct, EXCEPT:**

A) They have long tubular shafts.

B) The diaphysis contains a central medullary cavity.

C) The articular surfaces are covered by hyaline cartilage.

D) The shafts contain red bone marrow at puberty.

**14. Regarding the sesamoid bones, mark ONE CORRECT, answer:**

A) The carpal bones.

B) The patella.

C) The metacarpal bones.

D) The tarsal bones.

**15. Regarding the different types of joints, the following statements are correct EXCEPT:**

A) The sutures between bones of the skull form fibrous joints.

B) The joints between vertebral bodies are of synovial type.

C) In the synovial joints the articular surfaces are covered with hyaline cartilage.

D) The ligaments and tone of muscle around the joints play an important role in the stability of joints.

**16. The following are characteristics of the synovial joints, EXCEPT:**

A) The articular surfaces of bones are covered by hyaline cartilage.

B) Permit a great degree of freedom of movement.

C) All synovial joints have intra articular ligaments.

D) All synovial joints have fibrous capsule.

**17. The ellipsoid joint allows all of the following movements, EXCEPT:**

A) Abduction.

B) Rotation.

C) Flexion.

D) Extension.



**18. Regarding skeletal muscle, the following statements are correct, EXCEPT:**

- A) They are voluntary muscles.
- B) It is supplied by autonomic nerves.
- C) Maintain the body posture.
- D) Striated with peripheral nuclei.

**19. Regarding the classification of synovial joints, mark ONE WRONG answer:**

- A) The hinge joints.
- B) The ellipsoid joints.
- C) The sutural joints.
- D) The ball and socket joints.

**20. The bones of the forearm are:**

- A) Tibia and fibula.
- B) Ulna and radius.
- C) Humerus and radius.
- D) Clavicle and scapula.

**21. The joint between the tooth and its socket is:**

- A) Sutural joint.
- B) Ball and socket.
- C) Syndesmosis.
- D) Gomphosis.

**22. The intervertebral disc is considered as:**

- A) Fibrous joint.
- B) Primary cartilaginous joint.
- C) Secondary cartilaginous joint.
- D) Synovial joint.

**23. Regarding the varieties of the muscular tissue, the following are correct EXCEPT:**

- A) Cardiac muscle.
- B) Aponurosis.
- C) Skeletal muscles.
- D) Smooth muscles.

**24. The axial skeleton includes all of the following, EXCEPT:**

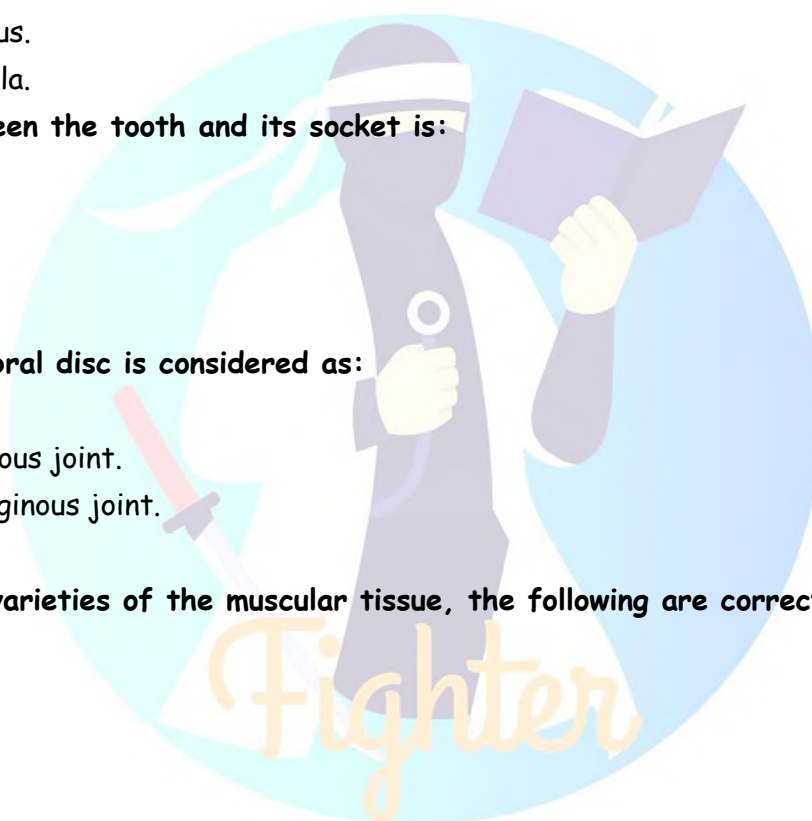
- A) The skull.
- B) The sternum.
- C) The clavicle.
- D) The vertebra.

**25. The hip bone is considered as:**

- A) Sesamoid bone.
- B) Short bone.
- C) Flat bone.
- D) Irregular bone.

**26. The lateral bone of the leg is:**

- A) Tibia.
- B) Fibula.





C) Ulna.

D) Radius.

**27. Pronation and supination of the forearm occurs at:**

A) Elbow joint.

B) Superior and inferior radio-ulnar joints.

C) Shoulder joint.

D) Radio-carpal joint.

**28. The actions of the skeletal muscle include the following, EXCEPT:**

A) Fixator.

B) Antagonist.

C) Synergist.

D) Peristalsis.

**29. The medial bone of the forearm is:**

A) Tibia.

B) Fibula.

C) Ulna.

D) Radius.

**30. The radius shares in the following joints, EXCEPT:**

A) Elbow joint.

B) Shoulder joint.

C) Superior radio-ulnar joint.

D) Inferior radio-ulnar joint.

**31. The sternum include the following parts, EXCEPT:**

A) Manubrium.

B) Body.

C) Transverse process.

D) Xiphoid process.

**32. The fibula shares in the following joints ,EXCEPT:**

A) Ankle joint.

B) Knee joint.

C) Superior tibiofibular joint.

D) Inferior tibiofibular joint.

**33. The sutures of the skull are considered as:**

A) Fibrous joints

B) Primary cartilaginous joints

C) Secondary cartilaginous joints

D) Synovial joints

**34. Which one of the following is a flat bone?**

A) Clavicle.

B) Humerus.

C) Scapula.

D) Vertebra.

**Answer Key**





|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-C  | 2-D  | 3-D  | 4-B  | 5-D  | 6-C  | 7-B  | 8-B  | 9-C  | 10-B |
| 11-B | 12-D | 13-D | 14-B | 15-B | 16-C | 17-B | 18-B | 19-C | 20-B |
| 21-D | 22-C | 23-B | 24-C | 25-D | 26-B | 27-B | 28-D | 29-C | 30-B |
| 31-C | 32-B | 33-A | 34-C |      |      |      |      |      |      |

**General anatomy in the previous examination - Give the definition of:**

1. Coronal plane
2. Eversion
3. Flexion
4. Horizontal section
5. Joint

**Write short note about:**

1. Characters of synovial joints
2. Functions of the skeleton
3. Shapes of the bone
4. Types of fibrous joints
5. Arterial supply of long bone
6. Differentiate between types of muscles
7. Forms of the skeletal muscles
8. Function of the muscle tone
9. Bone marrow in different stage of life





First year | Anatomy Department | Time: 1 hour | 29-11-2016 | Total marks: 15

**QUESTION (1) (5 MARKS, 1/2 marks for each space): Fill in the spaces:**

- a. Cardiac muscle is found in ..... and supplied by .....
- b. Bones containing air spaces are called .....
- c. .... is a structure essential for bone growth in length.
- d. The muscle which fibers converge from the origin to the insertion is called .....
- e. Elbow joint is ..... joint while symphysis pubis is ..... joint.
- f. The vertical plane dividing the body into anterior and posterior halves is called .....
- g. Forms of bone are ..... and .....

**QUESTION (2) (5 MARKS, 1/2 marks for each): Give a definition of:**

- a) Supination
- b) Adduction
- c) Ligament
- d) Hilton's law
- e) Fixator
- f) Suture
- g) Periosteal arteries
- h) Membranous ossification
- i) Primary cartilaginous joint
- j) Horizontal plane

**QUESTION (3) (5 MARKS, 1/2 marks for each): Choose the correct answer:**

**1. Which one of the following is a flat bone?:**

- A) Clavicle
- B) Humerus
- C) Scapula
- D) Vertebra

**2. One of the following is poly axial joint:**

- A) Shoulder joint
- B) Knee joint
- C) Elbow joint
- D) Wrist joint

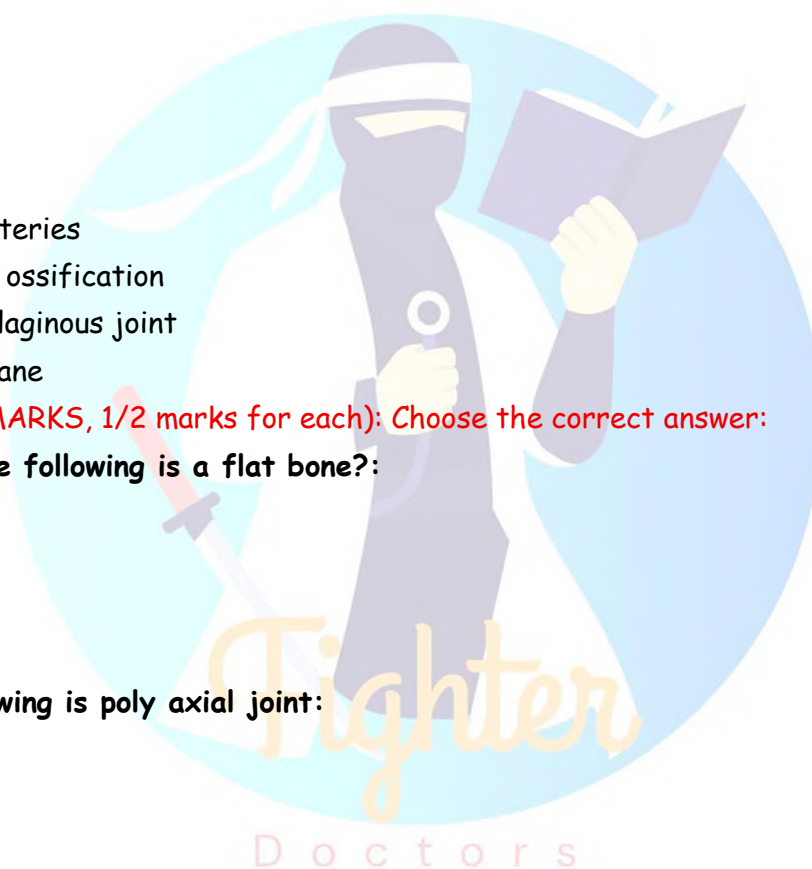
**3. Any muscle oppose the action of prime mover is named:**

- A) Antagonist
- B) Action of paradox
- C) Synergist
- D) Fixator

**4. Regarding the different types of joints, the following statements are correct EXCEPT:**

- A) The sutures between bones of the skull form fibrous joints.
- B) The joints between vertebral bodies are of synovial type.
- C) In the synovial joints the articular surfaces are covered with hyaline cartilage.
- D) The ligaments and tone of muscle around the joints play an important role in the stability of joints

**5. Regarding the smooth muscle, all of the following is true except:**





- A) consists of long spindle shaped cells
- B) controlled by autonomic nervous system
- C) have the property of spontaneous and rhythmic contraction
- D) found in the wall of the viscera and blood vessels

**6. Regarding the characters of the cartilage one is incorrect:**

- A) It is firm in consistency
- B) It is vascularised
- C) It form the temporary skeleton of the fetus
- D) It retained throughout the life in air passage

**7. One of the following about cardiac muscle is true :**

- A) It consists of non striated muscle fibers that branch
- B) It is voluntary muscle
- C) It has property for spontaneous contraction
- D) It has peripheral nucleus

**8. The following is true about the compact bone:**

- A) It form the interior of all bones
- B) It consists of numbers of Haversian system
- C) It is formed of trabeculae
- D) It is spongy

**9. Any muscle acts as anti gravity muscle is named:**

- A) Antagonist
- B) Action of paradox
- C) Synergist
- D) Fixator

**10. Palm of the hand means:**

- A) the sole of the foot
- B) the anterior surface of the hand
- C) the posterior surface of the hand
- D) the posterior surface of the foot

**Answer Key**

|     |     |     |     |     |     |     |     |     |      |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| 1-C | 2-A | 3-A | 4-B | 5-C | 6-B | 7-C | 8-B | 9-B | 10-B |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|





## Dr Rasha - Anatomy Question Bank for Dental Students

Source: *بنك د رشا محلول / بنك د رشا اسنان* (Dr Rasha Abdel Kader Elsisy, 1 January 2024) | same bank; answers taken from Dr Rasha's own green highlights in the solved copy

**General anatomy**

Note: Question numbers re-sequenced 1-89 (the source repeats number 32). Q54 and Q67 (source Q53 and Q66) were not highlighted in the source; answers added.

**1. One of the Irregular bones is :**

- A) Hip bone
- B) femur
- C) rib
- D) tarsal bone

**2. The axial skeleton includes one of the following:**

- A) The skull
- B) The clavicle
- C) The hip bone .
- D) The phalanges.

**3. patella is :**

- A) sesamoid bone
- B) short bone
- C) pneumatic bone
- D) irregular bone

**4. Regarding the yellow bone marrow , one statement is true :**

- A) It is present in the shaft of long bones in the adult
- B) It is present in the vertebra, skull and ribs in the adult
- C) It is concerned with the formation of blood cells
- D) The sternum is suitable for bone marrow aspiration

**5. In the triangular muscle :**

- A) the muscle fibers slope into the two sides of the central tendon
- B) The muscle fibers gaining two attachment sites
- C) the entire fibers slope into one side of the tendon
- D) The fibers converge from the origin to the insertion

**6. Regarding the characters of the cartilage one is incorrect:**

- A) it is firm in consistency
- B) it is vascularized
- C) it retained throughout the life in the ear
- D) it form the temporary skeleton of the fetus

**7. The anatomical position of the human body include the following except :**

- A) The person stands in erect position
- B) The palm faces anterior
- C) The arms are adjacent to the body
- D) The lower limbs are abducted

**8. connective tissue which connect the bone with bone is called :**

- A) tendon



- B) ligament
- C) aponeurosis
- D) fascia

**9. The carpal bones are:**

- A) Small long bones
- B) Irregular bones
- C) Short bones
- D) Sesamoid bones

**10. Any muscle has antigravity muscle is named:**

- A) antagonist
- B) action of paradox
- C) synergist
- D) fixator

**11. which of the following is the character of cardiac muscle ?:**

- A) It consists of long spindle shaped cells
- B) It has voluntary control
- C) have the property of spontaneous and rhythmic contraction
- D) found in the wall of the viscera and blood vessels

**12. Which is true about the parasagittal plane?**

- A) It divides the body into equal upper and lower halves
- B) It divides the body into un equal anterior and posterior halves
- C) It divides the body into equal right and left halves
- D) It divides the body into un equal right and left halves

**13. The term lateral means:**

- A) Toward the surface
- B) Near the attachment point
- C) On the same side
- D) Away from the midline

**14. Towards the front of the body means:**

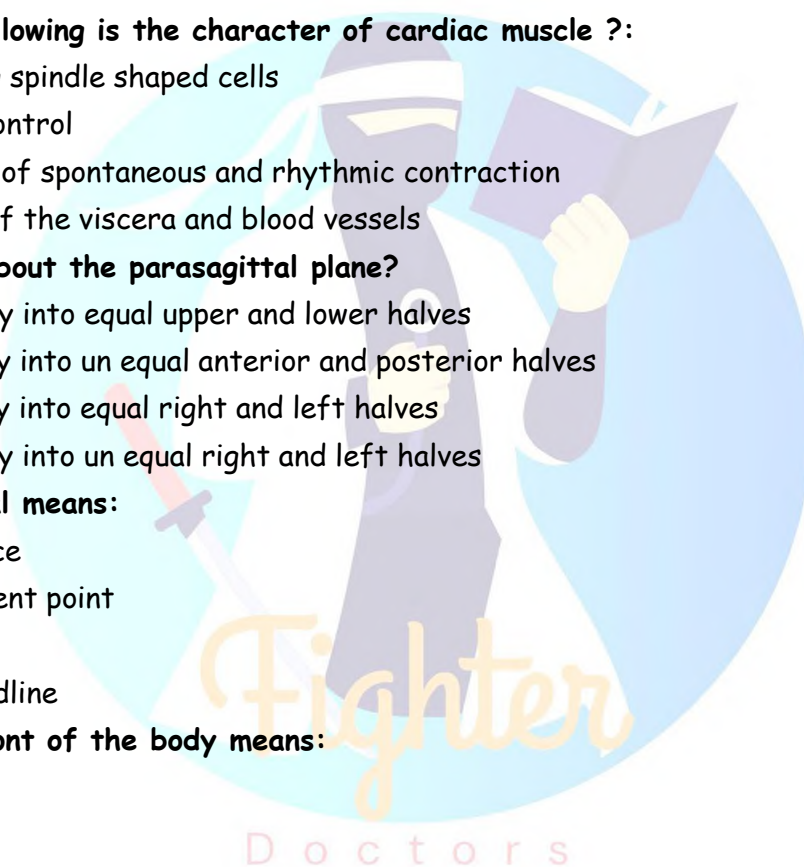
- A) Dorsal
- B) superior
- C) ventral
- D) inferior

**15. Which of the following terms means near the upper end?**

- A) caudal
- B) cranial
- C) superficial
- D) ventral

**16. Movement of a part of the body around its long axis means:**

- A) flexion
- B) extension
- C) rotation
- D) circumduction





**17. Elevation of the joint means:**

- A) move apart superiorly
- B) move the part forward
- C) moving the part anteriorly
- D) moving the part inferiorly

**18. Why the bones of the newborn are soft and flexible ?:**

- A) they contain low calcium
- B) they are composed mainly of cartilage
- C) they have less arteries
- D) they are composed mainly of spongy bone

**19. All the following cartilage remain in the adult except:**

- A) articular surfaces
- B) Costal cartilage
- C) growing plate
- D) nasal cartilage

**20. According to the vertebral column one statement is true:**

- A) It consists of 30 vertebrae
- B) It has 8 cervical vertebrae in the neck
- C) It has 12 thoracic vertebrae
- D) It has 5 sacral segments which are separates

**21. the last two ribs are :**

- A) floating ribs
- B) true ribs
- C) false ribs
- D) typical ribs

**22. Which muscle can eliminate unwanted movement on the proximal joint while the main muscle act on distal joint ?**

- A) antagonist
- B) action of paradox
- C) synergist
- D) fixator

**23. which of the following muscles has striated long fibers with voluntary control ?**

- A) muscles of the heart
- B) muscles of the upper limbs
- C) muscles of blood vessels
- D) muscles of the intestine

**24. The type of joint between the bodies of the vertebrae is:**

- A) Synovial joint.
- B) Fibrous joint.
- C) Primary cartilaginous joint.
- D) Secondary cartilaginous joint.

**25. The artery that supplies the outer compact bone is:**

- A) Periosteal artery.





- B) Epiphyseal artery.
- C) Metaphyseal artery.
- D) Nutrient artery.

**26. The joints that move freely are:**

- A) Fibrous joints.
- B) Primary cartilaginous joints.
- C) Synovial joints.
- D) Secondary cartilaginous joints.

**27. The joint between the tooth and its socket is:**

- A) Sutural joint.
- B) Ball and socket.
- C) Syndesmosis.
- D) Gomphosis

**28. The sutures of the skull are considered as:**

- A) Fibrous joints
- B) Primary cartilaginous joints
- C) Secondary cartilaginous joints
- D) Synovial joints

**29. Symphysis pubis is:**

- A) Secondary cartilaginous joint
- B) Primary cartilaginous joint
- C) Plane synovial joint
- D) Fibrous joint

**30. One of the following is poly axial joint:**

- A) Shoulder joint
- B) Knee joint
- C) Elbow joint
- D) Wrist joint

**31. the visceral muscles are :**

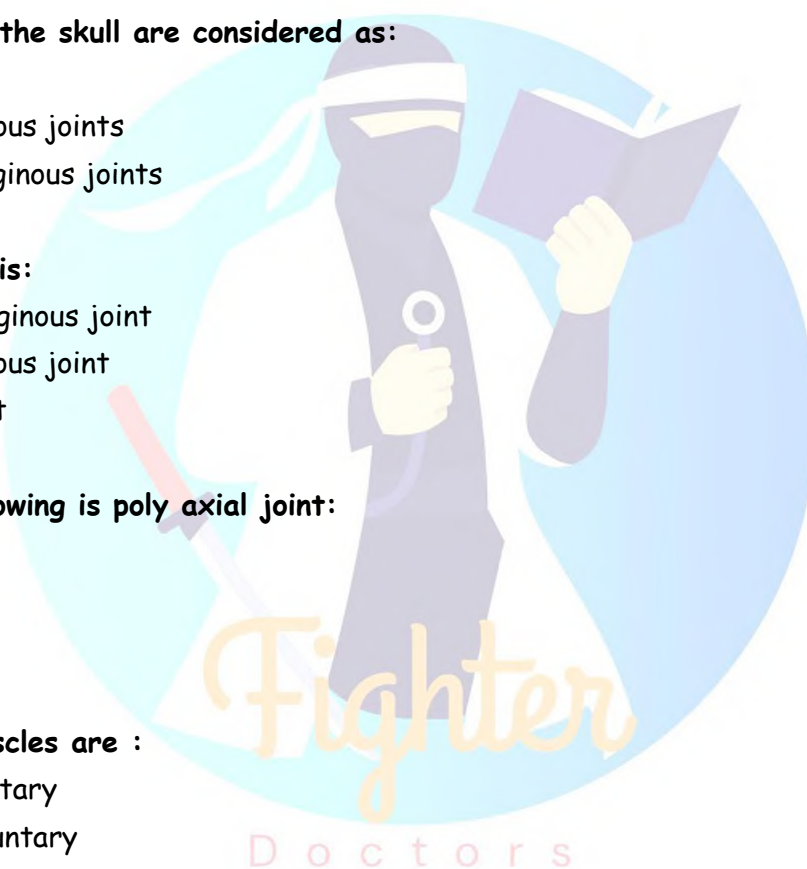
- A) striated and voluntary
- B) striated and involuntary
- C) non striated and involuntary
- D) non striated and involuntary

**32. moving the apart away from the mid line is called.....**

- A) flexion
- B) extension
- C) abduction
- D) adduction

**33. the muscle fibers which connected by intercalated disc is present in.... :**

- A) wall of the heart
- B) body wall
- C) wall of blood vessels
- D) upper limb





**34. Types of joints are classified on basis of**

- A) number of cartilages
- B) degree of movement
- C) weight strength
- D) number of bones

**35. Location or a point where two bones contact each other is called**

- A) joint
- B) rib
- C) vertebrae
- D) pectoral girdle

**36. Hip and shoulder joints allow movements in**

- A) One direction
- B) Two directions
- C) Three directions
- D) All directions

**37. which of the following muscles contracting to give the working muscles a firm base?**

- A) prime mover
- B) antagonist
- C) fixator
- D) synergist

**38. part of muscle attached with some movable bone is called**

- A) Origin
- B) Moveable muscles
- C) Insertion
- D) Muscle belly

**39. the scapula is .....:**

- A) long bone
- B) short bone
- C) flat bone
- D) sesamoid bone

**40. compact bone is characterized by:**

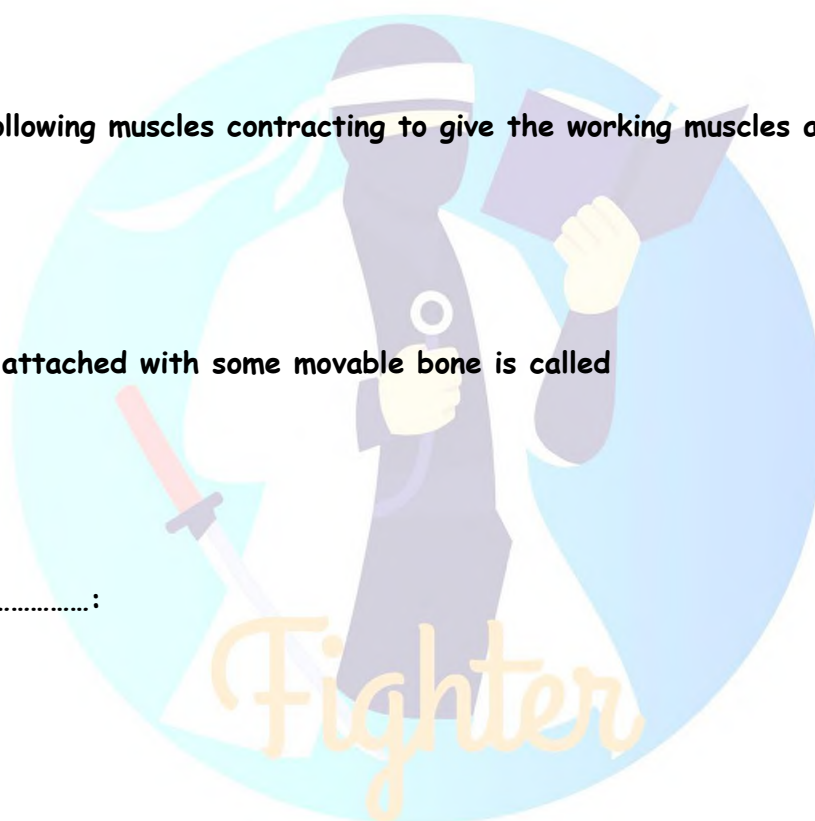
- A) It looks like a sponge which is usually filled with red bone marrow.
- B) It is formed of trabeculae , the lamellae are branching and anastomosing with each other
- C) It is found inside the hard bone
- D) It provides strength for weight bearing.

**41. straightening of the joint is called.....:**

- A) flexion
- B) extension
- C) adduction
- D) abduction

**42. The type of joint between the bodies of the vertebrae is:**

- A) Synovial joint.
- B) Fibrous joint.





C) Primary cartilaginous joint.

D) Secondary cartilaginous joint.

**43. The joints that move freely are:**

A) Fibrous joints.

B) Primary cartilaginous joints.

C) Synovial joints.

D) Secondary cartilaginous joints.

**44. The joint between the tooth and its socket is:**

A) Sutural joint.

B) Ball and socket.

C) Syndesmosis.

D) Gomphosis

**45. One of the following is poly axial joint:**

A) Shoulder joint

B) Knee joint

C) Elbow joint

D) d- Wrist joint

**46. the following is a fibrous joint in which the margins are serrated**

A) Syndesmosis

B) secondary cartilagenous joint

C) Sutures

D) Gomphosis

**47. Hip and shoulder joints allow movements in**

A) One direction

B) Two directions

C) Three directions

D) All directions

**48. One of the following is a primary cartilaginous joint..**

A) elbow joint

B) costochondral junction

C) syndesmosis

D) hip joint

**49. The sutures of the skull are considered as:**

A) Fibrous joints

B) Primary cartilaginous joints

C) Secondary cartilaginous joints

D) Synovial joints

**50. synovial fluid present in.....**

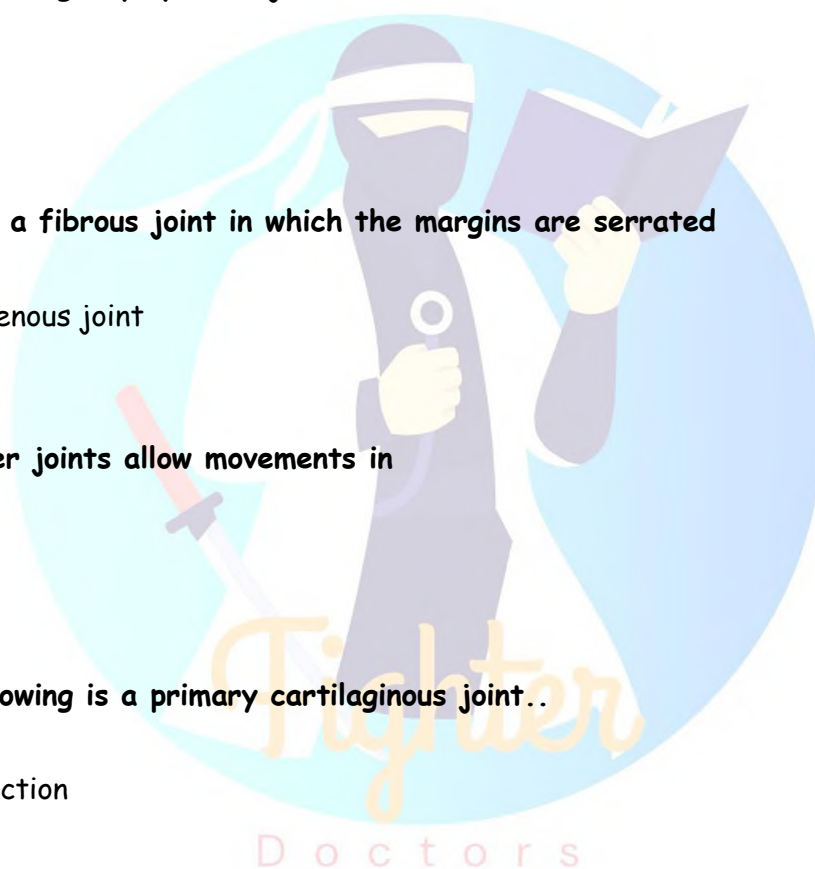
A) suture

B) gomphosis

C) symphysis pubis

D) elbow joint

**51. The following is true about the compact bone:**





- A) It form the interior of all bones
- B) It consists of numbers of Haversian system
- C) It is formed of trabeculae
- D) It is spongy

**52. Any muscle acts as anti gravity muscle is named:**

- A) Antagonist
- B) Action of paradox
- C) Synergist
- D) Fixator

**53. Planter surface means:**

- A) the anterior surface of the hand
- B) the posterior surface of the hand
- C) the posterior surface of the foot
- D) the sole of the foot

**54. The deep fascia is absent in:**

- A) abdomen
- B) palms
- C) leg
- D) Forearm

**55. The phalanges are:**

- A) Flat bones
- B) Irregular bones
- C) Long bones
- D) Short bones

**56. Which one of the following is irregular bone?:**

- A) Clavicle
- B) Humerus
- C) Scapula
- D) Vertebra

**57. Any muscle oppose the action of prime mover is named:**

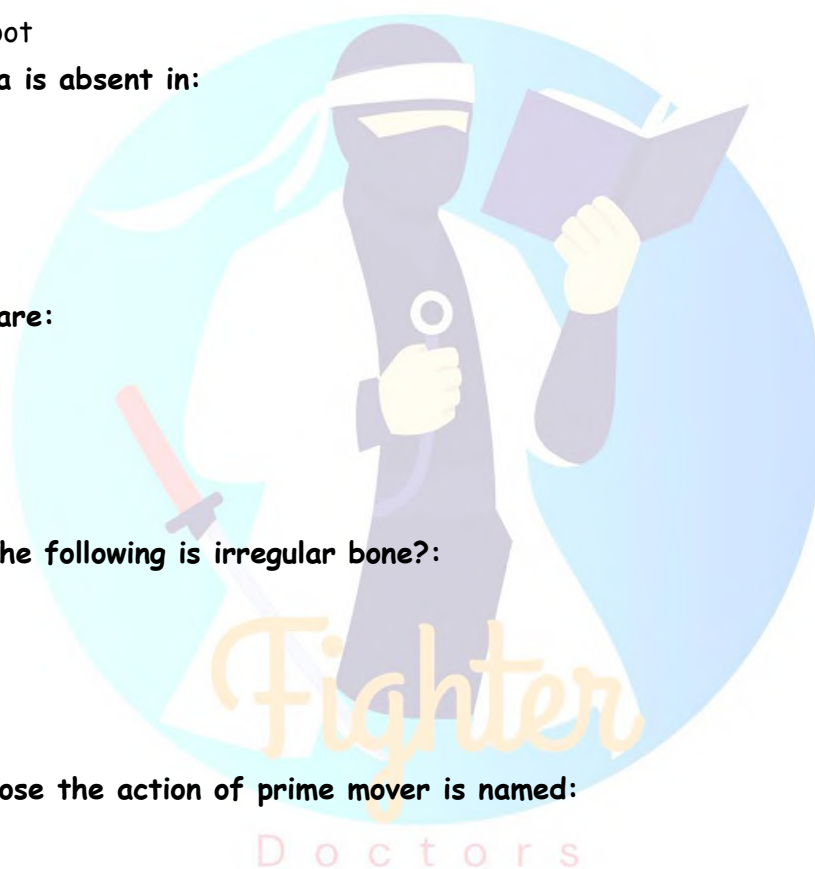
- A) Antagonist
- B) Action of paradox
- C) Synergist
- D) Fixator

**58. Regarding the smooth muscle, all of the following is true except:**

- A) consists of long spindle shaped cells
- B) controlled by autonomic nervous system
- C) has the property of spontaneous and rhythmic contraction
- D) found in the wall of the viscera and blood vessels

**59. Regarding the characters of the cartilage one is incorrect:**

- A) It is firm in consistency
- B) It is vascularised
- C) It form the temporary skeleton of the fetus





D) It retained throughout the life in air passage

**60. One of the following about cardiac muscle is true :**

A) It consists of non striated muscle fibers that branch

B) It is voluntary muscle

C) It has property for spontaneous contraction

D) It has peripheral nucleus

**61. Which one of the following is irregular bone?:**

A) Clavicle

B) Humerus

C) Scapula

D) Vertebra

**62. Any muscle oppose the action of prime mover is named:**

A) Antagonist

B) Synergist

C) Action of paradox

D) fixator

**63. Regarding the characters of the cartilage one is incorrect:**

A) It form the temporary skeleton of the fetus

B) It retained throughout the life in air passage

C) It is firm in consistency

D) It is vascularized

**64. The term which means near to the trunk is: Medial**

A) superficial

B) Proximal

C) distal

D)

**65. The following is true about the compact bone:**

A) It form the interior of all bones

B) It consists of numbers of Haversian system

C) It is formed of trabeculae

D) d. It is spongy

**66. Any muscle acts as anti gravity muscle is named:**

A) Antagonist

B) Synergist

C) Action of paradox

D) Fixator

**67. palmar surface means: the anterior surface of the hand**

A) the posterior surface of the foot

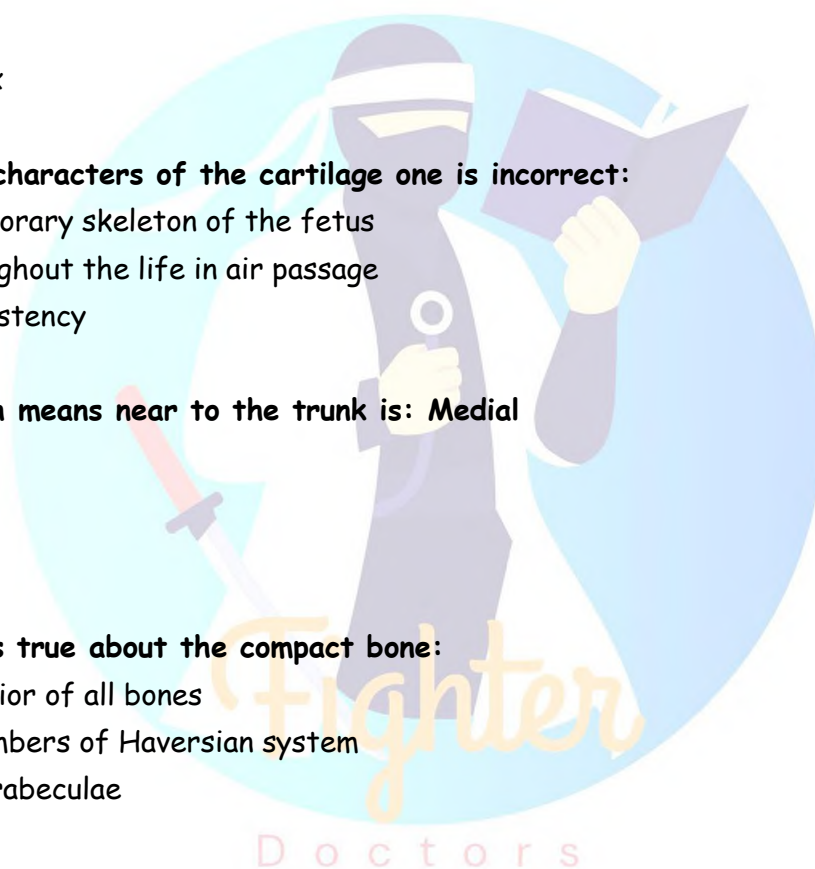
B) the posterior surface of the hand

C) the sole of the foot

D)

**68. The phalanges are: Flat bones**

A) Long bones





- B) Irregular bones
- C) Short bones
- D)

**69. Maxilla is :**

- A) Irregular bone
- B) flat bone
- C) pneumatic bone
- D) short bone

**70. Intervertebral disc is:**

- A) Secondary cartilaginous joint
- B) Primary cartilaginous joint
- C) Plane synovial joint
- D) Fibrous joint

**71. Regarding the different types of joints, the following statements are correct EXCEPT:**

- A) The sutures between bones of the skull form fibrous joints.
- B) The joints between vertebral bodies are of synovial type.
- C) In the synovial joints the articular surfaces are covered with hyaline cartilage.
- D) metacarpo-phalangeal joint is bi-axial synovial joint

**72. Regarding the characters of the cartilage one is incorrect:**

- A) It is firm in consistency
- B) It form the temporary skeleton of the fetus
- C) It retained throughout the life in air passage
- D) It is vascularised

**73. Superior radio-ulnar joint is :**

- A) Secondary cartilaginous joint
- B) Fibrous joint
- C) Plane synovial joint
- D) Pivot synovial joint

**74. The following is true about the compact bone:**

- A) It form the interior of all bones
- B) It consists of numbers of Haversian system
- C) It is formed of trabeculae
- D) It is spongy

**75. The term which means near to the trunk is:**

- A) Medial
- B) superficial
- C) Proximal
- D) distal

**76. Palmar surface means:**

- A) the sole of the foot
- B) the anterior surface of the hand
- C) the posterior surface of the hand
- D) the posterior surface of the foot



**77. Adduction of the fingers means:**

- A) Moving the finger away from the body
- B) Moving the finger away from the middle finger
- C) Moving the finger towards the middle finger
- D) Moving the finger towards the body

**78. Regarding the primary ossification center all are true except:**

- A) it appears in the diaphysis
- B) it form the shaft
- C) it appears during intrauterine life
- D) it form the epiphysis

**79. The muscle that perform a movement is called:**

- A) Prime mover
- B) Antagonist
- C) Fixator
- D) Synergic muscle

**80. Examples of bones classification according to their shape one is wrong:**

- A) Femur (long bone)
- B) Tarsal bones (irregular)
- C) Scapula (flat bone)
- D) Patella (sesamoid bone)
- E) Phalanges (short long)

**81. One of the following is not arrangement of muscle fibers:**

- A) Unipennate
- B) Multipennate
- C) Parallel
- D) Aponeurosis
- E) Strap

**82. Following are pneumatic (containing air) bones except:**

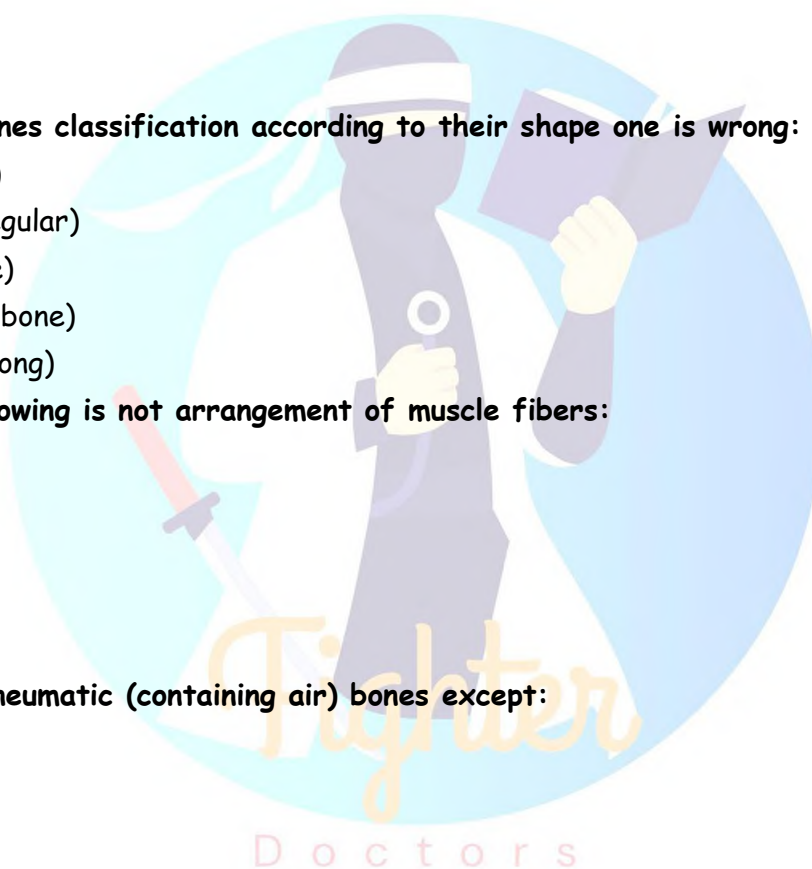
- A) Ethmoid
- B) Occipital
- C) Sphenoid
- D) Maxilla
- E) Frontal

**83. Regarding terms of movements one is wrong:**

- A) Flexion is forward movement except at knee joint
- B) Adduction is movement away from the midline
- C) Pivot joint is a uniaxial
- D) Shoulder joint is polyaxial

**84. Anatomical position means all the following except:**

- A) Body is erect
- B) Arms are adducted
- C) Palms of the hands are facing backwards
- D) Legs are close to each other





E) Face is directed forwards

**85. Regarding skeletal muscle action, following are true except:**

- A) Prime mover is a chief muscle responsible for particular movement
- B) Antagonist opposes the action of prime mover
- C) Fixator contracts isometrically
- D) Synergist relaxes to prevent unwanted movement
- E) All movements are the result of coordinated action of many muscles

**86. Following are characters of involuntary muscle, except:**

- A) Not controlled by will
- B) Under control of autonomic nervous system
- C) Attached to bones
- D) Their fibers are smooth, plane, not stripped

**87. Stability of joints depends on following except:**

- A) Shape of articular surfaces
- B) Ligaments
- C) Muscles around
- D) Synovial membrane
- E) Muscle tone

**88. Following are fibrous joints except:**

- A) Gomphosis
- B) Knee joint
- C) Interosseous membrane
- D) Sutures of the skull
- E) Inferior tibiofibular joint

**89. Regarding skeletal muscles, following are true except:**

- A) Attachment which moves the most is the insertion
- B) Its nerve supply is mainly sensory
- C) Degree of mobility between origin and insertion may be reversed
- D) Prime mover is the chief muscle responsible for a particular movement

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-A  | 2-A  | 3-A  | 4-A  | 5-D  | 6-B  | 7-D  | 8-B  | 9-C  | 10-D |
| 11-C | 12-D | 13-D | 14-C | 15-B | 16-C | 17-A | 18-B | 19-C | 20-C |
| 21-A | 22-C | 23-B | 24-D | 25-A | 26-C | 27-D | 28-A | 29-A | 30-A |
| 31-C | 32-C | 33-A | 34-B | 35-A | 36-D | 37-C | 38-C | 39-C | 40-D |
| 41-B | 42-D | 43-C | 44-D | 45-A | 46-C | 47-D | 48-B | 49-A | 50-D |
| 51-B | 52-B | 53-D | 54-A | 55-C | 56-D | 57-A | 58-C | 59-B | 60-C |
| 61-D | 62-A | 63-D | 64-B | 65-B | 66-C | 67-A | 68-A | 69-C | 70-A |
| 71-B | 72-D | 73-D | 74-B | 75-C | 76-B | 77-C | 78-D | 79-A | 80-E |
| 81-D | 82-B | 83-B | 84-C | 85-D | 86-C | 87-D | 88-B | 89-B |      |

**Scalp and face Quiz**

*Note: No highlighted answers exist in the source for this section; answers were solved. Q18 is cut off in the source (only option a is present).*

**1. Regarding Scalp, all are true Except:**

- A) Its wounds bleed profusely.
- B) Its wounds gape if the aponeurosis is cut.
- C) It has no sebaceous glands.
- D) Its veins communicate with the intracranial venous sinuses.
- E) It contains a muscle of facial expression.

**2. choose one correct statement regarding the scalp:**

- A) Its wounds gape if superficial to aponeurosis.
- B) Its blood vessels and nerves lie in the loose areolar connective tissue layer.
- C) Its wounds do not bleed easily.
- D) Bleeding beneath the periosteum usually extends under its whole extent.
- E) Its infections can spread through emissary veins to cranial cavity.

**3. Regarding the scalp the following statements are true Except: -**

- A) Infection from scalp wounds can spread to superior sagittal sinus.
- B) Its blood vessels run in the loose subaponeurotic space.
- C) Its skin, superficial fascia and aponeurosis move as one layer.
- D) Frontal bellies of occipito-frontalis can corrugate skin of forehead.
- E) Occipital bellies are supplied by facial nerve.

**4. The plane of movement of the scalp is between:**

- A) Skin and epicranial aponeurosis.
- B) Epicranial aponeurosis (galea aponeurotica) and pericranium.
- C) Skin and subcutaneous
- D) Pericranium and skull.
- E) None or the above.

**5. Bleeding beneath the pericranium of the scalp:**

- A) Spreads underneath all of the scalp.
- B) Makes a swelling that takes the shape of the underlying bone.
- C) Reaches to the upper eyelids.
- D) Reaches to the back of neck.
- E) Reaches the extradural space.

**6. All are true regarding orbicularis oculi Except:**

- A) Orbital part closes eye tightly.
- B) Palpebral part closes eye gently.
- C) Lacrimal part helps flow of tears.
- D) Palpebral part lies in the eyelid.
- E) Orbital part has no bony attachment medially.

**7. All are true regarding Buccinator muscle Except:**

- A) Its middle fibers decussate.
- B) Helps to form orbicularis oris.
- C) Keeps food within the grinding force of the molars.



D) If paralysed food dribbles from corner of mouth.

E) Supplied by buccal branch of mandibular nerve.

**8. Buccinator muscle is supplied by which nerve:**

A) Infraorbital.

B) Facial.

C) Mental.

D) Great auricular.

E) Auriculotemporal.

**9. Choose one correct combination: of a muscle a muscle and its action:**

A) Lateral pterygoid ----- opens mouth.

B) Medial pterygoid ----- depresses mandible.

C) Masseter ----- retracts mandible.

D) Mylohyoid ----- elevates mandible.

E) Temporalis ----- protracts mandible

**10. All are true regarding the upper eyelid Except:**

A) Is strengthened by the tarsus.

B) Is lowered by orbicularis oculi.

C) - Is elevated by levator palpebrae superioris.

D) Is supplied by sensory fibers from lacrimal nerve medially.

E) Contains part of the lacrimal gland.

**11. which of the following layers of the scalp does infection tend to spread:**

A) Skin.

B) Connective tissue layer.

C) Aponeurotic layer

D) Loose areolar tissue layer.

E) Periosteal layer.

**12. The large blood vessels of the scalp found in the:**

A) Skin.

B) Subcutaneous tissue.

C) Epicranial aponeurosis.

D) Subaponeurotic tissue.

E) Pericranium.

**13. Regarding muscles of facial expression all are true Except:**

A) Levator labii superioris can elevate lower lip.

B) Risorius is a dilator of lips.

C) Procerus produces horizontal wrinkles at root of nose.

D) frontalis produces horizontal wrinkles on skin on forehead.

E) Corrugator supercilii produces vertical wrinkles in skin of forehead.

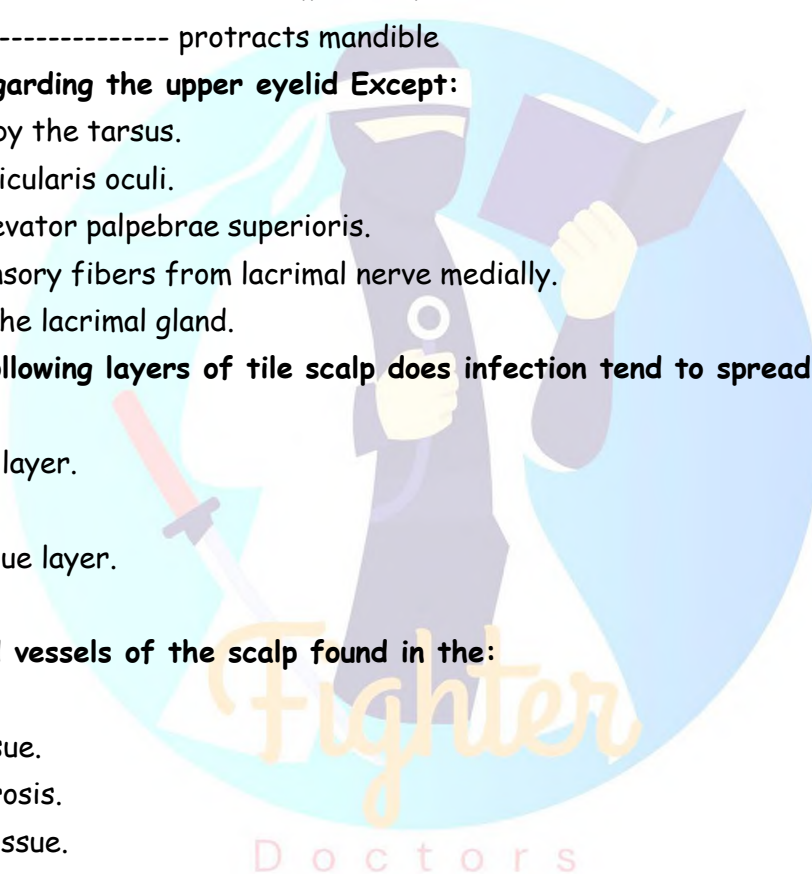
**14. all the following regarding orbicularis oris muscle is correct Except:**

A) It is a muscle of facial expression.

B) It is supplied by facial nerve.

C) Its fibers form complete circles around the mouth.

D) It can close the oral (labial) fissure.





E) It helps in speech.

### 15. Buccinator's

- A) Is one of the muscles of mastication.
- B) Is supplied by buccal branch of mandibular nerve.
- C) Arises from both mandible and maxilla.
- D) Is covered by investing layer of fascia.
- E) Accumulates food in vestibule of mouth.

### 16. Regarding buccinator's muscle, all the following statements are true Except:

- A) It arises from both maxilla and mandible.
- B) The pterygomandibular raphe is a common origin for this muscle and the middle constrictor.
- C) Its middle fibers decussate at the modiolus.
- D) It is covered by buccopharyngeal fascia.
- E) It compresses the cheeks against the teeth

### 17. Buccinator : -

- A) Is pierced by parotid duct.
- B) Lies at a more superficial plane than other muscles of facial expression.
- C) Arises from alveolar margin of mandible ,and maxilla only.
- D) Is covered by pharyngobasilar fascia.
- E) Its uppermost fibers decussate.

### Answer Key

|      |      |      |      |      |      |      |     |     |      |
|------|------|------|------|------|------|------|-----|-----|------|
| 1-C  | 2-E  | 3-B  | 4-B  | 5-B  | 6-E  | 7-E  | 8-B | 9-A | 10-D |
| 11-D | 12-B | 13-A | 14-C | 15-C | 16-B | 17-A |     |     |      |

### Dr Mona - General Embryology Questions

Source: General Embryology Questions by Dr Mona Abdel Atty Ahmed, Lecturer of Anatomy and Embryology | answers as printed under each question in the source

Note: Question numbers re-sequenced 1-32 (the source repeats number 14). Q31 and Q32 had no printed answer in the source; answers added.

### 1. The part of the sperm containing proteolytic enzymes to digest the zona pellucida is the:

- A) capicator
- B) head
- C) corona
- D) acrosome
- E) cumulus

### 2. The ovulated mammalian oocyte is arrested at:

- A) prophase of meiosis I
- B) metaphase of meiosis I
- C) prophase of meiosis II
- D) metaphase of meiosis II
- E) none of the above

### 3. The seven-day blastocyst:

- A) has a single layer of trophoblast at the embryonic pole
- B) has an amniotic cavity



- C) is attached to the endometrial epithelium
- D) is surrounded by a degenerating zona pellucida
- E) is called the hypoblast

**4. Haploid nuclei that fuse at fertilization are called:**

- A) homunculi
- B) mitotic figures
- C) centrioles
- D) nucleoli
- E) pronuclei

**5. The first week of human development is characterized by formation of the:**

- A) inner cell mass
- B) trophoblast
- C) blastocyst
- D) all of the above

**6. During implantation, the blastocyst:**

- A) implants in the endometrium
- B) usually attaches to endometrial epithelium at its embryonic pole
- C) usually implants in the posterior wall of the body of the uterus
- D) causes change in the endometrial tissues
- E) all of the above are correct

**7. Capacitation of the sperm:**

- A) is caused by the zona pellucida
- B) occurs in the male
- C) prevents polyspermy
- D) is essential for fertilization
- E) removes the head of the sperm

**8. The early stages of cleavage are characterized by:**

- A) formation of a hollow ball of cells
- B) formation of the zona pellucida
- C) increase in the size of the cells in the zygote
- D) increase in the number of cells in the zygote
- E) none of the above

**9. The most common site for implantation in ectopic pregnancy is:**

- A) internal os of the uterus
- B) mesentery
- C) ovary
- D) uterine tube
- E) other

**10. With the light microscope, the zona pellucida appears as a translucent membrane surrounding the:**

- A) primary oocyte
- B) zygote
- C) morula



- D) very early blastocyst
- E) all of the above are correct

**11. The amniotic cavity develops:**

- A) on the tenth day
- B) within the outer cell mass
- C) within the inner cell mass near the cytotrophoblast
- D) in extraembryonic mesoderm
- E) none of the above

**12. During the second week of development, the trophoblast differentiates into:**

- A) syncytiotrophoblast
- B) ectoderm
- C) intraembryonic mesoderm
- D) yolk sac (secondary)

**13. The first two intraembryonic germ layers to differentiate are the:**

- A) ectoderm and hypoblast
- B) epiblast and hypoblast
- C) ectoderm and endoderm
- D) ectoderm and mesoderm

**14. The blastocoele becomes the:**

- A) amniotic cavity
- B) extraembryonic coelom
- C) primary yolk sac
- D) chorionic cavity
- E) secondary cavity

**15. The bilaminar germ disc:**

- A) consists of epiblast and mesoblast
- B) is derived from the outer cells of the morula
- C) forms the embryo proper
- D) synthesizes human chorionic gonadotropin, HCG

**16. The primitive streak first appears at the beginning of the \_\_\_\_\_ week.**

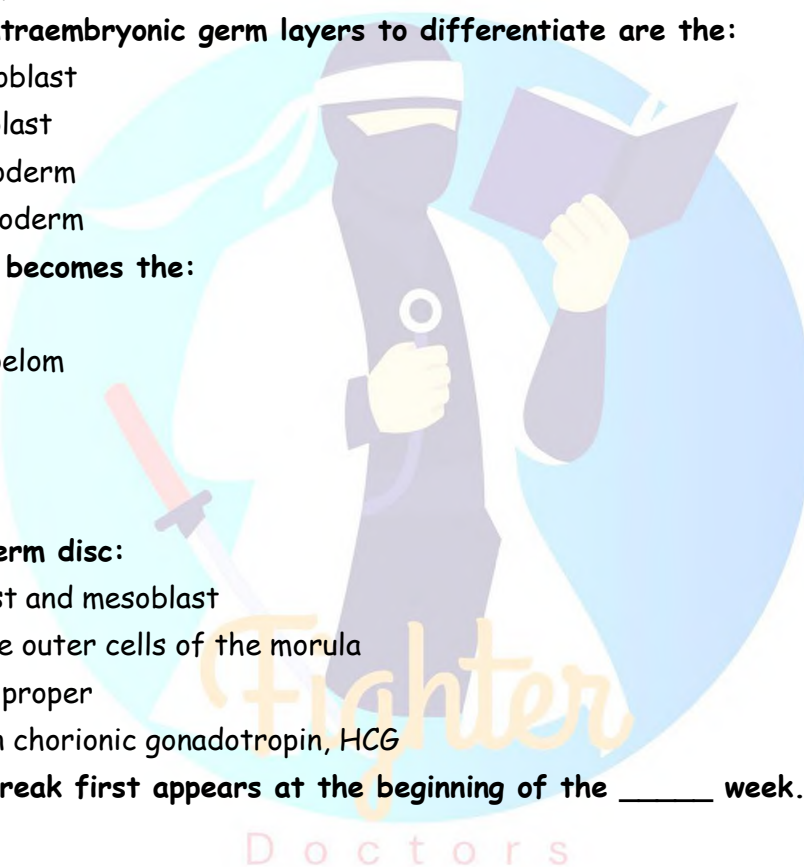
- A) first
- B) second
- C) third
- D) fourth
- E) fifth

**17. Cells from the primitive streak DO NOT become:**

- A) endoderm
- B) intermediate mesoderm
- C) paraxial mesoderm
- D) lateral plate mesoderm
- E) amnioblasts

**18. The primitive streak:**

- A) is derived from the outer cells of the morula





- B) is formed during the second week in development
- C) persists as the cloacal membrane
- D) is the site of involution of epiblast cells to form mesoderm

**19. In the third week of human embryonic development:**

- A) the amnion appears
- B) a bilaminar embryonic disc is formed
- C) the body stalk moves ventrally and joins with the yolk sac stalk to form the umbilical cord
- D) the neural plate is induced by the notochordal process and associated mesoderm
- E) the uteroplacental circulation is established

**20. During development, the notochordal process:**

- A) arises from involuting endodermal cells
- B) extends from the prochordal plate to the primitive node
- C) is involved in the induction of the primitive gut
- D) becomes the appendicular skeleton

**21. Somites:**

- A) differentiate into myotomes which give rise to skeletal muscle in trunk and limbs
- B) differentiate into sclerotomes which give rise to vertebrae
- C) arise from segmentation of the paraxial mesoderm
- D) differentiate into myotomes which give rise to skeletal muscle of the limbs
- E) all of the above are correct

**22. When the amount of amniotic fluid exceeds two liters, the condition is called:**

- A) oligohydramnios
- B) polyhydramnios or hydramnios
- C) amnionitis
- D) bag of waters
- E) hydrogravidia

**23. The wall of the chorionic sac is composed of:**

- A) cytotrophoblast and syncytiotrophoblast
- B) two layers of trophoblast lined by extraembryonic somatic mesoderm
- C) trophoblast and exocoelomic membrane
- D) extraembryonic splanchnic mesoderm and both layers of trophoblast
- E) none of the above

**24. The most distinctive characteristic of a primary chorionic villus is its:**

- A) cytotrophoblastic shell
- B) extraembryonic somatic mesodermal core
- C) bushy appearance
- D) cytotrophoblastic core

**25. Chorionic villi are designated as secondary chorionic villi when they:**

- A) contact the decidua basalis
- B) are covered by syncytiotrophoblast
- C) develop a mesenchymal core
- D) give rise to branch villi
- E) none of the above



26. When chorionic villi become vascularized they are called \_\_\_\_\_ villi.

- A) branch
- B) stem
- C) tertiary
- D) anchoring
- E) mature

27. The most important region of the decidua for the nourishment of the conceptus is the decidua

- A) frondosum
- B) capsularis
- C) parietalis
- D) basalis
- E) laeve

28. The intervillous space contains all of the following substances EXCEPT:

- A) oxygen
- B) carbon dioxide
- C) maternal blood cells
- D) fetal blood
- E) electrolytes

29. The portion of the decidua which does not survive until the end of pregnancy is the:

- A) capsularis
- B) basalis
- C) laeve
- D) parietalis
- E) frondosum

30. Which of the following is NOT a component of the mature placental barrier?

- A) the endothelial lining of fetal capillaries
- B) the cytotrophoblast
- C) the syncytiotrophoblast
- D) the basement membrane of fetal capillaries
- E) all of the above are part of the mature placental barrier

31. The notochord is replaced by the:

- A) ependyma
- B) nucleus pulposus
- C) spinal canal
- D) dorsal roots
- E) spinal cord

32. Implantation of human embryos typically occurs:

- A) about 1 day after fertilization
- B) about one week (i.e. 7 days) after fertilization
- C) about two weeks (i.e. 14 days) after fertilization
- D) after gastrulation
- E) at the same time as neural tube closure



F) NONE of the above

**33. During the development of the placenta:**

A) cytotrophoblast cells divide and their progeny fuse to form syncytiotrophoblast cells.

B) syncytiotrophoblast cells invade the uterine wall.

C) fetal blood vessels attach directly to maternal vessels such that there is mixing of fetal and maternal blood.

D) the placenta develops from the chorionic plate.

E) most of the placenta is derived from embryonic intermediate mesoderm.

F) NONE of the above

**Answer Key**

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-D  | 3-C  | 4-E  | 5-D  | 6-E  | 7-D  | 8-D  | 9-D  | 10-E |
| 11-C | 12-A | 13-B | 14-C | 15-C | 16-C | 17-E | 18-D | 19-D | 20-B |
| 21-E | 22-B | 23-B | 24-D | 25-C | 26-C | 27-D | 28-D | 29-A | 30-B |
| 31-B | 32-B | 33-A |      |      |      |      |      |      |      |

Questions 5-7 refer to the image below. For each of these questions, choose the SINGLE BEST answer.

**5. The label indicating the amniotic cavity is:**

A) A

B) B

C) C

D) D

**6. The label indicating a space lined by endoderm is:**

A) A

B) B

C) C

D) D

**7. The label indicating a space containing fluid that is produced by the kidneys in the developing fetus is**

A) A

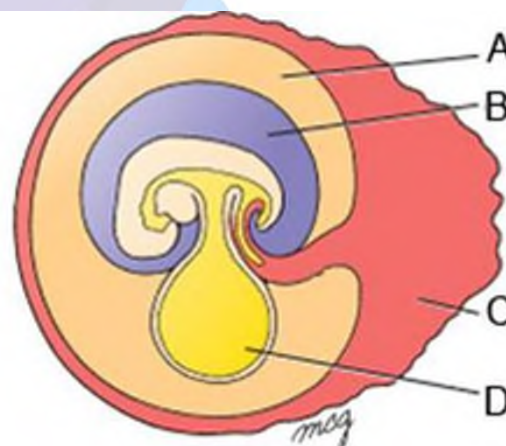
B) B

C) C

D) D

**Answer Key**

|     |     |     |  |  |  |  |  |  |  |
|-----|-----|-----|--|--|--|--|--|--|--|
| 5-B | 6-D | 7-B |  |  |  |  |  |  |  |
|-----|-----|-----|--|--|--|--|--|--|--|





## Essay questions

1. What is gastrulation?
2. How does the embryo turn into the neurula from gastrula?
3. How is the neural tube formed?
4. What is the embryonic origin of the nervous system in human embryo?
5. What is the notochord? How is this structure formed?
6. What are somites?
7. Define Fertilization
8. WHAT ARE THE STEPS OF FERTILIZATION
9. What are the results of fertilization
10. Discuss Cleavage of fertilized ovum
11. Discuss Implantation
12. Discuss Blastocyst formation
13. Define the Bilaminar embryonic disc
14. Discuss Differentiation of the intraembryonic mesoderm
15. Give definition to Capacitation
16. Describe maturation of spermatozoa
17. Give definition to Compaction in early embryo
18. Give definition OF gametogenesis
19. Steps of fertilization
20. Describe in detail the process of embedding of fertilized ovum
21. Ectopic pregnancy
22. Discuss the menstrual cycle and its hormonal control
23. Discuss placental function
24. Describe human placental structure
25. Anomalies of placenta
26. Placental barrier
27. Describe the development umbilical cord in mammals
28. Describe the developmental events taking place during the third week of gestation.
29. Derivatives of Neural crest cells

Doctors



## Fighter - Previous Exams of Histology (question bank)

Source: "previous exams of Histology" section of the Fighter Anatomy & Histology 2019-2024 file | answers reviewed against the Fighter key

**1. Which of the following is not a membranous organelles**

- A) Lysosomes
- B) Peroxisome
- C) Mitochondria
- D) Ribosomes

**2. Which organelle is considered the powerhouse of the cell**

- A) RER
- B) SER
- C) Golgi apparatus
- D) Mitochondria

**3. What are folds on the inner mitochondria membrane called**

- A) Plasmalemma
- B) Glycocalyx
- C) Protoplasm
- D) Cristae

**4. The cytoplasmic organelle which contains DNA&RNA**

- A) RER
- B) SER
- C) Lysosomes
- D) Mitochondria

**5. Membranous organelles is prominent in the cells that synthesize steroids hormones**

- A) RER
- B) SER
- C) Mitochondria
- D) Golgi apparatus

**6. Two organelles share in the formation of lysosomes**

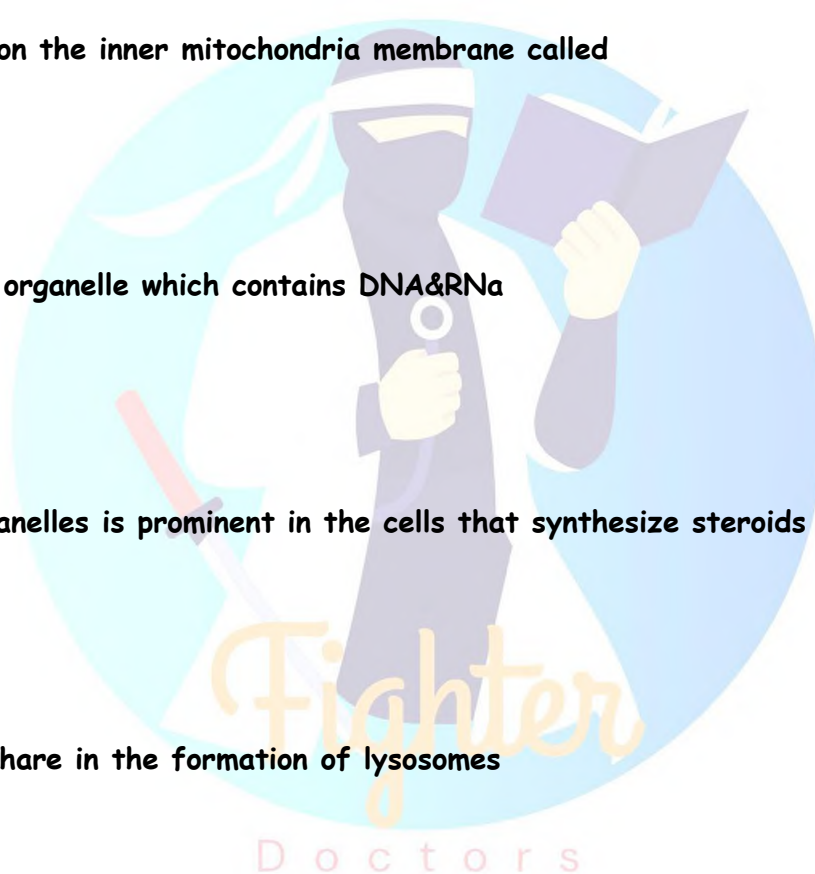
- A) RER&SER
- B) SER& Golgi
- C) RER& Golgi
- D) Golgi & cell membrane

**7. Which of these organelle is abundant in macro**

- A) Ribosomes
- B) RER
- C) Golgi apparatus
- D) Lysosomes

**8. When a primary lysosomes Fuses with pinocytic vesicles ,it became**

- A) Auto lysosomes
- B) Residual bodies
- C) Heterolysosomes
- D) Multivesicular bodies





**9. What process is nicknamed "cell drinking"**

- A) Endocytosis
- B) Exocytosis
- C) Pinocytosis
- D) Phagocytosis

**10. The wall of the centrioles is formed of..... Microtubules**

- A) 9
- B) 20
- C) 27
- D) 18

**11. The axoneme of the cilia has.... Microtubules**

- A) 9
- B) 18
- C) 20
- D) 27

**12. The flagella has ..... Microtubules**

- A) 9
- B) 18
- C) 20
- D) 27

**13. All of the below are exogenous pigments expect**

- A) Carotene
- B) Carbon
- C) Lipofuscin
- D) All of the above are exogenous pigments

**14. Fat can be demonstrated by using**

- A) Best's carmine
- B) PAS
- C) Sudan black stain
- D) None of the above

**15. Duplication of centrioles occurs at**

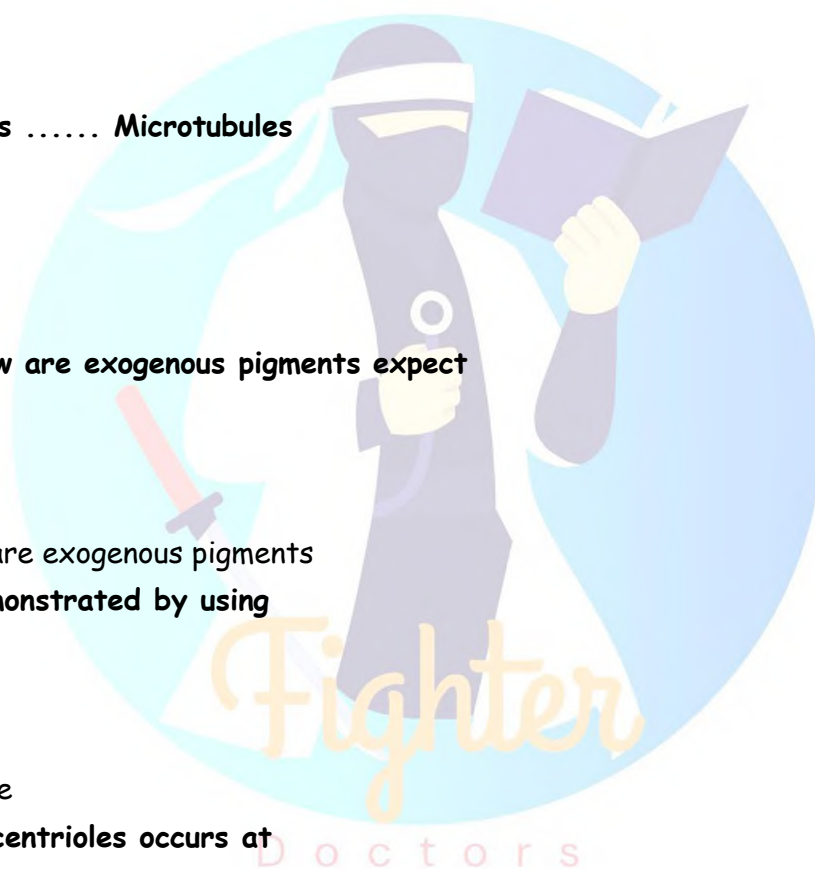
- A) G1 phase
- B) S phase
- C) G2 phase
- D) G0 phase

**16. Mitosis is characterized by the following**

- A) Exchange of genes
- B) Absence of crossing over
- C) Daughter cell contains n chromosome
- D) Daughter cell is not identical to mother cell

**17. Karyotyping of chromosomes are done in**

- A) Prophase
- B) Anaphase





C) Metaphase

D) Telophase

**18. Colchicine chromatids from separating during mitosis at**

A) Prophase

B) Anaphase

C) Metaphase

D) Telophase

**19. Tubulin is synthesized in**

A) G1 phase

B) G2 phase

C) G0 phase

D) S phase

**20. Philadelphia chromosome**

A) Duplication of chromosome

B) Terminal deletion of chromosome

C) Inversion affecting chromosome

D) Translocation between chromosomes

**21. What type of tissue lines the urinary bladder**

A) Simple squamous epi

B) Simple cuboidal epi

C) Simple columnar epi

D) Transitional epi

**22. What cell type makes up the epithelium of the gall bladder ?**

A) Simple squamous epi

B) Simple cuboidal epi

C) Simple columnar epi

D) Transitional epi

**23. What type of gland secretes its product through a duct or tube ?**

A) Endocrine gland

B) Multicellular gland

C) Exocrine gland

D) All of the above

**24. What is a gland called if it has an unbranched duct ?**

A) Simple gland

B) Compound gland

C) Tubular

D) Alveolar

**25. What is the surface modification seen on the cells of the epididymis**

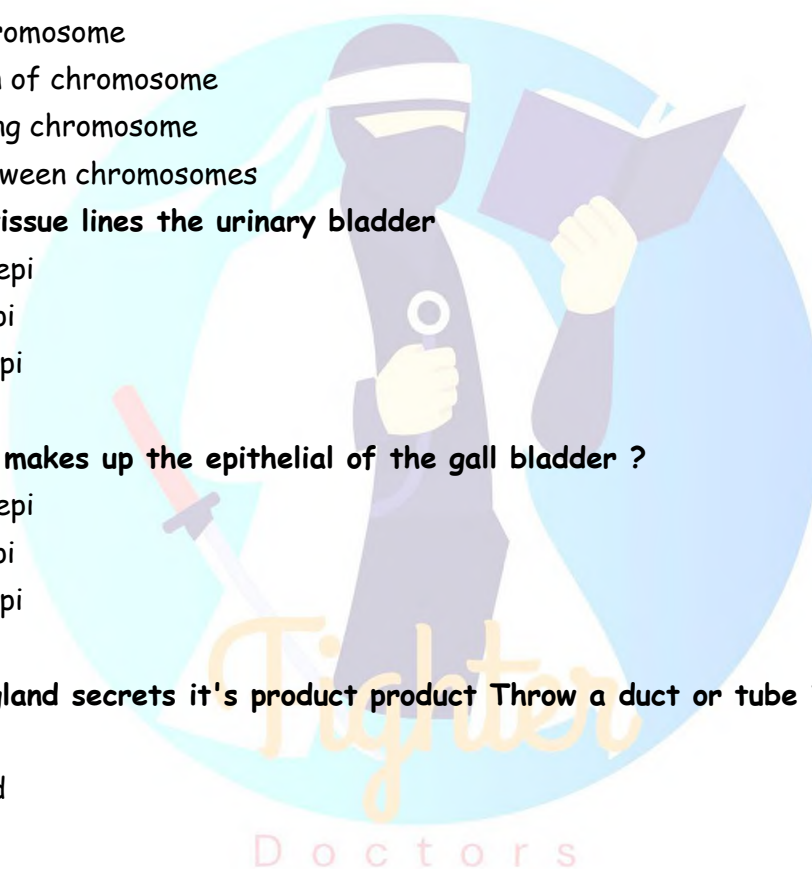
A) Microvilli

B) Stereocilia

C) Cilia

D) Both a & b

**26. What type of Epithelium forms epidermis**





- A) Simple cuboidal epi
- B) Simple columnar epi
- C) Stratified squamous epi
- D) Pseudostratified epi

**27. What do you call the simple squamous epi that lines abdominal cavity**

- A) Epitheloid tissue
- B) Mesothelium
- C) Endothelium
- D) Transitional

**28. What cell surface modification is made of microtubules ?**

- A) Microvilli
- B) Cilia
- C) Stereocilia
- D) Keratinization

**29. What type of tissue lines the esophagus**

- A) Simple squamous epi
- B) Simple columnar epi
- C) Stratified squamous epi
- D) Pseudostratified epi

**30. Which organ is not lined by transnational epi**

- A) Renal pelvis
- B) Ureter
- C) Uterus
- D) Urinary bladder

**31. What do you call the simple squamous epi that lines the serous cavities**

- A) Epitheloid tissue
- B) Mesothelium
- C) Endothelium
- D) Transitional

**32. What forms the brush border**

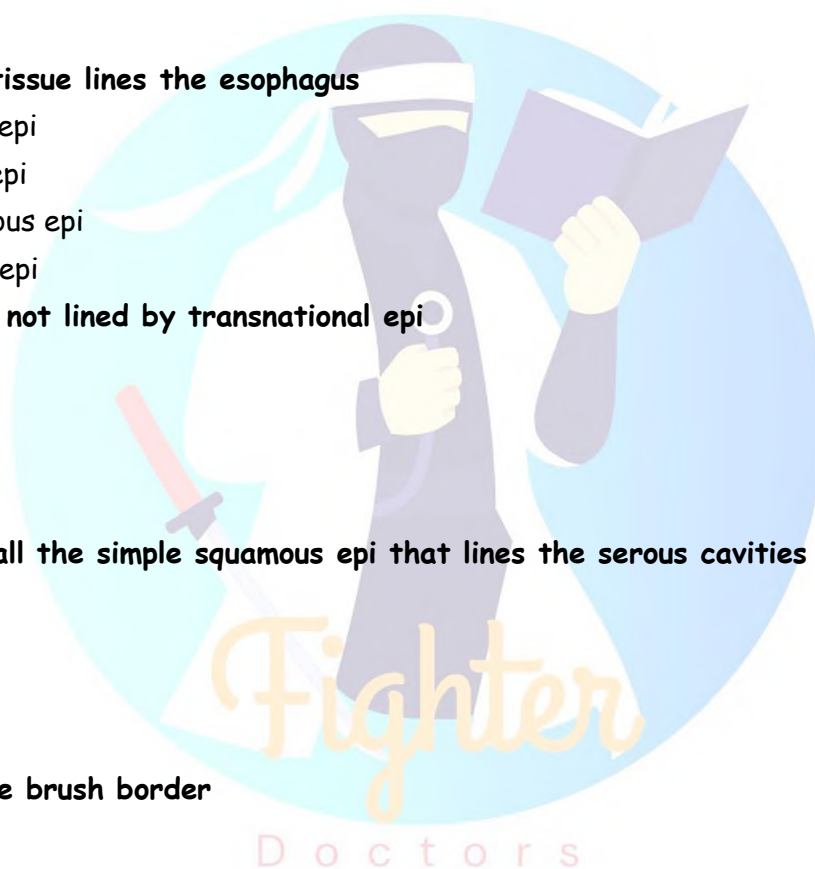
- A) Microvilli
- B) Stereocilia
- C) Cilia
- D) Both a and b

**33. What type of tissue lines most of GIT**

- A) Simple squamous epi
- B) Simple columnar epi
- C) Simple cuboidal epi
- D) Stratified squamous epi

**34. What are finger like projections of the surface of some cells called**

- A) Microvilli
- B) Stereocilia
- C) Cilia
- D) Both a and b





**35. Which one is the least common**

- A) Transitional
- B) Pseudostratified columnar ciliated
- C) Simple columnar
- D) Stratified cuboidal

**36. Regarding simple columnar epi which is least descriptive**

- A) May be keratinized
- B) May exhibit a microvillous border
- C) The height of each cell is typically greater than its width
- D) May have motile cilia
- E) May contain goblet cells

**37. Glands are classified based on**

- A) Their secretory product
- B) Mode of secretion
- C) Branching pattern of the ducts
- D) B and c are correct
- E) A,b and c are correct

**38. What type of tissue lines large intestine**

- A) Simple squamous epi
- B) Simple columnar epi
- C) Simple cuboidal epi
- D) Stratified squamous epi

**39. What type of tissue lines main pancreatic duct**

- A) Simple squamous epi
- B) Simple cuboidal epi
- C) Simple columnar epi
- D) Stratified squamous epi

**40. Which of the following is not a fiber found in C.T?**

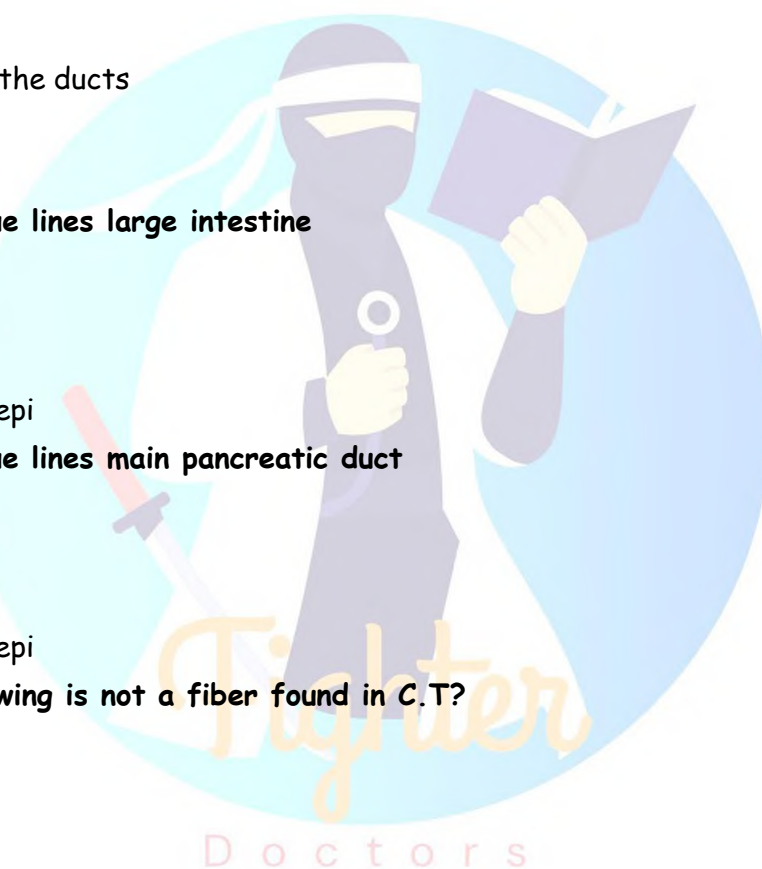
- A) Collagen fiber
- B) Elastic fiber
- C) Reticular fiber
- D) All of the above are fibers found in CT

**41. Which cell is a C.T macrophage**

- A) Kupffer cell
- B) Histiocyte
- C) Dust cell
- D) Microglia

**42. Which of the following can be classified as "specialized CT"**

- A) Mesenchyme
- B) Mucous CT
- C) Dense CT
- D) Blood





**43. Which of the following would be best suited to differentiate collagen fibers from other fibers ?**

- A) Hematoxylin and eosin stain
- B) Sudan stain
- C) Silver impregnation
- D) Masson's trichrome stain

**44. What type of tissue is wharton's jelly**

- A) Mucous CT
- B) Mesenchyme
- C) Loose irregular CT
- D) Dense irregular CT

**45. Which of the following would be best suited to visualize lipid**

- A) Hematoxylin and eosin stain
- B) Sudan stain
- C) Silver stain
- D) Masson's trichrome stain

**46. Which of the following would be best suited to visualize reticular fibers**

- A) Hematoxylin and eosin stain
- B) Sudan stain
- C) Silver impregnation
- D) Masson's trichrome stain

**47. Concerning plasma cell**

- A) Originates from T-lymphocytes
- B) Clock-face nucleus
- C) Small cell
- D) Few mitochondria

**48. Collagen fibers are formed by**

- A) U.M.C
- B) Mast cell
- C) Histiocyte
- D) Chondroblasts

**49. Elastic fibers are**

- A) Single ,thin
- B) Non-branching
- C) Basophilic
- D) Yellow with orcein stain

**50. The reticular CT**

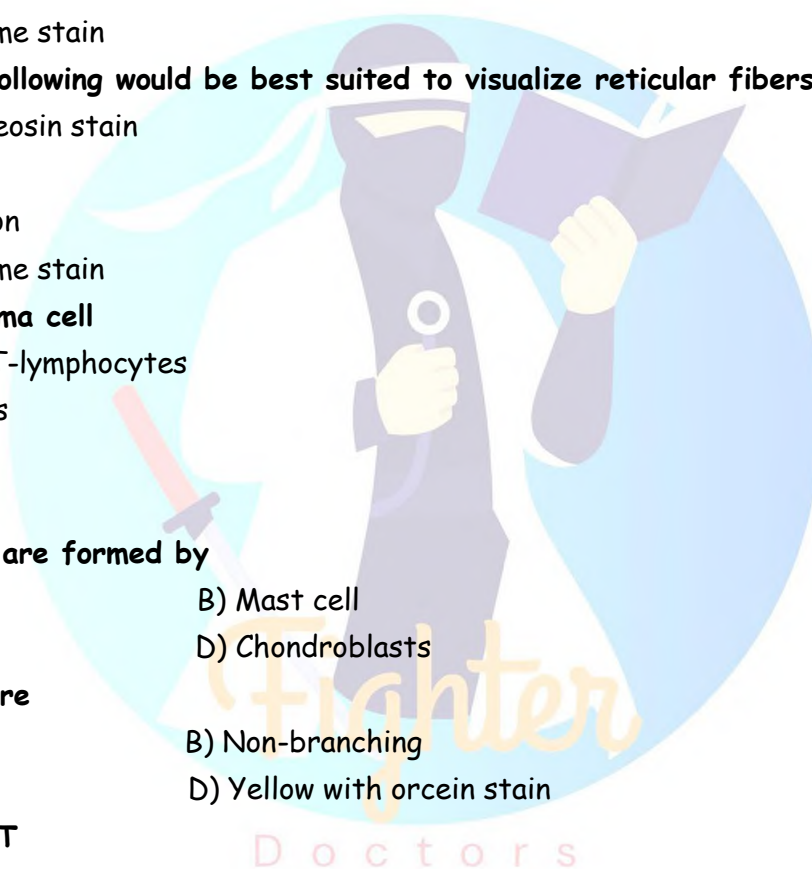
- A) Contains all types of CT cells
- B) Reticular fibers from branching bundles
- C) Stained with silver to demonstrate the reticular fibers
- D) Binds structures together

**51. Regular type of white fibrous CT is present in**

- A) The cornea under the epithelium
- B) Dermis of the skin
- C) Ligaments
- D) Sclera of the eye

**52. The yellow elastic CT**

- A) A loose type of CT
- B) Contain all types of CT cells
- C) Present in the wall of the aorta
- D) Present in the eustachian tube





**53. What does CT develop from**

- A) Mesothelium
- B) Mesoderm
- C) Mesentry
- D) All of them

**54. Concerning fibroblasts, all are true except**

- A) Most Popular type in CT proper
- B) Flat branched
- C) Few organelles
- D) Many RER

**55. In case of injury, pericytes can differentiate into all the following except**

- A) Smooth muscle fibers
- B) Fibroblasts
- C) Histiocytes
- D) Endothelial cells

**56. Which of the following is not associated with CT**

- A) Tightly packed cells
- B) Extracellular fibers
- C) Tissue fluid
- D) Ground substance

**57. Concerning reticular fibers all are true except**

- A) Form the stroma
- B) Synthesized by fibroblasts
- C) Brown with silver
- D) Yellow with PAS

**58. Which of the following would be best to differentiate Collagen fibers from other fibers**

- A) H & E
- B) Sudan stain
- C) Silver
- D) Mason's trichrome stain

**59. Which one of these cells is not found in loose CT**

- A) Fibroblasts
- B) Microglia
- C) Histiocytes
- D) Plasma cells

**60. What type of adipose tissue tends to increase as humans age**

- A) Brown adipose tissue
- B) White adipose tissue
- C) Multilocular adipose tissue
- D) Both a and c

**61. Concerning Collagen fibers**

- A) Wavy bundles formed of non-branching fibers
- B) Basophilic
- C) One type
- D) Yellow in colour when fresh

**62. Which of the following is not primarily composed of CT**

- A) Blood
- B) Bone
- C) Tendon
- D) Intervertebral disc
- E) Myometrium



63. Which of the following is not primarily composed of CT
- A) Bone  
B) Articular cartilage  
C) Heart  
D) Mesenchyme  
E) Fat
64. Which of the following can be classified as "specialized CT"
- A) Cartilage  
B) Loose CT  
C) Mesenchyme  
D) Dense CT  
E) Mucous CT
65. Which of the following can be classified as connective tissue proper
- A) Dense irregular CT  
B) Bone  
C) Blood  
D) Cartilage
66. Which of the following is a component of the ground substance
- A) Hyaluronic acid  
B) Proteoglycan  
C) Glycosaminoglycans  
D) Chondroitin sulfate  
E) All of the above
67. Which of the following can be classified as white fibrous CT
- A) Bone  
B) Blood  
C) Adipose tissue  
D) Irregular CT  
E) Cartilage
68. What type of tissue is a ligament composed of
- A) Mucous CT  
B) Mesenchyme  
C) Loose irregular CT  
D) Dense irregular CT  
E) Dense regular CT
69. Which CT cell type secreted heparin
- A) Fibroblast  
B) Myofibroblast  
C) Histiocytes  
D) Plasma cells  
E) Mast cells
70. Which one of the following cells would be best for the study of lysosomes
- A) Fibroblast  
B) Macrophages  
C) Mast cell  
D) Mesenchymal cells

## Answer Key

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1-D  | 2-D  | 3-D  | 4-D  | 5-B  | 6-C  | 7-D  | 8-D  | 9-C  | 10-C |
| 11-C | 12-C | 13-C | 14-C | 15-B | 16-B | 17-C | 18-C | 19-B | 20-D |
| 21-D | 22-C | 23-C | 24-A | 25-B | 26-C | 27-B | 28-B | 29-C | 30-C |
| 31-B | 32-A | 33-B | 34-A | 35-D | 36-A | 37-E | 38-B | 39-C | 40-D |
| 41-B | 42-D | 43-D | 44-A | 45-B | 46-C | 47-B | 48-D | 49-A | 50-C |
| 51-A | 52-C | 53-B | 54-C | 55-C | 56-A | 57-D | 58-D | 59-B | 60-B |
| 61-A | 62-E | 63-C | 64-A | 65-A | 66-E | 67-D | 68-E | 69-E | 70-B |